



**Web Service Interface –  
Northbound Open RESTful  
Integration Service (NORIS)  
API Version 7.0**

**Configuration**

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# About This Document

## Purpose

This document describes the interface between the outside world and the Management System through a Web Service Interface (Web API). It describes how the interface is to be accessed and what data in what format are exchanged.

## Scope

The document provides a description of the public API which can be used to access data and the functionality provided by the system platform. This document applies to the system version 7.0.

## Target Audience

- Testers of the public API
- Developers/Clients accessing the public API

## Cyber security disclaimer

Siemens products and solutions provide security functions to ensure the secure operation of building comfort, fire safety, security management and physical security systems. The security functions on these products and solutions are important components of a comprehensive security concept.

It is, however, necessary to implement and maintain a comprehensive, state-of-the-art security concept that is customized to individual security needs. Such a security concept may result in additional site-specific preventive action to ensure that the building comfort, fire safety, security management or physical security system for your site are operated in a secure manner. These measures may include, but are not limited to, separating networks, physically protecting system components, user awareness programs, defense in depth, etc.

For additional information on building technology security and our offerings, contact your Siemens sales or project department. We strongly recommend customers to follow our security advisories, which provide information on the latest security threats, patches and other mitigation measures.

<http://www.siemens.com/cert/en/cert-security-advisories.htm>

## Document Conventions

The following table lists conventions to help you use this document in a quick and efficient manner.

| Convention                                                                                                                                                                                                                                                                                                          | Examples                                                                                                                                                                                                                                                                                                                                                        |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Numbered Lists (1, 2, 3...) indicate a procedure with sequential steps.                                                                                                                                                                                                                                             | <ol style="list-style-type: none"> <li>1. Turn OFF power to the field panel.</li> <li>2. Turn ON power to the field panel.</li> <li>3. Open the panel.</li> </ol>                                                                                                                                                                                               |
| One-step procedures are indicated by a bullet point.                                                                                                                                                                                                                                                                | <ul style="list-style-type: none"> <li>● Expand the <b>Event List</b>.</li> </ul>                                                                                                                                                                                                                                                                               |
| <p>Conditions that you must complete or must be met before beginning a procedure are designated with a ►.</p> <p>Intermediate results (what will happen following the execution of a procedure step), are designated with an indented ⇨.</p> <p>Results, after completing a procedure, are designated with a ⇨.</p> | <p>► The report you want to print is open.</p> <ol style="list-style-type: none"> <li>1. Click <b>Print</b> .</li> <li>⇨ The <b>Print</b> dialog box displays.</li> <li>2. Select the printer and click <b>Print</b>.</li> <li>⇨ The print confirmation displays.</li> </ol> |
| <b>Bold</b> font indicates something you should type or select, or when a dialog box or window is specified.                                                                                                                                                                                                        | <p>Type <b>F</b> for field panels.</p> <p>Click <b>OK</b> to save changes and close the dialog box.</p> <p>The <b>Create a New Project</b> dialog box displays.</p>                                                                                                                                                                                             |
| Menu paths in procedures are indicated in <b>bold</b> .                                                                                                                                                                                                                                                             | Select <b>File &gt; Text, Copy &gt; Group</b> , which means from the <b>File</b> menu, select <b>Text, Copy</b> and then <b>Group</b> .                                                                                                                                                                                                                         |
| File paths containing placeholders display the placeholders in <i>italics</i> enclosed in square brackets.                                                                                                                                                                                                          | <i>[installation drive:]\[installation folder]\[project]\...</i>                                                                                                                                                                                                                                                                                                |
| Error and system messages are displayed in Courier New font.                                                                                                                                                                                                                                                        | The message <code>Report Definition successfully renamed</code> displays in the status bar.                                                                                                                                                                                                                                                                     |
| <i>Italics</i> are used to emphasize new or important terms.                                                                                                                                                                                                                                                        | The reaction processor continuously executes a user-defined set of instructions called the <i>control program</i> .                                                                                                                                                                                                                                             |
|                                                                                                                                                                                                                                  | This symbol signifies a Note. Notes provide additional information or helpful hints.                                                                                                                                                                                                                                                                            |
| Cross references to other information in printed material are indicated with an arrow and the page number, enclosed in brackets: [→ 92]                                                                                                                                                                             | For more information on creating flowcharts, see Flowcharts [→ 92].                                                                                                                                                                                                                                                                                             |

## Getting Help

For more information about our products, contact your local Siemens representative.

## Safety Messages According ANSI Z535.6

The following examples show the ANSI standard safety messages used in this document to draw the reader's attention to important information.

ANSI distinguishes between *personal injury* safety messages and *property damage* warning messages.

The personal injury safety messages have safety alert symbols and the following alert level labels: DANGER!, WARNING!, CAUTION!

The label for property damage messages is: NOTICE.

### Examples:

|                                                                                     |                                                                                                                                                    |
|-------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|
|    | <b>NOTICE</b>                                                                                                                                      |
|                                                                                     | <b>Property Damage Warning Message</b><br>Equipment damage or loss of data may occur if you do not follow a procedure or instruction as specified. |
|    | <b>CAUTION</b>                                                                                                                                     |
|                                                                                     | <b>Caution Safety Message</b><br>Minor or moderate injury may occur if you do not follow a procedure or instruction as specified.                  |
|  | <b>WARNING</b>                                                                                                                                     |
|                                                                                     | <b>Warning Safety Message</b><br>Personal injury or property damage may occur if you do not follow a procedure as specified.                       |
|  | <b>DANGER</b>                                                                                                                                      |
|                                                                                     | <b>Danger Safety Message</b><br>Electric shock, death, or severe property damage may occur if you do not perform a procedure as specified.         |



# Document Revision History

## Document Identification

The document ID is structured as follows:

ID\_Language(COUNTRY)\_ModificationIndex\_ProductVersionIndex

Example: A6Vnnnnnnnn\_en\_a\_02

| Document Revision History |              |                     |
|---------------------------|--------------|---------------------|
| Modification Index        | Edition Date | Brief Description   |
| a                         | 2019-04-11   | Release Edition 4.0 |
| b                         | 2019-07-09   | Edition 4.1         |
| c                         | 2019-07-30   | Edition 4.2         |
| d                         | 2020-09-29   | Edition 5.0         |
| e                         | 2021-06-30   | Edition 5.1         |
| f                         | 2022-04-30   | Edition 6.0         |
| g                         | 2023-04-21   | Edition 7.0         |

# 1 Introduction

## 1.1 References

### 1.1.1 Links

| Ref. | Abbreviation  | Link Description                                                                                                                            |
|------|---------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| [1]  | HAL           | <a href="http://stateless.co/hal_specification.html">http://stateless.co/hal_specification.html</a>                                         |
| [2]  | SignalR       | <a href="https://github.com/SignalR/SignalR/wiki">https://github.com/SignalR/SignalR/wiki</a>                                               |
| [3]  | Base64url     | <a href="http://www.ietf.org/rfc/rfc4648.txt">http://www.ietf.org/rfc/rfc4648.txt</a>                                                       |
| [4]  | OAuth         | <a href="https://tools.ietf.org/html/rfc6749">https://tools.ietf.org/html/rfc6749</a>                                                       |
| [5]  | Langauge-Tags | <a href="http://tools.ietf.org/html/rfc5646">http://tools.ietf.org/html/rfc5646</a>                                                         |
| [6]  | JSON:         | <a href="http://www.ietf.org/rfc/rfc4627.txt">http://www.ietf.org/rfc/rfc4627.txt</a>                                                       |
| [7]  | REST          | <a href="http://en.wikipedia.org/wiki/Representational_state_transfer">http://en.wikipedia.org/wiki/Representational_state_transfer</a>     |
| [8]  | URI           | <a href="https://tools.ietf.org/html/rfc3986">https://tools.ietf.org/html/rfc3986</a>                                                       |
| [9]  | EBNF          | <a href="http://en.wikipedia.org/wiki/Extended_Backus%E2%80%93Naur_Form">http://en.wikipedia.org/wiki/Extended_Backus%E2%80%93Naur_Form</a> |
| [10] | Media-Types   | <a href="http://www.iana.org/assignments/media-types/media-types.xhtml">http://www.iana.org/assignments/media-types/media-types.xhtml</a>   |
| [11] | Swagger       | <a href="http://swagger.io">http://swagger.io</a>                                                                                           |
| [12] | Mutual TLS    | <a href="https://datatracker.ietf.org/doc/html/rfc8705">https://datatracker.ietf.org/doc/html/rfc8705</a>                                   |

### 1.1.2 Documents

| Ref. | Name                                 | Version |
|------|--------------------------------------|---------|
| [20] | Regional Product Customization Guide |         |

### 1.1.3 Online Documentation

Web Service Interface also provides an interactive online documentation (web page). The documentation is based on Swagger[16] and is a runtime representation of the API. It requires a running Web Service Interface and can be accessed in two ways using the following URL:

i) Intranet :

[http\(s\)://\[hostname\]:\[port\]/swagger/ui/index](http(s)://[hostname]:[port]/swagger/ui/index)

*Example*

<https://myWebServiceHost:8443/swagger/ui/index>

ii) Internet (i.e via IIS)

Once Web Service Application is created in SMC, append `/swagger/ui/index` to WSI application URL shown.(in SMC)

*Example*

[https://\[hostname\]:\[websitePort\]/ApplicationName/swagger/ui/index](https://[hostname]:[websitePort]/ApplicationName/swagger/ui/index)



**NOTE:**

If you are using Internet Explorer make sure Compatibility View is turned off.

### 1.1.3.1 Exposure of Swagger via IIS is configurable

An advanced user can enable/disable the exposure of Swagger via IIS. To enable the swagger via IIS, user has to set the *enabled* flag to *true* for SwaggerRoutingRule under WSI web application web.config file. By default swagger via IIS is disabled. As shown below :

```
<rule name="SwaggerRoutingRule" enabled="false" stopProcessing="true">
  <match url="(/|^)(swagger/)(.*)" />
  <action type="Rewrite" url="http://localhost:8095/{R:2}{R:3}" />
</rule>
```

Path for WSI Web Application web.config :

Example :

<Drive:>\GMSProjects\WebSites\GmsWebsite\WsiWebApplicationName\

## 1.2 Definitions, Acronyms and Abbreviations, Conventions

### 1.2.1 Glossary

| Terms Used in Web Service Interface |                                                                                                                                                                                                                                                                                       |
|-------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Term                                | Description                                                                                                                                                                                                                                                                           |
| HAL                                 | Simple format that gives a consistent and easy way to hyperlink between resources in an API.                                                                                                                                                                                          |
| HTTP                                | Hypertext transfer protocol:<br><a href="http://en.wikipedia.org/wiki/Hypertext_Transfer_Protocol">http://en.wikipedia.org/wiki/Hypertext_Transfer_Protocol</a>                                                                                                                       |
| JSON                                | JavaScript Object Notation:<br><a href="http://en.wikipedia.org/wiki/JSON">http://en.wikipedia.org/wiki/JSON</a>                                                                                                                                                                      |
| OAuth 2.0                           | Open standard to authorization. OAuth provides client applications a 'secure delegated access' to server resources on behalf of a resource owner.                                                                                                                                     |
| Push Notification                   | In this document push notifications refer to real-time notifications (see real-time push below).<br>Push notifications to mobile devices (see wake-up push below) are not covered in this document. However, a dedicated extension module is available to provide this functionality. |
| REST                                | Representational state transfer:<br><a href="http://en.wikipedia.org/wiki/Representational_state_transfer">http://en.wikipedia.org/wiki/Representational_state_transfer</a>                                                                                                           |
| Real-time push                      | Push notification from a server (our Web Service in this case) to a client; used for notification of changes of values as long as client application is running. Notifies the client about events in the system, including COV, as long as the client application is running.         |
| Reverse Proxy                       | Retrieves resources on behalf of a client from a server. These resources are then returned to the client as though they originated from the server itself.                                                                                                                            |
| SignalR                             | Library for ASP.NET. SignalR allows bi-directional communication between server and client.                                                                                                                                                                                           |
| Wake-up push                        | Push notification to a mobile phone; notifies a user about an event in the system; takes advantage of third-party notification providers (Apple, Google, ...).                                                                                                                        |

### 1.2.2 Conventions in this Document

The product is developed for different regions and known under the following brands:

- Desigo CC
- Desigo DMS
- Cerberus DMS

In this document, we therefore use the neutral term *System* for each of these brands.

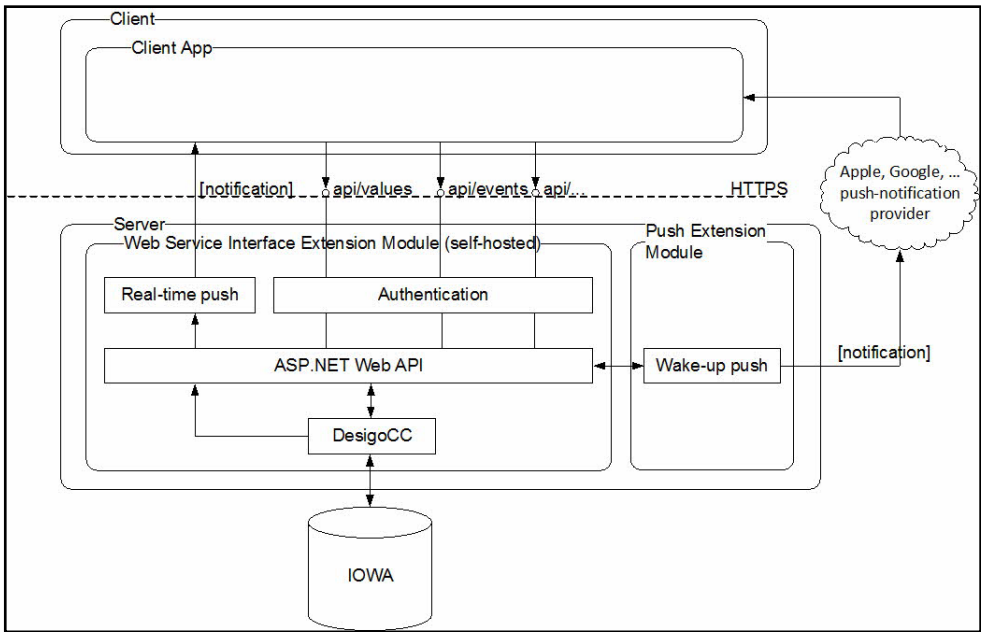
## 1.3 Open Issues in this Document Version

At the effective date of the current document version, the following open issues are identified:

- none

# 2 System Architecture

The following diagram shows the architecture for the Web Service Interface API.



## 2.1 System Limits

The following limits apply when using the Web Service Interface:

- Number of concurrent sessions: 100
- Delay for notifications: <1 second
- 5'000'000 value updates / day
- 20'000 value subscriptions
- 250'000 event notifications / day
- 250'000 event-counter notifications / day
- 1'000'000 trends / min

|          |                                                                                                                                                                                                                                                                                                                                                                                         |
|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>!</p> | <p><b>NOTICE</b></p> <p><b>IIS limits imposed by Operating System version</b></p> <p>The Client versions of Windows (e.g. Windows 7, Windows 8) have intentionally feature-limited versions of IIS. One limitation is the number of concurrent connections which can be as low as 10.</p> <p>This especially affects the usage of push notifications with SignalR (see [2] [→ 16]).</p> |
|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

## 3 Basic System Concepts

### 3.1 Glossary of the System

**Aggregator**

A parent object summarizing child objects, such as data points, displayed in the System Browser.

**Alarm**

See Event.

**Attribute**

The **Main** attributes belong to all existing objects visible in the System Browser. The **Details** attributes belong to each of the properties within an object defined in an Object Model. There are two kinds of attributes:

- Simple attribute, for example, unit or text group.
- Complex attribute, for example, commands or alarms.

**COV**

Change of Value. An updated value that represents the extent of change in a measured reading when change criteria have been met.

**Data point**

An input/output function consisting of all assigned information describing fully the data point's meaning.

**Discipline**

An area of building control, such as HVAC, lighting, blind control, fire or video. Building control disciplines can also contain subdisciplines.

**Event**

A change of state or value detected for processing and/or reporting.

**Function**

Is the semantic description for one or more data point types, for example, fire detector.

**Object**

A specific instance of an object type. While an object type is identified by a unique object type property, an object is identified by its object identifier property.

Project data points imported in the system database. Instances are customized with attributes, such as discipline or text group, during data import.

**Object Model**

A data point type and its attributes specific to a subsystem and protocol. Native presentation of subsystem objects in the system, for example, analog input.

**Property**

Each object has a different number of properties, depending on the Object Model or the defined Function. Properties show online values (COV or polling) and can have an operation function assigned to them, for example, Acknowledge.

Properties collect data from the field system and need more system resources than a static attribute in the database.

**Propagation**

A mechanism for relaying Common Status information up the building control system's hierarchical tree. Propagation is based on the parent-child-grandchild model. Each object in the system can have one or more parent objects and one or more child objects. Each child in the system can simultaneously propagate multiple active Common Status properties to a parent—for example, fire alarm, security fault, or technical excluded.

### System Manager

A system application that allows users to select and act on objects for monitoring and controlling a facility. The System Manager contains the following panes: System Browser (selection of data points), Primary, Secondary, and Contextual.

### System Browser

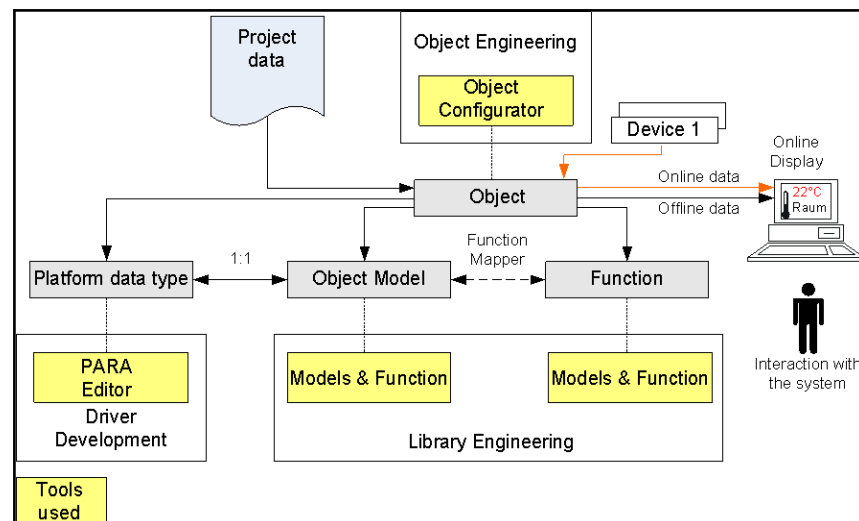
Shows the list of hierarchical objects, such as data points, in one or more views. Compiles objects into an aggregator object.

### View

In System Browser, there are two basic views: Management View and Application View. Additional project views can be defined as needed. The objects are displayed hierarchically, depending on the selected view. An object can be assigned to more than one view.

## 3.2 Basic Data Concept in the System

Instantiated objects for individual subsystems are always related to an Object Model and a Function. This concept ensures that the user experiences the same operating and monitoring philosophy for the same or similar Functions.



The system concept impacts the following main elements:

|                           |                                                                                                                                                                                                                                                                                                                                                                           |
|---------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Platform data type</b> | It forms the basis for the Object Model set up on the system. Platform tools are required in individual cases for driver development of additional subsystems.                                                                                                                                                                                                            |
| <b>Object Model</b>       | All system basic data types are designated Object Models. The corresponding default value is defined per subsystem for each Object Model.                                                                                                                                                                                                                                 |
| <b>Function</b>           | Function includes the semantic description (for example, fire detector or temperature sensor) for one or more specific data points.                                                                                                                                                                                                                                       |
| <b>Object</b>             | Imported project data can be modified specifically to the object using the Object Configurator.                                                                                                                                                                                                                                                                           |
| <b>Project data</b>       | <p>The project data contains the specific information on a data point or data point amount by the system and is responsible for system response. Collecting data using auto-discovery is possible if supported by the subsystem.</p> <p>The Function or Object Model supplements missing information on a project data point to ensure uniform functioning of system.</p> |

The tree structure ensures a consistent operating, commanding and alarming philosophy for the various integrated subsystems. The benefits of this concept include:

- Discipline-specific project data from fire, HVAC, intrusion, and so on can be configured using the same mechanism.
- Additional or new systems can easily be added.

### 3.3 Alarm Concepts

The system automatically controls and monitors building services plants. Events occur (such as a fault) that are important for the operator, requiring user intervention. The alarm system processes such events and reports them accordingly.

The system reacts automatically to a fault with a system reaction (for example, automatically locking a ventilation plant during a fire alarm).

For control deviations such as a dirty filter, a simple alarm is triggered and only an alarm message is sent.

The system switches to an alarm state for both cases and a corresponding alarm is issued. After trouble-shooting the cause of the alarm, the user resets the alarm state and the system returns to normal status.

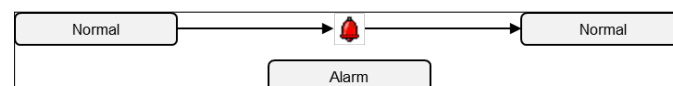
#### Alarm Types

- Field system alarms (alarm behavior configured in the field system)
- Management platform alarms
  - Discrete alarm type (for example, alarm range between 10 and 20 or between 30 and 40).
  - Continuous alarm type (for example, alarm range  $\leq 30$  or  $\geq 70$  ).

| Alarm Handling     |              |                                                                                      |
|--------------------|--------------|--------------------------------------------------------------------------------------|
| Alarm Type         | Name         | Alarm Behavior                                                                       |
| Alarm_NoAckNoReset | Simple       | No user action necessary                                                             |
| Alarm_NoReset      | Basic        | Acknowledge the alarm                                                                |
| Alarm              | Extended     | Acknowledge and reset the alarm                                                      |
| Alarm_Dynamic      | User defined | Acknowledge and reset the alarm<br>executing commands defined on<br>the alarm source |

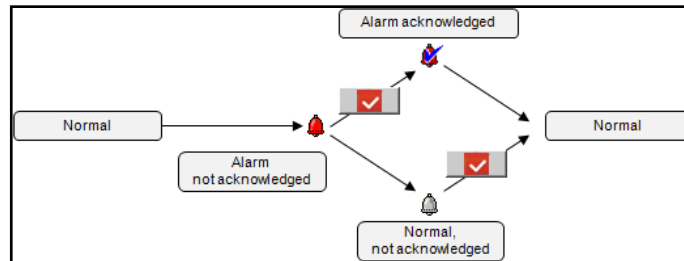
#### Simple Alarm

Simple alarms do not require you to acknowledge or reset, they simply return to the normal state when the monitored condition (for example, a dirty filter) returns to normal.



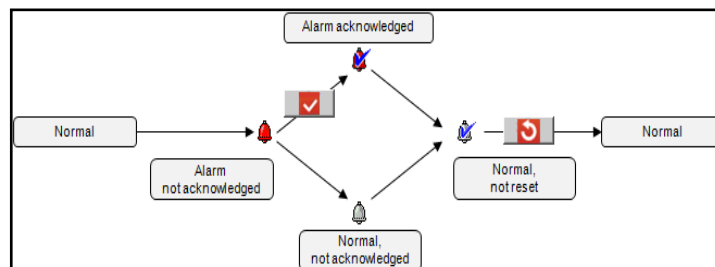
### Basic Alarm

More important alarms (for example, lack of water) must be acknowledged. After the alarm state is eliminated, the plant is once again released after acknowledgement.



### Extended Alarm

For an extended alarm (for example, velocity monitor), the reset returns the plant to normal. Provided the monitored state returns to the normal range. The idea is that important alarms are not overlooked when the alarm condition clears (Normal, not acknowledged situation). The plant must be reset to re-start operation.

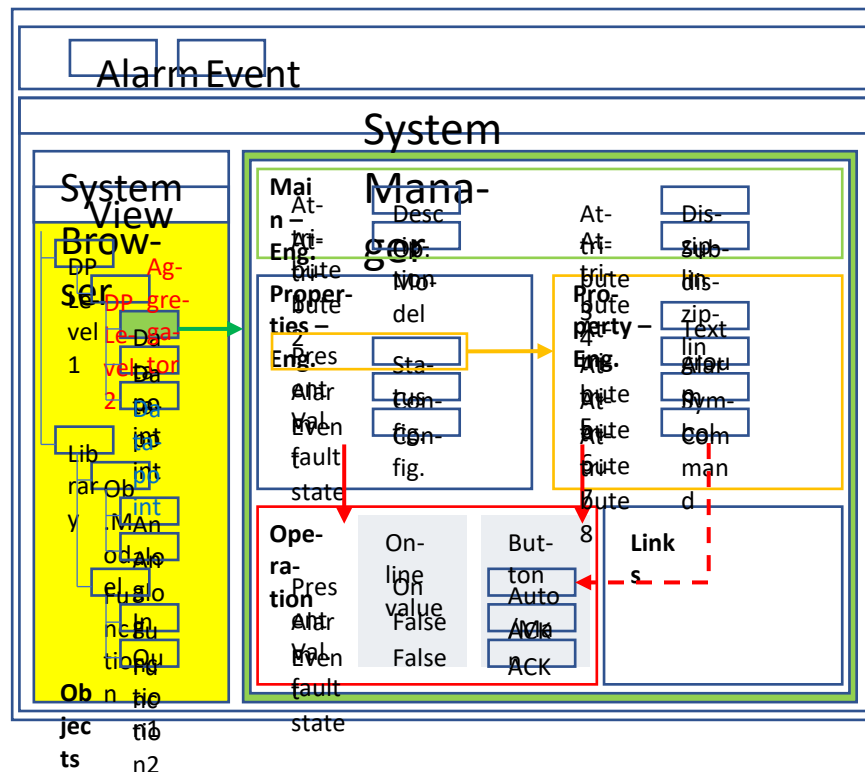


## 3.4 User interface Concepts

The user interface contains the information for:

- Displaying Alarms and Events in a list
- Object navigation in the System Browser tree
- Engineering dialogues to customize the objects and properties
- Operating the system

- Links to further information on an object



### Displaying Alarms and Events

Alarms or events are displayed in the Event Summary bar. From there, Alarms or events can – depending on their type - be acknowledged or reset.

### System Browser Navigation

If an object (green) is selected in the System Browser tree, all associated objects and properties (green frame) are displayed as part of the Engineering dialogue.

### Engineering Dialogue

Each object in System Browser has the same number of standard attributes, that are displayed in the **Main** section (light green frame). The number of properties can vary depending on the type of the object. When selecting a property, its attributes (blue frame) are displayed.

### Operating the System

Only properties that are essential to the operation of the system are displayed in the **Operation** tab (red frame). The value of a property is read from the system database or the field network (online value).

If a property has a command attribute, an interactive element is displayed (text field, list box or button). To change a value, you need to perform two steps:

- Set the value
- Send the value

## 3.5 Search Mechanism Behavior

The result of a search is different depend which commanding you are using:

- /properties/object.property
- /properties/object

### /properties/object.property

The result shows list all data points with the requested property information.



Example:  
XXX.YYY.Present\_Value  
XXX.YYY.Alarm.Fault  
XXX.YYY.Event\_State

|                   |                          |                                                                                                      |       |         |
|-------------------|--------------------------|------------------------------------------------------------------------------------------------------|-------|---------|
| readAllProperties | <input type="checkbox"/> | If true all the properties for the requested Object OR Property Id will be returned in the response. | query | boolean |
|-------------------|--------------------------|------------------------------------------------------------------------------------------------------|-------|---------|

**/properties/object**

The result shows list all data points with the default property information.

**NOTE:** The default property is defined for each Object Model as a main attribute.

Example:  
XXX.YYY.Present\_Value  
XXX.ZZZ.Fault  
YYY.ZZZ. Present\_Value

## 4 Using the Web Service Interface

### 4.1 Communication

The API can be accessed over HTTP or HTTPS. In a production environment it's highly recommended to use HTTPS because otherwise primary credentials are transmitted in plaintext.

#### 4.1.1 Authentication

For authentication we use OAuth 2.0 Resource Owner Password Credentials Grant [4] [→ 16]. In order to access a resource of the Web Service Interface the Client needs a valid access token and a valid session within the system. In case no token is presented or the token is invalid (e.g. expired) or the system session is not available the user will get a status code 401 (*Unauthorized*) and needs to request an access token. The access token can be requested through a dedicated resource (/token) by presenting the primary credentials. In case the credentials are provided, the system creates a new session and returns an access token.

1. Client tries to access a resource.
2. Server responses with status code 401 (*Unauthorized*).  
⇒ Client asks the end-user for credentials.
3. Client sends credentials to a dedicated resource.
4. Server returns an access token, the system creates an internal session.
5. Client again tries to access a resource and includes the access token in an authorization header.
6. Server returns content with status code 200 (*Succeeded*).

|   |                                                                                                                                                                                                                                                                                                         |
|---|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ! | <b>NOTICE</b>                                                                                                                                                                                                                                                                                           |
|   | <b>Session alive</b><br>The Client either needs to subscribe for notifications (e.g. change of values) or needs to access the API at least once every 10 minutes otherwise its system session expires. The API provides a dedicated resource (/heartbeat) just for the sake of keeping a session alive. |

#### 4.1.2 Client Certificate Authentication

Optionally, a client can authenticate to the server by providing a client certificate. The Web Service Interface expects the client certificate to be provided in a header called *X-ARR-ClientCert*. In case the interface is accessed through a reverse proxy (almost always the case in a production environment), the reverse proxy asks the client for an optional client certificate. In this case the reverse proxy passes the client certificate to the backend server as HTTP header with the default header configured as *X-ARR-ClientCert*.

#### 4.1.3 Subscriptions (Push Notifications with SignalR)

Optionally, for specified services the API provides not only data on request (pull) but also in case of an event determined by the respective service (push).

Any Client subscribing for any push notifications needs to support SignalR (see [2] [→ 16]). The Client first needs to connect to a dedicated SignalR hub and can then subscribe for notifications by providing a connection ID which it gets after a connection between Client and server is established. The Client then needs to implement a function (see signature in respective service chapters) which will be called from the server in case a notification is due.

Noris hosts signalr url at `/signalr` route.

By default Javascript & C# signalr client tries to connect to `/signalr` route if no route is specified. Please verify with signalr client package's documentation for setting up signalr url.

Signalr also generates hub proxy file for javascript clients, this can be accessed at `/signalr/hubs` route.

i) Intranet :

`http(s)://[hostname]:[port]/signalr`

`http(s)://[hostname]:[port]/signalr/hubs`

ii) Internet (i.e via IIS)

`https://[hostname]:[websitePort]/ApplicationName/signalr`

`https://[hostname]:[websitePort]/ApplicationName/signalr/hubs`

*Example*

```
//norisHub is name of our hub on the server side
var hub = $.connection.norisHub;
//register notifyEvents as a method on the client side for events
hub.client.notifyEvents = function (events) {
    //do something in case a notification arrives
}
//register notifySystems as a method on the client side for systems
hub.client.notifySystems = function (systems) {
    //do something in case a notification arrives
}
// start hub
$.connection.hub.start().done(function() {
    //do other stuff
});
```

#### 4.1.4 Latency Push Notification to Request-Response

In the workflow of subscription and notification it may happen that push notification from server is received before subscription success response. This latency will be experienced because of network communication delay of the messages on two independent channels of SignalR hub connection for push notification and response of subscription request.

In this kind of scenario Clients of WSI should consider received notification as a valid notification and design their solution accordingly.

### 4.1.5 SignalR Hubs

For signalR notification client has to setup hub connection with server.

Legacy implementation of Noris provides specific hub for almost all the subscription services respectively.

Noris also introduces singleHub solutions as NorisHub. Now client should create only one NorisHub connection and use it for all the services for subscription-notifications. It is the recommended singlanR connection approach.

Service specific list of Hubs are as below:

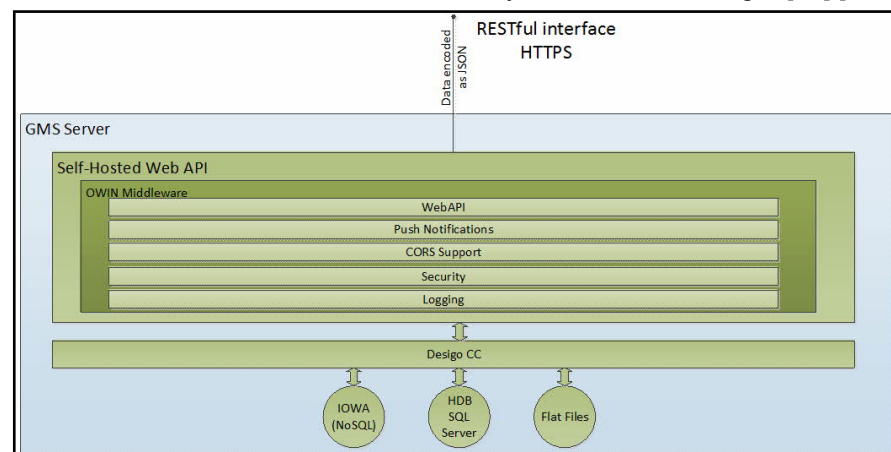
| Subscription Service             | Legacy Hub       | Recommended |
|----------------------------------|------------------|-------------|
| EventsSubscriptions              | eventsHub        | norisHub    |
| EventcountersSubscriptions       | eventCountersHub | norisHub    |
| SystemBrowserSubscriptions       | systemBrowserHub | norisHub    |
| ValuesSubscriptions              | valuesHub        | norisHub    |
| CommandsSubscriptions            | commandsHub      | norisHub    |
| EventCategorySoundsSubscriptions | eventSoundsHub   | norisHub    |
| SystemsSubscription              |                  | norisHub    |

## 4.2 Supported Client Environments

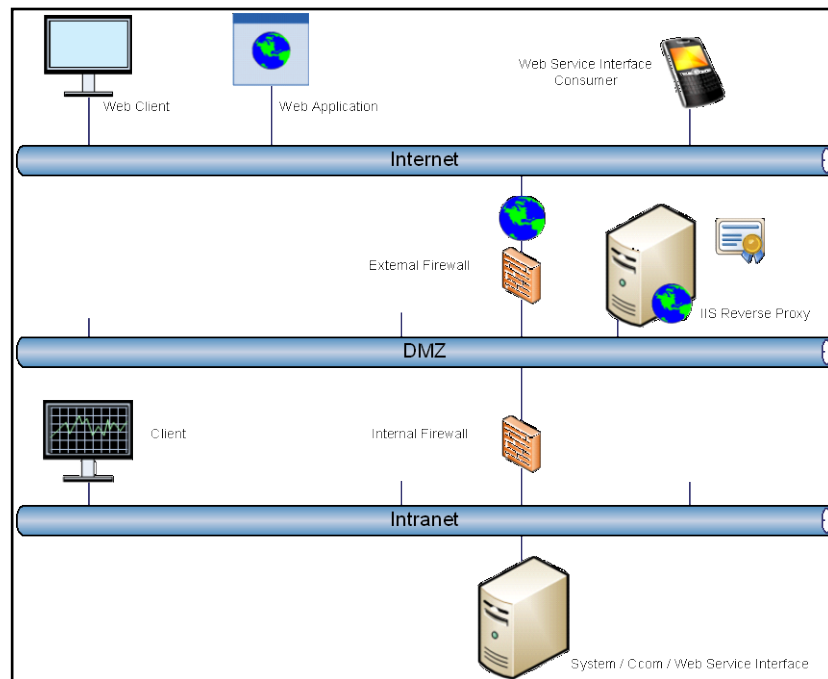
There are no Client dependencies except SignalR version 2.2.0 Client library (see [2] [→ 16] for available Client libraries) for Clients interested in push notifications.

## 4.3 Deployment

The Web Service Interface will be deployed as a self-hosted component. It's an executable and can be commanded as any other IOWA manager [12] [→ 16].



In a production environment the Web Service Interface (and System) will most likely only be accessible through a reverse proxy.



Network Topology

## 4.4 Configuring Web Service Interface in a Project



### NOTE:

A user has the same user rights as defined in the system. Therefore, you can check the user rights in advanced in the system bevroee using the Web Service Interface.

### 4.4.1 Stopping a Project

- ▷ At least one project is available under the **Projects** node and is started.
- 1. In the System Management Console tree, select the **Projects** node.
- 2. In the **Projects** tree, select the project that you want to stop.
- 3. Click the **Project Settings** tab, if not yet selected.
  - ⇒ **Project Settings** displays.
- 4. In the **Project Settings** toolbar, click **Stop** .
  - ⇒ A confirmation message displays.
- 5. Click **OK** to stop the started project.

### 4.4.2 Showing Settings

- ▷ The Extension module **Connectivity > Web Service Interface** is installed.
- ▷ At least one project is available under the **Projects** node and is started.
- 1. In the System Management Console tree, select the **Projects** node.
- 2. In the **Projects** tree, select the project.
- 3. Click the **WSI Settings** tab.
  - ⇒ **WSI Settings** displays and shows the used protocol type and port number.




---

**NOTE:**

If you want to change the settings, you must stop the project first.

---

### 4.4.3 Defining Protocol Type and Port

- ▷ The *Web Service Interface* extension module installed for the selected project.
- ▷ The project is stopped.
- ▷ IIS is installed according to the settings in the Installation Manual chapter 4.
- 1. Click **Edit** .
- 2. Click **Next** .
  - ⇒ The **Web Service Interface Settings** expander displays.
- 3. Select the protocol:
  - **Unsecured**
    - Under **Port**, enter the port number (Default: 8080).
  - **Secured**
    - Under **Port**, enter the port number (Default: 8443).
    - Under **Host Certificate**, click **Browse**.
    - Select the appropriate certificate and click **OK**.
- NOTE:** Additional information on certificates is available in the SMC Help.
- 4. Click **Save Project** .
  - ⇒ The protocol settings are saved.
- 5. Click **Start Project** .




---

**NOTE:**

HTTP is insecure and is vulnerable to man-in-the-middle and eavesdropping attacks, which can let attackers gain access to sensitive information.

Many organizations require certificates from third-party authorities instead of self-signed certificates to remain compliant with current regulations.

---

### 4.4.4 Creating a WSI on Server

- ▷ At least one Web site is created and available under the **Websites** root node in the System Management Console tree.
- 1. In the System Management Console tree, select the **Websites** node.
- 2. In the **Websites** tree, select the **Website**.
- 3. Click the **Management** tab.
  - ⇒ The **Details** expander displays.

4. In the **Website** toolbar, click **Web Service Interface**   
⇒ The **Web Application Details** expander displays.
5. In the **Web Application Details** expander, provide the Web application details as follows:
  - From the **Project Name** drop-down list, select a project that you want to link to the Web application.
  - In the **Name** field, type a unique name of the Web application.
  - Browse for the **Path** to save the Web application on the disk. The default path is `[installation drive:]\[installation folder]\[WebSites]\[Web site Name]`.
  - Browse for a web application **User** using the Select User dialog box. You can select a different user than the one you selected during Web site creation.  
**NOTE:** The Web application must be a member of the IIS\_IUSRS group. If you select a user that is not a member of the IIS\_IUSRS group, SMC prompts you to add it.
  - Enter the valid **Password** for the Web application user.
6. Click **Save**   
⇒ A confirm message displays.  
⇒ Click **OK** to initiate the Web application creation (or **Cancel** to abort).  
⇒ If confirmed, the data is validated on successful creation.  
⇒ The **Web Service Interface Settings** expanders displays the defined settings.

### 4.4.5 Starting a Project

- ▷ The project is stopped.
1. In the System Management Console tree, select the **Projects** node.
  2. In the **Projects** tree, select the project that you want to start.
  3. Click the **Project Settings** tab, if not yet selected.  
⇒ **Project Settings** displays.
  4. In the **Project Settings** toolbar, click **Start Project**   
⇒ A confirmation message displays.
  5. Click **OK** to start the stopped project

## 4.5 Service Category

Based on frequency of change in design, all of the services and APIs are categorized as below three types:

| Category         | Scope           | Description                                                                                                                                                           |
|------------------|-----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Public           | Service and API | Any change in APIs will not be breaking. Backward compatibility will always be supported.<br>Code: <b>Green</b>                                                       |
| Partial Internal | Service only    | Service contains both internal and Public APIs.<br>Code: <b>Yellow</b>                                                                                                |
| Internal         | Service and API | Service is developed strictly only for internal development. Interface might change with newer version. Backward compatibility is not guaranteed.<br>Code: <b>Red</b> |

### 4.5.1 Summary

For quick reference of service category refer below table

| S.No. | Service                          | Code | Service Category   |
|-------|----------------------------------|------|--------------------|
| 1     | Token                            |      | Public             |
| 2     | HeartBeat                        |      | Public             |
| 3     | Command                          |      | Public             |
| 4     | CommandSubscriptions             |      | Public             |
| 5     | Diagnostics                      |      | Public             |
| 6     | EventCategorySounds              |      | Public             |
| 7     | EventCategorySoundsSubscriptions |      | Public             |
| 8     | EventCounters                    |      | Public             |
| 9     | EventCountersSubscriptions       |      | Public             |
| 10    | EventsCommands                   |      | Public             |
| 11    | Events                           |      | Public             |
| 12    | EventsSubscriptions              |      | Public             |
| 13    | Files                            |      | Partially Internal |
| 14    | HistoryLog                       |      | Public             |
| 15    | Graphics                         |      | Internal           |
| 16    | Images                           |      | Internal           |
| 17    | Language                         |      | Public             |
| 18    | Products                         |      | Public             |
| 19    | Properties                       |      | Public             |
| 20    | PropertyValues                   |      | Public             |
| 21    | Settings                         |      | Internal           |
| 22    | SystemBrowser                    |      | Public             |
| 23    | SystemBrowserSubscriptions       |      | Public             |
| 24    | Systems                          |      | Public             |
| 25    | Tables                           |      | Partially Internal |
| 26    | TrendSeries                      |      | Public             |
| 27    | TrendSeriesInfo                  |      | Public             |
| 28    | Users                            |      | Internal           |
| 29    | Values                           |      | Public             |
| 30    | ValuesSubscriptions              |      | Public             |
| 31    | Systems                          |      | Public             |
| 32    | SystemsSubscriptions             |      | Public             |
| 33    | Scheduler                        |      | Internal           |
| 34    | Related Items                    |      | Public             |
| 35    | AccessRights                     |      | Internal           |
| 36    | AccessRightsSubscriptions        |      | Internal           |



## 4.6 Service mapping with Desigo CC(Basic reference)



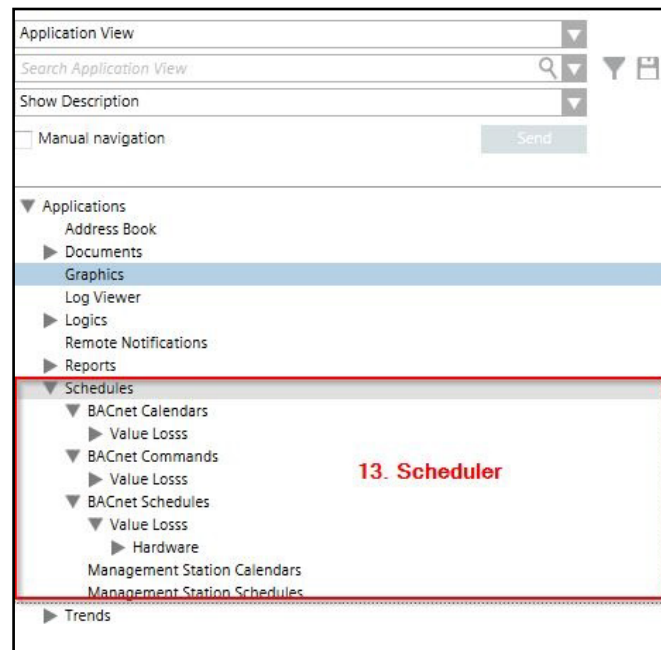
The screenshot displays the Siemens System Manager interface with several components highlighted by red boxes and numbered:

- 5. Event Counters**: Located in the top navigation bar.
- 6. System Details**: Located in the top right corner.
- 7. System Browser**: The left-hand tree view showing the project hierarchy.
- 8. Properties**: The central pane showing the properties of the selected object, 'Analog Input 1 [AI\_1]'.

The screenshot displays the Siemens System Manager interface with the following sections highlighted by red boxes and numbered:

- 9. Graphics**: The 'Graphics' folder in the left-hand tree view.
- 10. Reports**: The 'Reports' folder in the left-hand tree view.
- 11. Trends**: The 'Trends' folder in the left-hand tree view, and the 'Trend Log 1 [TL\_1]' view on the right showing a time-based data plot.

| Event List |                                                       |                                                                |
|------------|-------------------------------------------------------|----------------------------------------------------------------|
|            | Cause                                                 | Path                                                           |
|            | At least one service not running properly             | Project.Management System.Servers                              |
|            | BACnet Driver Not Communicating (Configuration Error) | Project.Management System.Servers.Main Server.Drivers          |
|            | BACnet Driver Not Communicating (Configuration Error) | Project.Management System.Servers.Main Server.Drivers          |
|            | History Database (Not Connected)                      | Project.Management System.Servers.Main Server.History Database |
|            | History Database (Not Connected)                      | Project.Management System.Servers.Main Server.History Database |
|            | Transaction Mode Simple                               | Project.Management System.Servers                              |



| S.No. | Workflow        | Services [43]                                             |
|-------|-----------------|-----------------------------------------------------------|
| 1     | Image Download  | Files[147]                                                |
| 2     | Login Support   | Token[43]                                                 |
| 3     | Change Password | Users[156]                                                |
| 4     | Product Details | Products[144]                                             |
| 5     | Event Counters  | EventCounters[54]                                         |
| 6     | Systeme Details | Systems[140]                                              |
| 7     | System Browser  | SystemBrowser[60]                                         |
| 8     | Properties      | 1. Value[71]<br>2. Properties[84]<br>3. PropertyValue[81] |
| 9     | Graphics        | Graphics[148]                                             |
| 10    | Reports         | Histry Logs[116]                                          |
| 11    | Trends          | Trend[107]                                                |
| 12    | Events          | 1. Event[47]<br>2. EventsCommands[52]                     |
| 13    | Scheduler       | 3. Scheduler[221]                                         |

## 4.7 Access rights for application services

Certain services in Web service interface can be related to applications in DesigoCC client. Application rights for such applications are also enforced service endpoints as well. Access to operation would be denied if application rights are not available with user.

| Service          | Endpoints                         | Application [ID- Name] | Operation [ID- Name] |
|------------------|-----------------------------------|------------------------|----------------------|
| Trend Service    | Delete TrendViewDefinition        | 54 - Trends            | 1730 - Configure     |
|                  | Delete TrendLogOnline             | 54 - Trends            | 1730 - Configure     |
|                  | Create/Update TrendViewDefinition | 54 - Trends            | 1730 - Configure     |
|                  |                                   |                        |                      |
| Schedule Service | Save Schedule                     | 46 - Schedules         | 1474 - Configure     |
|                  | Save Exceptions                   | 46 - Schedules         | 1474 - Configure     |
|                  | Save Schedule Options             | 46 - Schedules         | 1474 - Configure     |
|                  | Save Calendar                     | 46 - Schedules         | 1474 - Configure     |

## 4.8 WSI Configuration keys

Configuration keys in project config can be modified to change default behavior of NORIS. It follows Hierarchical configuration i.e. configuration defined at instance level can be used to override configuration at base configuration.

By default configuration at instance level is not defined and it will inherit config from base e.g. *MaxVirtualMemorySizeForProcess* is defined at base config only which can be overridden at instance level if certain instances require more memory for operation.

| Configuration Key              | Data Type | Default Value | Description                                                                                                                                                                                                                                                                                                                                                                                      |
|--------------------------------|-----------|---------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| WhiteListDomain                | String    |               | Update CORS policy to allow only whitelisted domains in access-control response headers.<br>Multiple domain names need to be separated by semi colon ( ; ).<br>Key is optional and default configuration allows all domains.<br>e.g. WhiteListDomain = <a href="http://xyz.com">http://xyz.com</a> ; <a href="https://xyz.com">https://xyz.com</a> ; <a href="https://abc.in">https://abc.in</a> |
| CachingDefaultMaxAge           | Unit      |               | Time duration in minutes to set Cache control max-age value in WSI response.<br>Key is optional and default configuration sets max-age to 30 mins.<br>Setting value as 0 will disable caching headers in response.<br>e.g. CachingDefaultMaxAge = 60                                                                                                                                             |
| CachingDefaultVaryHeaders      | String    |               | Header name to revalidate/refresh cache on browser side. This will set the Vary header in response if caching is enabled.<br>Multiple header names need to be separated by semi colon ( ; ).<br>Key is optional and default configuration sets vary to 'Authorization'.<br>e.g. CachingDefaultVaryHeaders = Header1;Header2;Header3                                                              |
| MaxSnapshotsForSingleSession   | String    | 5             | Maximum Snapshots time for single session                                                                                                                                                                                                                                                                                                                                                        |
| MaxSnapshotInactiveTime        | String    | 10            | Maximum Inactive Time for Snapshot                                                                                                                                                                                                                                                                                                                                                               |
| MaxVirtualMemorySizeForProcess | String    | 4294967296    | Maximum Virtual Memory Size For Process                                                                                                                                                                                                                                                                                                                                                          |
| maxBcmBufferSize               | String    | 100000        | Maximum Bcm Buffer Size                                                                                                                                                                                                                                                                                                                                                                          |
| DirectoriesAccess              | String    | libraries     | Directories Access                                                                                                                                                                                                                                                                                                                                                                               |
| SignalrMessageBuffer           | String    | 3000          | Signalr Message Buffer size                                                                                                                                                                                                                                                                                                                                                                      |
| SignalrKeepAlive               | String    | 6             | Signalr Keep Alive                                                                                                                                                                                                                                                                                                                                                                               |

|                               |        |                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                                                                             |
|-------------------------------|--------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| ExpiresUtcTicketForCodeGrant  | String | 5                                                                                                                                                                                                                                                                                                                                                                                                 | UtcTicket Expires time For Code Grant                                                                                                       |
| LicensedClient                | String | 40                                                                                                                                                                                                                                                                                                                                                                                                | Maximum number of clients of type ,LicensedClient' allowed to login.                                                                        |
| NonLicensedClient             | String | 10                                                                                                                                                                                                                                                                                                                                                                                                | Maximum number of clients of type , NonLicensedClient ' allowed to login.                                                                   |
| WebApiThrottlingConfiguration | String | <pre>[{"ApiRoute": "/api/token", "MaxRequestLimitPerSecond": 25}, {"ApiRoute": "/api/diagnostics", "MaxRequestLimitPerSecond": 25}, {"ApiRoute": "/api/Products", "MaxRequestLimitPerSecond": 25}, {"ApiRoute": "/api/systems", "MaxRequestLimitPerSecond": 25}, {"ApiRoute": "/api/users", "MaxRequestLimitPerSecond": 25}, {"ApiRoute": "/api/licenses", "MaxRequestLimitPerSecond": 25}]</pre> | WSI endpoints web api throttling configuration used for limiting maximum allowed request rate. For more details please refer section 4.8.1. |

### 4.8.1 Web Api Throttling Configuration

Web api throttling configuration can be given in quoted json format as shown in WSI Config keys table in section 4.8. This configuration supports three key-value pairs for each entry, details about which can be found in following table.

| Key                      | Value                                                                          |
|--------------------------|--------------------------------------------------------------------------------|
| ApiRoute                 | Route of the WSI endpoint for which throttling is to be configured.            |
| MaxRequestLimitPerSecond | Maximum number of allowed calls per second for endpoint mentioned in ApiRoute. |
| MaxRequestLimitPerMinute | Maximum number of allowed calls per minute for endpoint mentioned in ApiRoute. |

Default configuration for web api throttling is added to the project while creating new project and upgrading the project which doesn't have this configuration present in WSI Config section of the Project config file. If this configuration is already present in the config file, then it will not be altered while upgrading the project. When this configuration is present in config file, the mentioned endpoints will be monitored, and the maximum number of calls will be limited to the configured number, beyond which the IP will be denied

for more calls in that time interval. E.g., if for ApiRoute : “api/token”, MaxRequestLimitPerSecond is set to 25, then for every second only 25 calls are processed for any IP and the requests more than 25 are denied to process per second. Both MaxRequestLimitPerSecond and MaxRequestLimitPerMinute can be set for any ApiRoute, in this case both the limits will be checked before processing requests. E.g., if for ApiRoute : “api/token”, MaxRequestLimitPerSecond is set to 25 and MaxRequestLimitPerMinute is set to 200 then in this case the maximum number of allowed calls per second will limit to 25 and allowed calls per minute will be limited to 200, that means after processing 25 requests/second for 8 second, it will deny processing new requests till completion of one minute.

When requests to the api endpoints are denied due to frequent requests more than allowed limit, one will get message “Exceeded maximum number of allowed calls.” and 403 (Forbidden) as a response code.

## 4.9 Time sync

It could be scenario where NORIS server and Client application has some time difference. This time difference can result in inconsistent data – Event time represented incorrectly, during commanding incorrect data being sent to server. To sync the time between server and client , NTP time sync could be implemented - [Clock synchronization algorithm](#).

NORIS provides server's UTC datetime information as well as time difference between server time and client time if client pass client information to the server.

The information provided by Diagnostics Service [→127] can be used calculate time difference at client side.

## 5 Conventions

### 5.1 Case Sensitivity

Unless otherwise specified the system is case sensitive. This affects the API in terms of filters and search results.

### 5.2 Number Format

The API uses an invariant culture for the number formatting regardless of the logged in user. The decimal mark is a period.

### 5.3 Date/Time Format

Date and time representations under ISO-8601 acceptable formats are :

```
yyyy-MM-ddTHH:mm:sZ  
yyyy-MM-ddTHH:mm:ssZ  
yyyy-MM-ddTHH:mm:ss.FFFZ
```

**NOTE:**

Unless otherwise noted, the Zulu time zone is expected.

*Examples:*

```
2016-03-06T12:35:1Z  
2016-03-06T12:35:01Z  
2016-03-06T12:35:01.711Z
```

### 5.4 Terminology

| Terminology              |                                                                                    |
|--------------------------|------------------------------------------------------------------------------------|
| Term                     | Description                                                                        |
| Name <sup>1</sup>        | Literal; supposedly unreadable; usually not displayed in a UI.                     |
| Descriptor <sup>1</sup>  | Localized string in a readable form; supposed to be displayed to an end- user.     |
| Designation <sup>1</sup> | Unique address of a node; supposedly unreadable; usually not displayed in a UI.    |
| Location <sup>1</sup>    | Localized, fully expanded path of a node; supposed to be displayed to an end-user. |
| ObjectId                 | Identifies an object in the system.                                                |
| PropertyId               | Identifies a property and its object in the system.                                |
| AttributeId              | Identifies an attribute, its property and its object in the system.                |
| ObjectOrPropertyId       | An object ID or a property ID is expected.                                         |
| PropertyName             | Name of a property.                                                                |

<sup>1</sup> see also Naming [→ 231]

## 5.5 Maximum page size

Maximum page size which is supported in case of paging is 100000 records per page.

In case if page size is not specified then above mentioned size will be considered as page size. And if valid page size is specified then minimum from the both will be considered as page size.

## 5.6 Request per system

In a distributed environment (multiple systems) a request may only contain data points to a single system. If one needs to process data points of multiple systems multiple requests have to be invoked.



---

**NOTE:**

Unless explicitly stated, a service request may only affect a single system.

---



## 6 General API Specification

The REST API provides programmatic access to read, write and command the system data and functionality. The API identifies users using OAuth [4] [→ 16]; responses are available in JSON [6] [→ 16].

### 6.1 HTTP Methods

Following REST principles, one can call the following HTTP methods on the API resources: GET, POST, PUT, and DELETE.

| Method (HTTP Verb) | Description                                                                     |
|--------------------|---------------------------------------------------------------------------------|
| GET                | Retrieves a list of resources, or a specific resource.                          |
| POST               | Creates a resource, passes in a parameter, or performs an action on a resource. |
| PUT                | Updates an existing resource.                                                   |
| DELETE             | Deletes an existing resource or subscription.                                   |

### 6.2 Resources

The Web Service Interface API follows REST principles and therefore exposes data and functionality as resources [7] [→ 16].

Every resource is represented as an URI (see Appendix A in [8] [→ 16]). As a summary a resource looks like the following:

`scheme://domain:port/path?query`

| Part   | Description                                                                                                              |
|--------|--------------------------------------------------------------------------------------------------------------------------|
| scheme | Usually HTTPS; HTTP should only be used for testing                                                                      |
| domain | Hostname of host where manager is running                                                                                |
| port   | Usually 8080 for HTTP and 8443 for HTTPS                                                                                 |
| path   | Starts with api followed by the service-identifier and optional sub paths (e.g. api/languages/en-us); see sections below |
| query  | Optional; depends on resource (see sections below)                                                                       |

#### Example

`https://ch1w80012:8443/api/events?sorting=1`



#### NOTE:

In a production environment, the API runs most likely behind a reverse proxy. In this case, the URI might look differently and might hide some information from the Client.

#### Example

`https://www.example.com/example/api/events?sorting=1`

Therefore the general format of a URL is:

`{entry-point}/api{resource-address}`

Where **entry-point** is `http://[hostname]:[port]` by default or dependent on the deployment, and **resource-address** in EBNF is:

`resource-address = path [ "?" query ]`

`path = /root-resource | path "/" sub-resource`

`query = parameter { "&" parameter }`

`parameter = key [ "=" value ]`

## 6.3 Status Codes

### 6.3.1 HTTP Status Codes

| Status Code | Meaning      | Description                                                                                                                                                                                                                                                        |
|-------------|--------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 200         | OK           | Request processed.<br><b>NOTE:</b> Although the request has been processed as a whole, some of the sub results could still be invalid. See Error Code [→ 42] to check sub-results.                                                                                 |
| 204         | No Content   | The server successfully processed the request and is not returning any content.                                                                                                                                                                                    |
| 304         | Not Modified | Resources are not modified from last request.<br><b>NOTE:</b> This status code is returned only if Etag header is present in earlier response and request contains If-None-Match header with Etag value and resource is not modified. See caching support [→234] . |
| 400         | Bad Request  | Request rejected by application.                                                                                                                                                                                                                                   |
| 401         | Unauthorized | Invalid or expired token presented. User needs to login.                                                                                                                                                                                                           |
| 403         | Forbidden    | The server understood the request but refuses to authorize it.                                                                                                                                                                                                     |
| 404         | Not Found    | Resource couldn't be found for the provided URL and query parameters.                                                                                                                                                                                              |

### 6.3.2 Error Code

| Code | Description        |
|------|--------------------|
| 0    | No error occurred. |

## 6.4 Encoding

Data is encoded in UTF-8.

## 6.5 Array in Query Parameter

In case a query parameter represents an array, the array is passed as a JSON array [6] [→ 16].

*Example*

```
someurl?Ids=[1,2,3]
```

## 6.6 HTTP caching

Browser can create local cache of response data based on certain headers in HTTP headers, thus avoiding repetitive calls to server for same set of data .For more details on HTTP caching refer : <https://developer.mozilla.org/en-US/docs/Web/HTTP/Caching>

Details of implemnetation for HTTP caching in NORIS is mentioned in Support for Browser Caching[→9.4]

## 7 Services

### 7.1 Token Service

| Resource | Service Category | Requires Authentication | Requires SignalR (Push Notifications) |
|----------|------------------|-------------------------|---------------------------------------|
| /token   | Public           | x (Logout only)         |                                       |

**Note :** For clients of type **LicensedClient**, the maximum and default number of logins allowed is 40. For clients of other types, the maximum and default number of logins allowed is 10. The numbers can be configured in the project config file > 4.8. But maximum values are limited to 40 and 10 for LicensedClient and NonLicensedClient respectively.

#### 7.1.1 Logging with Username and Password

|               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                     |                                                                                                                                                               |
|---------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Resource Path | /token                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                     |                                                                                                                                                               |
| Category      | Public                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                     |                                                                                                                                                               |
| HTTP Method   | POST                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                     |                                                                                                                                                               |
| Content Type  | application/x-www-form-urlencoded                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                     |                                                                                                                                                               |
| Body          | <b>Property</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | <b>Content</b>      | <b>Description</b>                                                                                                                                            |
|               | grant_type                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | password            | grant_type=password                                                                                                                                           |
|               | username                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | [username]          | Username provided by user.                                                                                                                                    |
|               | password                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | [password]          | Password provided by user.                                                                                                                                    |
|               | client_id                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | [client identifier] | Optional : LicensedClient<br>Client identifier of application ,<br>certain services may require use of<br>license that are available with<br>license server . |
| Response      | HTTP 200 OK<br>A Login object (see Token Service Login [194])<br>HTTP 403 Forbidden<br>User password is expired and need to be changed first                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                     |                                                                                                                                                               |
| Examples      | <b>Request:</b><br>POST /WSI/api/token<br><b>Post Data:</b><br>grant_type=password&username=DefaultAdmin&password=Siemens@12345<br><b>Response:</b><br><pre>{   "access_token": "AQAANCMnd8BFdERjHoAwE_Cl-sBAAAAJbLmifXxrkmVwrHtNdOoBQAAAAACAAAAAAQZgAAAAEAAACAAAAAD1DER3xtPANmCUT41Ag_C0ms9Y6UL6NQxW5WZkUNyMAAAAAAOGAAAAIAACAAAAAbZ568TgfMVBn6qRpR-wtLTaxWHhjGF8dPPuv3AXH7ZABAABx6NBN9NVK-DxS0muRa7LnUlKUz8gxOaq3Nvd9jZptozEzQxSsdNs0JV5mXfuSDn2qIqjihwavx4i4ge7qyxQ_xIPuMW4cwWs3qSnqCdbJBku2WwqTT12zrZhvFvR4uKnXnPHdXV-_lqqpPgdtay9CPWR_fns4WyaY_Rj_CYmaWidZlUmVuJUZNkn96LeTD5nN2Df1EKKJ8g2L7DDg8ZLpCERChoCist6CW47w7hlhME1ZySDUiUbZ5GotMIUMSaJrMk9vQLPSn_HMVgyPYog79HWQC0xZelSeUYCAD0Pj5fzw6luzjYz8fQ20BjIw4hH22Ef9xQocYnjGI4S1hxsAt1Bx6w_T9FPczSjJ8OCHb51a9hmWE8mbIw7mGsJbn2PlY_adVec85qBwt_XEMQ3vMPWcfMOeQqTGO38NB04YsZPoeyR_Md_4p7jdz07qUHwqDR2tdzI5iM5tEVyMGIysuXr6pKAxj_9IT67g9FMFzePvUCb0qieCT2ksQVwrVC5F7ZtY1uQb-a0jNefC76cXQAAAACFzePjZimp2xZ RTPV-rwxk_cclpFGLU44GHRyt76ZrqWsIMgcQZKE8M5OySwVih10d5ukHFgcXa7</pre> |                     |                                                                                                                                                               |

|  |                                                                                                                                                                                                                                                                                                                                                                                                        |
|--|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | <pre> BQJ4M37ffI", "token_type":"bearer","expires_in":2591999, "user_name":"DefaultAdmin", "user_descriptor":"Default Administrator", "user_profile":"DEFAULT.ldl", "flex_user_profile":"DEFAULT", "user_inactivity_timeout":"0", "password_reminder_counter":"0", "personally_used":"false" } </pre>                                                                                                  |
|  | <p><b>Example: User password is expired</b></p> <p><b>Request:</b><br/>POST /WSI/TestClient/api/token</p> <p><b>Post Data:</b><br/>grant_type=password&amp;username=testuser&amp;password=123</p> <p><b>Response:</b><br/>HTTP 403 Forbidden</p> <pre> {"Id":2400003,"Error":"password_expired","Details":"Logon rejected. Password of this account is expired and need to get changed first."} </pre> |

## 7.1.2 SSO Log in

It is only available for internal gms application developers. Get Single sign on (SSO) token directly from Logon orchestrator and provide it to WSI using below interface.

|               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                    |                                                              |
|---------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------|--------------------------------------------------------------|
| Resource Path | /token                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                    |                                                              |
| Category      | Public                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                    |                                                              |
| HTTP Method   | POST                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                    |                                                              |
| Content Type  | application/x-www-form-urlencoded                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                    |                                                              |
| Body          | <b>Property</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | <b>Content</b>     | <b>Description</b>                                           |
|               | grant_type                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | client_credentials | grant_type= client_credentials<br>To login provide SSO token |
|               | Client_Id                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | SSO token          | Retrived from Logon Orch.                                    |
| Response      | HTTP 200 OK<br>A Login object (see Token Service Login [→87])                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                    |                                                              |
| Examples      | <p><b>Request:</b><br/>POST /WSI/TestClient/api/token</p> <p><b>Post Data:</b><br/>grant_type=client_credentials&amp;Client_Id=811d8bb9856598ff60c05ad525746bef40517ab51020faae303c28afe932a323d45469eb8b9e2ff85ef1c00dc7b4cfcf3e1bc33e6cd878e5ed6b63d2271b9d6ef98248544bd5eal6d6fdb72aacd960ad8611c98f01d7f85e8e924829edb7dc689fb248ba9fd5093faf6cb942587e9516a17bf0c13af994c0814e53dce98923a67a0fa02015f44580e0016e862aa600a8ecdd2a2597a3074bd1c5d8af3e2db5a2868de2176a1fa533e4dbba3082798dde4938cdc95c086055bedc6b4d923ebec2ce16d8eab3405955ae6c881d6d4f274796b166cc541e18a16039282de247b3b0083508e60674088a3f7d6fcff12c0d2144414faa09cdc376581c0e2ff1711d34</p> <p><b>Response:</b><br/>{ "access_token": "AQAAANCMnd8BFdERjHoAwE_C1-sBAAAAJbLmifXxrkmVwrHtNdOoBQAAAAACAAAAAAQZgAAAAEAACAAAAACNj5CbeMxGu7jEh6ZgK_sEnNOSLtVZfaE2SriUxy8FwAAAAAogAAAAIAACAAAABILv8nDmuJV57WEjOgUGyCH6ofwAI6RQyt7sgu6GFM1ZABAABEXjxXAqZnkyYVctuZdpqrEUHVqh5uc16PT4qC3EjVSK9cnq5qD2rtdLFaLJQoO-Nklg7fpSmQBF84Ts7FpKgmU_EYhq4lMgzJWW0kiXbWTJhG1Lj12W3wmyx3zfENU8chB uoKM4 nB57Kule77cYocsk0zIwaayq1TvLQ1sXbIiESoNGY</p> |                    |                                                              |

|  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|--|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | <pre> ExyHoEu-Cetv2zAw-00hAj6p9UG8rIQXfyUxYIDpkBYLdcrvqEt18GQ8- pjAWIQac46WjA_W7nYm7MZGJYmHgtiet5SXtQz5cm88y1JBMxrmgCMhDx ub7QZ- Xq6V6V1liPhMEDnoPnd5KIrlmdTDPGuD7XxNFbKOFdsxa9GQxN6KLxM6og WlwMP3R3edlTFP2KvF4C6V8o6R2fKFU71HaDatJdqLjyc5MdUpIs19daZz G-eW7XNM6_haihEtN9F-6Sxk2q-5O9wNbeYGUPHv_PbRxXEYF- bRauayKQnJ3XpO321zeppaXjyep7KtlezrtBaZ1E- 3WIf4QKFkjPnaxT84XMWYdXe- PQAAAAp5JIPc4l0Sk4D_ATLk7FQX000sKZINKr6JzQQCVQXMuK9qC9hQX WJSr0Ck3wVhUkSMgKz1xSvi2xRQ_V9T40k", "token_type": "bearer", "expires_in": 2591999, "user_name": "DefaultAdmin", "user_descriptor": "Default Administrator", "user_profile": "DEFAULT.ldl", "flex_user_profile": "DEFAULT", "user_inactivity_timeout": "0", "password_reminder_counter": "0", "personally_used": "false" } </pre> |
|--|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

### 7.1.3 Default Logon (Mutual TLS)

This method enables the client to login with a client certificate, on page load. GMS certificate should be configured beforehand. Login is successful if the certificate is valid.

|               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                    |                                                                        |
|---------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------|------------------------------------------------------------------------|
| Resource Path | /token                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                    |                                                                        |
| Category      | Public                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                    |                                                                        |
| HTTP Method   | POST                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                    |                                                                        |
| Content Type  | application/x-www-form-urlencoded                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                    |                                                                        |
| Body          | <b>Property</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | <b>Content</b>     | <b>Description</b>                                                     |
|               | grant_type                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | client_credentials | grant_type= client_credentials<br>To login, provide client certificate |
|               | token_endpoint_auth_method                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | tls_mutual_auth    | Fixed parameter to be passed for mutual tls. [12]                      |
|               | Client_Id                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | DefaultLogon       | Fixed string value.                                                    |
| Response      | HTTP 200 OK<br>A Login object (see Token Service Login [→87])                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                    |                                                                        |
| Examples      | <b>Request:</b><br>POST /WSI/api/token<br><b>Post Data:</b><br>grant_type=client_credentials&Client_Id=LicensedClient&token_endpoint_auth_method=tls_mutual_auth<br><b>Response:</b><br><pre> { "access_token": "AQAAANCMnd8BFdERjHoAwE_Cl- sBAAAAJbLmifXxrkmVwrHtNdOoBQAAAAACAAAAAAQZgAAAAE- AACAAAABpcv7QPhIBZ6ScpqqXVufFpHYKWCDEompqv9waXbL- BAAAAAAOgAAAAAIAACAAAAABrUNEhn- 4dsVwkv1LL6EMU5_Z6bE_5aVLn-TAszFNOy5ABAAB_ZYoK6ka- 3EVBzGM79LcR_Cq3fMULwX1Xki_h1oj3vEzb9pO- DrwWbWwF3NHIS4NAm5Ugf-S3oVm5y8-4n4tH86ydU- zoVVl_6wFYmuiBmDRI8nyEFJrz1U2O26sr0PeUDjTYx6kxTzsZL6w6 jQT95z5PTQh8SDzUnJJQO0i5zFtNH-b-0I6skazkjVzuEwVBKqSs- qkGA0DG64RNqbCUEhZMWDxnjtEdQi7D5_ws2TEhd- femBZ25XcgYu4xbMo8-uzBLKPO0EB82YplGJvUKgzMbeouqle2- AUMP3mv01J8Vn1RXls8IQprQP9NpHmK23sVz9zEgznYE- kATWU_3hc6_5a5pQA2Y4_DTMS7mWUNN6eUdlQT_2WbvbEzAld- TIJ430Zo_JraBnCjgaHm3FDMpRcHtji2AGlCm4cmMLPEsnk- PBZemSvUp4HR62VzK4n1yH0CXZBPA06YR07t7mHdq_GNIWks- BQH_QLeSws0EHstfGsgmPgqm0FpJibBW- </pre> |                    |                                                                        |

|  |                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|--|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | <pre>lvG41ldW7RzAVKkHZ6sLc03rQAAAAJ_25yeN- dodB63HNqT_0KWF6zbVVtX9H7tCtzzS1MPhxSSEGKw8tEp- MmQn65yb8n5SAzBBwDJmO8OEg7ruth3L8",  "token_type":"bearer","expires_in":2591999,  "user_name":"SWAccForDefaultLogon",  "user_descriptor":"Software account for default lo- gon",  "user_profile":"DEFAULT.1dl",  "flex_user_profile":"DEFAULT",  "user_inactivity_timeout":"0",  "password_reminder_counter":"0",  "personally_used":"false"  }</pre> |
|--|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

### 7.1.4 Logging out

| Logging out a User |                                                                                                          |
|--------------------|----------------------------------------------------------------------------------------------------------|
| Resource Path      | /token                                                                                                   |
| Category           | Public                                                                                                   |
| HTTP Method        | DELETE                                                                                                   |
| Response           | HTTP 200 OK                                                                                              |
| Examples           | <b>Request:</b><br>DELETE /NBWSITest/NorthboundWebApiClient/api/token<br><b>Response:</b><br>HTTP 200 OK |

## 7.2 Heartbeat Service

| Resource   | Service Category | Requires Authentication | Requires SignalR (Push Notifications) |
|------------|------------------|-------------------------|---------------------------------------|
| /heartbeat | Public           | x                       |                                       |

### 7.2.1 Extending the Lifetime of a Session and its Bearer Token

If a Client logs in, a session is created and bound to the bearer token that the Client receives as part of a successful login request. The session and its bearer token expire after some time of inactivity (default: 10 minutes). Inactivity is detected if no subscription is alive and no protected (authenticated) call is made within this timeout.

The **/heartbeat** resource can be called to artificially signal activity and to reset the timeout.

|                               |
|-------------------------------|
| <b>Lifetime of a Session.</b> |
|-------------------------------|

|               |                                                                                                            |
|---------------|------------------------------------------------------------------------------------------------------------|
| Resource Path | /heartbeat                                                                                                 |
| Category      | Public                                                                                                     |
| HTTP Method   | POST                                                                                                       |
| Response      | HTTP 200 OK                                                                                                |
| Examples      | <b>Request:</b><br>POST /NBWSITest/NorthboundWebApiClient/api/heartbeat<br><b>Response:</b><br>HTTP 200 OK |

## 7.3 Event Service

| Resource             | Service Category | Requires Authentication | Requires SignalR (Push Notifications) |
|----------------------|------------------|-------------------------|---------------------------------------|
| /events              | Public           | x                       |                                       |
| /eventssubscriptions | Public           | x                       | x                                     |

Events depend on the event type (see [20]). The current event type can be read with the following object ID: ManagementView\_System\_Settings.EventSchema.

### 7.3.1 Retrieving a List of Events



#### NOTE:

1. In order to retrieve grouped events right combination or unique identifier have to be provided in parameters
2. Wild card characters are supported in eventfilter search wherever applicable.
3. Events extension commands are also supported in response.(e.g. StartReno etc)

|               |                   |                 |                            |
|---------------|-------------------|-----------------|----------------------------|
| Resource Path | /events           |                 |                            |
| Category      | Public            |                 |                            |
| HTTP Method   | GET               |                 |                            |
| Parameters    | <b>Property</b>   | <b>Type</b>     | <b>Description</b>         |
|               | sorting           | int             | See EventSorting [→ 197]   |
|               | page              | int             |                            |
|               | size              | int             |                            |
|               | ids               | Array of string |                            |
|               | caseSensitive     | bool            |                            |
|               | categories        | Array of int    |                            |
|               | cause             | string          |                            |
|               | creationTime      | int             | See EventTimeFilter [→ 49] |
|               | creationTimeFrom  | DateTime        |                            |
|               | creationTimeTo    | DateTime        |                            |
|               | disciplines       | Array of int    |                            |
|               | Systemid          | UInt            | System Identifier          |
|               | states            | Array of string | See EventState [→ 196]     |
|               | designations      | Array of string |                            |
|               | sourceName        | string          |                            |
|               | sourceDescription | string          |                            |

|          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                 |                              |
|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|------------------------------|
|          | locationDescription                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | string          |                              |
|          | locationName                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | string          |                              |
|          | objectState                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Array of string | See ObjectState [→ 196]      |
|          | maintenance                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | string          | See MaintenanceState [→ 197] |
| Response | HTTP 200 OK<br>A Page<Events, Event>-object (see Page<Name, Type>) [→ 190]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                 |                              |
| Examples | <b>Request:</b><br>GET /api/events?sorting=0<br><b>Response:</b><br><pre>{   "Events": [     {       "Id": "System1:ManagementView_ManagementSystem_Servers_Server.SoftwareRunning:_alert_hdl.2._value~635742985171620000~0",       "EventId": 86,       "CategoryId": 9,       "CategoryDescriptor": "Fault",       "State": "Unprocessed",       "Cause": "Services running properly",       "SrcPropertyId": "System1:ManagementView_ManagementSystem_Servers_Server.SoftwareRunning:_alert_hdl.2._value",       "SrcObservedPropertyId": "System1:ManagementView_ManagementSystem_Servers_Server.SoftwareRunning:_alert_hdl.2._value",       "SrcState": "Quiet",       "SrcSystemId": 1,       "SrcViewName": "ManagementView",       "SrcViewDescriptor": "Management View",       "SrcDesignation": "ManagementView.ManagementSystem.Servers.Server",       "SrcLocation": "Project.Management System.Servers.Main Server",       "SrcName": "Server",       "SrcDescriptor": "Main Server",       "SrcDisciplineId": 0,       "SrcDisciplineDescriptor": "Management System",       "CreationTime": "2015-08-04T09:51:57.162Z",       "Direction": "None",       "SuggestedAction":WaitforCondition,       "NextCommand":NoActionPossible,       "InfoDescriptor": "At least one service not running properly",       "Commands": [         {           "Id": "Suspend",           "EventId": "System1:ManagementView_ManagementSystem_Servers_Server.SoftwareRunning:_alert_hdl.2._value~635742985171620000~0",           "Configuration":1           "_links": [             {               "Rel": "command",               "Href": "api/eventscmds/System1%3aManagementView_ManagementSystem_Servers_Server.SoftwareRunning%3a_alert_hdl.2._value~635742985171620000~0/Suspend",               "IsTemplated": false             }           ]         }       ]     }   ] }</pre> |                 |                              |



|  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|--|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | <pre>         }       ],     },     {       "Id": "Ack",       "EventId": "System1:ManagementView_ManagementSystem_Servers_Server.SoftwareRunning:_alert_hdl.2._value~635742985171620000~0",       "Configuration":1       "_links": [         {           "Rel": "command",           "Href": "api/eventscommands/System1%3aManagementView_ManagementSystem_Servers_Server.SoftwareRunning%3a_alert_hdl.2._value~635742985171620000~0/Ack",           "IsTemplated": false         }       ]     }   ],   "_links": [     {       "Rel": "category",       "Href": "api/tables/categories/9",       "IsTemplated": false     },     {       "Rel": "discipline",       "Href": "api/tables/disciplines/0",       "IsTemplated": false     }   ] } ], "Total": 1, "Page": 1, "Size": 100000, "_links": [   {     "Rel": "page",     "Href": "api/events?page={page}",     "IsTemplated": true   } ] } </pre> |
|--|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

### 7.3.2 Creating a Subscription for Events (Obsolete)

Obsolete: Use channelized subscription instead.

|               |                                       |        |                            |
|---------------|---------------------------------------|--------|----------------------------|
| Resource Path | /sr/eventssubscriptions:/connectionId |        |                            |
| Category      | Public                                |        |                            |
| HTTP Method   | POST                                  |        |                            |
| Parameters    | Property                              | Type   | Description                |
|               | connectionId                          | string | Identifier of subscription |

|          |                                                                                                                                               |     |                           |
|----------|-----------------------------------------------------------------------------------------------------------------------------------------------|-----|---------------------------|
|          |                                                                                                                                               |     | <b>Required</b>           |
|          | sorting                                                                                                                                       | int | See Event Sorting [→ 197] |
| Body     | EventFilter[→ 197]                                                                                                                            |     |                           |
| Response | HTTP 200 OK                                                                                                                                   |     |                           |
| Examples | <b>Request:</b><br>POST /api/sr/eventssubscriptions/983a0ade-7e7a-4812-bdf3-ea9ea59a7e9e?sorting=0<br><br><b>Response:</b><br>HTTP/1.1 200 OK |     |                           |

If subscription Post request submitted again then latest request will be considered and notifications will be pushed with the filter criteria of latest submitted request.

### 7.3.2.1 Push Notification

After subscribing for events, a list of current events will be sent in an initial notification. Afterwards, the Client will get notified about events according to the filter set (if any).

|          |                                                    |                |                                                                                         |
|----------|----------------------------------------------------|----------------|-----------------------------------------------------------------------------------------|
| Type     | Name (case sensitive)                              |                |                                                                                         |
| Hub      | 1. norisHub : recommended<br>2. eventsHub : legacy |                |                                                                                         |
| Function | notifyEvents (data)                                |                |                                                                                         |
|          | <b>Property</b>                                    | <b>Type</b>    | <b>Description</b>                                                                      |
|          | data                                               | Array of Event | See Event [→ 195]<br>Check the flag Deleted for checking whether the event got removed. |

### 7.3.3 Creating Channalized Events Subscription

If subscription status will be pused through notifySubscriptionStatus function available at client side. Events updates will be notified over notifyEvents [→5057].

|               |                                                                                                                              |             |                                                                                                                                                                                                                                                    |
|---------------|------------------------------------------------------------------------------------------------------------------------------|-------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Resource Path | /sr/eventssubscriptions/channelize/:requestId/:connectionId                                                                  |             |                                                                                                                                                                                                                                                    |
| Category      | Public                                                                                                                       |             |                                                                                                                                                                                                                                                    |
| HTTP Method   | POST                                                                                                                         |             |                                                                                                                                                                                                                                                    |
| Parameters    | <b>Property</b>                                                                                                              | <b>Type</b> | <b>Description</b>                                                                                                                                                                                                                                 |
|               | requestId                                                                                                                    | String      | Request Identifier of subscription hub connection.<br>RequestId is to be generated and provided by the client.<br>It is expected from client that client should use unique id for for each request.<br>Recommendation: Use GUID<br><b>Required</b> |
|               | connectionId                                                                                                                 | String      | Identifier of subscription hub connection<br><b>Required</b>                                                                                                                                                                                       |
|               | sorting                                                                                                                      | Int         | See Event Sorting [→ 197]                                                                                                                                                                                                                          |
| Body          | EventFilter[→ 197]                                                                                                           |             |                                                                                                                                                                                                                                                    |
| Response      | HTTP 200 OK                                                                                                                  |             |                                                                                                                                                                                                                                                    |
| Examples      | <b>Request:</b><br>POST /api/sr/eventssubscriptions/channelize/9565ca41-8556-4dbb-94cc-1b89451a5db5/983a0ade-7e7a-4812-bdf3- |             |                                                                                                                                                                                                                                                    |

|  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|--|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | <pre> ea9ea59a7e9e?sorting=0 <b>Response:</b> HTTP/1.1 200 OK <b>notifySubscriptionStatus:</b> {   "ErrorCode": 0,   "SortingOrder": 0,   "EventFilter": {     "Ids": [],     "Categories": [       1,       2,       3,       10     ],     "CreationTime": 6,     "Disciplines": [       0,       20,       100,       200     ],     "States": [       "Unprocessed",       "ReadyToBeReset",       "ReadyToBeClosed",       "WaitingOPCompletion",       "Aked"     ],     "systemID": 1   },   "RequestId": "9565ca41-8556-4dbb-94cc-1b89451a5db5" } </pre> |
|--|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

If subscription Post request submitted again then latest request will be considered and notifications will be pushed with the filter criteria of latest submitted request.

### 7.3.3.1 Subscription Status Notification

After subscribing for event counters, the subscription status will be sent only once.

| Type     | Name (case sensitive)                                                                                                                  |                                    |                                      |
|----------|----------------------------------------------------------------------------------------------------------------------------------------|------------------------------------|--------------------------------------|
| Hub      | 1. norisHub : recommended<br>2. eventsHub : legacy                                                                                     |                                    |                                      |
| Function | notifySubscriptionStatus(status)                                                                                                       |                                    |                                      |
|          | Property                                                                                                                               | Type                               | Description                          |
|          | Status                                                                                                                                 | EventSubscription<br>Status object | See<br>EventSubscriptionStatus[→198] |
| Example  | <pre> {   "ErrorCode": 0,   "SortingOrder": 0,   "EventFilter": {     "Ids": [],     "Categories": [       1,       2,       3, </pre> |                                    |                                      |

|  |                                                                                                                                                                                                                                                                                  |
|--|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | <pre> 10 ], "CreationTime": 6, "Disciplines": [ 0, 20, 100, 200 ], "States": [ "Unprocessed", "ReadyToBeReset", "ReadyToBeClosed", "WaitingOPCompletion", "Acked" ], "systemID": 1 }, "RequestId": "9565ca41-8556-4dbb-94cc-1b89451a5db5", "RequestFor": "notifyEvents" } </pre> |
|--|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

### 7.3.4 Modifying a Subscription for Events

Change an existing subscription for events: the same procedure as Creating a Subscription for Events [→ 49] but use PUT as HTTP method.

### 7.3.5 Deleting a Subscription for Events

Delete an existing subscription for events: the same procedure as Creating a Subscription for Events [→ 49] but use DELETE as HTTP method.

## 7.4 EventsCommands Service

| Resource        | Service Category | Requires Authentication | Requires SignalR (Push Notifications) |
|-----------------|------------------|-------------------------|---------------------------------------|
| /eventscommands | Public           | x                       |                                       |

### 7.4.1 Executing a Command on an Event

|               |                                     |                               |                                                                                                                |
|---------------|-------------------------------------|-------------------------------|----------------------------------------------------------------------------------------------------------------|
| Resource Path | /eventscommands/:eventId/:commandId |                               |                                                                                                                |
| Category      | Public                              |                               |                                                                                                                |
| HTTP Method   | POST                                |                               |                                                                                                                |
| Parameters    | <b>Property</b>                     | <b>Type</b>                   | <b>Description</b>                                                                                             |
|               | eventId                             | string                        | <b>Required</b>                                                                                                |
|               | commandId                           | string                        | See EventCommandId [→ 49]<br><b>Required</b>                                                                   |
|               | commentPasswordData                 | ValidationData<br>See [→ 199] | Event-command parameter details (comment ,password and supervisor details) sent from client if required.       |
|               | treatmenttype                       | string                        | Optional query parameter to specify type of treatment being performed.<br>This parameter is applicable only id |

|          |                                                                                                                                                                                                                                                                                                                                                                                                    |                    |
|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------|
|          |                                                                                                                                                                                                                                                                                                                                                                                                    | command is Select. |
| Response | HTTP 200 OK                                                                                                                                                                                                                                                                                                                                                                                        |                    |
| Examples | <b>Request:</b><br>POST<br>/api/eventscommands/System1%3aManagementView_ManagementSystem_Servers_Server.SoftwareRunning%3a_alert_hdl.2._value~635742985171620000~0/Ack<br><br><b>Request Body:</b><br>{"MultiLangComments":["comment1","comment2"],<br>"Password":"password","SuperName":"supervisorUserName",<br>"SuperPassword":"supervisorPassword"}<br><br><b>Response:</b><br>HTTP/1.1 200 OK |                    |

### 7.4.1.1 EventCommandId

:commandId must have one of the following values.

**Note** :Newly added commands:Events extension commands are also supported in response.(e.g. StartReno etc)

| :commandId | Description                                |
|------------|--------------------------------------------|
| select     | Signals operator is treating an event      |
| suspend    | Signals operator stopped treating an event |
| ack        | Acknowledge event                          |
| reset      | Reset event                                |
| silence    | Silence event                              |
| unsilenced | Unsilence event                            |
| close      | Close event                                |

### 7.4.2 Bulk Commanding on Events

|               |                                                                                  |                                       |                                                                                                                                |
|---------------|----------------------------------------------------------------------------------|---------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|
| Resource Path | /eventscommands/:commandId                                                       |                                       |                                                                                                                                |
| Category      | Public                                                                           |                                       |                                                                                                                                |
| HTTP Method   | POST                                                                             |                                       |                                                                                                                                |
| Parameters    | <b>Property</b>                                                                  | <b>Type</b>                           | <b>Description</b>                                                                                                             |
|               | commandId                                                                        | string                                | See EventCommandId [→ 49]<br><b>Required</b>                                                                                   |
|               | BatchEventsCommandExecution                                                      | BatchEventsCommandExecution See[→200] | Event-command parameter details (eventId,comment ,password and supervisor details) sent from client.<br><b>Required</b>        |
|               | treatmenttype                                                                    | string                                | Optional query parameter to specify type of treatment being performed. This parameter is applicable only id command is Select. |
| Response      | HTTP 200 OK                                                                      |                                       |                                                                                                                                |
| Examples      | <b>Request:</b><br>POST /api/eventscommands/Ack<br><br><b>Request Body:</b><br>{ |                                       |                                                                                                                                |

|  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|--|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | <pre> "ValidationInput": {   "MultiLangComments": [     "Comment en-US", "Comment de-DE"   ],   "Password": "currentUserPassword",   "SuperName": "Supervisor1",   "SuperPassword": "SupervisorPassword" }, "EventIds": [   "System1:ManagementView_ManagementSystem_Servers_Server.SoftwareRunning:_alert_hdl.2._value~635742985171620000~0" ] }  <b>Response:</b> [   {     "EventId":       "System1:ManagementView_ManagementSystem_Servers_Server.SoftwareRunning:_alert_hdl.2._value~635742985171620000~0"   } ]  HTTP/1.1 200 OK </pre> |
|--|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

## 7.5 EventCounter Service

| Resource                    | Service Category | Requires Authentication | Requires SignalR (Push Notifications) |
|-----------------------------|------------------|-------------------------|---------------------------------------|
| /eventcounters              | Public           | x                       |                                       |
| /eventcounterssubscriptions | Public           | x                       | x                                     |

### 7.5.1 Retrieving Event Counters

When retrieving Event counters two options are available with parameter categoryId which is optional. If categoryId is provided then Event counter for that particular categoryId will be returned as EventCounter object. And if categoryId is not provided then Event counters for all the categories in the system will be provided as an EventCounterList object. In both the cases response will be different depending on the request made. For more details see the examples below.

**Note :** Link has also made available to provide event category colors information. See [→201]

|               |                                                                                                                                                         |             |                        |
|---------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|------------------------|
| Resource Path | /eventcounters/:categoryId                                                                                                                              |             |                        |
| Category      | Public                                                                                                                                                  |             |                        |
| HTTP Method   | GET                                                                                                                                                     |             |                        |
| Parameters    | <b>Property</b>                                                                                                                                         | <b>Type</b> | <b>Description</b>     |
|               | categoryId                                                                                                                                              | int         | Identifier of category |
| Response      | HTTP 200 OK<br>1. When categoryId is provided: EventCounter object see [→ 52]<br>2. When categoryId is not provided: EventCounterList object see [→ 52] |             |                        |
| Examples      | <b>Request1: categoryId is provided</b>                                                                                                                 |             |                        |

|  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|--|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | <pre>GET api/eventcounters/9</pre> <p><b>Response1:</b></p> <pre>{   "CategoryId": 9,   "CategoryDescriptor": "Fault",   "TotalCount": 1,   "UnprocessedCount": 1,   "TotalSubsequentGrouping":0,   "UnprocessedSubsequentGrouping":0,   "_links": [     {       "Rel": "eventcounter",       "Href": "api/eventcounters/9",       "IsTemplated": false     },     {       "Rel": "categorycolor",       "Href": "api/tables/categories/9/subtables/colors",       "IsTemplated": false     }   ] }</pre> <p><b>Request2: categoryId is NOT provided</b><br/>GET api/eventcounters</p> <p><b>Response2:</b></p> <pre>{   "TotalCounters": 1,   "TotalUnprocessedCounters": 1,   "EventCategoryCounters": [     {       "CategoryId": 1,       "CategoryDescriptor": "Emergency",       "TotalCount": 0,       "UnprocessedCount": 0,       "TotalSubsequentGrouping":0,       "UnprocessedSubsequentGrouping":52,       "_links": [         {           "Rel": "eventcounter",           "Href": "api/eventcounters/1",           "IsTemplated": false         },         {           "Rel": "categorycolor",           "Href": "api/tables/categories/1/subtables/colors",           "IsTemplated": false         }       ]     },     {       "CategoryId": 2,       "CategoryDescriptor": "Life Safety",       "TotalCount": 0,       "UnprocessedCount": 0,       "TotalSubsequentGrouping":0,</pre> |
|--|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

|  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|--|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | <pre>         "UnprocessedSubsequentGrouping"=52,         "_links": [             {                 "Rel": "eventcounter",                 "Href": "api/eventcounters/2",                 "IsTemplated": false             },             {                 "Rel": "categorycolor",                 "Href": "api/tables/categories/2/subtables/colors",                 "IsTemplated": false             }         ],         {             "CategoryId": 3,             "CategoryDescriptor": "Security",             "TotalCount": 0,             "UnprocessedCount": 0,             "TotalSubsequentGrouping":20,             "UnprocessedSubsequentGrouping"=16,             "_links": [                 {                     "Rel": "eventcounter",                     "Href": "api/eventcounters/3",                     "IsTemplated": false                 },                 {                     "Rel": "categorycolor",                     "Href": "api/tables/categories/3/subtables/colors",                     "IsTemplated": false                 }             ]         }     ] }</pre> |
|--|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

## 7.5.2 Creating a Subscription for Event Counters (Obsolete)

Obsolete: Use channelized subscription instead.

|               |                                                                                                                                           |        |                                               |
|---------------|-------------------------------------------------------------------------------------------------------------------------------------------|--------|-----------------------------------------------|
| Resource Path | /sr/eventcounterssubscriptions/:connectionId                                                                                              |        |                                               |
| Category      | Public                                                                                                                                    |        |                                               |
| HTTP Method   | POST                                                                                                                                      |        |                                               |
| Parameters    | Property                                                                                                                                  | Type   | Description                                   |
|               | connectionId                                                                                                                              | string | Identifier of subscription<br><b>Required</b> |
| Response      | HTTP 200 OK                                                                                                                               |        |                                               |
| Examples      | <b>Request:</b><br>POST api/sr/eventcounterssubscriptions/448b3117-44ef-4d8f-baa2-7274aa3050af<br><br><b>Response:</b><br>HTTP/1.1 200 OK |        |                                               |



### 7.5.2.1 Push Notification

After subscribing for event counters, the current event counters will be sent in an initial notification. Afterwards, any change will be notified.

| Type     | Name (case sensitive)                                     |                  |                             |
|----------|-----------------------------------------------------------|------------------|-----------------------------|
| Hub      | 1. norisHub : recommended<br>2. eventCountersHub : legacy |                  |                             |
| Function | notifyEventCounters( data )                               |                  |                             |
|          | Property                                                  | Type             | Description                 |
|          | data                                                      | EventCounterList | See EventCounterList [→ 52] |

### 7.5.3 Creating Channelized Event Counters Subscription

If subscription status will be pushed through notifySubscriptionStatus function available at client side. EventCounter changes will be notified over notifyEventCounters [→57].

| Resource Path | /sr/eventcounterssubscriptions/channelize/:requestId/:connectionId                                                                                                                                                                                                                                                                                         |        |                                                                                                                                                                                                                                                    |
|---------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Category      | Public                                                                                                                                                                                                                                                                                                                                                     |        |                                                                                                                                                                                                                                                    |
| HTTP Method   | POST                                                                                                                                                                                                                                                                                                                                                       |        |                                                                                                                                                                                                                                                    |
| Parameters    | Property                                                                                                                                                                                                                                                                                                                                                   | Type   | Description                                                                                                                                                                                                                                        |
|               | requestId                                                                                                                                                                                                                                                                                                                                                  | String | Request Identifier of subscription hub connection.<br>RequestId is to be generated and provided by the client.<br>It is expected from client that client should use unique id for for each request.<br>Recommendation: Use GUID<br><b>Required</b> |
|               | connectionId                                                                                                                                                                                                                                                                                                                                               | String | Identifier of subscription hub connection<br><b>Required</b>                                                                                                                                                                                       |
| Response      | HTTP 200 OK                                                                                                                                                                                                                                                                                                                                                |        |                                                                                                                                                                                                                                                    |
| Examples      | <b>Request:</b><br>POST<br>api/sr/eventcounterssubscriptions/channelize/9565ca41-8556-4dbb-94cc-1b89451a5db5/448b3117-44ef-4d8f-baa2-7274aa3050af<br><b>Response:</b><br>HTTP/1.1 200 OK<br><b>notifySubscriptionStatus:</b><br>{<br>"ErrorCode":0,<br>"RequestId": "9565ca41-8556-4dbb-94cc-1b89451a5db5",<br>"RequestFor":"notifyEventCounters"<br>}<br> |        |                                                                                                                                                                                                                                                    |

### 7.5.3.1 Subscription Status Notification

After subscribing for event counters, the subscription status will be sent only once.

| Type     | Name (case sensitive)                                                                                                       |                       |                               |
|----------|-----------------------------------------------------------------------------------------------------------------------------|-----------------------|-------------------------------|
| Hub      | 1. norisHub : recommended<br>2. eventCountersHub : legacy                                                                   |                       |                               |
| Function | notifySubscriptionStatus(status)                                                                                            |                       |                               |
|          | Property                                                                                                                    | Type                  | Description                   |
|          | Status                                                                                                                      | Subscription Response | See Subscription Status [201] |
| Example  | <pre>{   "ErrorCode":0,   "RequestId": "9565ca41-8556-4dbb-94cc-1b89451a5db5",   "RequestFor":"notifyEventCounters" }</pre> |                       |                               |

### 7.5.4 Deleting a Subscription for Event Counters

Delete an existing subscription for event counters: the same procedure as creating event counters subscription [→ 32] but use DELETE as HTTP method.

### 7.5.5 Retrieving Event Suppressed objects count

Event Suppressed object returns the current count of all the suppressed object. For more details see the examples below.

|               |                                                                                                                                                    |
|---------------|----------------------------------------------------------------------------------------------------------------------------------------------------|
| Resource Path | /sr/eventcounters/suppressedobjects                                                                                                                |
| Category      | Public                                                                                                                                             |
| HTTP Method   | GET                                                                                                                                                |
| Response      | HTTP 200 OK                                                                                                                                        |
| Examples      | <b>Request:</b><br>GET<br>api/eventcounters/suppressedobjects?api_key=defaultadmin%3Aa<br><b>Response:</b><br><pre>{   "SuppressedCount":3 }</pre> |

### 7.5.6 Creating Channelized Event Suppressed Objects Subscription

It will subscribe the Suppressed Objects. Subscription status will be available at client side through notifySuppressedObjects function.

| Resource Path | /sr/eventcounterssubscriptions/channelize/suppressedobjects/{requestId}/{connectionId} |        |                                                                                                 |
|---------------|----------------------------------------------------------------------------------------|--------|-------------------------------------------------------------------------------------------------|
| Category      | Public                                                                                 |        |                                                                                                 |
| HTTP Method   | POST                                                                                   |        |                                                                                                 |
| Parameters    | Property                                                                               | Type   | Description                                                                                     |
|               | requestId                                                                              | String | Request Identifier of subscription hub connection.<br>RequestId is to be generated and provided |

|          |                                                                                                                                                                                                                                                                                                                                                                                    |        |                                                                                                                                                |
|----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|------------------------------------------------------------------------------------------------------------------------------------------------|
|          |                                                                                                                                                                                                                                                                                                                                                                                    |        | by the client.<br>It is expected from client that client should use unique id for each request.<br>Recommendation: Use GUID<br><b>Required</b> |
|          | connectionId                                                                                                                                                                                                                                                                                                                                                                       | String | Identifier of subscription hub connection<br><b>Required</b>                                                                                   |
| Response | HTTP 200 OK                                                                                                                                                                                                                                                                                                                                                                        |        |                                                                                                                                                |
| Examples | <b>Request:</b><br>POST<br>api/sr/eventcounterssubscriptions/channelize/suppressedobjects/0db77356-bc3f-0275-7262-c28ff19cfb73/8468d443-17b8-41c0-9e41-b12f45a07d22<br><b>Response:</b><br>HTTP/1.1 200 OK<br><br><b>notifySuppressedObjects:</b><br>{<br>"ErrorCode":0,<br>"RequestId":"c8ead472-10b8-0f85-b20e-eabd58bc7054",<br>"RequestFor":"notifySuppressedObjects"<br>}<br> |        |                                                                                                                                                |

### 7.5.6.1 Suppressed Objects Subscription Status Notification

After subscribing for Suppressed object, the subscription status will be sent only once.

|          |                                                                                                                            |                       |                               |
|----------|----------------------------------------------------------------------------------------------------------------------------|-----------------------|-------------------------------|
| Type     | Name (case sensitive)                                                                                                      |                       |                               |
| Hub      | norisHub : recommended                                                                                                     |                       |                               |
| Function | notifySuppressedObjectsSubscription (status)                                                                               |                       |                               |
|          | <b>Property</b>                                                                                                            | <b>Type</b>           | <b>Description</b>            |
|          | Status                                                                                                                     | Subscription Response | See Subscription Status [201] |
| Example  | {<br>"ErrorCode":0,<br>"RequestId": "9565ca41-8556-4dbb-94cc-1b89451a5db5",<br>"RequestFor":"notifyEventCounters"<br>}<br> |                       |                               |

### 7.5.7 Deleting a Subscription for Suppressed Objects

Deleting Suppressed objects subscriptions.

|               |                                                                 |             |                                               |
|---------------|-----------------------------------------------------------------|-------------|-----------------------------------------------|
| Resource Path | /sr/eventcounterssubscriptions/suppressedobjects/{connectionId} |             |                                               |
| Category      | Public                                                          |             |                                               |
| HTTP Method   | DELETE                                                          |             |                                               |
| Parameters    | <b>Property</b>                                                 | <b>Type</b> | <b>Description</b>                            |
|               | connectionId                                                    | string      | Identifier of subscription<br><b>Required</b> |

|          |                                                                                                                                                           |
|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|
| Response | HTTP 200 OK                                                                                                                                               |
| Examples | <b>Request:</b><br>DELETE api/sr/eventcounterssubscriptions/suppressedobjects/7cb4f8c2-d524-4532-9305-8fac31db175e<br><b>Response:</b><br>HTTP/1.1 200 OK |

## 7.6 System Browser Service

| Resource                    | Service Category | Requires Authentication | Requires SignalR (Push Notifications) |
|-----------------------------|------------------|-------------------------|---------------------------------------|
| /systembrowser              | Public           | x                       |                                       |
| /systembrowsersubscriptions | Public           | x                       | x                                     |

### 7.6.1 Retrieving a List of Views of a System

Retrieves list of Views in the current system. This interface have two optional query parameters systemId and viewId. If systemId is received as zero or if it is not provided then current systems id will be taken else received systemId will be taken for further operations. If viewId is provided then only the view, which have viewId same as that of in the request will be returned in the response. If viewId is not provided then all the views in the system will be returned in the response.

|               |                                                                                                                                                                                                                                                                          |             |                                                                                                                                     |
|---------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|-------------------------------------------------------------------------------------------------------------------------------------|
| Resource Path | /systembrowser                                                                                                                                                                                                                                                           |             |                                                                                                                                     |
| Category      | Public                                                                                                                                                                                                                                                                   |             |                                                                                                                                     |
| HTTP Method   | GET                                                                                                                                                                                                                                                                      |             |                                                                                                                                     |
| Parameters    | <b>Property</b>                                                                                                                                                                                                                                                          | <b>Type</b> | <b>Description</b>                                                                                                                  |
|               | systemId                                                                                                                                                                                                                                                                 | uint        | ID of a system<br>0 or Not Provided: All system will be considered<br>>= 1: Specific System will be considered                      |
|               | viewId                                                                                                                                                                                                                                                                   | uint        | ID of specific View to be retrieved<br>Not Provided : All views will be returned<br>> 0: Views will be returned for provided ViewId |
| Response      | HTTP 200 OK<br>An array of View objects see [→ 55]                                                                                                                                                                                                                       |             |                                                                                                                                     |
| Examples      | <b>Request:</b><br>GET api/systembrowser?systemId=1&viewId=10<br><b>Response:</b><br>[ { "SystemId": 1, "ViewId": 10, "Descriptor": "Application View", "Designation": "System1.ApplicationView", "Name": "ApplicationView", "SystemName": "System1", "ViewType": 0, } ] |             |                                                                                                                                     |

|  |                                                                                                                                                                                                                                 |
|--|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | <pre>         "_links": [           {             "Rel": "systembrowser",             "Href": "api/systembrowser/1/10/System1.ApplicationView",             "IsTemplated": false           }         ]       }     ]   ] </pre> |
|--|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

## 7.6.2 Retrieving a List of Browser Objects

Retrieve a list of browser objects for a specific view. To retrieve child objects of an object pass Designation of the parent object as node parameter.

|               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |             |                                                                          |
|---------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|--------------------------------------------------------------------------|
| Resource Path | /systembrowser/:systemId/:viewId/:node                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |             |                                                                          |
| Category      | Public                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |             |                                                                          |
| HTTP Method   | GET                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |             |                                                                          |
| Parameters    | <b>Property</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | <b>Type</b> | <b>Description</b>                                                       |
|               | systemId                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | uint        | ID of a system<br><b>Required</b>                                        |
|               | viewId                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | uint        | ID of a view<br><b>Required</b>                                          |
|               | node                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | string      | Designation of a parent browser object (Node or View)<br><b>Required</b> |
| Response      | HTTP 200 OK<br>An array of BrowserObject objects see [→ 55]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |             |                                                                          |
| Examples      | <b>Request:</b><br>GET api/systembrowser/1/9/System1.ManagementView<br><b>Response:</b><br><pre> [   {     "HasChild": true,     "SystemId": 1,     "ViewId": 9,     "Name": "ManagementView",     "Descriptor": "Project",     "Designation": "System1.ManagementView:ManagementView",     "ObjectId": "System1:GMS_Aggregator_5",     "Attributes": {       "Alias": "Project Alias",       "DefaultProperty": "StatusPropagation.AggregatedSummaryStatus",       "ObjectId": "GMS_Aggregator_5",       "DisciplineDescriptor": "Management System",       "DisciplineId": 0,       "SubDisciplineDescriptor": "System Settings",       "SubDisciplineId": 4,       "TypeDescriptor": "View Element",       "TypeId": 8000,       "SubTypeDescriptor": "System Folder",       "SubTypeId": 8014,       "ManagedType": 56     }   } ] </pre> |             |                                                                          |

|  |                                                                                                                                                                                                                                                                                                                                                                                                                           |
|--|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | <pre>     },     "Location": "System1.Management View:Project",     "_links": [       {         "Rel": "systembrowser",         "Href": "api/systembrowser/1/9/System1.ManagementView%3aManagement View",         "IsTemplated": false       },       {         "Rel": "properties",         "Href": "api/properties/System1%3aGMS_Aggregator_5?requestType=0",         "IsTemplated": false       }     ]   } } ] </pre> |
|--|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

### 7.6.3 Searching for Browser Objects

Search for browser objects in a provided system.

|               |                                  |             |                                                                                                                              |
|---------------|----------------------------------|-------------|------------------------------------------------------------------------------------------------------------------------------|
| Resource Path | /systembrowser/:systemId:/viewId |             |                                                                                                                              |
| Category      | Public                           |             |                                                                                                                              |
| HTTP Method   | GET                              |             |                                                                                                                              |
| Parameters    | <b>Property</b>                  | <b>Type</b> | <b>Description</b>                                                                                                           |
|               | systemId                         | uint        | ID of a system<br><b>Required</b>                                                                                            |
|               | viewId                           | uint        | ID of the view in which search needs to be performed                                                                         |
|               | searchString                     | string      | String to be searched for<br><b>Required</b>                                                                                 |
|               | searchOption                     | int         | See SearchOption [→ 37]                                                                                                      |
|               | caseSensitive                    | bool        | Only valid for search options 0 or 1                                                                                         |
|               | groupByParent                    | bool        | If true, group search results by parent node                                                                                 |
|               | size                             | int         | Number of browser objects per page                                                                                           |
|               | page                             | int         | Current page number (default: 1)                                                                                             |
|               | disciplineFilter                 | string      | Discipline and subdiscipline filters<br>See Discipline- and Objecttype Filters [→ 657]                                       |
|               | objecttypeFilter                 | string      | Type and subtype filters<br>See Discipline- and Objecttype Filters [→ 657]                                                   |
|               | aliasFilter                      | string      | Case-sensitive alias filter                                                                                                  |
|               | alarmSuppression                 | Boolean     | When a Boolean is provided, will search for objects where the alarm suppression is set to the Boolean value (true or false). |
|               | outOfScan                        | Boolean     | When a Boolean is provided, will search for objects where the out of scan flag is set to the Boolean value (true or false).  |

|          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Response | <p>HTTP 200 OK</p> <p>A Page&lt;Nodes, BrowserObject&gt;-object (see Page&lt;Name, Type [→ 50]&gt;)</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Examples | <p><b>Request::</b></p> <pre>GET /api/systembrowser/1?searchstring=*view*&amp;searchOption=1&amp;caseSensitive=false&amp;groupByParent=false</pre> <p><b>Response:</b></p> <pre>{   "Total": 1,   "Page": 1,   "Size": 100000,   "Nodes": [     {       "SystemId": 1,       "ViewId": 9,       "Name": "TxG_BA_BACnetAkedTransitionsEnum",       "Descriptor": "(0000) BACnet Aaked Transitions Enum list",       "Designation": "System1.ManagementView:ManagementView.SystemSettings.Libr aries.Headquarter (HQ).BA.Device.BA_Device_PX_HQ_2.TextGrou ps.TxG_BA_BACnetAkedTransitionsEnum",       "ObjectId": "System1:TxG_BA_BACnetAkedTransitionsEnum",       "Attributes": {         "DefaultProperty": "Read_Only",         "ObjectId": "TxG_BA_BACnetAkedTransitionsEnum",         "DisciplineDescriptor": "Management System",         "DisciplineId": 0,         "SubDisciplineDescriptor": "Unassigned",         "SubDisciplineId": 0,         "TypeDescriptor": "Library Element",         "TypeId": 4600,         "SubTypeDescriptor": "Text Group",         "SubTypeId": 4609,         "ManagedType": 20       },       "Location": "System1.Management View:Project.System Settings.Libraries.Headquarter.BA.Device.Desigo PX.Text Groups.(0000) BACnet Aaked Transitions Enum list",       "_links": []     }   ],   "_links": [     {       "Rel": "page",       "Href": "api/systembrowser/1?searchString=*view*&amp;searchOption=1&amp;ca seSensitive=False&amp;groupByParent=False&amp;size=100000&amp;page={pa ge}&amp;disciplineFilter=null&amp;objectTypeFilter=null&amp;aliasFilde r=",       "IsTemplated": true     }   ] }</pre> |

### 7.6.3.1 Searching with Wildcards

Two wildcard characters are supported in System Browser—the asterisk (\*) and the question mark (?). Each functions differently. The asterisk wildcard serves as a placeholder for zero or more characters. The question mark wildcard serves as a placeholder for exactly one character only. Therefore, each wildcard serves different purposes.

\*: Allows you to add zero or more characters to your search criteria. For example, "a\*" matches and displays, "a", "ab", "abc", and "abcd".

?: Allows you to add one character to your search. For example, "ab?" matches and returns "abc" but does not match or return "ab" and "abcd".



### 7.6.3.2 SearchOption

**searchOption** must have one of the following values.

| searchOption | Description                                |
|--------------|--------------------------------------------|
| 0            | Search in full name (Designation); default |
| 1            | Search in full description (Location)      |
| 2            | Search in ObjectId                         |
| 3            | Search in Alias                            |

### 7.6.3.3 Discipline and Objecttype Filters

This JSON string has two level filtering options—, group level (e.g. disciplines/types) and subgroup level (e.g. subdisciplines/subtypes). In case if SubGroupID/SubObjectTypeID is not provided along with GroupID/ObjectTypeID respectively then all the SubGroups/SubObjectTypes from the metioned Group/ObjectType will be taken for filtering.

- **Example**

```
{ "<GroupID>": [ <SubGroupID>, . . . , <SubGroupID> ], "<GroupID>": [ <SubGroupID>, . . . , <SubGroupID> ] }
```

- **Concrete example**

```
{ "0": [ 0, 1, 2, 4 ], "20": [ 0, 21, 22 ] }
```

### 7.6.4 Search multiple Object Ids

Along with normal searches available, This API also provides bulk search option for searching multiple Object Ids.

|               |                                                                                                                                                                                                                                                  |          |                                                      |
|---------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|------------------------------------------------------|
| Resource Path | /systembrowser/:systemId:viewId                                                                                                                                                                                                                  |          |                                                      |
| Category      | Public                                                                                                                                                                                                                                           |          |                                                      |
| HTTP Method   | POST                                                                                                                                                                                                                                             |          |                                                      |
| Parameters    | systemId                                                                                                                                                                                                                                         | uint     | ID of a system<br><b>Required</b>                    |
|               | viewId                                                                                                                                                                                                                                           | uint     | ID of the view in which search needs to be performed |
|               | groupByParent                                                                                                                                                                                                                                    | bool     | If true, group search results by parent node         |
| Body          | Array of ObjectId                                                                                                                                                                                                                                | String[] | Array of object ids to be searched                   |
| Response      | HTTP 200 OK Array of Multiple Object Ids search Browser Objects see [→ 55]<br>An array of searched browser objects                                                                                                                               |          |                                                      |
| Examples      | <b>Request:</b><br>GET<br>api/systembrowser/1/9?searchOption=2&caseSensitive=false&groupByParent=false<br><br><b>Post Data:</b><br>[<br>"System1:ManagementView_System_Settings",<br>"System1:JournalingRootFolder"<br>]<br><br><b>Response:</b> |          |                                                      |

```
[
  {
    "ErrorCode": 0,
    "ObjectId": "System1:ManagementView_System_Settings",
    "Nodes": [
      {
        "SystemId": 1,
        "ViewId": 9,
        "Name": "SystemSettings",
        "Descriptor": "System Settings",
        "Designation":
"System1.ManagementView:ManagementView.SystemSettings",
        "ObjectId":
"System1:ManagementView_System_Settings",
        "Attributes": {
          "DefaultProperty":
"StatusPropagation.AggregatedSummaryStatus",
          "ObjectId": "ManagementView_System_Settings",
          "DisciplineDescriptor": "Management System",
          "DisciplineId": 0,
          "SubDisciplineDescriptor": "System Settings",
          "SubDisciplineId": 4,
          "TypeDescriptor": "View Element",
          "TypeId": 8000,
          "SubTypeDescriptor": "System Folder",
          "SubTypeId": 8014,
          "ManagedType": 153
        },
        "Location": "System1.Management
View:Project.System Settings",
        "_links": []
      }
    ]
  },
  {
    "ErrorCode": 0,
    "ObjectId": "System1:JournalingRootFolder",
    "Nodes": [
      {
        "SystemId": 1,
        "ViewId": 9,
        "Name": "JournalingRootFolder",
        "Descriptor": "Journaling",
        "Designation":
"System1.ManagementView:ManagementView.SystemSettings.Jour
nalingRootFolder",
        "ObjectId": "System1:JournalingRootFolder",
        "Attributes": {
          "DefaultProperty":
"StatusPropagation.AggregatedSummaryStatus",
          "ObjectId": "JournalingRootFolder",
          "DisciplineDescriptor": "Management System",
          "DisciplineId": 0,
          "SubDisciplineDescriptor": "System Settings",
          "SubDisciplineId": 4,
          "TypeDescriptor": "Journaling",
          "TypeId": 4500,
          "SubTypeDescriptor": "Journaling Printer",
          "SubTypeId": 4501,

```

|  |                                                                                                                                                                 |
|--|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | <pre>         "ManagedType": 63       },       "Location": "System1.Management View:Project.System Settings.Journaling",       "_links": []     }   ] } ]</pre> |
|--|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|

### 7.6.5 Creating a Subscription for Changes in the System Browser views (Obsolete)

Obsolete: Use channelized subscription instead.

In System browser views subscription, client can provide one or many designations for which client is interested for changes. This basically provides the feature of subtree subscription. All the nodes matching (or underneath) the provided designation will be subscribed for the changes. And any change happening under the provided designation will be notified to the subscribed clients.

|               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |          |                                                                 |
|---------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|-----------------------------------------------------------------|
| Resource Path | /sr/SystemBrowserSubscriptions/:connectionId/                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |          |                                                                 |
| Category      | Public                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |          |                                                                 |
| HTTP Method   | POST                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |          |                                                                 |
| Parameters    | connectionId                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | string   | Identifier of subscription<br><b>Required</b>                   |
| Body          | designationsList                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | String[] | JSON Array of designation(s) for which client want to subscribe |
| Response      | SystemBrowserSubscriptionKeyRepresentation object See [→ 78]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |          |                                                                 |
| Examples      | <p><b>Request:</b></p> <p>POST /api/sr/SystemBrowserSubscriptions/01ac873d-23ae-49cb-86bc-6764d0b38904</p> <p><b>Post Data:</b></p> <p>["System1.ApplicationView", "System1.ManagementView"]</p> <p><b>Response:</b></p> <pre> {   "Designations": [     "System1.ApplicationView",     "System1.ManagementView"   ],   "Key": 2,   "ErrorCode": 0,   "_links": [     {       "Rel": "unsubscribe",       "Href": "api/sr/SystemBrowserSubscriptions/01ac873d-23ae-49cb-86bc-6764d0b38904?subscriptionKey=%5b2%5d",       "IsTemplated": false     }   ] }</pre> |          |                                                                 |

### 7.6.5.1 Push Notification

After subscribing for changes in the System Browser views, for any changes in the System Browser views a notification will be sent.

| Type     | Name (case sensitive)                                     |                           |                                               |
|----------|-----------------------------------------------------------|---------------------------|-----------------------------------------------|
| Hub      | 1. norisHub : recommended<br>2. systemBrowserHub : legacy |                           |                                               |
| Function | notifySystemBrowserChanges( data )                        |                           |                                               |
|          | Property                                                  | Type                      | Description                                   |
|          | data                                                      | SystemBrowserSubscription | See System Browser subscription object [→ 78] |

### 7.6.5.2 Distributed System support for subscription

- For distributed systems subscription support, A filter "\*" is explicitly supported for the System part. That means client can specify paths 'for all systems' using \* as shown below.  
 "\*.ApplicationView" – this will subscribe Application View of all the systems added in the distributed system.
- And if value of designation array is not provided then, no such filtering as described above will be applied. In case of NULL, client will receive notifications of all the changes in the all of the systems.

## 7.6.6 Creating Channelized Systembrowser Subscription

If subscription status will be pused through notifySubscriptionStatus function available at client side. Rest of the push notificaitons will be notified over notifySystemBrowserChanges.

| Resource Path | /sr/SystemBrowserSubscriptions /channelize/:requestId/:connectionId                                                                                                    |        |                                                                                                                                                                                                                                                    |
|---------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Category      | Public                                                                                                                                                                 |        |                                                                                                                                                                                                                                                    |
| HTTP Method   | POST                                                                                                                                                                   |        |                                                                                                                                                                                                                                                    |
| Parameters    | Property                                                                                                                                                               | Type   | Description                                                                                                                                                                                                                                        |
|               | requestId                                                                                                                                                              | String | Request Identifier of subscription hub connection.<br>RequestId is to be generated and provided by the client.<br>It is expected from client that client should use unique id for for each request.<br>Recommendation: Use GUID<br><b>Required</b> |
|               | connectionId                                                                                                                                                           | String | Identifier of subscription hub connection<br><b>Required</b>                                                                                                                                                                                       |
| Response      | HTTP 200 OK                                                                                                                                                            |        |                                                                                                                                                                                                                                                    |
| Examples      | <b>Request:</b><br>POST<br>api/sr/SystemBrowserSubscriptions/channelize/9565ca41-8556-4dbb-94cc-1b89451a5db5/448b3117-44ef-4d8f-baa2-7274aa3050af<br><b>Post Data:</b> |        |                                                                                                                                                                                                                                                    |

|  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|--|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | ["System1.ApplicationView","System1.ManagementView"]<br><br><b>Response:</b><br>HTTP/1.1 200 OK<br><br><b>notifySubscriptionStatus :</b><br><pre>{   "Designations": [     "System1.ApplicationView",     "System1.ManagementView"   ],   "Key": 2,   "ErrorCode": 0,   "RequestId": "9565ca41-8556-4dbb-94cc-1b89451a5db5",   "_links": [     {       "Rel": "unsubscribe",       "Href": "api/sr/SystemBrowserSubscriptions/01ac873d-23ae-49cb-86bc-6764d0b38904?subscriptionKey=%5b2%5d",       "IsTemplated": false     }   ] }</pre> |
|--|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

### 7.6.6.1 Subscription Status Notification

After subscribing for system browser, the subscription status will be sent only once.

| Type     | Name (case sensitive)                                                                                                                                                                                                                                                                                                                                                                                                                       |                       |                                            |
|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|--------------------------------------------|
| Hub      | 1. norisHub : recommended<br>2. systemBrowserHub : legacy                                                                                                                                                                                                                                                                                                                                                                                   |                       |                                            |
| Function | notifySubscriptionStatus(status)                                                                                                                                                                                                                                                                                                                                                                                                            |                       |                                            |
|          | Property                                                                                                                                                                                                                                                                                                                                                                                                                                    | Type                  | Description                                |
|          | Status                                                                                                                                                                                                                                                                                                                                                                                                                                      | Subscription Response | System Browser subscription object [→ 204] |
| Example  | <pre>{   "Designations": [     "System1.ApplicationView",     "System1.ManagementView"   ],   "Key": 2,   "ErrorCode": 0,   "RequestId": "9565ca41-8556-4dbb-94cc-1b89451a5db5",   "RequestFor": "notifySystemBrowserChanges",   "_links": [     {       "Rel": "unsubscribe",       "Href": "api/sr/SystemBrowserSubscriptions/01ac873d-23ae-49cb-86bc-6764d0b38904?subscriptionKey=%5b2%5d",       "IsTemplated": false     }   ] }</pre> |                       |                                            |

## 7.6.7 Deleting a Subscription for Changes in the System Browser views

### 7.6.7.1 Passing subscription keys in query parameter

|               |                                                                                                                                                                                                              |                  |                                                                        |
|---------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|------------------------------------------------------------------------|
| Resource Path | /sr/systemBrowsersubscriptions/:connectionId                                                                                                                                                                 |                  |                                                                        |
| Category      | Public                                                                                                                                                                                                       |                  |                                                                        |
| HTTP Method   | DELETE                                                                                                                                                                                                       |                  |                                                                        |
| Parameters    | <b>Property</b>                                                                                                                                                                                              | <b>Type</b>      | <b>Description</b>                                                     |
|               | connectionId                                                                                                                                                                                                 | String           | Identifier of subscription<br><b>Required</b>                          |
|               | subscriptionKey                                                                                                                                                                                              | Array of integer | Querystring parameter, which holds the JSON array of subscription keys |
| Response      | UnsubscriptionStatus See [→ 72]<br>Unsubscribe object which contains Key and ErrorCode based on whether the request is successful or not.                                                                    |                  |                                                                        |
| Examples      | <b>Request:</b><br>DELETE /api/sr/SystemBrowserSubscriptions/01ac873d-23ae-49cb-86bc-6764d0b38904?subscriptionKey=%5b2%5d<br><br><b>Response:</b><br><pre>[   {     "ErrorCode": 0,     "Key": 2   } ]</pre> |                  |                                                                        |

### 7.6.7.2 Passing subscription keys in request body

|               |                                                                                                                                                                                         |             |                                               |
|---------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|-----------------------------------------------|
| Resource Path | /sr/systemBrowsersubscriptions/withbody/:connectionId                                                                                                                                   |             |                                               |
| Category      | Public                                                                                                                                                                                  |             |                                               |
| HTTP Method   | DELETE                                                                                                                                                                                  |             |                                               |
| Parameters    |                                                                                                                                                                                         |             |                                               |
| Body          | Array of Subscription keys                                                                                                                                                              |             |                                               |
| Query         | <b>Property</b>                                                                                                                                                                         | <b>Type</b> | <b>Description</b>                            |
|               | connectionId                                                                                                                                                                            | String      | Identifier of subscription<br><b>Required</b> |
| Response      | UnsubscriptionStatus See [→ 72]<br>Unsubscribe object which contains Key and ErrorCode based on whether the request is successful or not.                                               |             |                                               |
| Examples      | <b>Request:</b><br>DELETE /api/sr/SystemBrowserSubscriptions/01ac873d-23ae-49cb-86bc-6764d0b38904<br><br><b>Body Data:</b><br><pre>[   "2" ]</pre> <b>Response:</b><br><pre>[   {</pre> |             |                                               |

|  |                                                                                                   |
|--|---------------------------------------------------------------------------------------------------|
|  | <pre>                 "ErrorCode": 0,                 "Key": 2             }         ]     </pre> |
|--|---------------------------------------------------------------------------------------------------|

## 7.6.8 Update CNS Node Description

|               |                                                                                                                                             |                                  |                                                                                                                                                                    |
|---------------|---------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Resource Path | /src/systembrowser/{designation}                                                                                                            |                                  |                                                                                                                                                                    |
| Category      | Internal                                                                                                                                    |                                  |                                                                                                                                                                    |
| HTTP Method   | GET                                                                                                                                         |                                  |                                                                                                                                                                    |
| Parameters    | <b>Property</b>                                                                                                                             | <b>Type</b>                      | <b>Description</b>                                                                                                                                                 |
|               | designation                                                                                                                                 | string                           | Full view name;<br>Designation id of node in systembrowser tree<br>(Recommendation: Double UrlEncode parameter prior to making it to part of URL.) <b>Required</b> |
|               | cnsinformation                                                                                                                              | System-BrowserEditRepresentation | Descriptor text for node Object                                                                                                                                    |
| Response      | HTTP 200 OK                                                                                                                                 |                                  |                                                                                                                                                                    |
| Examples      | <b>Request:</b><br>api/systembrowser/System1.ApplicationView%3AAApplicationView.Documents?api_key=defaultadmin%3Aa'<br><b>Response: 200</b> |                                  |                                                                                                                                                                    |

## 7.7 Value Service

| Resource             | Service Category | Requires Authentication | Requires SignalR (Push Notifications) |
|----------------------|------------------|-------------------------|---------------------------------------|
| /values              | Public           | x                       |                                       |
| /valuessubscriptions | Public           | x                       | x                                     |

### 7.7.1 Retrieving a Value of an Object or Property

If no property is specified but only an object, the value of the default property is returned.

|               |                              |             |                                                                                                                                                                                                                                            |
|---------------|------------------------------|-------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Resource Path | /values/{objectOrPropertyId} |             |                                                                                                                                                                                                                                            |
| Category      | Public                       |             |                                                                                                                                                                                                                                            |
| HTTP Method   | GET                          |             |                                                                                                                                                                                                                                            |
| Parameters    | <b>Property</b>              | <b>Type</b> | <b>Description</b>                                                                                                                                                                                                                         |
|               | objectOrPropertyId           | string      | Address of object or property<br><b>Required</b>                                                                                                                                                                                           |
|               | readMaxAge                   | int         | Optional query parameter with default value as null;<br><br>When the value is 'older' than this number of milliseconds, it will first be re-read at the driver (0 implies always re-read). A negative value disables driver re-reading and |

|          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |  |                                              |
|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|----------------------------------------------|
|          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |  | always returns the value from process image. |
| Response | HTTP 200 OK<br>An array of ValueDetails objects [→62]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |  |                                              |
| Examples | <p><b>Request 1: Object Id +Property</b></p> <p>GET /api/values/GmsDevice_2_1504_4194306.Present_Value</p> <p><b>Response:</b></p> <pre>[   {     "DataType": "BasicFloat",     "Value": {       "Value": "35",       "Quality": "9439544818971182337",       "QualityGood": true,       "Timestamp": "2017-12-20T08:56:06.904Z"     },     "OriginalObjectOrPropertyId": "GmsDevice_2_1504_4194306.Present_Value",     "ObjectId": "System1:GmsDevice_2_1504_4194306",     "PropertyName": "Present_Value",     "AttributeId": "System1:GmsDevice_2_1504_4194306.Present_Value:online.._value",     "ErrorCode": 0,     "IsArray": false   } ]</pre> <p><b>Request 2: Designation + Property</b></p> <p>GET /api/values/System1.ManagementView:ManagementView.FieldNetworks.TestNet.Hardware.Devices_1504.Local_IO.AO_2;.Present_Value</p> <p><b>Response:</b></p> <pre>[   {     "DataType": "BasicFloat",     "Value": {       "Value": "35",       "Quality": "9439544818971182337",       "QualityGood": true,       "Timestamp": "2017-12-20T08:58:05.037Z"     },     "OriginalObjectOrPropertyId": "System1.ManagementView:ManagementView.FieldNetworks.TestNet.Hardware.Devices_1504.Local_IO.AO_2;.Present_Value",     "ObjectId": "System1:GmsDevice_2_1504_4194306",     "PropertyName": "Present_Value",     "AttributeId": "System1:GmsDevice_2_1504_4194306.Present_Value:online.._value",     "ErrorCode": 0,     "IsArray": false   } ]</pre> <p><b>Request 3: Object Id +Property</b></p> <p>GET /api/values/GmsDevice_2_1504_4194306.Present_Value?readMaxAge=-1</p> <p><b>Response:</b></p> <pre>[   {</pre> |  |                                              |



|  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|--|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | <pre>         "DataType": "BasicFloat",         "Value": {             "Value": "35",             "Quality": "9439544818971182337",             "QualityGood": true,             "Timestamp": "2017-12-20T08:56:06.904Z"         },         "OriginalObjectOrPropertyId": "GmsDevice_2_1504_4194306.Present_Value",         "ObjectId": "System1:GmsDevice_2_1504_4194306",         "PropertyName": "Present_Value",         "AttributeId": "System1:GmsDevice_2_1504_4194306.Present_Value:_online..._value",         "ErrorCode": 0,         "IsArray": false     } ]</pre> |
|--|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

## 7.7.2 Retrieving Values for a List of Objects or Properties

|               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |     |                                                                                                                                                                                                                                                                                      |
|---------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Resource Path | /values                                                                                                                                                                                                                                                                                                                                                                                                                                                               |     |                                                                                                                                                                                                                                                                                      |
| Category      | Public                                                                                                                                                                                                                                                                                                                                                                                                                                                                |     |                                                                                                                                                                                                                                                                                      |
| HTTP Method   | POST                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |     |                                                                                                                                                                                                                                                                                      |
| Parameters    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |     |                                                                                                                                                                                                                                                                                      |
| Body          | Array of ObjectOrPropertyId                                                                                                                                                                                                                                                                                                                                                                                                                                           |     |                                                                                                                                                                                                                                                                                      |
| Query         | readMaxAge                                                                                                                                                                                                                                                                                                                                                                                                                                                            | int | <p>Optional parameter: default value as null;</p> <p>When the value is 'older' than this number of milliseconds, it will first be re-read at the driver (0 implies always re-read). A negative value disables driver re-reading and always returns the value from process image.</p> |
| Response      | <p>HTTP 200 OK</p> <p>An array of ValueDetails objects [→62]</p>                                                                                                                                                                                                                                                                                                                                                                                                      |     |                                                                                                                                                                                                                                                                                      |
| Examples      | <p><b>Request:</b></p> <p>POST /api/values</p> <p><b>Post Data:</b></p> <pre>[     "GmsDevice_2_1504_4194306.Present_Value",     "System1:GmsDevice_2_1504_4194306.Present_Value",     "System1.ManagementView:ManagementView.FieldNetworks.TestNet.Hardware.Devices_1504.Local_IO.AO_2;.Present_Value"     //Designation+Property ]</pre> <p><b>Response:</b></p> <pre>[     {         "DataType": "BasicFloat",         "Value": {             "Value": "35",</pre> |     |                                                                                                                                                                                                                                                                                      |

```

        "Quality": "9439544818971181313",
        "QualityGood": true,
        "Timestamp": "2017-12-20T08:31:40.132Z"
    },
    "OriginalObjectOrPropertyId": "GmsDevice_2_1504_4194306.Present_Value",
    "ObjectId": "System1:GmsDevice_2_1504_4194306",
    "PropertyName": "Present_Value",
    "AttributeId": "System1:GmsDevice_2_1504_4194306.Present_Value:_online.._value",
    "ErrorCode": 0,
    "IsArray": false
},
{
    "DataType": "BasicFloat",
    "Value": {
        "Value": "35",
        "Quality": "9439544818971181313",
        "QualityGood": true,
        "Timestamp": "2017-12-20T08:31:40.132Z"
    },
    "OriginalObjectOrPropertyId": "System1:GmsDevice_2_1504_4194306.Present_Value",
    "ObjectId": "System1:GmsDevice_2_1504_4194306",
    "PropertyName": "Present_Value",
    "AttributeId": "System1:GmsDevice_2_1504_4194306.Present_Value:_online.._value",
    "ErrorCode": 0,
    "IsArray": false
},
{
    "DataType": "BasicFloat",
    "Value": {
        "Value": "35",
        "Quality": "9439544818971181313",
        "QualityGood": true,
        "Timestamp": "2017-12-20T08:31:40.132Z"
    },
    "OriginalObjectOrPropertyId": "System1.ManagementView:ManagementView.FieldNetworks.TestNet.Hardware.Devices_1504.Locall_IO.AO_2;.Present_Value",
    "ObjectId": "System1:GmsDevice_2_1504_4194306",
    "PropertyName": "Present_Value",
    "AttributeId": "System1:GmsDevice_2_1504_4194306.Present_Value:_online.._value",
    "ErrorCode": 0,
    "IsArray": false
}
}
]

```

**Request 2:**

POST /api/values?readMaxAge=-1

**Post Data:**

```

[
    "GmsDevice_2_1504_4194306.Present_Value"
]

```

**Response:**

```

[
    {

```

|  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|--|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | <pre>         "DataType": "BasicFloat",         "Value": {             "Value": "35",             "Quality": "9439544818971182337",             "QualityGood": true,             "Timestamp": "2017-12-20T08:56:06.904Z"         },         "OriginalObjectOrPropertyId": "GmsDevice_2_1504_4194306.Present_Value",         "ObjectId": "System1:GmsDevice_2_1504_4194306",         "PropertyName": "Present_Value",         "AttributeId": "System1:GmsDevice_2_1504_4194306.Present_Value:_online..._value",         "ErrorCode": 0,         "IsArray": false     } ]</pre> |
|--|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

### 7.7.3 Creating a Subscription for a Change of a Value (Obsolete)

Obsolete: Use channelized subscription instead.

|               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |             |                                               |
|---------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|-----------------------------------------------|
| Resource Path | /sr/valuessubscriptions/:connectionId                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |             |                                               |
| Category      | Public                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |             |                                               |
| HTTP Method   | POST                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |             |                                               |
| Parameters    | <b>Property</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | <b>Type</b> | <b>Description</b>                            |
|               | connectionId                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | string      | Identifier of subscription<br><b>Required</b> |
|               | detailsRequired                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | bool        |                                               |
| Body          | Array of ObjectOrPropertyId                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |             |                                               |
| Response      | HTTP 200 OK<br>An array of Subscription objects [→56]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |             |                                               |
| Examples      | <p><b>Request:</b><br/>                     POST /api/sr/valuessubscriptions/977a6389-980c-41ca-84ab-2518fd0ca41a</p> <p><b>Post Data:</b><br/> <pre>[     "GmsDevice_2_1504_4194306.Present_Value",     "System1:GmsDevice_2_1504_4194306.Present_Value",     "System1.ManagementView:ManagementView.FieldNetworks.TestNet.Hardware.Devices_1504.Local_IO.AO_2;.Present_Value"//Designation+Property ]</pre></p> <p><b>Response:</b><br/> <pre>[     {         "Key": 1,         "PropertyId": "System1:GmsDevice_2_1504_4194306.Present_Value",         "ErrorCode": 0,         "_links": [             {</pre></p> |             |                                               |

|  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|--|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | <pre>                 "Rel": "unsubscribe",                 "Href": "api/sr/valuessubscriptions/4999fe9e-05c9-41a8-bc82-9ade768c3ced?subscriptionKey=%5b1%5d",                 "IsTemplated": false             }         ],         {             "Key": 2,             "PropertyId": "System1:GmsDevice_2_1504_4194306.Present_Value",             "ErrorCode": 0,             "_links": [                 {                     "Rel": "unsubscribe",                     "Href": "api/sr/valuessubscriptions/4999fe9e-05c9-41a8-bc82-9ade768c3ced?subscriptionKey=%5b2%5d",                     "IsTemplated": false                 }             ]         },         {             "Key": 3,             "PropertyId": "System1:GmsDevice_2_1504_4194306.Present_Value",             "ErrorCode": 0,             "_links": [                 {                     "Rel": "unsubscribe",                     "Href": "api/sr/valuessubscriptions/4999fe9e-05c9-41a8-bc82-9ade768c3ced?subscriptionKey=%5b3%5d",                     "IsTemplated": false                 }             ]         }     ] </pre> |
|--|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

### 7.7.3.1 Push Notification

After subscribing for changes of a value, the current value(s) will be sent in an initial notification. Afterwards, any change will be notified.

| Type     | Name (case sensitive)                              |                       |                        |
|----------|----------------------------------------------------|-----------------------|------------------------|
| Hub      | 1. norisHub : recommended<br>2. valuesHub : legacy |                       |                        |
| Function | notifyValues( data )                               |                       |                        |
|          | Property                                           | Type                  | Description            |
|          | data                                               | Array of ValueDetails | See ValueDetails [→62] |

## 7.7.4 Creating Channelized Value Subscription

If channelized subscription is done ,subscription status will be pushed through notifySubscriptionStatus function available at client side. Value changes will be notified over notifyValues[→76].

|               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |             |                                                                                                                                                                                                                                                    |
|---------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Resource Path | /sr/valuessubscriptions/channelize/:requestId/:connectionId                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |             |                                                                                                                                                                                                                                                    |
| Category      | Public                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |             |                                                                                                                                                                                                                                                    |
| HTTP Method   | POST                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |             |                                                                                                                                                                                                                                                    |
| Parameters    | <b>Property</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | <b>Type</b> | <b>Description</b>                                                                                                                                                                                                                                 |
|               | requestId                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | String      | Request Identifier of subscription hub connection.<br>RequestId is to be generated and provided by the client.<br>It is expected from client that client should use unique id for for each request.<br>Recommendation: Use GUID<br><b>Required</b> |
|               | connectionId                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | String      | Identifier of subscription hub connection<br><b>Required</b>                                                                                                                                                                                       |
| Response      | HTTP 200 OK                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |             |                                                                                                                                                                                                                                                    |
| Examples      | <p><b>Request:</b></p> <pre>POST /api/sr/valuessubscriptions/channelize/9565ca41-8556-4dbb-94cc-1b89451a5db5/977a6389-980c-41ca-84ab-2518fd0ca41a</pre> <p><b>Post Data:</b></p> <pre>[   "GmsDevice_2_1504_4194306.Present_Value",   "System1:GmsDevice_2_1504_4194306.Present_Value",   "System1.ManagementView:ManagementView.FieldNetworks.TestNet.Hardware.Devices_1504.Local_IO.AO_2;.Present_Value"   //Designation+Property ]</pre> <p><b>Response:</b></p> <pre>HTTP/1.1 200 OK</pre> <p><b>notifySubscriptionStatus :</b></p> <p>Notification 1:</p> <pre>{   "Key":1,   "PropertyId":"System1:GmsDevice_2_1504_4194306.Present_Value",   "ErrorCode":0,   objectOrPropertyId: "GmsDevice_2_1504_4194306.Present_Value",   "RequestId": "9565ca41-8556-4dbb-94cc-1b89451a5db5",   "_links":[     {       "Rel":"unsubscribe",       "Href":"/api/sr/valuessubscriptions/4999fe9e-05c9-41a8-bc82-9ade768c3ced?subscriptionKey=%5b1%5d",       "IsTemplated":false     }   ] }</pre> |             |                                                                                                                                                                                                                                                    |

|  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|--|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | <pre> Notification 2: {   "Key":2,   "PropertyId":"System1:GmsDevice_2_1504_4194306.Present_Value",   "ErrorCode":0,   objectOrPropertyId: "System1:GmsDevice_2_1504_4194306.Present_Value",   "RequestId": "9565ca41-8556-4dbb-94cc-1b89451a5db5",   "_links":[     {       "Rel":"unsubscribe",       "Href":"api/sr/valuessubscriptions/4999fe9e-05c9-41a8-bc82-9ade768c3ced?subscriptionKey=%5b2%5d",       "IsTemplated":false     }   ] }  Notification 3: {   "Key":3,   "PropertyId":"System1:GmsDevice_2_1504_4194306.Present_Value",   objectOrPropertyId:   "System1.ManagementView:ManagementView.FieldNetworks.TestNet.Hardware.Devices_1504.Local_IO.AO_2;.Present_Value",   "RequestId": "9565ca41-8556-4dbb-94cc-1b89451a5db5"   "ErrorCode":0,   "_links":[     {       "Rel":"unsubscribe",       "Href":"api/sr/valuessubscriptions/4999fe9e-05c9-41a8-bc82-9ade768c3ced?subscriptionKey=%5b3%5d",       "IsTemplated":false     }   ] }  <b>notifyValues:</b> Notification 1: [[   {"DataType":"BasicFloat",   "Value":     {"Value":"35",     "Quality":"9439544818971181313",     "QualityGood":true,     "Timestamp":"2019-04-10T10:21:02.527Z"},   "ErrorCode":0,   "IsArray":false,   "SubscriptionKey":1   }]  Notification 2: [[   {"DataType":"BasicFloat",   "Value":     {"Value":"40",     "Quality":"9439544818971181313", </pre> |
|--|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

|  |                                                                                                                                                                                                                                                                                                                                                           |
|--|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | <pre>"QualityGood":true, "Timestamp":"2019-04-10T10:21:02.527Z"}, "ErrorCode":0, "IsArray":false, "SubscriptionKey":2 }]  Notification 3: [[ {"DataType":"BasicFloat", "Value": {"Value":"90", "Quality":"9439544818971181313", "QualityGood":true, "Timestamp":"2019-04-10T10:21:02.527Z"}, "ErrorCode":0, "IsArray":false, "SubscriptionKey":3 }]</pre> |
|--|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

7.7.4.1 Subscription Status Notification

After subscribing for Values, for each key subscription status will be sent through notifySubscriptionStatus.

|          |                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                       |                             |
|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|-----------------------------|
| Type     | Name (case sensitive)                                                                                                                                                                                                                                                                                                                                                                                                                                       |                       |                             |
| Hub      | 1. norisHub : recommended<br>2. valuesHub : legacy                                                                                                                                                                                                                                                                                                                                                                                                          |                       |                             |
| Function | notifySubscriptionStatus(status)                                                                                                                                                                                                                                                                                                                                                                                                                            |                       |                             |
|          | Property                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Type                  | Description                 |
|          | Status                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Subscription Response | Subscription objects [→193] |
| Example  | <pre>{   "Key":1,   "PropertyId":"System1:GmsDevice_2_1504_4194306.Present_Value",   "ErrorCode":0,   objectOrPropertyId: "GmsDevice_2_1504_4194306.Present_Value",   "RequestId": "9565ca41-8556-4dbb-94cc-1b89451a5db5",   "RequestFor":"notifyValues",   "_links":[     {       "Rel":"unsubscribe",       "Href":"api/sr/valuessubscriptions/4999fe9e-05c9-41a8-bc82-9ade768c3ced?subscriptionKey=%5b1%5d",       "IsTemplated":false     }   ] }</pre> |                       |                             |

## 7.7.1 Deleting a Subscription for a Change of a Value

### 7.7.1.1 Passing subscription keys in query parameters

|               |                                                                                                                                                                                                                                           |              |                                               |
|---------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|-----------------------------------------------|
| Resource Path | /sr/valuessubscriptions/:connectionId                                                                                                                                                                                                     |              |                                               |
| Category      | Public                                                                                                                                                                                                                                    |              |                                               |
| HTTP Method   | DELETE                                                                                                                                                                                                                                    |              |                                               |
| Parameters    | <b>Property</b>                                                                                                                                                                                                                           | <b>Type</b>  | <b>Description</b>                            |
|               | connectionId                                                                                                                                                                                                                              | String       | Identifier of subscription<br><b>Required</b> |
|               | subscriptionkey                                                                                                                                                                                                                           | Array of int | List of subscription-keys                     |
| Response      | HTTP 200 OK<br>An array of unsubscription objects. See Link [→57]                                                                                                                                                                         |              |                                               |
| Examples      | <b>Request:</b><br>DELETE /api/sr/valuessubscriptions/9d4211db-a740-44c3-93b8-015df1a529ca?subscriptionKey=[2,3]<br><b>Response:</b><br><pre>[   {     "ErrorCode": 0,     "Key": 2   },   {     "ErrorCode": 0,     "Key": 3   } ]</pre> |              |                                               |

### 7.7.1.2 Passing subscription keys in request body

|               |                                                                                                                                                                                                                 |             |                                               |
|---------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|-----------------------------------------------|
| Resource Path | /sr/valuessubscriptions/withbody/:connectionId                                                                                                                                                                  |             |                                               |
| Category      | Public                                                                                                                                                                                                          |             |                                               |
| HTTP Method   | DELETE                                                                                                                                                                                                          |             |                                               |
| Parameters    |                                                                                                                                                                                                                 |             |                                               |
| Body          | Array of Subscription keys                                                                                                                                                                                      |             |                                               |
| Query         | <b>Property</b>                                                                                                                                                                                                 | <b>Type</b> | <b>Description</b>                            |
|               | connectionId                                                                                                                                                                                                    | String      | Identifier of subscription<br><b>Required</b> |
| Response      | HTTP 200 OK<br>An array of unsubscription objects. See Link [→57]                                                                                                                                               |             |                                               |
| Examples      | <b>Request:</b><br>DELETE /api/sr/valuessubscriptions/9d4211db-a740-44c3-93b8-015df1a529ca<br><b>Body Data:</b><br>["2","3"]<br><b>Response:</b><br><pre>[   {     "ErrorCode": 0,     "Key": 2   },   { </pre> |             |                                               |



|  |                                          |
|--|------------------------------------------|
|  | <pre> "ErrorCode": 0, "Key": 3 } ]</pre> |
|--|------------------------------------------|

## 7.8 Property Value Service

| Resource        | Service Category | Requires Authentication | Requires SignalR (Push Notifications) |
|-----------------|------------------|-------------------------|---------------------------------------|
| /propertyvalues | Public           | x                       |                                       |

### 7.8.1 Retrieving Detailed Values for Object or Property Id

Retrieves properties with runtime values for an Object OR Property Id. This gives properties and attributes associated with the requested Object OR Property Id. If input is actually an Object Id then default property for the object id along with the attributes is returned in the response. If input is a property id then that specific property and attributes of that object id will be returned in the response. One optional Boolean type query string parameter readAllProperties is available in both the cases. If readAllProperties is set to True then all the properties along with the attributes are returned in the response for the requested Object OR Property Id

|               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |             |                                                                                                                             |
|---------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|-----------------------------------------------------------------------------------------------------------------------------|
| Resource Path | /propertyvalues/objectOrPropertyId                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |             |                                                                                                                             |
| Category      | Public                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |             |                                                                                                                             |
| Category      | Public                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |             |                                                                                                                             |
| HTTP Method   | GET                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |             |                                                                                                                             |
| Parameters    | <b>Property</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | <b>Type</b> | <b>Description</b>                                                                                                          |
|               | objectOrPropertyId                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | string      | ObjectOrPropertyId<br><b>Required</b>                                                                                       |
|               | readAllProperties                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | bool        | If True then all the properties for the requested Object OR Property Id will be returned in the response. Default is False. |
| Response      | HTTP 200 OK<br>The Object<Type> see [→ 62]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |             |                                                                                                                             |
| Examples      | <b>Request 1:</b><br>GET api/propertyvalues/System1:HDB_ArchiveGroup_1<br><b>Response:</b><br><pre> {   "ErrorCode": 0,   "ObjectId": "System1:HDB_ArchiveGroup_1",   "Attributes": {     "DefaultProperty": "StatusPropagation.AggregatedSummaryStatus",     "ObjectId": "HDB_ArchiveGroup_1",     "DisciplineDescriptor": "Management System",     "DisciplineId": 0,     "SubDisciplineDescriptor": "Unassigned",     "SubDisciplineId": 0,     "TypeDescriptor": "Server Software",     "TypeId": 6900,     "SubTypeDescriptor": "History Database",     "SubTypeId": 6903,</pre> |             |                                                                                                                             |

|  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|--|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | <pre> "ManagedType": 14, "ManagedTypeName": "Station" }, "Properties": [ { "Order": 0,  "PropertyName": "StatusPropagation.AggregatedSummaryStatus", "Descriptor": "Summary Status", "Type": "ExtendedEnum", "Usage": 7, "Value": { "Value": "0", "DisplayValue": "Normal", "Quality": "9439544818969084161", "QualityGood": true, "Timestamp": "2015-05-21T05:08:56.772Z" }, "Resolution": 0, "PropertyAbsent": false, "IsArray": false, "TextTable": TxG_PropagationSummaryStatus } ] } </pre> <p><b>Request 2: with scaled information</b></p> <p>GET<br/>api/propertyvalues/System2:GmsDevice_1_20811_1.Present_Value?</p> <p><b>Response :</b></p> <pre> { "ErrorCode": 0, "ObjectId": "System2:GmsDevice_1_20811_1", "Attributes": { "DefaultProperty": "Present_Value", "ObjectId": "System2:GmsDevice_1_20811_1", "DisciplineDescriptor": "Building Automation", "DisciplineId": 50, "SubDisciplineDescriptor": "Unassigned", "SubDisciplineId": 0, "TypeDescriptor": "Sensor", "TypeId": 6700, "SubTypeDescriptor": "Unassigned", "SubTypeId": 0, "ManagedType": 80, "ManagedTypeName": "BACnetPoint", "ObjectModelName": "GMS_BACNET_EO_BA_AI_1", "ValidationRules": { "CommentRule": "Mandatory", "ReAuthentication": "NoNeed", "Configuration": 1, "IsFourEyesEnabled": false, "_links": [] } }, "Properties": [ { "Order": 2, </pre> |
|--|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

|  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|--|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | <pre>"PropertyName": "Present_Value", "Descriptor": "Present Value", "Type": "ExtendedReal", "Usage": 7, "Value": {   "Value": "25.5555555555556",   "DisplayValue": "25.5555555555556",   "Quality": "9439544818975376641",   "QualityGood": true,   "Timestamp": "2019-06-11T12:41:00.067Z" }, "Min": -572.777777777778, "Max": 537.222222222222, "UnitDescriptor": "°C", "UnitId": 62, "Resolution": 2, "PropertyAbsent": false, "IsArray": false, "ScaledInformation": {   "Factor": 0.555555555555556,   "Index": 62,   "Offset": -17.7777777777778 } } ]</pre> <p><b>Request 3: using Designation</b></p> <p>GET<br/>/api/propertyvalues/System1.ManagementView:ManagementView.Fie<br/>ldNetworks.TestNet.Hardware.Devices_1504.Local_IO.AO_2;?readA<br/>llProperties=false</p> <p><b>Response:</b></p> <pre>{   "ErrorCode":0,   "ObjectId":"System1:GmsDevice_2_1504_4194306",   "Attributes":{     "DefaultProperty":"Present_Value",     "ObjectId":"System1:GmsDevice_2_1504_4194306",     "DisciplineDescriptor":"Building Automation",     "DisciplineId":50,     "SubDisciplineDescriptor":"Unassigned",     "SubDisciplineId":0,     "TypeDescriptor":"Command",     "TypeId":1100,     "SubTypeDescriptor":"Analog",     "SubTypeId":1101,     "ManagedType":80,     "ManagedTypeName":"BACnetPoint"   },   "Properties":[     {       "Order":0,       "PropertyName":"Present_Value",       "Descriptor":"Present Value",       "Type":"ExtendedReal",       "Usage":7,       "Value":{</pre> |
|--|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

|  |                                                                                                                                                                                                                                                           |
|--|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | <pre> "Value": "35", "DisplayValue": "35", "Quality": "9439544818971182337", "QualityGood": true, "Timestamp": "2017-12-20T09:15:11.96Z" }, "UnitDescriptor": "°F", "UnitId": 64, "Resolution": 0, "PropertyAbsent": false, "IsArray": false } ] } </pre> |
|--|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

## 7.9 Properties Service

| Resource    | Service Category | Requires Authentication | Requires SignalR (Push Notifications) |
|-------------|------------------|-------------------------|---------------------------------------|
| /properties | Public           | x                       |                                       |

### 7.9.1 Retrieving Detailed Values for Object or Property Id

Retrieves properties without any runtime values for an Object OR Property Id. If input is actually an Object Id then default property for the object id will be returned in the response. If input is a property id then that specific property will be returned in the response. Two optional query string parameters are available in this interface. First is `requestType` of type `int`, which specifies whether the request is to read Property Names, Properties, Attributes, Properties and Attributes. Second parameter is `readAllProperties` of type `bool`, if this is `True` then all the properties for the requested Object OR Property Id will be returned in the response.

|               |                                            |                    |                                                                                                                             |
|---------------|--------------------------------------------|--------------------|-----------------------------------------------------------------------------------------------------------------------------|
| Resource Path | /properties/:objectOrPropertyId            |                    |                                                                                                                             |
| Category      | Public                                     |                    |                                                                                                                             |
| HTTP Method   | GET                                        |                    |                                                                                                                             |
| Parameters    | <b>Property</b>                            | <b>Type</b>        | <b>Description</b>                                                                                                          |
|               | objectOrPropertyId                         | string             | ObjectOrPropertyId<br><b>Required</b>                                                                                       |
|               | requestType                                | int                | Specifies the request type. See [→ 44]. Default is 2.                                                                       |
|               | readAllProperties                          | bool               | If True then all the properties for the requested Object OR Property Id will be returned in the response. Default is False. |
| Response      | HTTP 200 OK<br>The Object<Type> see [→ 62] |                    |                                                                                                                             |
|               | <b>requestType</b>                         | <b>Return Type</b> | <b>Description</b>                                                                                                          |
|               | 0                                          | See [→ 67]         | Property collections contain list of all property-names                                                                     |
|               | 1                                          | Object<null>       | Property collections empty; object contains                                                                                 |

|          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                         |                                                            |
|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------|------------------------------------------------------------|
|          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                         | attributes                                                 |
|          | 2 (default)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Object<PropertyDetails> | Property collections populated; no attribute field         |
|          | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Object<PropertyDetails> | Property collections populated; object contains attributes |
| Examples | <p><b>Request1: Get Property Names</b></p> <p>GET api/properties/System1:HDB_ArchiveGroup_1?requestType=0<br/>OR<br/>GET<br/>/api/properties/System1.ManagementView:ManagementView.ManagementSystem.Servers.Server.HistoryDatabase.ArchiveSchedules.ActivityLog;?requestType=0 //using Designation</p> <p><b>Response1:</b></p> <pre>{   "ObjectId": "System1:HDB_ArchiveGroup_1",   "Properties": [     "Export.Activated",     "Max_Retention.Activated",     "Max_Retention.Delay_Span",     "Max_Retention.Delay_Unit",     "Max_Retention.FirstDate",     "Max_Retention.Invocation_Period",     "Max_Retention.Invocation_Unit",     "Max_Retention.Max_Count",     "Min_Retention.Delay_Span",     "Min_Retention.Delay_Unit",     "Min_Retention.Min_Count",     "StatusPropagation.AggregatedSummaryStatus"   ],   "FunctionProperties": [] }</pre> <p><b>Request2: Get Properties and Attributes</b></p> <p>GET api/properties/System1:HDB_ArchiveGroup_1?requestType=3<br/>OR<br/>GET<br/>/api/properties/System1.ManagementView:ManagementView.ManagementSystem.Servers.Server.HistoryDatabase.ArchiveSchedules.ActivityLog;?requestType=3 //using Designation</p> <p><b>Response2:</b></p> <pre>[   {     "ErrorCode": 0,     "ObjectId": "System1:HDB_ArchiveGroup_1",     "Attributes": {       "DefaultProperty": "StatusPropagation.AggregatedSummaryStatus",       "ObjectId": "HDB_ArchiveGroup_1",       "DisciplineDescriptor": "Management System",       "DisciplineId": 0,       "SubDisciplineDescriptor": "Unassigned",       "SubDisciplineId": 0,       "TypeDescriptor": "Server Software",       "TypeId": 6900,       "SubTypeDescriptor": "History Database",       "SubTypeId": 6903,       "ManagedType": 14,</pre> |                         |                                                            |

```

    "ManagedTypeName" : "Station"
  },
  "Properties": [
    {
      "Order": 0,
      "PropertyName":
"StatusPropagation.AggregatedSummaryStatus",
      "Descriptor": "Summary Status",
      "Type": "ExtendedEnum",
      "Usage": 7,
      "Resolution": 0,
      "PropertyAbsent": false,
      "IsArray": false
    }
  ]
}
]

```

### Request 3: with scaled information

```

GET
api/properties/System2:GmsDevice_1_20811_1.Present_Value?requestType=3

```

### Response 3 :

```

{
  "ErrorCode": 0,
  "ObjectId": "System2:GmsDevice_1_20811_1",
  "Attributes": {
    "DefaultProperty": "Present_Value",
    "ObjectId": "System2:GmsDevice_1_20811_1",
    "DisciplineDescriptor": "Building Automation",
    "DisciplineId": 50,
    "SubDisciplineDescriptor": "Unassigned",
    "SubDisciplineId": 0,
    "TypeDescriptor": "Sensor",
    "TypeId": 6700,
    "SubTypeDescriptor": "Unassigned",
    "SubTypeId": 0,
    "ManagedType": 80,
    "ManagedTypeName": "BACnetPoint",
    "ObjectModelName": "GMS_BACNET_EO_BA_AI_1",
    "ValidationRules": {
      "CommentRule": "Mandatory",
      "ReAuthentication": "NoNeed",
      "Configuration": 1,
      "IsFourEyesEnabled": false,
      "_links": []
    }
  },
  "Properties": [
    {
      "Order": 2,
      "PropertyName": "Present_Value",
      "Descriptor": "Present Value",
      "Type": "ExtendedReal",
      "Usage": 7,
      "Value": {
        "Value": "25.55555555555556",

```

|  |                                                                                                                                                                                                                                                                                                                                                                                                                    |
|--|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | <pre> "DisplayValue": "25.5555555555556", "Quality": "9439544818975376641", "QualityGood": true, "Timestamp": "2019-06-11T12:41:00.067Z" }, "Min": -572.777777777778, "Max": 537.222222222222, "UnitDescriptor": "°C", "UnitId": 62, "Resolution": 2, "PropertyAbsent": false, "IsArray": false, "ScaledInformation": {   "Factor": 0.555555555555556,   "Index": 62,   "Offset": -17.7777777777778 } } ] } </pre> |
|--|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

### 7.9.1.1 Detail Request Type

**requestType** must have one of the following values.

| requestType | Description                                                           |
|-------------|-----------------------------------------------------------------------|
| 0           | Return property names only.                                           |
| 1           | Return attributes of object only.                                     |
| 2           | Return properties with detailed information; default.                 |
| 3           | Return properties with detailed information and attributes of object. |

## 7.9.2 Retrieving Detailed Values for multiple Object or Property Ids bulk interface

This interface is same as that of 7.9.1 only thing is this provides bulk interface to read properties and attributes of multiple Object OR Property Ids in the same request using HTTP POST. Array of Object OR Property Ids is passed in the POST data. Rest all the configuration is same as that of 7.9.1.

|               |                                            |             |                                                                                                                             |
|---------------|--------------------------------------------|-------------|-----------------------------------------------------------------------------------------------------------------------------|
| Resource Path | /properties                                |             |                                                                                                                             |
| Category      | Public                                     |             |                                                                                                                             |
| HTTP Method   | POST                                       |             |                                                                                                                             |
| Parameters    | <b>Property</b>                            | <b>Type</b> | <b>Description</b>                                                                                                          |
|               | requestType                                | Int         | Specifies the request type. See [→ 44]. Default is 2.                                                                       |
|               | readAllProperties                          | bool        | If True then all the properties for the requested Object OR Property Id will be returned in the response. Default is False. |
| Body          | Array of Object OR Property Ids            |             |                                                                                                                             |
| Response      | HTTP 200 OK<br>The Object<Type> see [→ 62] |             |                                                                                                                             |

|          | requestType                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Return Type             | Description                                                |
|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------|------------------------------------------------------------|
|          | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Object<null>            | Property collections empty; object contains attributes     |
|          | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Object<PropertyDetails> | Property collections populated; no attribute field         |
|          | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Object<PropertyDetails> | Property collections populated; object contains attributes |
| Examples | <p><b>Request:</b><br/> Post api/properties?requestType=3</p> <p><b>Request Body:</b><br/> ["System1:HDB_ArchiveGroup_1.Min_Retention.Delay_Span", "System1:HDB_ArchiveGroup_1.Min_Retention.Delay_Unit"]</p> <p><b>OR</b><br/> //Using Designation + Property<br/> [<br/> "System1.ManagementView:ManagementView.ManagementSystem.Servers.Server.HistoryDatabase.ArchiveSchedules.ActivityLog;.Min_Retention.Delay_Span",<br/> "System1.ManagementView:ManagementView.ManagementSystem.Servers.Server.HistoryDatabase.ArchiveSchedules.ActivityLog;.Min_Retention.Delay_Unit"<br/> ]<br/> </p> <p><b>Response:</b><br/> [<br/> {<br/> "ErrorCode": 0,<br/> "ObjectId": "System1:HDB_ArchiveGroup_1",<br/> "Attributes": {<br/> "DefaultProperty":<br/> "StatusPropagation.AggregatedSummaryStatus",<br/> "ObjectId": "HDB_ArchiveGroup_1",<br/> "DisciplineDescriptor": "Management System",<br/> "DisciplineId": 0,<br/> "SubDisciplineDescriptor": "Unassigned",<br/> "SubDisciplineId": 0,<br/> "TypeDescriptor": "Server Software",<br/> "TypeId": 6900,<br/> "SubTypeDescriptor": "History Database",<br/> "SubTypeId": 6903,<br/> "ManagedType": 14,<br/> "ManagedTypeName": "Station"<br/> },<br/> },<br/> "Properties": [<br/> {<br/> "Order": 3,<br/> "PropertyName": "Min_Retention.Delay_Span",<br/> "Descriptor": "Min Retention Span",<br/> "Type": "ExtendedInt",<br/> "Usage": 11,<br/> "Min": 1,<br/> "Resolution": 0,<br/> "PropertyAbsent": false,<br/> "IsArray": false<br/> }<br/> ]<br/> ]<br/> </p> |                         |                                                            |



|  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|--|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | <pre>] }, {   "ErrorCode": 0,   "ObjectId": "System1:HDB_ArchiveGroup_1",   "Attributes": {     "DefaultProperty": "StatusPropagation.AggregatedSummaryStatus",     "ObjectId": "HDB_ArchiveGroup_1",     "DisciplineDescriptor": "Management System",     "DisciplineId": 0,     "SubDisciplineDescriptor": "Unassigned",     "SubDisciplineId": 0,     "TypeDescriptor": "Server Software",     "TypeId": 6900,     "SubTypeDescriptor": "History Database",     "SubTypeId": 6903,     "ManagedType": 14,     "ManagedTypeName": "Station"   },   "Properties": [     {       "Order": 4,       "PropertyName": "Min_Retention.Delay_Unit",       "Descriptor": "Min Retention Unit",       "Type": "ExtendedEnum",       "Usage": 11,       "Max": 7,       "Resolution": 0,       "PropertyAbsent": false,       "IsArray": false     }   ] } ]</pre> |
|--|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

### 7.9.1 Retrieving an Image for a Property

Retrieve an icon for a property. A couple of important notes:

- 1) If the property has not been assigned an image the return will still be OK (200) but the return will be an empty string.
- 2) The only supported format for the image is Base64-encoded, PNG with a size of 32 pixels wide by 32 pixels high. This may change in the future.

|               |                                                               |             |                                                                                                                                                       |
|---------------|---------------------------------------------------------------|-------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|
| Resource Path | /properties/:propertyId/icon                                  |             |                                                                                                                                                       |
| Category      | Public                                                        |             |                                                                                                                                                       |
| HTTP Method   | GET                                                           |             |                                                                                                                                                       |
| Parameters    | <b>Property</b>                                               | <b>Type</b> | <b>Description</b>                                                                                                                                    |
|               | propertyId                                                    | string      | The fully-qualified property name, which includes the system information. For example:<br>System1:GmsDevice_1_8166_1.Present_Value<br><b>Required</b> |
| Response      | HTTP 200 OK<br>The icon as a Base64 encoded PNG of size 32x32 |             |                                                                                                                                                       |

|          |                                                                                                                                                                                                                                                                         |
|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Examples | <b>Request:</b><br>GET<br>api/properties/System1:GmsDevice_1_8166_1.Present_Value/icon<br><b>Response:</b><br>iVBORw0KGgoAAAANSUhEUgAAAAEAAAABCAyAAAAFfcSJAAXNSR0IArs4c6QAAARnQU1BAACxjwv8YQUAAAJcEhZcwAADsIAAA7CARUoSoAAAAANSURVBVhXY/j//78GAakAyb6sRXrAAAAE1FTkSuQmCC |
|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

## 7.10 Command Service

| Resource               | Service Category | Requires Authentication | Requires SignalR (Push Notifications) |
|------------------------|------------------|-------------------------|---------------------------------------|
| /commands              | Public           | x                       |                                       |
| /commandssubscriptions | Public           | x                       | x                                     |

### 7.10.1 Retrieving a List of Commands for a Provided Property

Some commands might be non executable at this moment.

|               |                                                                                                                                                                                                                                                                                                                                                            |             |                                                                                  |
|---------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|----------------------------------------------------------------------------------|
| Resource Path | /commands/:propertyId                                                                                                                                                                                                                                                                                                                                      |             |                                                                                  |
| Category      | Public                                                                                                                                                                                                                                                                                                                                                     |             |                                                                                  |
| HTTP Method   | GET                                                                                                                                                                                                                                                                                                                                                        |             |                                                                                  |
| Parameters    | <b>Property</b>                                                                                                                                                                                                                                                                                                                                            | <b>Type</b> | <b>Description</b>                                                               |
|               | propertyId                                                                                                                                                                                                                                                                                                                                                 | string      | PropertyId<br><b>Required</b>                                                    |
|               | commandId                                                                                                                                                                                                                                                                                                                                                  | string      | Optional filter for a specific commandId                                         |
|               | enabledCommandsOnly                                                                                                                                                                                                                                                                                                                                        | bool        | If set to true only currently enabled commands will be returned (default false). |
|               | clientType                                                                                                                                                                                                                                                                                                                                                 | string      | If set commands can be filtered for specific clients (All, Headless, Headful)    |
| Response      | HTTP 200 OK<br>An array of PropertyCommand objects See [→ 52]                                                                                                                                                                                                                                                                                              |             |                                                                                  |
| Examples      | <b>Request 1:</b><br>GET<br>api/commands/System1%3AHDB_ArchiveGroup_1.Min_Retention.Delay_Span<br><b>Response:</b><br>[ { "PropertyId": "System1:HDB_ArchiveGroup_1.Min_Retention.Delay_Span", "ErrorCode": 0, "Commands": [ { "Configuration":2 "PropertyId": "System1:HDB_ArchiveGroup_1.Min_Retention.Delay_Span", "Descriptor": "Set", "Parameters": [ |             |                                                                                  |

|  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|--|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | <pre>{   "Name": "Value",   "DataType": "ExtendedInt",   "DefaultValue": "0",   "Order": 0,   "Descriptor": "",   "Min": 1,   "Application": 0,   "EnumerationTexts": [] } ], "Id": "Write", "IsDefault": false, "_links": [   {     "Rel": "self",     "Href": "api/commands/System1%3aHDB_ArchiveGroup_1.Min_Retention.D elay_Span/Write",     "IsTemplated": false   } ] } ] }</pre> <p><b>Request 2: using Designation + Property</b></p> <p>GET<br/>/api/commands/System1.ManagementView:ManagementView.Manage<br/>mentSystem.Servers.Server.HistoryDatabase.ArchiveSchedules<br/>.ActivityLog;.Min_Retention.Delay_Span</p> <p><b>Response:</b></p> <pre>[ {   "PropertyId":"System1.ManagementView:ManagementView.Manage mentSystem.Servers.Server.HistoryDatabase.ArchiveSchedules .ActivityLog;.Min_Retention.Delay_Span",   "ErrorCode":0,   "Commands":[     {       "PropertyId":"System1.ManagementView:ManagementView.Manage mentSystem.Servers.Server.HistoryDatabase.ArchiveSchedules .ActivityLog;.Min_Retention.Delay_Span",       "Descriptor":"Set",       "Parameters":[         {           "Name":"Value",           "DataType":"ExtendedInt",           "DefaultValue":"4",           "Order":0,           "Descriptor":"",           "Min":1,           "Application": 0,           "EnumerationTexts":[]         }       ]     },     {       "Id":"Write",       "IsDefault": false,       "Configuration":0, </pre> |
|--|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

|  |                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|--|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | <pre>                 "_links": [                     {                         "Rel": "self",                         "Href": "api/commands/System1.ManagementView%3aManagementVi ew.ManagementSystem.Servers.Server.HistoryDatabase.Archive Schedules.ActivityLog%3b.Min_Retention.Delay_Span/Write",                         "IsTemplated": false                     }                 ]             }         ]     } } </pre> |
|--|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

## 7.10.2 Retrieving Lists of Commands for a List of Properties

|               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |             |                                                                                  |
|---------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|----------------------------------------------------------------------------------|
| Resource Path | /commands                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |             |                                                                                  |
| Category      | Public                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |             |                                                                                  |
| HTTP Method   | POST                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |             |                                                                                  |
| Parameters    | <b>Property</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | <b>Type</b> | <b>Description</b>                                                               |
|               | propertyId                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | string      | PropertyId<br><b>Required</b>                                                    |
|               | commandId                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | string      | Optional filter for a specific commandId                                         |
|               | enabledCommandsOnly                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | bool        | If set to true only currently enabled commands will be returned (default false). |
|               | clientType                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | string      | If set commands can be filtered for specific clients (All, Headless, Headful)    |
| Body          | Array of PropertyId                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |             |                                                                                  |
| Response      | HTTP 200 OK<br>An array of PropertyCommand objects See [→ 52]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |             |                                                                                  |
| Examples      | <p><b>Request 1:</b><br/>Post: /api/commands/</p> <p><b>Request Body:</b></p> <pre> [     "System1:HDB_ArchiveGroup_1.Min_Retention.Delay_Span",     "System1:HDB_ArchiveGroup_1.Min_Retention.Delay_Unit" ] </pre> <p><b>Response:</b><br/>List of Commands fetched for two different PropertyId:</p> <pre> [     {         "PropertyId":         "System1:HDB_ArchiveGroup_1.Min_Retention.Delay_Span",         "ErrorCode": 0,         "Commands": [             {                 "Configuration": 2                 "PropertyId":                 "System1:HDB_ArchiveGroup_1.Min_Retention.Delay_Span", </pre> |             |                                                                                  |

|  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|--|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | <pre>"Descriptor": "Set", "Parameters": [   {     "Name": "Value",     "DataType": "ExtendedInt",     "DefaultValue": "0",     "Order": 0,     "Descriptor": "",     "Min": 1,     "Application": 0,     "EnumerationTexts": []   } ], "Id": "Write", "IsDefault": false, "_links": [   {     "Rel": "self",     "Href": "api/commands/System1%3aHDB_ArchiveGroup_1.Min_Retention.D elay_Span/Write",     "IsTemplated": false   } ] }, {   "PropertyId": "System1:HDB_ArchiveGroup_1.Min_Retention.Delay_Unit",   "ErrorCode": 0,   "Commands": [     {       "Configuration":1       "PropertyId": "System1:HDB_ArchiveGroup_1.Min_Retention.Delay_Unit",       "Descriptor": "Set",       "Parameters": [         {           "Name": "Value",           "DataType": "ExtendedEnum",           "DefaultValue": "0",           "Order": 0,           "Descriptor": "",           "Max": 7,           "Application": 0,           "EnumerationTexts": [             {               "Descriptor": "seconds",               "Value": 0             },             {               "Descriptor": "minutes",               "Value": 1             }           ]         }       ]     }   ] }, ],</pre> |
|--|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

```

      "Id": "Write",
      "IsDefault": false,
      "_links": [
        {
          "Rel": "self",
          "Href":
"api/commands/System1%3aHDB_ArchiveGroup_1.Min_Retention.Delay_Unit/Write",
          "IsTemplated": false
        }
      ]
    }
  ]
}
]

```

#### Request 2: Using Designation + Property

Post: /api/commands/

#### Request Body:

```

[
  "System1.ManagementView:ManagementView.ManagementSystem.Servers.Server.HistoryDatabase.ArchiveSchedules.ActivityLog;.Min_Retention.Delay_Span",
  "System1.ManagementView:ManagementView.ManagementSystem.Servers.Server.HistoryDatabase.ArchiveSchedules.ActivityLog;.Min_Retention.Delay_Unit"
]

```

#### Response:

```

[
  {
    "PropertyId": "System1.ManagementView:ManagementView.ManagementSystem.Servers.Server.HistoryDatabase.ArchiveSchedules.ActivityLog;.Min_Retention.Delay_Span",
    "ErrorCode": 0,
    "Commands": [
      {
        "PropertyId": "System1.ManagementView:ManagementView.ManagementSystem.Servers.Server.HistoryDatabase.ArchiveSchedules.ActivityLog;.Min_Retention.Delay_Span",
        "Descriptor": "Set",
        "Parameters": [
          {
            "Name": "Value",
            "DataType": "ExtendedInt",
            "DefaultValue": "4",
            "Order": 0,
            "Descriptor": "",
            "Min": 1,
            "Application": 0,
            "EnumerationTexts": []
          }
        ],
        "Id": "Write",
        "IsDefault": false,
        "Configuration": 0,
        "_links": [
          {
            "Rel": "self",
            "Href": "api/commands/System1.ManagementView%3aManagementView.ManagementSystem.Servers.Server.HistoryDatabase.ArchiveSchedules.ActivityLog%3b.Min_Retention.Delay_Span/Write",

```

|  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|--|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | <pre>        "IsTemplated":false       }     ]   } ] }, { "PropertyId": "System1.ManagementView:ManagementView.Manag ementSystem.Servers.Server.HistoryDatabase.ArchiveSchedule s.ActivityLog;.Min_Retention.Delay_Unit",   "ErrorCode":0,   "Commands":[ { "PropertyId": "System1.ManagementView:ManagementView.Manag ementSystem.Servers.Server.HistoryDatabase.ArchiveSchedule s.ActivityLog;.Min_Retention.Delay_Unit",   "Descriptor": "Set",   "Parameters": [     {       "Name": "Value",       "DataType": "ExtendedEnum",       "DefaultValue": "4",       "Order": 0,       "Descriptor": "",       "Max": 7,       "Application": 0,       "EnumerationTexts": [         {           "Descriptor": "seconds",           "Value": 0         },         {           "Descriptor": "minutes",           "Value": 1         },         {           "Descriptor": "hours",           "Value": 2         },         {           "Descriptor": "days",           "Value": 3         },         {           "Descriptor": "months",           "Value": 4         },         {           "Descriptor": "years",           "Value": 5         },         {           "Descriptor": "week",           "Value": 6         },         {           "Descriptor": "quarter",           "Value": 7         }       ]     }   ] } }</pre> |
|--|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

|  |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|--|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | <pre>         ],         "Id": "Write",         "IsDefault": false,         "Configuration": 0,         "_links": [             { "Rel": "self",               "Href": "api/commands/System1.ManagementView%3aManagementVi                 ew.ManagementSystem.Servers.Server.HistoryDatabase.Archive                 Schedules.ActivityLog%3b.Min_Retention.Delay_Unit/Write",               "IsTemplated": false             }         ]     } } ] ] </pre> |
|--|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

### 7.10.3 Executing a Command

|               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                                            |                                                                            |
|---------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|----------------------------------------------------------------------------|
| Resource Path | /commands/:propertyId/:commandId                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                            |                                                                            |
| Category      | Public                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                            |                                                                            |
| HTTP Method   | POST                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                            |                                                                            |
| Parameters    | <b>Property</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | <b>Type</b>                                                                                | <b>Description</b>                                                         |
|               | propertyId                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | string                                                                                     | PropertyId<br><b>Required</b>                                              |
|               | commandId                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | string                                                                                     | Command identifier<br><b>Required</b>                                      |
|               | commandInput                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | CommandInputForExecutionValidateParam<br><br>See [ → <b>Error! Bookmark not defined.</b> ] | command parameter details (Name, Value pair) sent from client if required. |
| Body          | An array of NameValue objects of Type(CommandInputForExecutionValidateParam)                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                            |                                                                            |
| Response      | HTTP 200 OK                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                            |                                                                            |
| Examples      | <b>Request:</b><br>POST<br>/api/commands/System1:HDB_ArchiveGroup_1.Min_Retention.Delay_Span/Write<br><b>OR</b><br>//Using Designation + Property<br>POST<br>/api/commands/System1.ManagementView:ManagementView.ManagementSystem.Servers.Server.HistoryDatabase.ArchiveSchedules.ActivityLog;.Min_Retention.Delay_Span/Write<br><b>Request Body:</b><br><pre> [   {     "Name": "Value",     "DataType": "ExtendedInt",     "Value": "3",     "MultiLangComments": [       "comment1",       "comment2",       "comment3"     ]   } ] </pre> |                                                                                            |                                                                            |



|  |                                                                                                                                                              |
|--|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | <pre> ], "Password": "password", "SuperName": "supervisorusername", "SuperPassword": "supervisorpassword" } ] </pre> <p><b>Response:</b><br/>HTTP 200 OK</p> |
|--|--------------------------------------------------------------------------------------------------------------------------------------------------------------|

## 7.10.4 Bulk Commanding

|                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                        |                                                                                                                          |
|-----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------|--------------------------------------------------------------------------------------------------------------------------|
| Resource Path   | /commands/:commandId                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                        |                                                                                                                          |
| Category        | Public                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                        |                                                                                                                          |
| HTTP Method     | POST                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                        |                                                                                                                          |
| Parameters      | <b>Property</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | <b>Type</b>                            | <b>Description</b>                                                                                                       |
|                 | commandId                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | string                                 | Command identifier<br><b>Required</b>                                                                                    |
| Body            | bulkCommandInput                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | BatchCommand Execution<br>See [ → 209] | command parameter details (Name, Value pair) sent from client if required along with array of property ids of same type. |
| User Validation | User Validation is at parameter level.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                        |                                                                                                                          |
| Response        | HTTP 200 OK<br>An array of BatchExecution objects See Link [ → 209]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                        |                                                                                                                          |
| Examples        | <p><b>Request:</b><br/>POST : /api/commands/Write</p> <p><b>Request Body:</b></p> <pre> {   "CommandInputForExecution": [     {       "Name": "Value",       "DataType": "ExtendedInt",       "Value": "3",       "MultiLangComments": [         "comment1",         "comment2",         "comment3"       ],       "Password": "password1",       "SuperName": "supervisorusername",       "SuperPassword": "supervisorpassword"     }   ],   "PropertyIds": [     "System1:HDB_ArchiveGroup_1.Min_Retention.Delay_Span",     "System1:HDB_ArchiveGroup_2.Min_Retention.Delay_Span",     "System1.ManagementView:ManagementView.ManagementSystem.Servers.Server.HistoryDatabase.ArchiveSchedules.ActivityLog; .Min_Retention.Delay_Span", //Designation + Property     "System1.ManagementView:ManagementView.ManagementSystem.Servers.Server.HistoryDatabase.ArchiveSchedules.EventLog; .Min_Retention.Delay_Span" //Designation + Property   ] } </pre> <p><b>Response:</b></p> |                                        |                                                                                                                          |

|  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|--|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | <pre> HTTP 200 OK [   {     "PropertyId":     "System1:HDB_ArchiveGroup_1.Min_Retention.Delay_Span",     "ErrorCode": 0   },   {     "PropertyId":     "System1:HDB_ArchiveGroup_2.Min_Retention.Delay_Span",     "ErrorCode": 0   },   {     "PropertyId":     "System1.ManagementView:ManagementView.ManagementSystem.Servers.Server.HistoryDatabase.ArchiveSchedules.ActivityLog;Min_Retention.Delay_Span",     "ErrorCode": 0   },   {     "PropertyId":     "System1.ManagementView:ManagementView.ManagementSystem.Servers.Server.HistoryDatabase.ArchiveSchedules.EventLog;Min_Retention.Delay_Span",     "ErrorCode": 0   } ] </pre> |
|--|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

### 7.10.5 Bulk Commanding

|                 |                                                                                                                                                                                                                                                            |                                                   |                                                                                                                          |
|-----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------|
| Resource Path   | /validateproperty/commands/:commandId                                                                                                                                                                                                                      |                                                   |                                                                                                                          |
| Category        | Public                                                                                                                                                                                                                                                     |                                                   |                                                                                                                          |
| HTTP Method     | POST                                                                                                                                                                                                                                                       |                                                   |                                                                                                                          |
| Parameters      | <b>Property</b>                                                                                                                                                                                                                                            | <b>Type</b>                                       | <b>Description</b>                                                                                                       |
|                 | commandId                                                                                                                                                                                                                                                  | string                                            | Command identifier<br><b>Required</b>                                                                                    |
| Body            | bulkCommandInput                                                                                                                                                                                                                                           | BatchCommandExecutionValidateProperty See [→ 209] | command parameter details (Name, Value pair) sent from client if required along with array of property ids of same type. |
| User Validation | User Validation is at Property level.                                                                                                                                                                                                                      |                                                   |                                                                                                                          |
| Response        | HTTP 200 OK<br>An array of BatchCommandExecutionValidateProperty objects See Link [→ 209]                                                                                                                                                                  |                                                   |                                                                                                                          |
| Examples        | <b>Request:</b><br>POST : api/commands/validateproperty/write<br><br><b>Request Body:</b><br><pre> {   "CommandInputForExecution": [     {       "Name": "Value",       "DataType": "ExtendedInt",       "Value": "3"     }   ],   "PropertyIds": [ </pre> |                                                   |                                                                                                                          |

|  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|--|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | <pre> "System1:HDB_ArchiveGroup_1.Min_Retention.Delay_Span", "System1:HDB_ArchiveGroup_2.Min_Retention.Delay_Span",  "System1.ManagementView:ManagementView.ManagementSystem.Servers.Server.HistoryDatabase.ArchiveSchedules.ActivityLog;.Min_Retention.Delay_Span", //Designation + Property  "System1.ManagementView:ManagementView.ManagementSystem.Servers.Server.HistoryDatabase.ArchiveSchedules.EventLog;.Min_Retention.Delay_Span" //Designation + Property ], "Comments": {   "MultiLangText": [     "comment1",     "comment2",     "comment3"   ] }, "Password": "password1", "SuperName": "supervisorusername", "SuperPassword": "supervisorpassword" } </pre> <p><b>Response:</b><br/>HTTP 200 OK</p> <pre> [   {     "PropertyId": "System1:HDB_ArchiveGroup_1.Min_Retention.Delay_Span",     "ErrorCode": 0   },   {     "PropertyId": "System1:HDB_ArchiveGroup_2.Min_Retention.Delay_Span",     "ErrorCode": 0   },   {     "PropertyId": "System1.ManagementView:ManagementView.ManagementSystem.Servers.Server.HistoryDatabase.ArchiveSchedules.ActivityLog;.Min_Retention.Delay_Span",     "ErrorCode": 0   },   {     "PropertyId": "System1.ManagementView:ManagementView.ManagementSystem.Servers.Server.HistoryDatabase.ArchiveSchedules.EventLog;.Min_Retention.Delay_Span",     "ErrorCode": 0   } ] </pre> |
|--|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

### 7.10.6 Bulk Commanding with different pair of property values and command id

|               |                                |                      |                                                                  |
|---------------|--------------------------------|----------------------|------------------------------------------------------------------|
| Resource Path | /commands/propertycommand      |                      |                                                                  |
| Category      | Public                         |                      |                                                                  |
| HTTP Method   | POST                           |                      |                                                                  |
| Body          | An array of propertyCommandInp | PropertyCommandInput | Array of property command object. Each property command contains |

|                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                          |                                                                 |
|-----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------|-----------------------------------------------------------------|
|                 | ut                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Execution<br>See [ →210] | commandid ,propertyid and optional<br>command parameter details |
| User Validation | User Validaiton is at parameter level.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                          |                                                                 |
| Response        | HTTP 200 OK<br>An array of BatchExecution objects See Link [→ 209]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                          |                                                                 |
| Examples        | <b>Request:</b><br>POST : /api/commands/propertycommand<br><br><b>Request Body:</b><br><pre>[   {     "CommandInputForExecution": [       {         "Name": "Value",         "DataType": "ExtendedUint",         "Value": "1500"       }     ],     "PropertyId": "LogViewer.ViewSize",     "CommandId": "Write"   },   {     "PropertyId": "LogViewer.SuppressAlarms",     "CommandId": "Enable"   } ]</pre> <b>Response:</b><br>HTTP 200 OK<br><pre>[   {     "PropertyId": "LogViewer.ViewSize",     "ErrorCode": 0   },   {     "PropertyId": "LogViewer.SuppressAlarms",     "ErrorCode": 0   } ]</pre> |                          |                                                                 |

### 7.10.7 Bulk Commanding with different pair of property values and command id

|                 |                                                                          |                                                  |                                                                                                                                        |
|-----------------|--------------------------------------------------------------------------|--------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------|
| Resource Path   | /validateproperty/commands/propertycommand                               |                                                  |                                                                                                                                        |
| Category        | Public                                                                   |                                                  |                                                                                                                                        |
| HTTP Method     | POST                                                                     |                                                  |                                                                                                                                        |
| Body            | An array of<br>propertyCommandInput<br>ut                                | PropertyCommandInput<br>Execution<br>See [ →210] | Array of property command object.<br>Each property command contains<br>commandid ,propertyid and optional<br>command parameter details |
| User Validation | User Validaiton is at Property level.                                    |                                                  |                                                                                                                                        |
| Response        | HTTP 200 OK<br>An array of BatchExecution objects See Link [→ 209]       |                                                  |                                                                                                                                        |
| Examples        | <b>Request:</b><br>POST : /api/commands/validateproperty/propertycommand |                                                  |                                                                                                                                        |

|  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|--|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | <p><b>Request Body:</b></p> <pre>[   {     "CommandInputForExecution": [       {         "Name": "Value",         "DataType": "ExtendedUInt",         "Value": "1500"       }     ],     "PropertyId": "LogViewer.ViewSize",     "CommandId": "Write"     "Comments": {       "CommonText": "Common comment for all the languages"     },     "Password": "loggedinuserpassword",     "SuperName": "SuperUserName",     "SuperPassword": "SuperUserPassword"   },   {     "CommandInputForExecution": [       {         "Name": "Value",         "DataType": "ExtendedUInt",         "Value": "1500"       }     ],     "PropertyId": "LogViewer.SuppressAlarms",     "CommandId": "Enable"     "Comments": {       "CommonText": "Common comment for all the languages"     },     "Password": "loggedinuserpassword",     "SuperName": "SuperUserName",     "SuperPassword": "SuperUserPassword"   } ]</pre> <p><b>Response:</b></p> <p>HTTP 200 OK</p> <pre>[   {     "PropertyId": "LogViewer.ViewSize",     "ErrorCode": 0   },   {     "PropertyId": "LogViewer.SuppressAlarms",     "ErrorCode": 1,     "ErrorMessage": "command execution failed!"   } ]</pre> |
|--|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

7.10.8 Creating a Subscription for a Change of a

## Command (Obsolete)

Obsolete: Use channelized subscription instead.

|               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |          |                                                                                              |
|---------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|----------------------------------------------------------------------------------------------|
| Resource Path | /sr/commandssubscriptions/:connectionId                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |          |                                                                                              |
| Category      | Public                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |          |                                                                                              |
| HTTP Method   | POST                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |          |                                                                                              |
| Parameters    | Property                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Type     | Description                                                                                  |
|               | connectionId                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | string   | Identifier of subscription<br><b>Required</b>                                                |
|               | propertyIdList                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | String[] | Array of Properties from POST data, which will be subscribed for Command Change notification |
| Body          | Array of PropertyId                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |          |                                                                                              |
| Response      | HTTP 200 OK<br>An array of Subscription objects See Link [→ 46]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |          |                                                                                              |
| Examples      | <p><b>Request:</b></p> <pre>Post :api/sr/commandssubscriptions/87a73827-f25b-41e8-9225-2f08316ea0fb</pre> <p><b>Request Body:</b></p> <pre>[   "System1:HDB_ArchiveGroup_2.Min_Retention.Delay_Span",   "System1.ManagementView:ManagementView.ManagementSystem.Servers.Server.HistoryDatabase.ArchiveSchedules.ActivityLog;.Min_Retention.Delay_Span" //Designation + Property ]</pre> <p><b>Response:</b></p> <pre>[   {     "Key":5,     "PropertyId":"System1:HDB_ArchiveGroup_2.Min_Retention.Delay_Span",     "ErrorCode":0,     "_links":[       {         "Rel":"unsubscribe",         "Href":"api/sr/commandssubscriptions/aa82215b-b635-40a7-92a3-4fcc80a677d1?subscriptionKey=%5b5%5d",         "IsTemplated":false       }     ]   },   {     "Key":6,     "PropertyId":"System1.ManagementView:ManagementView.ManagementSystem.Servers.Server.HistoryDatabase.ArchiveSchedules.ActivityLog;.Min_Retention.Delay_Span",     "ErrorCode":0,     "_links":[       {         "Rel":"unsubscribe",         "Href":"api/sr/commandssubscriptions/aa82215b-b635-40a7-92a3-4fcc80a677d1?subscriptionKey=%5b6%5d",         "IsTemplated":false       }     ]   } ]</pre> |          |                                                                                              |

|  |   |
|--|---|
|  | ] |
|--|---|

## 7.10.9 Push Notification

After subscribing for changes of a command(s) **all** command(s) will be sent in an initial notification. Afterwards any change will be notified.

| Type     | Name (case sensitive)                                |                          |                             |
|----------|------------------------------------------------------|--------------------------|-----------------------------|
| Hub      | 1. norisHub : recommended<br>2. commandsHub : legacy |                          |                             |
| Function | notifyCommands( data )                               |                          |                             |
|          | Property                                             | Type                     | Description                 |
|          | data                                                 | Array of PropertyCommand | See PropertyCommand [→ 207] |

### 7.10.9.1 Creating Channalized Command Subscription

If subscription status will be pushed through notifySubscriptionStatus function available at client side. Command changes will be notified over notifyCommands.

| Resource Path | /sr/commandssubscriptions/channelize/:connectionId                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                                                                                              |
|---------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|----------------------------------------------------------------------------------------------|
| Category      | Public                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |          |                                                                                              |
| HTTP Method   | POST                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |          |                                                                                              |
| Parameters    | Property                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Type     | Description                                                                                  |
|               | connectionId                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | string   | Identifier of subscription<br><b>Required</b>                                                |
|               | propertyIdList                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | String[] | Array of Properties from POST data, which will be subscribed for Command Change notification |
| Body          | Array of PropertyId                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |          |                                                                                              |
| Response      | HTTP 200 OK                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |          |                                                                                              |
| Examples      | <p><b>Request:</b></p> <pre>Post :api/sr/commandssubscriptions/ channelize/9565ca41-8556-4dbb-94cc-1b89451a5db5/87a73827-f25b-41e8-9225-2f08316ea0fb</pre> <p><b>Request Body:</b></p> <pre>[   "System1:HDB_ArchiveGroup_2.Min_Retention.Delay_Span",   "System1.ManagementView:ManagementView.ManagementSystem.Servers.Server.HistoryDatabase.ArchiveSchedules.ActivityLog; .Min_Retention.Delay_Span" //Designation + Property ]</pre> <p>HTTP/1.1 200 OK</p> <p><b>notifySubscriptionStatus :</b></p> <p>Notification 1:</p> <pre>{   "Key":5,   "PropertyId":"System1:HDB_ArchiveGroup_2.Min_Retention.Delay_Span",   "SubscriptionId":"System1:HDB_ArchiveGroup_2.Min_Retention.Delay_Span",   "ErrorCode":0,   "RequestId": "9565ca41-8556-4dbb-94cc-1b89451a5db5",</pre> |          |                                                                                              |

|  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|--|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | <pre> "RequestFor": "notifyCommands",   "_links": [     {       "Rel": "unsubscribe",       "Href": "api/sr/commandssubscriptions/aa82215b- b635-40a7-92a3-4fcc80a677d1?subscriptionKey=%5b5%5d",       "IsTemplated": false     }   ] } Notification 2:{   "Key": 6,   "PropertyId": "System1.ManagementView:ManagementView.Manage mentSystem.Servers.Server.HistoryDatabase.ArchiveSchedules .ActivityLog;.Min_Retention.Delay_Span",   "SubscriptionId": "System1.ManagementView:ManagementView.Ma nagementSystem.Servers.Server.HistoryDatabase.ArchiveSched ules.ActivityLog;.Min_Retention.Delay_Span",   "RequestId": "9565ca41-8556-4dbb-94cc-1b89451a5db5",   "RequestFor": "notifyCommands",   "ErrorCode": 0,   "_links": [     {       "Rel": "unsubscribe",       "Href": "api/sr/commandssubscriptions/aa82215b- b635-40a7-92a3-4fcc80a677d1?subscriptionKey=%5b6%5d",       "IsTemplated": false     }   ] } ] </pre> |
|--|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

### 7.10.9.2 Subscription Status Notification

After subscribing for Commands, for each key subscription status will be sent through notifySubscriptionStatus.

| Type     | Name (case sensitive)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                       |                                                         |
|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|---------------------------------------------------------|
| Hub      | 1. norisHub : recommended<br>2. commandsHub : legacy                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                       |                                                         |
| Function | notifySubscriptionStatus(status)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                       |                                                         |
|          | Property                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Type                  | Description                                             |
|          | Status                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Subscription Response | Subscription objects See Link [ <a href="#">→ 192</a> ] |
| Example  | <pre> {   "Key": 6,   "PropertyId": "System1.ManagementView:ManagementView.Managem entSystem.Servers.Server.HistoryDatabase.ArchiveSchedules.ActivityLog;.Mi n_Retention.Delay_Span",   "SubscriptionId": "System1.ManagementView:ManagementView.Managem entSystem.Servers.Server.HistoryDatabase.ArchiveSchedules.ActivityLog;. Min_Retention.Delay_Span",   "RequestId": "9565ca41-8556-4dbb-94cc-1b89451a5db5",   "RequestFor": "notifyCommands",   "ErrorCode": 0,   "_links": [     {       "Rel": "unsubscribe", </pre> |                       |                                                         |



|  |                                                                                                                                                     |
|--|-----------------------------------------------------------------------------------------------------------------------------------------------------|
|  | <pre> "Href": "api/sr/commandssubscriptions/aa82215b-b635-40a7-92a3-4fcc80a677d1?subscriptionKey=%5b6%5d", "IsTemplated": false     }   ] } ]</pre> |
|--|-----------------------------------------------------------------------------------------------------------------------------------------------------|

### 7.10.9.3 Deleting a Subscription for a Change of a Command.

### 7.10.9.4 Passing subscription keys in query parameter

|               |                                                                                                                                                                                                                          |              |                                               |
|---------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|-----------------------------------------------|
| Resource Path | /sr/commandssubscriptions/:connectionId                                                                                                                                                                                  |              |                                               |
| Category      | Public                                                                                                                                                                                                                   |              |                                               |
| HTTP Method   | DELETE                                                                                                                                                                                                                   |              |                                               |
| Parameters    | <b>Property</b>                                                                                                                                                                                                          | <b>Type</b>  | <b>Description</b>                            |
|               | connectionId                                                                                                                                                                                                             | String       | Identifier of subscription<br><b>Required</b> |
|               | subscriptionkey                                                                                                                                                                                                          | Array of int | List of subscription keys                     |
| Response      | HTTP 200 OK<br>An array of unsubscription objects. See Link <a href="#">[→56]</a>                                                                                                                                        |              |                                               |
| Examples      | <b>Request: Delete</b><br>api/sr/commandssubscriptions/a43e4363-296e-405d-977b-36dec9ddbafc?subscriptionKey=[1,2]<br><b>Response:</b><br><pre> {   "ErrorCode": 0,   "Key": 1 }, {   "ErrorCode": 0,   "Key": 2 } </pre> |              |                                               |

### 7.10.9.5 Passing subscription keys in request body

|               |                                                                                                                                               |             |                                               |
|---------------|-----------------------------------------------------------------------------------------------------------------------------------------------|-------------|-----------------------------------------------|
| Resource Path | /sr/commandssubscriptions/withbody/:connectionId                                                                                              |             |                                               |
| Category      | Public                                                                                                                                        |             |                                               |
| HTTP Method   | DELETE                                                                                                                                        |             |                                               |
| Parameters    |                                                                                                                                               |             |                                               |
| Body          | Array of Subscription keys                                                                                                                    |             |                                               |
| Query         | <b>Property</b>                                                                                                                               | <b>Type</b> | <b>Description</b>                            |
|               | connectionId                                                                                                                                  | String      | Identifier of subscription<br><b>Required</b> |
| Response      | HTTP 200 OK<br>An array of unsubscription objects. See Link <a href="#">[→56]</a>                                                             |             |                                               |
| Examples      | <b>Request:</b><br>api/sr/commandssubscriptions/a43e4363-296e-405d-977b-36dec9ddbafc<br><b>Body Data:</b><br>[ "1", "2" ]<br><b>Response:</b> |             |                                               |

|  |                                                                                                           |
|--|-----------------------------------------------------------------------------------------------------------|
|  | <pre>{<br/>  "ErrorCode": 0,<br/>  "Key": 1<br/>},<br/>{<br/>  "ErrorCode": 0,<br/>  "Key": 2<br/>}</pre> |
|--|-----------------------------------------------------------------------------------------------------------|

## 7.11 Trend Service

| Resource         | Service Category | Requires Authentication | Requires SignalR (Push Notifications) |
|------------------|------------------|-------------------------|---------------------------------------|
| /trendseriesinfo | Public           | x                       |                                       |
| /trendseries     | Public           | x                       |                                       |

### 7.11.1 Retrieving a List of All Trend Collector Objects

Retrieve a list of all Trend Collector objects of a given system. It includes all Online and Offline objects as following

1. Trendlog
2. Trendlog Multiple
3. Trend Log Definitions
4. Trended property with VL flag set to TRUE

|               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |             |                                   |
|---------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|-----------------------------------|
| Resource Path | /trendseriesinfo:/systemId                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |             |                                   |
| Category      | Public                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |             |                                   |
| HTTP Method   | GET                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |             |                                   |
| Parameters    | <b>Property</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | <b>Type</b> | <b>Description</b>                |
|               | systemId                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | uint        | ID of a system<br><b>Required</b> |
| Response      | HTTP 200 OK<br>An array of TrendCollector objects <a href="#">[-64]</a>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |             |                                   |
| Examples      | <b>Request:</b><br>GET /api/trendseriesinfo/1<br><b>Response:</b><br><pre>[   {     "ObjectId": "System1:GmsDevice_1_29129_4194305",     "PropertyName": "Present_Value",     "CollectorObjectOrPropertyId": "System1:GmsDevice_1_29129_113246209",     "TrendseriesId": "System1:GmsDevice_1_29129_113246209.general.Data:_origina l.2._value"   },   {     "ObjectId": "System1:GmsDevice_1_29129_4194305",     "PropertyName": "Present_Value",     "CollectorObjectOrPropertyId": "System1:GmsDevice_1_29129_113246210",     "TrendseriesId": "System1:GmsDevice_1_29129_113246210.general.Data:_origina l.2._value"   } ]</pre> |             |                                   |

### 7.11.2 Retrieving a List of Trend Collector Objects

Retrieve a list of Trend Collector objects. An object ID of a trend object (e.g. TrendLog/TrendLogMultiple) or a property ID of a trended object is to be provided.

|               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |             |                                       |
|---------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|---------------------------------------|
| Resource Path | /trendseriesinfo/:objectOrPropertyId                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |             |                                       |
| Category      | Public                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |             |                                       |
| HTTP Method   | GET                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |             |                                       |
| Parameters    | <b>Property</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | <b>Type</b> | <b>Description</b>                    |
|               | objectOrPropertyId                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | string      | ObjectOrPropertyId<br><b>Required</b> |
| Response      | HTTP 200 OK<br>An array of TrendCollector objects <a href="#">[→64]</a>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |             |                                       |
| Examples      | <b>Request:</b><br>GET /api/trendseriesinfo/System1:GmsDevice_2_1500_83886081<br><b>OR</b><br>//Using Designation<br>GET<br>/api/trendseriesinfo/System1.ApplicationView:ApplicationVi<br>ew.Trends.OfflineTrends.TestNet.Hardware.Devices_1500.Loca<br>l_IO.TL_1;<br><b>Response:</b><br><pre>[   {     "ObjectId": "System1:GmsDevice_2_1500_1",     "PropertyIndex": 1,     "PropertyName": "Present_Value",     "CollectorObjectOrPropertyId": "System1:GmsDevice_2_1500_83886081",     "TrendseriesId": "System1:GmsDevice_2_1500_83886081.general.Data:_offline.._value"   } ]</pre> |             |                                       |

### 7.11.3 Retrieving TrendViewDefinition object details

Retrieve a TrendViewDefition object details. An object ID of a trend view definition.

|               |                                                                                                                                                                                                                                                                  |             |                                                             |
|---------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|-------------------------------------------------------------|
| Resource Path | /trendseriesinfo/tvd/:objectId                                                                                                                                                                                                                                   |             |                                                             |
| Category      | Public                                                                                                                                                                                                                                                           |             |                                                             |
| HTTP Method   | GET                                                                                                                                                                                                                                                              |             |                                                             |
| Parameters    | <b>Property</b>                                                                                                                                                                                                                                                  | <b>Type</b> | <b>Description</b>                                          |
|               | objectId                                                                                                                                                                                                                                                         | string      | ObjectId of Trend View Definition object<br><b>Required</b> |
| Response      | HTTP 200 OK<br>A TrendViewDefinitionRepresentation object <a href="#">[→212]</a>                                                                                                                                                                                 |             |                                                             |
| Examples      | <b>Request:</b><br>GET<br>api/trendseriesinfo/tvd/ApplicationView_Trends_TrendViewDe<br>finitions_sampletvd<br><b>OR</b><br>GET<br>api/trendseriesinfo/tvd/System1.ApplicationViewApplication<br>View.Trends.TrendViewDefinitions.sampletvd;<br><b>Response:</b> |             |                                                             |

|  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|--|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | <pre>{   "AxisX": {     "AutoScale": true,     "AxisType": 0,     "ScaleMax": 100,     "ScaleMin": 0,     "ShowTitle": false,     "ShowZeroLine": false,     "Title": ""   },   "AxisYLeft": {     "AutoScale": true,     "AxisType": 1,     "ScaleMax": 100,     "ScaleMin": 0,     "ShowTitle": false,     "ShowZeroLine": false,     "Title": ""   },   "AxisYRight": {     "AutoScale": true,     "AxisType": 2,     "ScaleMax": 100,     "ScaleMin": 0,     "ShowTitle": false,     "ShowZeroLine": false,     "Title": ""   },   "DescriptionTvd": "",   "ObjectIdOfTvd": "System1:ApplicationView_Trends_TrendViewDefinitions_sampletvd",   "IdTvd": "a6393de5-454d-4f41-8e13-043c2f01dfdb",   "NameTvd": "sampletvd",   "NumberDisplayedSamplesPerTrendSerie": 100,   "RemoveOnlineTrendLogOfDeletedTrendSerie": false,   "TimeRange": {     "ValidTimeRange": 1,     "AbsoluteTimeRange": {       "From": "2019-06-24T10:07:33.075Z",       "To": "2019-06-24T10:12:33.075Z"     },     "RelativeTimeRange": {       "AnchorMode": 1,       "AnchorTime": "2019-06-24T10:12:30.986Z",       "NumberOfTimeUnits": 1,       "TimeUnit": 3     }   },   "TitleLeft": "",   "TitleRight": "",   "TitleTop": "",   "TsdCollectionInfo": {     "TrendSeriesDefinitions": [       {         "ReservedDateTime": "1777-07-07T07:07:07Z",         "AccessDenied": false,</pre> |
|--|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

|  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|--|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | <pre> "ArchiveOn": true, "AxisAttachment": "ToLeft", "ChartLineStyle": "Full", "ChartLineType": "Line", "Color": "ffff0000", "DescriptionTlField": "", "DescriptionTlGms": "Log Viewer-ViewSize", "DescriptionTo": "", "DescriptionToProperty": "View Size", "ObjectIdOfTrendLog": "", "ObjectIdOfTrendLogOnline": "System1:TLO_LogViewer_ViewSize", "ObjectIdOfTrendedObject": "System1:LogViewer.ViewSize:_original.._value", "Index": -1, "IsLineVisible": true, "LineWidth": 1, "MaxTimeStamp": "0001-01-01T00:00:00Z", "MinTimeStamp": "9999-12-31T18:29:59.999Z", "ShowDataLabels": false, "ShowMarkers": false, "ShowQualityIcons": false, "Type": "Online", "UnitText": "", "ValueType": "Float"     }     ]     },     "TvdType": 0,     "TvCovType": "Auto",     "_links": []     } </pre> |
|--|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

#### 7.11.4 Create/Update TrendViewDefinition

Api to Modify Existing Tvd or Create new tvd. If Existing tvd is to be modified, then the TrendViewDefinitionUpdateRepresentation TvdObjectId must be set. If new Tvd is to be created, then TvdObjectId must be empty and CNSDescription and Location parameters must be set.

|               |                                                                                 |                                         |                                                        |
|---------------|---------------------------------------------------------------------------------|-----------------------------------------|--------------------------------------------------------|
| Resource Path | /trendseriesinfo/tvd                                                            |                                         |                                                        |
| Category      | Public                                                                          |                                         |                                                        |
| HTTP Method   | PUT                                                                             |                                         |                                                        |
| Parameters    | <b>Property</b>                                                                 | <b>Type</b>                             | <b>Description</b>                                     |
|               | tvdUpdate                                                                       | TrendViewDefinitionUpdateRepresentation | Trend View Definition Update object<br><b>Required</b> |
| Response      | HTTP 200 OK<br>A TrendViewDefinition object <a href="#">[↗212]</a>              |                                         |                                                        |
| Examples      | <b>Request:</b><br>PUT api/trendseriesinfo/tvd<br><br><b>Request Body:</b><br>{ |                                         |                                                        |

```

"AxisX": {
  "AutoScale": true,
  "ScaleMax": 100,
  "ScaleMin": 0,
  "ShowTitle": false,
  "ShowZeroLine": false,
  "Title": ""
},
"AxisYLeft": {
  "AutoScale": true,
  "ScaleMax": 500,
  "ScaleMin": 0,
  "ShowTitle": false,
  "ShowZeroLine": false,
  "Title": ""
},
"AxisYRight": {
  "AutoScale": true,
  "ScaleMax": 100,
  "ScaleMin": 0,
  "ShowTitle": false,
  "ShowZeroLine": false,
  "Title": ""
},
"ObjectIdOfTvd": "",
"CNSDescription": "tvd4",
"Location":
[System1.ApplicationView:ApplicationView.Trends.TrendViewDefinitions,
  "NumberDisplayedSamplesPerTrendSerie": 200,
  "RemoveOnlineTrendLogOfDeletedTrendSerie": false,
  "TimeRange": {
    "ValidTimeRange": 2,
    "AbsoluteTimeRange": {
      "From": "2020-01-23T09:23:49.969Z",
      "To": "2020-02-07T10:01:48.329Z"
    },
    "RelativeTimeRange": {
      "AnchorMode": 1,
      "AnchorTime": "2020-01-27T09:23:13Z",
      "NumberOfTimeUnits": 1,
      "TimeUnit": 3
    }
  },
  "TitleLeft": "Title for Left",
  "TitleRight": "Title for Right",
  "TitleTop": "Top name",

  "TsdCollectionInfo": {
    "TrendSeriesDefinitions": [
      {
        "ArchiveOn": false,
        "AxisAttachment": "ToLeft",
        "ChartLineStyle": "Full",
        "ChartLineType": "Line",
        "Color": "ffff00f0",
        "ObjectIdOfTrendLog":
[System1.GmsDevice_1_90_83886081",

```

|  |                                                                                                                                                                                                                                                                                                                                                                                                                         |
|--|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | <pre>       "ObjectIdOfTrendLogOnline": "",       "ObjectIdOfTrendedObject": "System1:GmsDevice_1_90_1",       "IsLineVisible": true,       "LineWidth": 1,       "ShowDataLabels": false,       "ShowMarkers": true,       "ShowQualityIcons": false,       "MarkerType": "Circle",       "Type": "Offline"     }   ] },   "TvCovType": "Auto" } </pre> <p><b>Response:</b><br/>A TrendViewDefinition Object[↗212]</p> |
|--|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

### 7.11.5 Delete TrendViewDefinition

|               |                                                                                                                                                                            |             |                                                             |
|---------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|-------------------------------------------------------------|
| Resource Path | /trendseriesinfo/tvd/:objectId                                                                                                                                             |             |                                                             |
| Category      | Public                                                                                                                                                                     |             |                                                             |
| HTTP Method   | DELETE                                                                                                                                                                     |             |                                                             |
| Parameters    | <b>Property</b>                                                                                                                                                            | <b>Type</b> | <b>Description</b>                                          |
|               | objectId                                                                                                                                                                   | string      | ObjectId of Trend View Definition object<br><b>Required</b> |
| Response      | HTTP 200 OK<br>A TrendViewDefinition object [↗212]                                                                                                                         |             |                                                             |
| Examples      | <p><b>Request:</b></p> <pre>DELETE api/trendseriesinfo/tvd/ System1:ApplicationView_Trends_TrendViewDefinitions_tvd42</pre> <p><b>Response:</b></p> HTTP 200 OK No Content |             |                                                             |

### 7.11.6 Delete TrendLogOnline

|               |                                                                                                                                        |             |                                                 |
|---------------|----------------------------------------------------------------------------------------------------------------------------------------|-------------|-------------------------------------------------|
| Resource Path | /trendseriesinfo/tsdOnline/:objectId                                                                                                   |             |                                                 |
| Category      | Public                                                                                                                                 |             |                                                 |
| HTTP Method   | DELETE                                                                                                                                 |             |                                                 |
| Parameters    | <b>Property</b>                                                                                                                        | <b>Type</b> | <b>Description</b>                              |
|               | objectId                                                                                                                               | string      | ObjectId of Trend Log Online<br><b>Required</b> |
| Response      | HTTP 200 OK                                                                                                                            |             |                                                 |
| Examples      | <p><b>Request:</b></p> <pre>DELETE api/trendseriesinfo/tvd/ System1:TLO_bac1_Hardware_proj1device_90_Local_IO_AI_1_Present_Value</pre> |             |                                                 |



|  |                                            |
|--|--------------------------------------------|
|  | <b>Response:</b><br>HTTP 200 OK No Content |
|--|--------------------------------------------|

### 7.11.7 Retrieving Borders of a Trend Series

|               |                                                                                                                                                                                                                                      |             |                                            |
|---------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|--------------------------------------------|
| Resource Path | /trendseries/:trendseriesId/borders                                                                                                                                                                                                  |             |                                            |
| Category      | Public                                                                                                                                                                                                                               |             |                                            |
| HTTP Method   | GET                                                                                                                                                                                                                                  |             |                                            |
| Parameters    | <b>Property</b>                                                                                                                                                                                                                      | <b>Type</b> | <b>Description</b>                         |
|               | trendseriesId                                                                                                                                                                                                                        | string      | Trend series identifier<br><b>Required</b> |
| Response      | HTTP 200 OK<br>A TrendBorder object <a href="#">→64</a>                                                                                                                                                                              |             |                                            |
| Examples      | <b>Request:</b><br>GET<br>api/trendseries/System1%3AGmsDevice_1_1546_113246209.general.Data%3A_original.1._value/borders<br><b>Response:</b><br><pre>{   "From": "2015-05-13T12:00:34.37Z",   "To": "2015-05-13T12:06:57.9Z" }</pre> |             |                                            |

### 7.11.8 Retrieving a List of Trends

Retrives historical time-series, consisting of values, quality attributes and time-stamps.

|               |                                                                                                                                                                                                                                               |             |                                                                            |
|---------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|----------------------------------------------------------------------------|
| Resource Path | /trendseries/:trendseriesId                                                                                                                                                                                                                   |             |                                                                            |
| Category      | Public                                                                                                                                                                                                                                        |             |                                                                            |
| HTTP Method   | GET                                                                                                                                                                                                                                           |             |                                                                            |
| Parameters    | <b>Property</b>                                                                                                                                                                                                                               | <b>Type</b> | <b>Description</b>                                                         |
|               | trendseriesId                                                                                                                                                                                                                                 | string      | Trend series identifier<br><b>Required</b>                                 |
|               | from                                                                                                                                                                                                                                          | DateTime    | Start time of trend series<br><b>Required</b>                              |
|               | to                                                                                                                                                                                                                                            | DateTime    | End time of trend series<br><b>Required</b>                                |
|               | intervals                                                                                                                                                                                                                                     | ushort      | If provided, the result contains the min. and max. value per interval.     |
|               | addDescriptor                                                                                                                                                                                                                                 | bool        | If true, adds localized value (if applicable) to result.<br>Default: false |
| Response      | HTTP 200 OK<br>A TrendSeries object <a href="#">→64</a>                                                                                                                                                                                       |             |                                                                            |
| Examples      | <b>Request:</b><br>GET<br>api/trendseries/System1:GmsDevice_1_29129_83886081.general.Data:_offline.._value?from=2015-05-13T12:05:53.000Z&to=2015-05-13T12:05:54.000Z&intervals=5&description=true<br><b>Response:</b><br><pre>{   "Id":</pre> |             |                                                                            |

```

[System1:GmsDevice_1_29129_83886081.general.Data:_offline.
._value",
  "SeriesPropertyId":
[System1:GmsDevice_1_29129_16777217.Present_Value",
  "Series": [
    {
      "Value": "1",
      "DisplayValue": "ACTIVE",
      "Quality": "8589934592",
      "QualityGood": true,
      "Timestamp": "2015-05-13T12:05:52.59Z"
    },
    {
      "Value": "1",
      "DisplayValue": "INACTIVE",
      "Quality": "8589934592",
      "QualityGood": true,
      "Timestamp": "2015-05-13T12:05:55.59Z"
    }
  ]
}

```

### 7.11.9 Retrieving a List of Trend Aggregates

For good samples of history data retrieves mathematical aggregations for Standard Deviation and Least Square Regression.

**Trend aggregation is not Distribution enabled. It only serves for local system only.**

|               |                                                                    |             |                                                                            |
|---------------|--------------------------------------------------------------------|-------------|----------------------------------------------------------------------------|
| Resource Path | /trendseries/:trendseriesId/:aggregateUnit                         |             |                                                                            |
| Category      | Public                                                             |             |                                                                            |
| HTTP Method   | GET                                                                |             |                                                                            |
| Parameters    | <b>Property</b>                                                    | <b>Type</b> | <b>Description</b>                                                         |
|               | trendseriesId                                                      | string      | Trend series identifier<br><b>Required</b>                                 |
|               | from                                                               | DateTime    | Start time of trend series<br><b>Required</b>                              |
|               | to                                                                 | DateTime    | End time of trend series<br><b>Required</b>                                |
|               | aggregateUnit                                                      | ushort      | Unit on which aggregation will take place. See [115].<br>Default is 7      |
|               | length                                                             | int         | Default Value = 1000                                                       |
|               | qualityMask                                                        | long        | Mask tells you which bits you have to take into account                    |
|               | qualityValue                                                       | long        | Value tells you what value the bits in consideration need to have          |
|               | correctedValues                                                    | bool        | Flag that signals use of corrected values in Trends (Default Value=True)   |
|               | addDescriptor                                                      | bool        | If true, adds localized value (if applicable) to result.<br>Default: false |
| Response      | HTTP 200 OK<br>A TrendAggregateSeries object <a href="#">[211]</a> |             |                                                                            |

|          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Examples | <p><b>Request:</b></p> <pre>GET api/trendseries/System1%3AGmsDevice_1_29128_83886081.general.Data%3A_offline._value/7?from=2015-10-27T06:01:04.510Z&amp;to=2015-11-27T11:44:41.660Z&amp;length=1000&amp;qualityMask=0&amp;qualityValue=0&amp;addDescriptor=false</pre> <p><b>Response:</b></p> <pre>{   "Id": "System1:GmsDevice_1_29128_83886081.general.Data:_offline._value",   "SeriesPropertyId": "System1:GmsDevice_1_29128_1.Present_Value",   "Series": [     {       "AvgX": 1445945963123.153,       "AvgY": 65.28828828828829,       "Count": 333,       "FromValueDescriptor": "65",       "FromValue": 65,       "FromTime": "2015-10-27T06:01:05.12Z",       "Min": 45,       "Max": 80,       "StDevX": 287308.05418783723,       "StDevY": 8.742276612722005,       "SumX": 481500005720010,       "SumXX": 6.962229895146511e+26,       "SumXY": 481549949961870,       "SumY": 83,       "SumYY": 1444883,       "ToTime": "2015-10-27T06:17:41.13Z",       "ToValueDescriptor": "61",       "ToValue": 61,       "VarX": 82545918001.2012,       "VarY": 76.42740037334613     },     .....   ] }</pre> |
|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

### 7.11.9.1 Aggregate Unit

**Aggregate Unit** must have one of the following values.

| Id | Description |
|----|-------------|
| 0  | Year        |
| 1  | Quarter     |
| 2  | Month       |
| 3  | Week        |
| 4  | Day         |
| 5  | Hour        |
| 6  | Minute      |
| 7  | Second      |
| 8  | Millisecond |

## 7.12 History Logs Service

This service provides a way to retrieve data for available log tables.

| Resource     | Service Category | Requires Authentication | Requires SignalR (Push Notifications) |
|--------------|------------------|-------------------------|---------------------------------------|
| /historylogs | Public           | x                       |                                       |

### 7.12.1 Retrieving Log Tables

Provides a list of all available log tables with other details which can be used in subsequent requests for data retrieval and filtering. This method returns tables for a specified system.

|               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |             |                                   |
|---------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|-----------------------------------|
| Resource Path | /historylogs/:systemId                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |             |                                   |
| Category      | Public                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |             |                                   |
| HTTP Method   | GET                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |             |                                   |
| Parameters    | <b>Property</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | <b>Type</b> | <b>Description</b>                |
|               | systemId                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | uint        | ID of a system<br><b>Required</b> |
| Response      | HTTP 200 OK<br>List of LogTable objects. See [→215]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |             |                                   |
| Examples      | <b>Request:</b><br>GET api/historylogs/1<br><b>Response:</b><br><pre>[   {     "Name": "ActivityLogTable",     "Descriptor": "Activities",     "Description": "Logs Activity Data",     "HasChildColumns": false,     "HasConditionFilter": true,     "HasNameFilter": true,     "HasTimeFilter": true,     "HasTimeFilterDefinedBySource": false,     "HasAndOperatorOnly": true,     "HasMaxAge": false,     "HasSingleNameFilter": true,     "SupportedTypes": [       "*"     ]   },   {     "Name": "AlarmLogTable",     "Descriptor": "Events",     "Description": "Logs Alarm Data",     "HasChildColumns": false,     "HasConditionFilter": true,     "HasNameFilter": true,     "HasTimeFilter": true,     "HasTimeFilterDefinedBySource": false,     "HasAndOperatorOnly": true,     "HasMaxAge": false,     "HasSingleNameFilter": true,     "SupportedTypes": [</pre> |             |                                   |

|  |                                              |
|--|----------------------------------------------|
|  | <pre>         " * "       ]     }   ] </pre> |
|--|----------------------------------------------|

## 7.12.2 Retrieving Columns of a Log Table

Provides a list columns for a requested log table from the specified system. Each column is associated with other details which can be used in subsequent requests for displaying and filtering data.

|               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |             |                                                                                                                                                    |
|---------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|----------------------------------------------------------------------------------------------------------------------------------------------------|
| Resource Path | /historylogs/:systemId/:tableName                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |             |                                                                                                                                                    |
| Category      | Public                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |             |                                                                                                                                                    |
| HTTP Method   | GET                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |             |                                                                                                                                                    |
| Parameters    | <b>Property</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | <b>Type</b> | <b>Description</b>                                                                                                                                 |
|               | systemId                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | uint        | ID of a system<br><b>Required</b>                                                                                                                  |
|               | tableName                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | string      | Name of table for which columns need to be retrieved<br><b>Required</b>                                                                            |
|               | isParentColumns                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | bool        | If TRUE then parent table columns will be retrieved for the requested table. Otherwise child table columns will be retrieved.<br>(Default is TRUE) |
| Response      | HTTP 200 OK<br>List of LogColumn objects. See [→216]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |             |                                                                                                                                                    |
| Examples      | <b>Request:</b><br>GET api/historylogs/1/ActivityLogTable<br><b>Response:</b><br><pre> [   {     "Name": "UserName",     "Descriptor": "User Name",     "DataType": "String",     "ErrorSupport": false,     "isArray": false,     "IsDefault": true,     "IsHidden": false,     "IsSortable": true,     "IsEnum": false,     "IsFilterable": true   },   {     "Name": "Attachment",     "Descriptor": "Attachment",     "DataType": "String",     "ErrorSupport": false,     "isArray": false,     "IsDefault": true,     "IsHidden": false,     "IsSortable": true,     "IsEnum": false,     "IsFilterable": false   } ] </pre> |             |                                                                                                                                                    |

### 7.12.3 Retrieving Condition Filter Operators for a Column

Provides a list of Condition Filter operators supported by the requested column. This operators then can be used to compose a Condition Filter.

|               |                                                                                                                                                                                                                              |             |                                                                                          |
|---------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|------------------------------------------------------------------------------------------|
| Resource Path | /historylogs/:systemId/:tableName/operators/:columnName                                                                                                                                                                      |             |                                                                                          |
| Category      | Public                                                                                                                                                                                                                       |             |                                                                                          |
| HTTP Method   | GET                                                                                                                                                                                                                          |             |                                                                                          |
| Parameters    | <b>Property</b>                                                                                                                                                                                                              | <b>Type</b> | <b>Description</b>                                                                       |
|               | systemId                                                                                                                                                                                                                     | uint        | ID of a system<br><b>Required</b>                                                        |
|               | tableName                                                                                                                                                                                                                    | string      | Name of the Table<br><b>Required</b>                                                     |
|               | columnName                                                                                                                                                                                                                   | string      | Name of column for which Condition Filter operators will be retrieved<br><b>Required</b> |
| Response      | HTTP 200 OK<br>LogConditionFilter object. See [→216]                                                                                                                                                                         |             |                                                                                          |
| Examples      | <b>Request:</b><br>GET api/historylogs/1/ActivityLogTable/operators/LogType<br><b>Response:</b> <pre>{   "Operators": [     "=",     "&lt;&gt;"   ],   "ErrorInfo": [     "This is Error1",     "This is Error2"   ] }</pre> |             |                                                                                          |

### 7.12.4 Retrieving Enum Values for a Column

Provides a list of enum values which are associated with the requested column if column is of type enum. These enum values then can be used to compose a Condition Filter. This list will be a global list from all systems i.e. union of all the values from all systems.

|               |                                                          |             |                                                                           |
|---------------|----------------------------------------------------------|-------------|---------------------------------------------------------------------------|
| Resource Path | /historylogs/:systemId/:tableName/enumvalues/:columnName |             |                                                                           |
| Category      | Public                                                   |             |                                                                           |
| HTTP Method   | GET                                                      |             |                                                                           |
| Parameters    | <b>Property</b>                                          | <b>Type</b> | <b>Description</b>                                                        |
|               | systemId                                                 | uint        | ID of a system<br><b>Required</b>                                         |
|               | tableName                                                | string      | Name of the table<br><b>Required</b>                                      |
|               | columnName                                               | string      | Name of column for which enum values will be retrieved<br><b>Required</b> |
| Response      | HTTP 200 OK<br>LogEnumValues object. See [→216]          |             |                                                                           |

|          |                                                                                                                                                                                                                                                  |
|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Examples | <b>Request:</b><br>GET api/historylogs/1/ActivityLogTable/enumvalues/LogType<br><b>Response:</b><br><pre>{   "EnumValues": [     "Unknown",     "Reno",     "Trend"   ],   "ErrorInfo": [     "This is Error1",     "This is Error2"   ] }</pre> |
|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

### 7.12.5 Retrieving Data for Table

Provides page wise data for the requested table, based on specified filters and sorting criteria. User also needs to send a list of interested columns. If no columns are provided data for all columns will be returned.

A call without a snapshotId creates a snapshot with all filters and sorting parameters applied. Call with a snapshotId returns data of this snapshot without applying new filters and sorting criteria.

|               |                                   |             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|---------------|-----------------------------------|-------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Resource Path | /historylogs/:systemId/:tableName |             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Category      | Public                            |             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| HTTP Method   | POST                              |             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Parameters    | <b>Property</b>                   | <b>Type</b> | <b>Description</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|               | systemId                          | uint        | ID of a system<br><b>Required</b>                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|               | tableName                         | string      | Name of table<br><b>Required</b>                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|               | snapshotId                        | string      | User defined Snapshot id.<br>Provided Snapshot Id can be used to cancel report processing. If SnapshotId is provided by client then same Id will be received in snapshotId as response else client will receive system generated snapshotId                                                                                                                                                                                                                                    |
|               | sortColumnData                    | string      | This is a JSON serialized string of Array of LogSortInfo objects. For LogSortInfo see [→216]<br>This array conveys 3 different information for sorting. <ol style="list-style-type: none"> <li>1. ColumnName on which sorting needs to be performed</li> <li>2. LogSortType either Ascending or Descending See [→216]</li> <li>3. Order in which elements are present in the array will be considered as sort priority. i.e. first element has the highest priority</li> </ol> |
|               | nameFilter                        | string      | This is JSON serialized string of a string array. Each element in the                                                                                                                                                                                                                                                                                                                                                                                                          |

|          |                                                                                                                                                                                                                                                                                                                                                     |            |                                                                                                                                                                                                                                         |
|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|          |                                                                                                                                                                                                                                                                                                                                                     |            | array is a Name Filter which will be used to filter data. See [→121]                                                                                                                                                                    |
|          | conditionFilter                                                                                                                                                                                                                                                                                                                                     | string     | This is Condition Filter string which will be used to filter data. See [→122]                                                                                                                                                           |
|          | fromDate                                                                                                                                                                                                                                                                                                                                            | DateTime   | From date time for the Time Filter<br>Note: fromDate and toDate both should be present to Time filter the data                                                                                                                          |
|          | toDate                                                                                                                                                                                                                                                                                                                                              | DateTime   | To date time for the Time Filter<br>Note: fromDate and toDate both should be present to Time filter the data                                                                                                                            |
|          | isDefinedBySourceTimeFilter                                                                                                                                                                                                                                                                                                                         | bool       | Flag which denotes whether Time Filter defined by source should be applied<br>If TRUE then Time Filter received in the request will be ignored and instead Time Filter defined by source will be used to filter data (Default is FALSE) |
|          | clientTimeSpan                                                                                                                                                                                                                                                                                                                                      | string     | Client's timezone UTC offset converted to timespan.<br>Used to localize datetime data as per client's time zone.                                                                                                                        |
|          | page                                                                                                                                                                                                                                                                                                                                                | int        | Current page number (Default is 1)                                                                                                                                                                                                      |
|          | size                                                                                                                                                                                                                                                                                                                                                | int        | Number of records per page                                                                                                                                                                                                              |
|          | isChildDataRequired                                                                                                                                                                                                                                                                                                                                 | bool       | Flag which denotes that user is also interested in child data, if supported by the requested table (Default is FALSE)                                                                                                                   |
| Body     | columnList                                                                                                                                                                                                                                                                                                                                          | ColumnList | JSON serialized object of type LogColumnList. See [→217]                                                                                                                                                                                |
| Response | HTTP 200 OK<br>LogTabData object. See [→217]                                                                                                                                                                                                                                                                                                        |            |                                                                                                                                                                                                                                         |
| Examples | <b>Request:</b><br>Post : api/historylogs/1/ActivityLogTable?size=2&isChildDataRequired=true<br><br><b>Request Body:</b><br><pre>{   "Parent": [     "UserName",     "Workstation",     "DpeName1",     "Action",     "MessageText"   ],   "Child": [     "ParentId",     "RDbId",     "RowNum",     "DpeName",     "Id",     "LogType"   ] }</pre> |            |                                                                                                                                                                                                                                         |



|  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|--|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | <p><b>Response:</b></p> <pre>{   "TableName": "ActivityLogTable",   "Size": 2,   "Page": 1,   "Result": [     {       "UserName": "DefaultAdmin",       "Workstation": "_#WEBCLIENT#_",       "DpeName1": "System1:Gms_Security_Users",       "Action": "Login"     },     {       "UserName": "root",       "Workstation": "AAEINPU742011D",       "DpeName1": "System1:Gms_Security_Users",       "Action": "Automatic Logoff"     }   ],   "SnapshotId": "ceffdb59-6b5d-4b2a-a3e5-e127867ed299:fe671773-1d76-4f13-8630-a338263565ca",   "Total": 2919,   "_links": [     {       "Rel": "discardsnapshot",       "Href": "api/historylogs/1/ActivityLogTable/ceffdb59-6b5d-4b2a-a3e5-e127867ed299%3afe671773-1d76-4f13-8630-a338263565ca",       "IsTemplated": false     },     {       "Rel": "page",       "Href": "api/historylogs/1/ActivityLogTable?snapShotId=ceffdb59-6b5d-4b2a-a3e5-e127867ed299%3afe671773-1d76-4f13-8630-a338263565ca&amp;page=1&amp;size=2",       "IsTemplated": false     },     {       "Rel": "next",       "Href": "api/historylogs/1/ActivityLogTable?snapShotId=ceffdb59-6b5d-4b2a-a3e5-e127867ed299%3afe671773-1d76-4f13-8630-a338263565ca&amp;page=2&amp;size=2",       "IsTemplated": false     }   ],   "ErrorInfo": [     "This is error 1",     "This is error 2"   ] }</pre> |
|--|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

### 7.12.5.1 Name Filter

- Name filter allows to filter data on the basis of Designation of the object
- If no Name filter is specified, data will be retrieved for all objects of current system.

- While specifying Name filter, following wild cards are supported: \* and ?
- To retrieve child nodes of an object, Name filter string should end with .\* (<dot><star>)

## 7.12.5.2 Condition Filter

- Condition filter can be provided on multiple columns
- Multiple conditions on same or different columns can be ANDed
- If Condition filter is available on invalid column, data is zero records
- Condition filter specification formats for different types of columns:

### 1. String column

```
<'Column Descriptor'>[space]<Operator>[space]<"Value">
e.g. 'Location' = "\"text\""
```

Column Descriptor is in single quotes and Value is in double quotes. If an input string contains double quotes (""), it needs to be escaped with backslash(\).

### 2. Integer/float/double column

```
<'Column Descriptor'>[space]<Operator>[space]<Value>
```

Column Descriptor is in single quotes and Value is NOT in double quotes.

### 3. Date Time column

```
<'Column Descriptor'>[space]<Operator>[space]<Value>
```

Column Descriptor is in single quotes and Value is NOT in double quotes. Here date time value format should be based on client/logged in User's culture.

### 4. Enum (Test Group) column (type of String)

```
<'Column Descriptor'>[space]<Operator>[space]<"Value">
```

Column Descriptor is in single quotes and Value is in double quotes.

### 5. GmsDuration column

```
<'Column Descriptor'>[space]<Operator>[space]<"Value">
```

Column Descriptor is in single quotes and Value is in double quotes. Gms duration is special kind of case for which value should be specified in specific format. Its fixed format is DD:HH:MM:SS.fff

- Specifying multiple values for a column:

```
<'Column Descriptor'>[space]<Operator>[space]{<Value 1>; [<value 2>; [<value N>]}
```

e.g., Discipline = {Fire ; HVAC}

This is nothing but OR operator supported between conditions for a single column. This format is supported only for equal-to(=) operator.

- Condition filter specification with Array values:  
Some columns support display of array data. To specify condition filter on these columns, <-(IN) operator shall be used.

## 7.12.6 Discarding a Snapshot of a Log Table

Discards a snapshot of log table from memory. The data of this snapshot cannot be accessed afterwards.

|               |                                                                                                                                                                                                                                                                                                                                              |             |                                                            |
|---------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|------------------------------------------------------------|
| Resource Path | /historylogs/:systemId/:tableName/:snapshotId                                                                                                                                                                                                                                                                                                |             |                                                            |
| Category      | Public                                                                                                                                                                                                                                                                                                                                       |             |                                                            |
| HTTP Method   | DELETE                                                                                                                                                                                                                                                                                                                                       |             |                                                            |
| Parameters    | <b>Property</b>                                                                                                                                                                                                                                                                                                                              | <b>Type</b> | <b>Description</b>                                         |
|               | systemId                                                                                                                                                                                                                                                                                                                                     | uint        | ID of a system<br><b>Required</b>                          |
|               | tableName                                                                                                                                                                                                                                                                                                                                    | string      | Name of table<br><b>Required</b>                           |
|               | snapshotId                                                                                                                                                                                                                                                                                                                                   | string      | Id of snapshot which is to be discarded<br><b>Required</b> |
| Response      | HTTP 200 OK<br>LogDiscardedSnapshot object. See [→217]                                                                                                                                                                                                                                                                                       |             |                                                            |
| Examples      | <b>Request:</b><br>DELETE api/historylogs/1/ActivityLogTable/ceffdb59-6b5d-4b2a-a3e5-e127867ed299%3afe671773-1d76-4f13-8630-a338263565ca<br><b>Response:</b><br><pre>{   "TableName": "ActivityLogTable",   "SnapshotId": "c4005c4b-2d00-4a9c-b22a-750d55fba609:591ce21f-11a2-42e6-b875-21ec59abea70",   "IsSnapshotDiscarded": true }</pre> |             |                                                            |

### 7.12.6.1 Discard Snapshot strategy used by the WSI manager

Every Snapshot can consists of millions of records from the log table, it's a very heavy object to maintain. Maintaining multiple such snapshots for every session can intern increase the load of WSI in terms of Memory and Performance. So WSI have to have its own strategy to discard any snapshots which are not in use.

For this following configurable parameters will be used:

| Property                       | Value      | Description                                                                                                                                                                                                         |
|--------------------------------|------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| MaxSnapshotsForSingleSession   | 5          | Maximum number of Snapshots that a single session can have at any time                                                                                                                                              |
| MaxSnapshotInactiveTime        | 10 minutes | Maximum time in Minutes for which snapshot will be maintained by WSI if it is not used. After 10 minutes of unused time Snapshot will be discarded                                                                  |
| MaxVirtualMemorySizeForProcess | bytes      | Default Value: 4294967296 bytes (4 GB)<br>Memory band for WSI manager. If WSI is running out of this memory band then before creating a new snapshot, an existing Snapshot from the system will be discarded first. |

For discarding Snapshots following strategy will be used:

**Greedy or fair share:**

1. First check if current session is having any snapshots associated with it. If YES then the Snapshot with the oldest LastUsedTime will be identified to discard.
2. If current session is not having any Snapshot associated with it. Then the session for which maximum number of Snapshots are present will be identified first. And from that session Snapshot with the oldest LastUsedTime will be identified to discard.

Below are the 4 different cases in which force discard of snapshot will be called:

1. Request to create a new Snapshot and WSI manager is running out of decided Memory band
  - In this case till the time sufficient memory is not available and WSI runs within the decided memory band one by one Snapshots will be discarded by applying above mentioned strategy.
2. Request to create a new Snapshot and limit for maximum number of Snapshots is reached for that session
  - In this case applying above strategy one snapshot will be discarded first and after that request to create a new Snapshot will be processed
3. Snapshot is idle (not used) for 10 minutes
  - Any Snapshot for any session will be discarded if it is not in use for 10 minutes
4. User logs out/ Session timeout
  - If user logs out from WSI or if session timeout is called then all the snapshots for that session if any will be discarded.

## 7.12.7 Validating Condition Filter

Validates the Condition Filter which will be used to filter the data. If filter is not valid then a to-do message is provided.

|               |                                                                                                                                                                                                            |             |                                                        |
|---------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|--------------------------------------------------------|
| Resource Path | /historylogs/:systemId/:tableName/validateconditionfilter/:conditionFilter                                                                                                                                 |             |                                                        |
| Category      | Public                                                                                                                                                                                                     |             |                                                        |
| HTTP Method   | GET                                                                                                                                                                                                        |             |                                                        |
| Parameters    | <b>Property</b>                                                                                                                                                                                            | <b>Type</b> | <b>Description</b>                                     |
|               | systemId                                                                                                                                                                                                   | uint        | ID of a system<br><b>Required</b>                      |
|               | tableName                                                                                                                                                                                                  | string      | Name of the Table<br><b>Required</b>                   |
|               | conditionFilter                                                                                                                                                                                            | string      | Condition filter string to validate<br><b>Required</b> |
| Response      | HTTP 200 OK<br>LogConditionFilterValidation object. See [→217]                                                                                                                                             |             |                                                        |
| Examples      | <b>Request:</b><br>GET<br>api/historylogs/1/EventLogTable/validateconditionfilter/'Action'%20=%20%22Add%20Camera<br><b>Response:</b><br>{<br>"ErrorPosition": 11,<br>"IsValidConditionFilter": false,<br>} |             |                                                        |

|  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|--|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | <pre> "ErrorMessageDescriptor": "The symbol/term could not be recognized: \"Add Camera\", "ToDoMessageDescriptor": "Correct the condition filter expression as per rules:\r\nCondition filter name: 'HdbId', 'HRdbId', 'Event ID', 'Event Category', 'Event Time', ..., \r\nCondition filter operator: =, &lt;&gt;, &lt;, &lt;=, &gt;, &gt;=, &lt;-, \r\nCondition filter data: textual data (e.g. '\\"Any text\\"'), numeric data (e.g. '42'), date time data (e.g. '07/05/1976 14:07:00'), #COM, N/A, #ERR, null, \r\nBoolean operator: AND, OR, NOT, ;, \r\nSymbol: (, ), {, }, " } </pre> |
|--|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

## 7.12.8 Retrieving Child Data for a Parent Table

Retrieves child data for a parent table. To retrieve child data a call 'Retrieving Data for Table' with isChildDataRequired set to true is required first. User also needs to send a list of interested columns for child data. If no columns are provided data for all columns will be returned. See [→**Error! Bookmark not defined.**]

|               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |             |                                                                         |
|---------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|-------------------------------------------------------------------------|
| Resource Path | /historylogs/:systemId/:tableName/:snapshotId/:parentId                                                                                                                                                                                                                                                                                                                                                                                                                                    |             |                                                                         |
| Category      | Public                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |             |                                                                         |
| HTTP Method   | POST                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |             |                                                                         |
| Parameters    | <b>Property</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | <b>Type</b> | <b>Description</b>                                                      |
|               | systemId                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | uint        | ID of a system<br><b>Required</b>                                       |
|               | tableName                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | string      | Name of table<br><b>Required</b>                                        |
|               | snapshotId                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | string      | Snapshot Id for which child data will be retrieved<br><b>Required</b>   |
|               | parentId                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | string      | Parent Id for which child data needs to be retrieved<br><b>Required</b> |
| Body          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |             |                                                                         |
| Response      | HTTP 200 OK<br>LogChildData object. See [→217]                                                                                                                                                                                                                                                                                                                                                                                                                                             |             |                                                                         |
| Examples      | <p><b>Request:</b></p> <pre>POST api/historylogs/1/EventLogTable/78b71424-bcce-44e1-9d7e-ffdfc11a0179:f6139cbd-5854-467f-be0e-227cab6b88fa/235</pre> <p><b>Request Body:</b></p> <p><b>Response:</b></p> <pre>{   "Result": [     {       "RowNum": "2",       "DpeName": "System1:ManagementView_ManagementSystem_Servers_Server.SoftwareRunning",       "Id": "235"     },     {       "RowNum": "1",       "DpeName": "System1:ManagementView_ManagementSystem_Servers_Server.So </pre> |             |                                                                         |

|  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|--|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | <pre> ftwareRunning",     "Id": "354",     "LogType": "Property Changes"   } ], "Total": 2, "Page": 1, "Size": 100000, "_links": [   {     "Rel": "discardsnapshot",     "Href": "api/historylogs/1/EventLogTable/78b71424- bcce-44e1-9d7e-ffdfc11a0179%3af6139cbd-5854-467f-be0e- 227cab6b88fa",     "IsTemplated": false   },   {     "Rel": "page",     "Href": "api/historylogs/1/EventLogTable/78b71424- bcce-44e1-9d7e-ffdfc11a0179%3af6139cbd-5854-467f-be0e- 227cab6b88fa/235?page=1&amp;size=100000",     "IsTemplated": false   } ], "ErrorInfo":[   "This is error 1",   "This is error 2" ] } </pre> |
|--|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

### 7.12.9 Cancel Report Processing

Cancels in-process report generation for the SnapshotId that was provided in Retrive Data Pagewise Request [119]. If cancellation is successful then Retrive Data Pagewise Request will return Http status Bad Request(400) with below error:

Id":2402098

Error:"htlog\_cancelled"

Details:"Request preliminary canceled by the user."}

|               |                                                                                                                                                                                                                                                   |             |                                                         |
|---------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|---------------------------------------------------------|
| Resource Path | /historylogs/:systemId/:snapshotId                                                                                                                                                                                                                |             |                                                         |
| Category      | Public                                                                                                                                                                                                                                            |             |                                                         |
| HTTP Method   | DELETE                                                                                                                                                                                                                                            |             |                                                         |
| Parameters    | <b>Property</b>                                                                                                                                                                                                                                   | <b>Type</b> | <b>Description</b>                                      |
|               | systemId                                                                                                                                                                                                                                          | uint        | ID of a system<br><b>Required</b>                       |
|               | snapshotId                                                                                                                                                                                                                                        | string      | Id of snapshot which is to be Cancel<br><b>Required</b> |
| Response      | HTTP 200 OK<br>LogDiscardedSnapshot object. See [→217]                                                                                                                                                                                            |             |                                                         |
| Examples      | <b>Request:</b><br>DELETE<br><a href="https://aaepun001661.ad001.siemens.net:444/WSIApp/api/historylogs/1/sdfsdfsdf3333">https://aaepun001661.ad001.siemens.net:444/WSIApp/api/historylogs/1/sdfsdfsdf3333</a><br><b>Response:</b><br>HTTP 200 OK |             |                                                         |

## 7.13 Diagnostics Service

| Resource     | Service Category | Requires Authentication | Requires SignalR (Push Notifications) |
|--------------|------------------|-------------------------|---------------------------------------|
| /diagnostics | Public           |                         |                                       |

### 7.13.1 Test Reachability of the Web Service Interface

A test whether or not the Web Service Interface replies to requests. This service also returns Server's UTC datetime as json response.

|               |                                                                                                              |             |                                           |
|---------------|--------------------------------------------------------------------------------------------------------------|-------------|-------------------------------------------|
| Resource Path | /diagnostics                                                                                                 |             |                                           |
| Category      | Public                                                                                                       |             |                                           |
| HTTP Method   | GET                                                                                                          |             |                                           |
| Parameters    | <b>Property</b>                                                                                              | <b>Type</b> | <b>Description</b>                        |
|               | clientDatetime                                                                                               | string      | Client's UTC datetime.<br><b>Optional</b> |
| Response      | HTTP 200 OK <a href="#">[→228]</a>                                                                           |             |                                           |
| Examples      | <b>Request:</b><br>GET /NBWSITest/NorthboundWebApiClient/api/diagnostics/<br><b>Response:</b><br>HTTP 200 OK |             |                                           |

## 7.14 Language Service

| Resource   | Service Category | Requires Authentication | Requires SignalR (Push Notifications) |
|------------|------------------|-------------------------|---------------------------------------|
| /languages | Public           | x                       |                                       |

### 7.14.1 Retrieving the Language of the Logged in User

|               |                                                                                                                                                                    |  |  |
|---------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|--|
| Resource Path | /languages                                                                                                                                                         |  |  |
| Category      | Public                                                                                                                                                             |  |  |
| HTTP Method   | GET                                                                                                                                                                |  |  |
| Response      | HTTP 200 OK<br>A Language object <a href="#">[→64]</a>                                                                                                             |  |  |
| Examples      | <b>Request:</b><br>GET /api/Languages?api_key=username:password<br><b>Response:</b><br><pre>{   "Descriptor": "English (United States)",   "Code": "en-US" }</pre> |  |  |

## 7.15 Image Service

| Resource | Service Category | Requires Authentication | Requires SignalR (Push Notifications) |
|----------|------------------|-------------------------|---------------------------------------|
| /images  | Internal         | x                       |                                       |

## 7.15.1 Retrieving an Image


**NOTE:**

In order to successfully process this request, the image needs to be present in a feasible format and size in the system.

|               |                                                                                                                                                                                                                                                                                                                                               |                                              |                                                                                                                                                            |
|---------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Resource Path | /images:/imageId                                                                                                                                                                                                                                                                                                                              |                                              |                                                                                                                                                            |
| Category      | Internal                                                                                                                                                                                                                                                                                                                                      |                                              |                                                                                                                                                            |
| HTTP Method   | GET                                                                                                                                                                                                                                                                                                                                           |                                              |                                                                                                                                                            |
| Parameters    | <b>Property</b>                                                                                                                                                                                                                                                                                                                               | <b>Type</b>                                  | <b>Description</b>                                                                                                                                         |
|               | imageId                                                                                                                                                                                                                                                                                                                                       | string                                       | Image name (file name without extension). Note that while XAML files can contain multiple images, only one image per file is supported.<br><b>Required</b> |
|               | path                                                                                                                                                                                                                                                                                                                                          | string                                       | Path within a library. As of this writing, the path must not include the file extension: only XAML files are supported.<br><b>Required</b>                 |
|               | format                                                                                                                                                                                                                                                                                                                                        | string                                       | Image format (png or svg)<br><b>Required</b>                                                                                                               |
|               | width                                                                                                                                                                                                                                                                                                                                         | int                                          | Image width in pixels                                                                                                                                      |
|               | height                                                                                                                                                                                                                                                                                                                                        | int                                          | Image height in pixels                                                                                                                                     |
|               | encodeAsBase64                                                                                                                                                                                                                                                                                                                                | bool                                         | If true, the image is returned as a base64-encoded string.                                                                                                 |
| Response      | HTTP 200 OK<br>The returned data depends on encodeAsBase64.                                                                                                                                                                                                                                                                                   |                                              |                                                                                                                                                            |
|               | <b>encodeAsBase64</b>                                                                                                                                                                                                                                                                                                                         | <b>Return Type</b>                           | <b>Description</b>                                                                                                                                         |
|               | false (default)                                                                                                                                                                                                                                                                                                                               | image<br>(image media type; see [10] [→ 16]) | Image in the requested format                                                                                                                              |
|               | true                                                                                                                                                                                                                                                                                                                                          | string                                       | base64-encoded string                                                                                                                                      |
| Examples      | <b>Request:</b><br>/api/images/Evt_Discp_Infrastr_None_001?format=png&width=1&height=1&path=System1:libraries%5CGlobal_Base_HQ_1%5Cicons&encodeAsBase64=true<br><br><b>Response:</b><br>iVBORw0KGgoAAAANSUgAAAAEAAAABCAyAAAAfFcSJAAXNSR0IArs4c6QAAARnQU1BAACxjwv8YQUAAAJcEhZcwAADsIAAA7CARUoSoAAAAANSURBVhXY/j//78GAakAyb6sRXrAAAAE1FTkSuQmCC |                                              |                                                                                                                                                            |

## 7.16 Tables Service

| Resource | Service Category   | Requires Authentication | Requires SignalR (Push Notifications) |
|----------|--------------------|-------------------------|---------------------------------------|
| /tables  | Partially Internal | x                       |                                       |



## 7.16.1 Retrieving a Table or an Entry of a Table

|               |                                                                                                                                                                                                                                                                                                                                                                                               |                                      |                                                           |
|---------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------|-----------------------------------------------------------|
| Resource Path | /tables/:tableName/:tableEntryId                                                                                                                                                                                                                                                                                                                                                              |                                      |                                                           |
| Category      | Public                                                                                                                                                                                                                                                                                                                                                                                        |                                      |                                                           |
| HTTP Method   | GET                                                                                                                                                                                                                                                                                                                                                                                           |                                      |                                                           |
| Parameters    | <b>Property</b>                                                                                                                                                                                                                                                                                                                                                                               | <b>Type</b>                          | <b>Description</b>                                        |
|               | tableName                                                                                                                                                                                                                                                                                                                                                                                     | string                               | Table identifier (table name)<br>[129]<br><b>Required</b> |
|               | id                                                                                                                                                                                                                                                                                                                                                                                            | string                               | Table entry identifier (key of row of table)              |
| Response      | HTTP 200 OK<br>The returned data depends whether or not id is provided.                                                                                                                                                                                                                                                                                                                       |                                      |                                                           |
|               | <b>id</b>                                                                                                                                                                                                                                                                                                                                                                                     | <b>Return Type</b>                   | <b>Description</b>                                        |
|               | Not provided                                                                                                                                                                                                                                                                                                                                                                                  | Array of<br>KeyValue<keyname,string> | List of all key-value pairs in this table                 |
|               | Provided                                                                                                                                                                                                                                                                                                                                                                                      | string                               | Localized value of this column                            |
| Examples      | <b>Request1: id Not provided</b><br>GET api/tables/disciplines<br><b>Response1:</b><br><pre>{   "0": "Management System",   "20": "Building Infrastructure",   "50": "Building Automation",   "100": "Fire",   "150": "Security",   "200": "Energy Management",   "250": "Notification" }</pre> <b>Request2: id provided</b><br>GET api/tables/disciplines/100<br><b>Response2:</b><br>"Fire" |                                      |                                                           |

### 7.16.1.1 Table Names

**TableId** must have one of the following Name:

| Name        | Description                               |
|-------------|-------------------------------------------|
| Categories  | Categories defined for the System         |
| Disciplines | Disciplines supported by Manage System    |
| ObjectTypes | Object types used in Subsystems           |
| Timefilter  | Timefilter supported in Management System |
| Units       | Supported units in Management System      |

## 7.16.2 Retrieving a Subtable of a Table

|               |                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                  |                                                                         |
|---------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------|-------------------------------------------------------------------------|
| Resource Path | /tables/:tableName/subtables/:subtableName                                                                                                                                                                                                                                                                                                                                                                                                             |                                                                                  |                                                                         |
| Category      | Public                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                  |                                                                         |
| HTTP Method   | GET                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                  |                                                                         |
| Parameters    | <b>Property</b>                                                                                                                                                                                                                                                                                                                                                                                                                                        | <b>Type</b>                                                                      | <b>Description</b>                                                      |
|               | tableName                                                                                                                                                                                                                                                                                                                                                                                                                                              | string                                                                           | Table identifier (table name)<br><b>Required</b>                        |
|               | subtableName                                                                                                                                                                                                                                                                                                                                                                                                                                           | string                                                                           | Subtable identifier (limited to "icons" or "colors")<br><b>Required</b> |
|               | systemId                                                                                                                                                                                                                                                                                                                                                                                                                                               | Int                                                                              | System Identifier (query parameter)<br><b>Optional</b>                  |
| Response      | HTTP 200 OK<br>The returned data depends on the subtable id. Note that the example below is for the "icons" subtable id.                                                                                                                                                                                                                                                                                                                               |                                                                                  |                                                                         |
|               | <b>Subtable Id</b>                                                                                                                                                                                                                                                                                                                                                                                                                                     | <b>Return Type</b>                                                               | <b>Description</b>                                                      |
|               | icons                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Key-value pairs (int, string) containing the index and relative path to the icon | Icons are only available from the Disciplines table                     |
|               | colors                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Key-value pairs (int, Dictionary<int, Color>) indicating the valid colors        | Colors are only available from the Categories table                     |
| Examples      | <b>Request:</b><br>GET api/tables/disciplines/subtables/icons<br><b>Response:</b><br><pre>{   "0":     "System1:libraries\\Global_Base_HQ_1\\icons\\Evt_Discp_System_None_001.xml Evt_Discp_System_None_001",   "20":     "System1:libraries\\Global_Base_HQ_1\\icons\\Evt_Discp_Infrastr_None_001.xml Evt_Discp_Infrastr_None_001",   "50":     "System1:libraries\\Global_Base_HQ_1\\icons\\Evt_Discp_BA_None_001.xml Evt_Discp_BA_None_001" }</pre> |                                                                                  |                                                                         |

## 7.16.3 Retrieving a Subtable of a Table for given row of table

|               |                                                                                                                           |             |                                                                 |
|---------------|---------------------------------------------------------------------------------------------------------------------------|-------------|-----------------------------------------------------------------|
| Resource Path | /tables/:tableName/:id/subtables/:subtableName                                                                            |             |                                                                 |
| Category      | Public                                                                                                                    |             |                                                                 |
| HTTP Method   | GET                                                                                                                       |             |                                                                 |
| Parameters    | <b>Property</b>                                                                                                           | <b>Type</b> | <b>Description</b>                                              |
|               | tableName                                                                                                                 | string      | Table identifier (table name)<br><b>Required</b>                |
|               | id                                                                                                                        | int         | Table entry identifier (key of row of table)<br><b>Required</b> |
|               | subtableName                                                                                                              | string      | Subtable Identifier<br><b>Required</b>                          |
| Response      | HTTP 200 OK<br>The returned data depends on the subtable id. Note that the example below is for the "colors" subtable id. |             |                                                                 |

|          | Subtable Id                                                                                                                                                                                                                                                                                                                                                                                                  | Return Type                                                                      | Description                                                                      |
|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------|----------------------------------------------------------------------------------|
|          | icons                                                                                                                                                                                                                                                                                                                                                                                                        | Key-value pairs (int, string) containing the index and relative path to the icon | Icons are only available from the Disciplines table                              |
|          | colors                                                                                                                                                                                                                                                                                                                                                                                                       | Key-value pairs (int, Dictionary<int, Color>) indicating the valid colors        | Colors are only available from the Categories table<br><b>See Object [→ 201]</b> |
| Examples | <b>Request:</b><br>GET api/tables/categories/1/subtables/colors<br><br><b>Response:</b><br><pre>{   "1": "255, 255, 0",   "2": "255, 255, 0",   "3": "255, 255, 0",   "4": "0, 0, 0",   "5": "255, 255, 0",   "6": "239, 105, 38",   "7": "201, 43, 39",   "8": "167, 73, 27",   "9": "147, 40, 16",   "10": "152, 150, 154",   "11": "90, 93, 96",   "12": "230, 148, 113",   "13": "250, 189, 155" }</pre> |                                                                                  |                                                                                  |

### 7.16.4 Retrieving Subgroups with filter

This method help retrieves subgroups of the Disciplines OR ObjectTypes. Filter is also available for retrieving specific Disciplines, SubDisciplines OR ObjectTypes, SubObjectTypes. This filter parameter is query string parameter. If no SubDisciplines/SubObjectTypes are specified while requesting Disciplines/ObjectTypes respectively then all the SubDisciplines/SubObjectTypes which comes under the requested Discipline/ObjectType will be returned. Otherwise only the requested SubDisciplines/SubObjectTypes will be returned.

|               |                                                                                                                                                                                                                     |             |                                                                                                                             |
|---------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|-----------------------------------------------------------------------------------------------------------------------------|
| Resource Path | /tables/:tableName/subgroups                                                                                                                                                                                        |             |                                                                                                                             |
| Category      | Public                                                                                                                                                                                                              |             |                                                                                                                             |
| HTTP Method   | GET                                                                                                                                                                                                                 |             |                                                                                                                             |
| Parameters    | <b>Property</b>                                                                                                                                                                                                     | <b>Type</b> | <b>Description</b>                                                                                                          |
|               | tableName                                                                                                                                                                                                           | string      | Table identifier (table name)<br><b>Required</b>                                                                            |
|               | filter                                                                                                                                                                                                              | string      | This is query string parameter.<br>Filter for Disciplines, SubDisciplines OR ObjectTypes, SUBObjectTypes<br><b>Required</b> |
| Response      | HTTP 200 OK<br>Based on the request is for Disciplines OR ObjectTypes response will be returned. <ol style="list-style-type: none"> <li>1. Disciplines : See [→ 63]</li> <li>2. ObjectTypes : See [→ 63]</li> </ol> |             |                                                                                                                             |
| Examples      | <b>Request1: Disciplines and SubDisciplines request</b><br>GET<br>api/tables/disciplines/subgroups?filter={"0": [0,1], "100": [101                                                                                  |             |                                                                                                                             |

```
,102]]
```

**Response1:**

```
[
  {
    "DisciplineId": 0,
    "DisciplineDescriptor": "Management System",
    "SubDisciplines": [
      {
        "Id": 0,
        "Descriptor": "Unassigned"
      },
      {
        "Id": 1,
        "Descriptor": "Applications"
      }
    ]
  },
  {
    "DisciplineId": 100,
    "DisciplineDescriptor": "Fire",
    "SubDisciplines": [
      {
        "Id": 101,
        "Descriptor": "Evacuation"
      },
      {
        "Id": 102,
        "Descriptor": "Extinguishing"
      }
    ]
  }
]
```

**Request2: ObjectTypes and SubObjectTypes request**

GET

```
api/tables/objecttypes/subgroups?filter={"1000":[1001,1002],"1100":[1101,1102]}
```

**Response2 :**

```
[
  {
    "ObjectId": 1000,
    "ObjectDescriptor": "Client Software",
    "SubObjectTypes": [
      {
        "Id": 1001,
        "Descriptor": "Control Room"
      },
      {
        "Id": 1002,
        "Descriptor": "Mobile"
      }
    ]
  },
  {
    "ObjectId": 1100,
    "ObjectDescriptor": "Command",
    "SubObjectTypes": [
```

|  |                                                                                                          |
|--|----------------------------------------------------------------------------------------------------------|
|  | <pre> {   "Id": 1101,   "Descriptor": "Analog" }, {   "Id": 1102,   "Descriptor": "Binary" } ] } ]</pre> |
|--|----------------------------------------------------------------------------------------------------------|

### 7.16.5 Retrieving Image from Global Text Table


**NOTE:**

If the request is for a PNG image, the image file must be in a format that is either PNG or can be converted to PNG. If the request is for an SVG image, no conversion can be made, so the image file must be SVG.

|               |                                                                                                                                                                                                                                                                                                                                                            |             |                                                                                                                                                                                                                                                             |
|---------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Resource Path | /tables/global/:tableName/icons/:entryId                                                                                                                                                                                                                                                                                                                   |             |                                                                                                                                                                                                                                                             |
| Category      | Internal                                                                                                                                                                                                                                                                                                                                                   |             |                                                                                                                                                                                                                                                             |
| HTTP Method   | GET                                                                                                                                                                                                                                                                                                                                                        |             |                                                                                                                                                                                                                                                             |
| Parameters    | <b>Property</b>                                                                                                                                                                                                                                                                                                                                            | <b>Type</b> | <b>Description</b>                                                                                                                                                                                                                                          |
|               | tableName                                                                                                                                                                                                                                                                                                                                                  | string      | Well-known global text table name [→ 229]<br><b>Required</b>                                                                                                                                                                                                |
|               | entryId                                                                                                                                                                                                                                                                                                                                                    | int         | Entry within the table; only values greater than or equal to zero are valid.<br><b>Required</b>                                                                                                                                                             |
|               | format                                                                                                                                                                                                                                                                                                                                                     | string      | Image format(s) in the order in which to attempt to load them – separated by a comma (currently only “png” and “svg” are supported). The default is “svg,png” but if only one format is acceptable, the string should be “png” or “svg”.<br><b>Optional</b> |
|               | width                                                                                                                                                                                                                                                                                                                                                      | int         | Image width in pixels. The default is 32.<br><b>Optional</b>                                                                                                                                                                                                |
|               | height                                                                                                                                                                                                                                                                                                                                                     | int         | Image height in pixels. The default is 32.<br><b>Optional</b>                                                                                                                                                                                               |
| Response      | HTTP 200 OK.                                                                                                                                                                                                                                                                                                                                               |             |                                                                                                                                                                                                                                                             |
| Examples      | <b>Request:</b><br>/api/tables/global/Disciplines/icons/0<br><b>Response:</b><br>One of the following:<br>1) Base64 encoded PNG: with the specified dimensions.<br>2) SVG: the entire contents of the SVG file<br>3) Blank: there is no icon assigned to the entry, it could not be loaded, or the specified icon was not in one of the supported formats. |             |                                                                                                                                                                                                                                                             |

### 7.16.5.1 Well-Known Global Text Table Names

Global text tables are tables that are found on all online systems in a distributed environment. Within DCC, there is a limited number of tables that are considered “well-known global text tables:” they are specified below.

| Name           | Description                                                                                  |
|----------------|----------------------------------------------------------------------------------------------|
| Disciplines    | Disciplines supported by the system.                                                         |
| Subdisciplines | Subdisciplines supported by the system; a subdiscipline will normally refer to a discipline. |
| ObjectTypes    | Object types used in the system.                                                             |
| ObjectSubtypes | Object subtypes used in the system; a subtype will normally refer to an object type.         |
| Categories     | Categories defined for the system.                                                           |

### 7.16.6 Retrieving Text From Global Text Table

Note that Color is an optional property of a text table entry and will only appear in the JSON string if a color has been assigned to that entry. The sample response below includes the color property on some (but not all) of the entries. See [→ 219] for further discussion of the color property.

|               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |             |                                                              |
|---------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|--------------------------------------------------------------|
| Resource Path | /tables/global/:tableName/text                                                                                                                                                                                                                                                                                                                                                                                                                                                   |             |                                                              |
| Category      | Internal                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |             |                                                              |
| HTTP Method   | GET                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |             |                                                              |
| Parameters    | <b>Property</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | <b>Type</b> | <b>Description</b>                                           |
|               | tableName                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | string      | Well-known global text table name [→ 229]<br><b>Required</b> |
| Response      | HTTP 200 OK<br>A list of objects containing value and text properties [→ 218]                                                                                                                                                                                                                                                                                                                                                                                                    |             |                                                              |
| Examples      | <b>Request:</b><br>/api/tables/global/Disciplines/text<br><b>Response:</b><br><pre>[   {     "Value": 0,     "Color": 4290764270,     "Text": "Management System"   },   {     "Value": 20,     "Text": "Building Infrastructure"   },   {     "Value": 50,     "Color": 4290764270,     "Text": "Building Automation"   },   {     "Value": 100,     "Color": 4290764270,     "Text": "Fire"   },   {     "Value": 150,     "Text": "Security"   },   {     "Value": 200,</pre> |             |                                                              |

|  |                                                                                                                                      |
|--|--------------------------------------------------------------------------------------------------------------------------------------|
|  | <pre>         "Text": "Energy Management"       },       {         "Value": 250,         "Text": "Notification"       }     ] </pre> |
|--|--------------------------------------------------------------------------------------------------------------------------------------|

### 7.16.7 Retrieving all Language Text from Global Text Table

The returned table will contain text for every language that is installed; in the example below there are three languages (en-US, de-DE, fr-CA).

Note that Color is an optional property of a text table entry and will only appear in the JSON string if a color has been assigned to that entry. The sample response below includes the color property on some (but not all) of the entries. See [→ 219] for further discussion of the color property.

|               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |             |                                                              |
|---------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|--------------------------------------------------------------|
| Resource Path | /tables/global/:tableName/langText                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |             |                                                              |
| Category      | Internal                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |             |                                                              |
| HTTP Method   | GET                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |             |                                                              |
| Parameters    | <b>Property</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | <b>Type</b> | <b>Description</b>                                           |
|               | tableName                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | string      | Well-known global text table name [→ 229]<br><b>Required</b> |
| Response      | HTTP 200 OK<br>An array of objects containing a value, text for each language, and optionally a color [→ 218]                                                                                                                                                                                                                                                                                                                                                                                                                   |             |                                                              |
| Examples      | <b>Request:</b><br>/api/tables/global/Categories/langText<br><b>Response:</b><br><pre> [   {     "Value": 1,     "Color": 4290764270,     "LangText": [       "Status",       "Zustand",       "État",       "Status"     ]   },   {     "Value": 2,     "Color": 4292203223,     "LangText": [       "Fault",       "Störung",       "Défaut",       "Fault"     ]   },   {     "Value": 3,     "Color": 4294501476,     "LangText": [       "Trouble",       "Problem",       "Problème",       "Trouble"     ]   } ], </pre> |             |                                                              |

### 7.16.8 Retrieving Nested Text From Global Text Table

The returned table will contain entries from the specified text table and any subtext. The example below is of the Disciplines table: the Subdisciplines table considers the Disciplines table as a 'parent' and an entry in the Subdisciplines table normally refers to an entry in the Disciplines table.

Note that Color is an optional property of a text table entry and will only appear in the JSON string if a color has been assigned to that entry. The sample response below includes the color property on some (but not all) of the entries. See [→ 219] for further discussion of the color property.

|               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |        |                                                              |
|---------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|--------------------------------------------------------------|
| Resource Path | /tables/global/: tableName /nestedText                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |        |                                                              |
| Category      | Internal                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |        |                                                              |
| HTTP Method   | GET                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |        |                                                              |
| Parameters    | Property                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Type   | Description                                                  |
|               | tableName                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | string | Well-known global text table name [→ 229]<br><b>Required</b> |
| Response      | HTTP 200 OK<br>The return is an array of TextTableEntry objects [→ 218]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |        |                                                              |
| Examples      | <b>Request:</b><br>/api/tables/global/Disciplines/nestedText<br><b>Response:</b><br><pre>[   {     "Value": 0,     "Text": "Management System",     "Color": 4290764270,     "SubText": [       {         "Value": 0,         "Color": 4290764270,         "Text": "Unassigned"       },       {         "Value": 1,         "Text": "Applications"       },       {         "Value": 2,         "Color": 4290764270,         "Text": "Communication"       },       {         "Value": 3,         "Text": "Supervision"       },       {         "Value": 4,         "Color": 4290764270,         "Text": "System Settings"       }     ]   },   {     "Value": 20,     "Text": "Building Infrastructure",     "SubText": [</pre> |        |                                                              |



|  |                                                                                                                                                                                                                                                                                                                  |
|--|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | <pre> {   "Value": 0,   "Text": "Unassigned" }, {   "Value": 21,   "Color": 4290764270,   "Text": "Elevators" }, {   "Value": 22,   "Color": 4290764270,   "Text": "Escalators" }, {   "Value": 23,   "Text": "Data Network" }, {   "Value": 24,   "Text": "I/O Monitoring" } ] { ... more entries ... } ]</pre> |
|--|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

### 7.16.9 Retrieving Text From a Local Text Table

Note that Color is an optional property of a text table entry and will only appear in the JSON string if a color has been assigned to that entry. The sample response below includes the color property on some (but not all) of the entries. See [→ 219] for further discussion of the color property.

|               |                                                                                                                                                                                                        |             |                                                                                                                                                              |
|---------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Resource Path | /tables/local/:systemId/:tableName/text                                                                                                                                                                |             |                                                                                                                                                              |
| Category      | Internal                                                                                                                                                                                               |             |                                                                                                                                                              |
| HTTP Method   | GET                                                                                                                                                                                                    |             |                                                                                                                                                              |
| Parameters    | <b>Property</b>                                                                                                                                                                                        | <b>Type</b> | <b>Description</b>                                                                                                                                           |
|               | systemId                                                                                                                                                                                               | int         | The id of the system from which the text table is to be retrieved; zero is not a legal id but can be used to indicate the "local" system.<br><b>Required</b> |
|               | tableName                                                                                                                                                                                              | string      | Local text table name<br><b>Required</b>                                                                                                                     |
| Response      | HTTP 200 OK<br>A list of objects containing value, text, and optionally color properties. [→ 218]                                                                                                      |             |                                                                                                                                                              |
| Examples      | <b>Request:</b><br>/api/tables/local/0/TxG_Discipline/text<br><b>Response:</b><br><pre> [   {     "Value": 0,     "Color": 4290764270,     "Text": "Management System"   },   {     "Value": 20,</pre> |             |                                                                                                                                                              |

|  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|--|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | <pre>         "Text": "Building Infrastructure"       },       {         "Value": 50,         "Color": 4290764270,         "Text": "Building Automation"       },       {         "Value": 100,         "Color": 4290764270,         "Text": "Fire"       },       {         "Value": 150,         "Text": "Security"       },       {         "Value": 200,         "Text": "Energy Management"       },       {         "Value": 250,         "Text": "Notification"       }     ] </pre> |
|--|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

### 7.16.10 Retrieving All Language Text From a Local Text Table

The returned table will contain text for every language that is installed; in the example below there are three languages (en-US, de-DE, fr-CA).

Note that if a color has been assigned to the text table entry it will be included in the returned JSON string. The sample response below includes color in some of the entries to illustrate this fact. See [→ 219] for further discussion of the color property.

|               |                                                                                                                                                                     |             |                                                                                                                                                              |
|---------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Resource Path | /tables/local/:systemId/: tableName /langText                                                                                                                       |             |                                                                                                                                                              |
| Category      | Internal                                                                                                                                                            |             |                                                                                                                                                              |
| HTTP Method   | GET                                                                                                                                                                 |             |                                                                                                                                                              |
| Parameters    | <b>Property</b>                                                                                                                                                     | <b>Type</b> | <b>Description</b>                                                                                                                                           |
|               | systemId                                                                                                                                                            | int         | The id of the system from which the text table is to be retrieved; zero is not a legal id but can be used to indicate the "local" system.<br><b>Required</b> |
|               | tableName                                                                                                                                                           | string      | Local text table name<br><b>Required</b>                                                                                                                     |
| Response      | HTTP 200 OK<br>A list of objects that includes value, text for every installed language, and optionally a color (if one has been assigned to the entry [→ 218])     |             |                                                                                                                                                              |
| Examples      | <b>Request:</b><br>/api/tables/local/0/TxG_Categories/langText<br><b>Response:</b><br><pre> [   {     "Value": 1,     "Color": 4290764270,     "LangText": [ </pre> |             |                                                                                                                                                              |

|  |                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|--|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | <pre>         "Status",         "Zustand",         "État",         "Status"       ],     },     {       "Value": 2,       "Color": 4292203223,       "LangText": [         "Fault",         "Störung",         "Défaut",         "Fault"       ]     },     {       "Value": 3,       "Color": 4294501476,       "LangText": [         "Trouble",         "Problem",         "Problème",         "Trouble"       ]     }   ], }, </pre> |
|--|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

### 7.16.11 Retrieving an Image for an Entry of a Table

|               |                                                                                                                                                                                                        |             |                                                                              |
|---------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|------------------------------------------------------------------------------|
| Resource Path | /tables/local/:systemId/:tableName/icons/:tableEntryId                                                                                                                                                 |             |                                                                              |
| Category      | Internal                                                                                                                                                                                               |             |                                                                              |
| HTTP Method   | GET                                                                                                                                                                                                    |             |                                                                              |
| Parameters    | <b>Property</b>                                                                                                                                                                                        | <b>Type</b> | <b>Description</b>                                                           |
|               | tableName                                                                                                                                                                                              | string      | Identifies a text table (such as "TxG_ValidationProfile")<br><b>Required</b> |
|               | tableEntryId                                                                                                                                                                                           | uint        | Table entry identifier (key of row of table)                                 |
| Response      | HTTP 200 OK<br>The returned data will be an empty string if no image has been assigned to the entry.                                                                                                   |             |                                                                              |
| Examples      | <b>Request:</b><br>GET api/tables/local/0/TxG_ObjectType/icons/100<br><b>Response:</b><br>iVBORw0KGgoAAAANSUheUgAAACAAAAAgCAYAAABzenr0AAAAAXNSR0IArs4c6QAAARnQU1BAACxjwv8YQUAAAAJcEhZcwAADsMAAA7DAcdvq |             |                                                                              |

### 7.16.12 Retrieving an Entry of a Table

Note that if a color has been assigned to the text table entry it will be included in the returned JSON string. The sample response below includes color to illustrate this fact, but color is not guaranteed to appear in all cases. See [→ 219] for further discussion of the color property.

|               |                                                       |             |                                                                              |
|---------------|-------------------------------------------------------|-------------|------------------------------------------------------------------------------|
| Resource Path | /tables/local/:systemId/:tableName/text/:tableEntryId |             |                                                                              |
| Category      | Internal                                              |             |                                                                              |
| HTTP Method   | GET                                                   |             |                                                                              |
| Parameters    | <b>Property</b>                                       | <b>Type</b> | <b>Description</b>                                                           |
|               | tableName                                             | string      | Identifies a text table (such as "TxG_ValidationProfile")<br><b>Required</b> |
|               | tableEntryId                                          | uint        | Table entry identifier (key of row of table)                                 |

|          |                                                                                                                                                                        |
|----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Response | HTTP 200 OK<br>An object containing the value, text, and optionally a color. [→ 218]                                                                                   |
| Examples | <b>Request:</b><br>GET api/tables/local/0/TxG_ValidationProfile/text/0<br><b>Response:</b><br><pre>{   "Value": 0,   "Text": "Disabled",   "Color": 4278190080 }</pre> |

## 7.17 Systems Service

| Resource              | Service Category | Requires Authentication | Requires SignalR (Push Notifications) |
|-----------------------|------------------|-------------------------|---------------------------------------|
| /systems              | Public           | x                       |                                       |
| /systemsSubscriptions | Public           | x                       | x                                     |

### 7.17.1 Retrieving list of installed languages (same order) as installed in project

|               |                                                                                                                                                                                                                                                                                                                                |
|---------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Resource Path | /systems/languages                                                                                                                                                                                                                                                                                                             |
| Category      | Public                                                                                                                                                                                                                                                                                                                         |
| HTTP Method   | GET                                                                                                                                                                                                                                                                                                                            |
| Response      | HTTP 200 OK<br>A SystemsLanguages object [→ 219]                                                                                                                                                                                                                                                                               |
| Examples      | <b>Request:</b><br>GET /api/systems/languages<br><b>Response:</b><br><pre>{   "ArrayIndex": 0,   "Descriptor": "English (United States)",   "Code": "en-US" }, {   "ArrayIndex": 1,   "Descriptor": "French (France)",   "Code": "fr-FR" }, {   "ArrayIndex": 2,   "Descriptor": "German (Germany)",   "Code": "de-DE" }</pre> |

### 7.17.2 Retrieving list of all available systems with list of installed languages.

|               |          |
|---------------|----------|
| Resource Path | /systems |
| Category      | Public   |
| HTTP Method   | GET      |

|          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Response | HTTP 200 OK<br>A SystemsRepresentation object [→ 219]                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Examples | <b>Request:</b><br>GET / api/systems<br><b>Response:</b><br><pre>{   "Systems": [     {       "Name": "System14",       "Id": 14,       "IsOnline": true     },     {       "Name": "System15",       "Id": 15,       "IsOnline": true     }   ],   "Languages": [     {       "ArrayIndex": 0,       "Descriptor": "English (United States)",       "Code": "en-US"     },     {       "ArrayIndex": 1,       "Descriptor": "German (Germany)",       "Code": "de-DE"     }   ],   "IdLocal": 14,   "IsDistributed": true }</pre> |

### 7.17.3 Retrieving information of a specific system

|               |                                                                                                                                                                                                                           |
|---------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Resource Path | /systems/{systemid}                                                                                                                                                                                                       |
| Category      | Public                                                                                                                                                                                                                    |
| HTTP Method   | GET                                                                                                                                                                                                                       |
| Response      | HTTP 200 OK<br>A SystemsRepresentation object [→ 219]                                                                                                                                                                     |
| Examples      | <b>Request:</b><br>GET / api/systems/1<br><b>Response:</b><br><pre>{   "Systems": [     {       "Name": "System14",       "Id": 14,       "IsOnline": true     }   ],   "Languages": [     {       "ArrayIndex": 0,</pre> |

|  |                                                                                                                                                           |
|--|-----------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | <pre> "Descriptor": "English (United States)", "Code": "en-US" }, {   "ArrayIndex": 1,   "Descriptor": "German (Germany)",   "Code": "de-DE" } ] } </pre> |
|--|-----------------------------------------------------------------------------------------------------------------------------------------------------------|

### 7.17.4 Retrieving information of a local system

|               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|---------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Resource Path | /systems/local                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Category      | Public                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| HTTP Method   | GET                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Response      | HTTP 200 OK<br>A SystemsRepresentation object [→ 219]                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| Examples      | <b>Request:</b><br>GET /api/systems/local<br><b>Response:</b><br><pre> {   "Systems": [     {       "Name": "System2",       "Id": 2,       "IsOnline": true     }   ],   "Languages": [     {       "ArrayIndex": 0,       "Descriptor": "English (United States)",       "Code": "en-US"     },     {       "ArrayIndex": 1,       "Descriptor": "German (Germany)",       "Code": "de-DE"     },     {       "ArrayIndex": 2,       "Descriptor": "French (France)",       "Code": "fr-FR"     }   ] } </pre> |

### 7.17.5 Get service status

This will return the current status ('isConnected') of all services. If user has scope of multiple system, it will return the service statuses of multiple systems.

Note: Currently, only the status for Report Manager (id 58) is returned.

|               |                   |
|---------------|-------------------|
| Resource Path | /systems/services |
| Category      | Public            |

|             |                                                                                                                                                           |
|-------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|
| HTTP Method | GET                                                                                                                                                       |
| Response    | HTTP 200 OK<br>A List< <a href="#">ServicesNotificationRepresentation</a> > object                                                                        |
| Examples    | <b>Request:</b><br>GET / api/systems/services<br><b>Response:</b><br><pre>[   {     "SystemId":4,     "ServiceId":58,     "IsConnected":false   } ]</pre> |

## 7.17.6 Create channelize Systems subscription

If subscription response and notifications are needed on signalr channel.

Subscription response will be pushed through notifySubscriptionStatus function available at client side. Sound changes will be notified via notifySounds.

|               |                                                                                                                                                                                                                                                                                                       |             |                                                                                                                                                                                                                                                 |
|---------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Resource Path | /api/sr/systemssubscriptions/channelize/:requestId/:connectionId                                                                                                                                                                                                                                      |             |                                                                                                                                                                                                                                                 |
| Category      | Public                                                                                                                                                                                                                                                                                                |             |                                                                                                                                                                                                                                                 |
| HTTP Method   | POST                                                                                                                                                                                                                                                                                                  |             |                                                                                                                                                                                                                                                 |
| Parameters    | <b>Property</b>                                                                                                                                                                                                                                                                                       | <b>Type</b> | <b>Description</b>                                                                                                                                                                                                                              |
|               | requestId                                                                                                                                                                                                                                                                                             | String      | Request Identifier of subscription hub connection. RequestId is to be generated and provided by the client.<br>It is expected from client that client should use unique id for for each request.<br>Recommendation: Use GUID<br><b>Required</b> |
|               | connectionId                                                                                                                                                                                                                                                                                          | String      | Identifier of subscription hub connection<br><b>Required</b>                                                                                                                                                                                    |
| Response      | HTTP 200 OK                                                                                                                                                                                                                                                                                           |             |                                                                                                                                                                                                                                                 |
| Examples      | <b>Request:</b><br>POST api/sr/systemssubscriptions/channelize/9565ca41-8556-4dbb-94cc-1b89451a5db5/c6c95ece-133b-480b-af1f-067a173e0f5b<br><b>notifySubscriptionStatus:</b><br><pre>{   "ErrorCode":0,   "RequestId": "9565ca41-8556-4dbb-94cc-1b89451a5db5",   "RequestFor":"notifySystems" }</pre> |             |                                                                                                                                                                                                                                                 |

## 7.17.7 Delete Subscription for a Systems

|               |                                               |             |                                               |
|---------------|-----------------------------------------------|-------------|-----------------------------------------------|
| Resource Path | /api/sr/ systemssubscriptions /(connectionId) |             |                                               |
| Category      | Public                                        |             |                                               |
| HTTP Method   | DELETE                                        |             |                                               |
| Parameters    | <b>Property</b>                               | <b>Type</b> | <b>Description</b>                            |
|               | connectionId                                  | String      | Identifier of subscription<br><b>Required</b> |
| Response      | HTTP 200 OK.                                  |             |                                               |

|          |                                                                                                                                    |
|----------|------------------------------------------------------------------------------------------------------------------------------------|
| Examples | <b>Request:</b><br>DELETE /api/sr/systemssubscriptions/e3d90a35-1f7a-4ec2-8b3b-2fba71b76bc7<br><b>Response:</b><br>HTTP/1.1 200 OK |
|----------|------------------------------------------------------------------------------------------------------------------------------------|

## 7.17.8 Push Notification

After subscribing for event sounds, the current sound file with a link to download it will be sent in an initial notification. Afterwards, any change will be notified.

| Type     | Name (case sensitive)  |                       |                        |
|----------|------------------------|-----------------------|------------------------|
| Hub      | norisHub               |                       |                        |
| Function | notifySystems ( data ) |                       |                        |
|          | Property               | Type                  | Description            |
|          | data                   | SystemsRepresentation | System details [→ 219] |



**NOTE:**

If in JSON response the value of “FileName” and “\_links” are empty then no sound has to be played.

### 7.17.8.1 Subscription Status Notification

After subscribing for event sounds, the subscription status will be sent only once.

| Type     | Name (case sensitive)                                                                                                 |                       |                               |
|----------|-----------------------------------------------------------------------------------------------------------------------|-----------------------|-------------------------------|
| Hub      | 1. norisHub<br>2. eventSoundsHub (lagacy)                                                                             |                       |                               |
| Function | notifySubscriptionStatus(status)                                                                                      |                       |                               |
|          | Property                                                                                                              | Type                  | Description                   |
|          | Status                                                                                                                | Subscription Response | See Subscription Status [201] |
| Example  | <pre>{   "ErrorCode":0,   "RequestId": "9565ca41-8556-4dbb-94cc-1b89451a5db5",   "RequestFor":"notifySystems" }</pre> |                       |                               |

## 7.17.9 Create channelize Services subscription

This allows subscription of services according to system. If the system id is within the scope of the logged in user and if the service id is valid, then user will be subscribed to that service id for that particular system and will get notifications when the state of the service changes.

**Currently, the only supported Service ID is 58 (for Report Manager).** Other services will not be subscribe.

|               |                                                                            |
|---------------|----------------------------------------------------------------------------|
| Resource Path | /api/sr/systemssubscriptions/channelize/:requestId:/connectionId/services/ |
|---------------|----------------------------------------------------------------------------|



|             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                              |                                                                                                                                                                                                                                                    |
|-------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Category    | Public                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                              |                                                                                                                                                                                                                                                    |
| HTTP Method | POST                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                              |                                                                                                                                                                                                                                                    |
| Parameters  | <b>Property</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | <b>Type</b>                                                  | <b>Description</b>                                                                                                                                                                                                                                 |
|             | requestId                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | String                                                       | Request Identifier of subscription hub connection.<br>RequestId is to be generated and provided by the client.<br>It is expected from client that client should use unique id for for each request.<br>Recommendation: Use GUID<br><b>Required</b> |
|             | connectionId                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | String                                                       | Identifier of subscription hub connection<br><b>Required</b>                                                                                                                                                                                       |
| Body        | Services                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | <a href="#">ServicesRequestSubscriptionsRepresentation[]</a> | Contains a list of the systems and their corresponding services which the client wants to subscribe to.<br><b>Required</b>                                                                                                                         |
| Response    | HTTP 200 OK as <a href="#">List&lt;ServicesSubscriptionsRepresentation&gt;</a>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                              |                                                                                                                                                                                                                                                    |
| Examples    | <b>Request:</b><br>POST api/sr/systemssubscriptions/channelize/9565ca41-8556-4dbb-94cc-1b89451a5db5/c6c95ece-133b-480b-af1f-067a173e0f5b/services<br><b>Body: serviceIds</b><br>[{"SystemId":1,"Ids":[58,60,78]},{"SystemId":4,"Ids":[58,60]},{"SystemId":5,"Ids":[58,61,62]}]<br><b>Response:</b><br>[{"SystemId":1,"ServiceId":58,"ErrorCode":1}, {"SystemId":1,"ServiceId":60,"ErrorCode":1}, {"SystemId":1,"ServiceId":78,"ErrorCode":1}, {"SystemId":4,"ServiceId":60,"ErrorCode":1}, {"SystemId":5,"ServiceId":61,"ErrorCode":1}, {"SystemId":5,"ServiceId":62,"ErrorCode":1}, {"SystemId":4,"ServiceId":58,"ErrorCode":0}, {"SystemId":5,"ServiceId":58,"ErrorCode":0}]<br><br>notifySubscriptionStatus:<br><br>ErrorCode: 0, RequestFor: "notifyServices", RequestId: "9565ca41-8556-4dbb-94cc-1b89451a5db5" |                                                              |                                                                                                                                                                                                                                                    |

Note: For a particular connectionId, subscription request can be sent only once. If more systems or services are to be included in the subscription, then unsubscribe and subscribe again.

If no valid combination of systemid and service id is passed in the request, it will return an error with 400 code.

### 7.17.9.1 Services Notification

After successful subscription, an **initial notification** will be generated with the current statuses of the subscribed services. After that, a notification will be generated whenever the state of the subscribed service id(s) **changes**.

|      |                       |
|------|-----------------------|
| Type | Name (case sensitive) |
| Hub  | 1. norisHub           |

|          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                                                                                                                 |                                                        |
|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------|
| Function | notifyServices (status)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                 |                                                        |
|          | Property                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Type                                                                                                                                                                            | Description                                            |
|          | Status                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | List<ServicesNotificationRepresentation> in case of initial notification. ServicesNotificationRepresentation in case of change in service status (for subsequent notifications) | See <a href="#">ServicesNotificationRepresentation</a> |
| Example  | <pre> ▼{C: "d-165E5373-B,0 C,7 D,1", M: [{H: "NorisHub", M: "notifyServices", A: [,...]}]}   C: "d-165E5373-B,0 C,7 D,1"   ▼M: [{H: "NorisHub", M: "notifyServices", A: [,...]}]     ▼0: {H: "NorisHub", M: "notifyServices", A: [,...]}       ▼A: [,...]         ▼0: [{SystemId: 4, ServiceId: 58, IsConnected: false}, {SystemId: 5, ServiceId: 58, IsConnected: false}]           ►0: {SystemId: 4, ServiceId: 58, IsConnected: false}           ►1: {SystemId: 5, ServiceId: 58, IsConnected: false}         H: "NorisHub"         M: "notifyServices" </pre> |                                                                                                                                                                                 |                                                        |

### 7.17.10 Unsubscribe Services

This will unsubscribe all services which have been subscribed for a particular connection id.

|               |                                                                                                                                             |        |                                               |
|---------------|---------------------------------------------------------------------------------------------------------------------------------------------|--------|-----------------------------------------------|
| Resource Path | /api/sr/systemssubscriptions/services/{connectionId}                                                                                        |        |                                               |
| Category      | Public                                                                                                                                      |        |                                               |
| HTTP Method   | DELETE                                                                                                                                      |        |                                               |
| Parameters    | Property                                                                                                                                    | Type   | Description                                   |
|               | connectionId                                                                                                                                | String | Identifier of subscription<br><b>Required</b> |
| Response      | HTTP 200 OK.                                                                                                                                |        |                                               |
| Examples      | <b>Request:</b><br>DELETE /api/sr/systemssubscriptions/services/e3d90a35-1f7a-4ec2-8b3b-2fba71b76bc7<br><b>Response:</b><br>HTTP/1.1 200 OK |        |                                               |

## 7.18 Products Service

| Resource  | Service Category | Requires Authentication | Requires SignalR (Push Notifications) |
|-----------|------------------|-------------------------|---------------------------------------|
| /Products | Public           | x                       |                                       |

## 7.18.1 Retrieving Product Information

Retrieve a list of all Trend Collector objects of a given system. It includes all Online

|               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|---------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Resource Path | /Products                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Category      | Public                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| HTTP Method   | GET                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| Response      | HTTP 200 OK<br>ProductInfo object <a href="#">[→ 220Error! Bookmark not defined.]</a>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Examples      | <b>Request:</b><br>GET /api/Products<br><b>Response:</b><br><pre>{   "ProductNameShort": "Desigo CC",   "ProductName": "Desigo™ CC",   "VendorName": "Siemens Switzerland Ltd.",   "ProductCopyRights": "© Siemens Switzerland Ltd, 2016",   "ProductVersion": "V3.0",   "ServerVersion": "1.0.0.0",   "SupportLink": "www.siemens.com/SBT",   "_links": [     {       "Rel": "self",       "Href": "api/products",       "IsTemplated": false     },     {       "Rel": "downloadfile",       "Href": "api/repository/ProductPictures/files?names=entrypage_image.JPG",       "IsTemplated": false     },     {       "Rel": "downloadfile",       "Href": "api/repository/ProductPictures/files?names=SplashScreen-Image.jpg",       "IsTemplated": false     }   ] }</pre> |

## 7.19 Files Service

| Resource                             | Service Category   | Requires Authentication | Requires SignalR (Push Notifications) |
|--------------------------------------|--------------------|-------------------------|---------------------------------------|
| /files/{fileType}                    | Partially Internal |                         |                                       |
| /files/{systemId}/{relativeFilePath} |                    | X                       |                                       |

### 7.19.1 Retrieving Files of specific type

|               |                   |                                           |                                                                                                 |
|---------------|-------------------|-------------------------------------------|-------------------------------------------------------------------------------------------------|
| Resource Path | /files/{fileType} |                                           |                                                                                                 |
| Category      | Internal          |                                           |                                                                                                 |
| HTTP Method   | GET               |                                           |                                                                                                 |
| Parameters    | <b>Property</b>   | <b>Type</b>                               | <b>Description</b>                                                                              |
|               | fileType          | BasicFiles<br>See <a href="#">[→ 220]</a> | Supported file types<br><b>Required</b>                                                         |
|               | names             | String[]                                  | This is query string parameter.<br>Contains only the name of the files. or array of file names. |
| Response      | HTTP 200 OK       |                                           |                                                                                                 |
| Examples      | <b>Request:</b>   |                                           |                                                                                                 |

|  |                                                                                                                             |
|--|-----------------------------------------------------------------------------------------------------------------------------|
|  | GET /api/files/0?names=entrypage_image.JPG<br><b>Response:</b><br>Returns stream content down to the client, chunk by chunk |
|--|-----------------------------------------------------------------------------------------------------------------------------|

### 7.19.2 Retrieve Files from system

Currently access is provided for "libraries" folder path

|               |                                                                                                                                                                      |             |                                                      |
|---------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|------------------------------------------------------|
| Resource Path | /files/{systemId}/{relativeFilePath}                                                                                                                                 |             |                                                      |
| Category      | Public                                                                                                                                                               |             |                                                      |
| HTTP Method   | GET                                                                                                                                                                  |             |                                                      |
| Parameters    | <b>Property</b>                                                                                                                                                      | <b>Type</b> | <b>Description</b>                                   |
|               | systemId                                                                                                                                                             | uint        | System identifier<br><b>Required</b>                 |
|               | relativeFilePath                                                                                                                                                     | String      | Relative file path.e.g. see below<br><b>Required</b> |
| Response      | HTTP 200 OK                                                                                                                                                          |             |                                                      |
| Examples      | <b>Request:</b><br>GET / api/files/1/libraries\BA_Cooling_HQ_1\Symbols\DYN_2D_Chiller_None_Analog_Central_001.ccs<br><b>Response:</b><br>Returns byte array content. |             |                                                      |

### 7.19.3 Retrieve Documents from system

Documents can be retrieved by passing designation of Systembrowser tree nodes

|               |                                                                                                                                                                 |             |                                                                                                                                                    |
|---------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|----------------------------------------------------------------------------------------------------------------------------------------------------|
| Resource Path | /files/documents/{designation}                                                                                                                                  |             |                                                                                                                                                    |
| Category      | Public                                                                                                                                                          |             |                                                                                                                                                    |
| HTTP Method   | GET                                                                                                                                                             |             |                                                                                                                                                    |
| Parameters    | <b>Property</b>                                                                                                                                                 | <b>Type</b> | <b>Description</b>                                                                                                                                 |
|               | designation                                                                                                                                                     | string      | Designation id of node in systembrowser tree<br>(Recommendation: Double UrlEncode parameter prior to making it to part of URI.)<br><b>Required</b> |
| Response      | HTTP 200 OK<br>Content-Type header set to particular filetype                                                                                                   |             |                                                                                                                                                    |
| Examples      | <b>Request:</b><br>GET / api/files/documents/System1.ApplicationView%253AApplicationView.Documents.pdfSample<br><b>Response:</b><br>Returns byte array content. |             |                                                                                                                                                    |

## 7.20 Graphics Service

| Resource                              | Service Category | Requires Authentication | Requires SignalR (Push Notifications) |
|---------------------------------------|------------------|-------------------------|---------------------------------------|
| /graphics/{objectId}                  | Internal         | x                       |                                       |
| /graphics/itemIds/{objectDesignation} | Internal         | x                       |                                       |
| /graphics/items/{objectId}            | Internal         | x                       |                                       |
| /children/itemIds/{objectDesignation} | Internal         | x                       |                                       |

This service provides the following functionality:

- To determine whether an object is relevant to a client that processes graphics. By relevant we mean: (a) the object itself is one of the known graphic types (such as a graphic or a graphic viewport), (b) the object is related to a graphic type (such as an object that is found in a graphic), or (c) one of the object's ancestors is either a graphic type or is related to a graphic type. This functionality is generally referred to as the "preselect" operation.
- To retrieve the list of graphic objects related to this object. If the object is one of the graphic types it will be returned; if it is not one of the graphic types, the list will contain all graphic types related to the object and all graphic types related to all ancestors of the object. This functionality is generally referred to as the "select" operation.
- To retrieve a specific graphic object, such as a graphic or graphic viewport. This functionality is generally referred to as the "load" operation.
- To retrieve the first child (of a node in a view) that is either graphical or has a related graphics.

Note that the term *ancestor* in this context means objects higher in a hierarchy, up to and including the root object. During a preselect or a select operation all hierarchies on the system where the object is defined are checked.

### 7.20.1 Check if object is graphical

Determine if an object, an object it references, or one of its ancestors is graphical in nature. For example, an object will be considered "graphical in nature" if it is a DCC graphic, template, or viewport (within a graphic). The object itself may not be graphical but can refer to another object that is graphical (such as a non-graphical object that appears on a graphic).

|               |                                                                                                                                                                                                                                    |             |                                                                    |
|---------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|--------------------------------------------------------------------|
| Resource Path | /graphics/{objectId}                                                                                                                                                                                                               |             |                                                                    |
| Category      | Internal                                                                                                                                                                                                                           |             |                                                                    |
| HTTP Method   | GET                                                                                                                                                                                                                                |             |                                                                    |
| Parameters    | <b>Property</b>                                                                                                                                                                                                                    | <b>Type</b> | <b>Description</b>                                                 |
|               | objectId                                                                                                                                                                                                                           | String      | Object identifier, including system information<br><b>Required</b> |
| Response      | HTTP 200 Ok<br>HTTP 204 No Content                                                                                                                                                                                                 |             |                                                                    |
| Examples      | <b>Request:</b><br>GET /api/graphics2/System1:ObjectName<br><b>Response:</b><br>Returns status code 200 if the object (or something it references) or one of its ancestors is graphical in nature; otherwise code 204 is returned. |             |                                                                    |

### 7.20.2 Retrieving graphic item ids

Retrieve all graphic items ids related to an object or its ancestors: this may include the object itself. Note a couple of things about the graphic items:

- 1) Graphic items that reference a graphical template will include the designation of the 'owning' object (see 'Context' in example below).
- 2) There will be no duplicate references to a graphic.
- 3) There will be no duplicate references to a template for the same 'owning' object. For example: if an object is determined to be an ancestor in two or more situations, only one of the situations will be reflected in the results.

|               |                                       |             |                    |
|---------------|---------------------------------------|-------------|--------------------|
| Resource Path | /graphics/itemIds/{objectDesignation} |             |                    |
| Category      | Internal                              |             |                    |
| HTTP Method   | GET                                   |             |                    |
| Parameters    | <b>Property</b>                       | <b>Type</b> | <b>Description</b> |

|          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |        |                                                                                                                                                                                                                                        |
|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|          | objectDesignation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | String | The “fully assembled” name of an object within a hierarchy. This name is also commonly referred to as the “designation.” The name specifies the system, hierarchy, and position of the object within the hierarchy.<br><b>Required</b> |
| Response | HTTP 200 OK (one or more GraphicalItems [→ 221] will be returned)<br>HTTP 204 No Content (there are no graphic items associated with the object)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |        |                                                                                                                                                                                                                                        |
| Examples | <b>Request:</b><br>This example is a loop object found within a BACnet device (8166) on a BACnet network (“bacnetNetwork”).<br>GET /api/graphics/itemIds/System1:ManagementView:ManagementView.FieldNetworks.bacnetNetwork.Hardware.8166.Local_IO.LP_1<br><br><b>Response:</b><br>Returns a list of related graphic items, possibly including the object itself (if it is graphical in nature).<br><pre>[   {     "Context": "System1:ManagementView:ManagementView.FieldNetworks.bacnetNetwork.Hardware.8166.Local_IO.LP_1",     "ObjectId": "System1:GraphicsTemplate_TEM_Object_Loop_None_None_001",     "ManagedType": "Graphic Template",     "DisplayName": "TEM_Object_Loop_None_None_001"   },   {     "Context": "System1:ManagementView:ManagementView.FieldNetworks.bacnetNetwork.Hardware.8166.Local_IO.LP_1",     "ObjectId": "System1:Graphics_AnotherGraphic",     "ManagedType": "Graphic",     "DisplayName": "SomeGraphic"   } ]</pre> |        |                                                                                                                                                                                                                                        |

### 7.20.3 Retrieving a graphical item

Retrieve the content for a graphical item. The content will be in the form of XML containing SVG with two sections: <graphic> and <libraries>. The <graphic> section will include the definition of the graphical item, while the <libraries> section will contain any symbols referenced by the item – note that a symbol may refer to other symbols, and these will also be in this section.

|               |                                                                                                                                                                                                                                                                                                                                                                                                                                            |             |                                                                                                                  |
|---------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|------------------------------------------------------------------------------------------------------------------|
| Resource Path | /graphics/items/{objectId}                                                                                                                                                                                                                                                                                                                                                                                                                 |             |                                                                                                                  |
| Category      | Internal                                                                                                                                                                                                                                                                                                                                                                                                                                   |             |                                                                                                                  |
| HTTP Method   | GET                                                                                                                                                                                                                                                                                                                                                                                                                                        |             |                                                                                                                  |
| Parameters    | <b>Property</b>                                                                                                                                                                                                                                                                                                                                                                                                                            | <b>Type</b> | <b>Description</b>                                                                                               |
|               | objectId                                                                                                                                                                                                                                                                                                                                                                                                                                   | String      | Object identifier; this must include the system information. For example: “System1:MyGraphic”<br><b>Required</b> |
| Response      | HTTP 200 OK<br>HTTP 204 No Content                                                                                                                                                                                                                                                                                                                                                                                                         |             |                                                                                                                  |
| Examples      | <b>Request:</b><br>GET /api/graphics/items/System1:MyGraphicalObject<br><b>Response:</b><br><pre>&lt;?xml version="1.0" encoding="utf-8"?&gt; &lt;!DOCTYPE svg PUBLIC "-//W3C//DTD SVG 1.1//EN" ""&gt; &lt;graphicItems version="0.9"&gt;   &lt;graphic&gt;     &lt;svg&gt;       ... graphical item is defined here ...     &lt;/svg&gt;   &lt;/graphic&gt; &lt;/libraries&gt; &lt;svg&gt;   ... symbol definition ... &lt;/svg&gt;</pre> |             |                                                                                                                  |

|  |                                                                                                                                 |
|--|---------------------------------------------------------------------------------------------------------------------------------|
|  | <pre> &lt;/svg&gt; &lt;svg&gt;   ... another symbol definition ... &lt;/svg&gt; &lt;/libraries&gt; &lt;/graphicItems&gt; </pre> |
|--|---------------------------------------------------------------------------------------------------------------------------------|

### 7.20.4 Retrieving a child item

Retrieve the first “interesting” child of a node, where “interesting” means:

- 1) The child is a graphical object itself
- 2) The child has a related template
- 3) The child has a related graphical object (such as being in a graphic)

Note that the method requires a fully qualified view node name and that the children of that node are searched for one that is interesting.

|               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |        |                                                                                                                                                                                                                                                                                        |
|---------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Resource Path | /graphics/children/itemIds/{objectDesignation}                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |        |                                                                                                                                                                                                                                                                                        |
| Category      | Internal                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |        |                                                                                                                                                                                                                                                                                        |
| HTTP Method   | GET                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |        |                                                                                                                                                                                                                                                                                        |
| Parameters    | objectDesignation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | String | Object designation (a fully qualified node name). For example: "System.View:View.Level"<br><b>Required</b>                                                                                                                                                                             |
|               | generations                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | UInt32 | The number of generations to search (downward). The range is 1 to 20; if the value is not provided a default of 1 is used.<br><b>Optional</b>                                                                                                                                          |
|               | searchOptions                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | UInt64 | Search options as a bitmask. Currently only one option (value 0x1) has been defined: to return a single object (the alternative is to return all interesting children). As of right now this parameter is ignored and only the first interesting child is returned.<br><b>Optional</b> |
|               | types                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | UInt32 | The types of objects to consider 'graphical.' As of right now this parameter is ignored; the types to be considered 'graphical' for this method are custom graphics and templates.<br><b>Optional</b>                                                                                  |
| Response      | HTTP 200 OK (a browser object will be returned)<br>HTTP 204 No Content (no interesting child was found)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |        |                                                                                                                                                                                                                                                                                        |
| Examples      | <b>Request:</b><br>GET /api/graphics/children/itemIds/System.View:View.Level<br><b>Response:</b><br><pre> {   "HasChild": false,   "ViewId": 11,   "ViewType": 4,   "Attributes": {     "DefaultProperty": "Present_Value",     "ObjectId": "System1:GmsDevice_1_8166_1",     "DisciplineDescriptor": "Building Automation",     "DisciplineId": 50,     "SubDisciplineDescriptor": "Unassigned",     "SubDisciplineId": 0,     "TypeDescriptor": "Sensor",     "TypeId": 6700,     "SubTypeDescriptor": "Unassigned",     "SubTypeId": 0,     "ManagedType": 80,     "ManagedTypeName": "BACnetPoint",     "ObjectModelName": "GMS_BACNET_EO_BA_AI_1",     "ValidationRules": {       "CommentRule": "Optional",       "ReAuthentication": "NoNeed",       "Configuration": 0, </pre> |        |                                                                                                                                                                                                                                                                                        |

```

        "IsFourEyesEnabled": false,
        "_links": []
    },
    },
    "Location": "System.View:View.Level.Analog Input 1",
    "SystemId": 1,
    "Name": "AI_1",
    "Descriptor": "Analog Input 1",
    "Designation": "System.View:View.Level.AI_1",
    "ObjectId": "System1:GmsDevice_1_8166_1",
    "_links": [
        {
            "Rel": "properties",
            "Href": "api/properties/System1%253AGmsDevice_1_8166_1?requestType=0",
            "IsTemplated": false
        }
    ]
}

```

## 7.21 EventCategorySound Service

| Resource                                | Service Category | Requires Authentication | Requires SignalR (Push Notifications) |
|-----------------------------------------|------------------|-------------------------|---------------------------------------|
| /api/eventcategorysounds                | Public           | x                       |                                       |
| api/sr/eventcategorysoundssubscriptions | Public           | x                       | x                                     |

### 7.21.1 Get current event sound



**NOTE:**

If in JSON response the value of "FileName" and "\_links" are empty then no sound has to be played.

|               |                                                                                                                                                                                                                                                                                                                                                                                             |                |                                                                                                                                             |
|---------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| Resource Path | api/eventcategorysounds                                                                                                                                                                                                                                                                                                                                                                     |                |                                                                                                                                             |
| Category      | Public                                                                                                                                                                                                                                                                                                                                                                                      |                |                                                                                                                                             |
| HTTP Method   | GET                                                                                                                                                                                                                                                                                                                                                                                         |                |                                                                                                                                             |
| Parameters    | <b>Property</b>                                                                                                                                                                                                                                                                                                                                                                             | <b>Type</b>    | <b>Description</b>                                                                                                                          |
|               | disableCategories                                                                                                                                                                                                                                                                                                                                                                           | query:(string) | Filter to disable categories.<br>List of disable category. i.e. category Ids for which sound information is not required<br><b>Optional</b> |
| Response      | HTTP 200 OK<br>File details object See [→221]                                                                                                                                                                                                                                                                                                                                               |                |                                                                                                                                             |
| Examples      | <b>Request:</b><br>GET / api/eventcategorysounds?disableCategories=[1,2]<br><b>Response:</b><br><pre> {   "FileName": "0dfe9579-2c20-4cad-9a07-7bc14f4164a5.wav",   "_links": [     {       "Rel": "DownloadSoundFile",       "Href": "api/files/1/Libraries%5cGlobal_Events_HQ_1%5cMediaLibrary%5c0dfe9579-2c20-4cad-9a07-7bc14f4164a5.wav",       "IsTemplated": false     }   ] } </pre> |                |                                                                                                                                             |



|  |   |
|--|---|
|  | } |
|  | } |
|  | } |

## 7.21.2 Subscribe for event sound

|               |                                                                                                                                             |        |                                               |
|---------------|---------------------------------------------------------------------------------------------------------------------------------------------|--------|-----------------------------------------------|
| Resource Path | /api/sr/eventcategorysoundssubscriptions/{connectionId}                                                                                     |        |                                               |
| Category      | Public                                                                                                                                      |        |                                               |
| HTTP Method   | POST                                                                                                                                        |        |                                               |
| Parameters    | Property                                                                                                                                    | Type   | Description                                   |
|               | connectionId                                                                                                                                | string | Identifier of subscription<br><b>Required</b> |
| Response      | HTTP 200 OK                                                                                                                                 |        |                                               |
| Examples      | <b>Request:</b><br>POST api/sr/eventcategorysoundssubscriptions/c6c95ece-133b-480b-af1f-067a173e0f5b<br><b>Response:</b><br>HTTP/1.1 200 OK |        |                                               |

## 7.21.3 Delete Subscription for a event sounds changes.

|               |                                                                                                                                                |        |                                               |
|---------------|------------------------------------------------------------------------------------------------------------------------------------------------|--------|-----------------------------------------------|
| Resource Path | /api/sr/eventcategorysoundssubscriptions/{connectionId}                                                                                        |        |                                               |
| Category      | Public                                                                                                                                         |        |                                               |
| HTTP Method   | DELETE                                                                                                                                         |        |                                               |
| Parameters    | Property                                                                                                                                       | Type   | Description                                   |
|               | connectionId                                                                                                                                   | String | Identifier of subscription<br><b>Required</b> |
| Response      | HTTP 200 OK.                                                                                                                                   |        |                                               |
| Examples      | <b>Request:</b><br>DELETE /api/sr/eventcategorysoundssubscriptions/e3d90a35-1f7a-4ec2-8b3b-2fba71b76bc7<br><b>Response:</b><br>HTTP/1.1 200 OK |        |                                               |

### 7.21.3.1 Push Notification

After subscribing for event sounds, the current sound file with a link to download it will be sent in an initial notification. Afterwards, any change will be notified.

|          |                                                         |                    |                           |
|----------|---------------------------------------------------------|--------------------|---------------------------|
| Type     | Name (case sensitive)                                   |                    |                           |
| Hub      | 1. norisHub : recommended<br>2. eventSoundsHub : lagacy |                    |                           |
| Function | notifySounds ( data )                                   |                    |                           |
|          | Property                                                | Type               | Description               |
|          | data                                                    | FileRepresentation | File details . See [→221] |

**NOTE:**

If in JSON response the value of "FileName" and "\_links" are empty then no sound has to be played.

## 7.21.4 Create channelize event sound subscription

If subscription response and notifications are needed on signalr channel.

Subscription response will be pushed through notifySubscriptionStatus function available at client side. Sound changes will be notified via notifySounds.

|               |                                                                                                                                                                                                                                                                                                                       |             |                                                                                                                                                                                                                                                     |
|---------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Resource Path | /api/sr/eventcategorysoundssubscriptions/channelize/:requestId:/connectionId                                                                                                                                                                                                                                          |             |                                                                                                                                                                                                                                                     |
| Category      | Public                                                                                                                                                                                                                                                                                                                |             |                                                                                                                                                                                                                                                     |
| HTTP Method   | POST                                                                                                                                                                                                                                                                                                                  |             |                                                                                                                                                                                                                                                     |
| Parameters    | <b>Property</b>                                                                                                                                                                                                                                                                                                       | <b>Type</b> | <b>Description</b>                                                                                                                                                                                                                                  |
|               | requestId                                                                                                                                                                                                                                                                                                             | String      | Request Identifier of subscription hub connection. RequestId is to be generated and provided by the client.<br><br>It is expected from client that client should use unique id for for each request.<br>Recommendation: Use GUID<br><b>Required</b> |
|               | connectionId                                                                                                                                                                                                                                                                                                          | String      | Identifier of subscription hub connection<br><b>Required</b>                                                                                                                                                                                        |
| Response      | HTTP 200 OK                                                                                                                                                                                                                                                                                                           |             |                                                                                                                                                                                                                                                     |
| Examples      | <b>Request:</b><br>POST api/sr/eventcategorysoundssubscriptions/channelize/9565ca41-8556-4dbb-94cc-1b89451a5db5/c6c95ece-133b-480b-af1f-067a173e0f5b<br><br><b>notifySubscriptionStatus:</b><br><pre>{   "ErrorCode":0,   "RequestId": "9565ca41-8556-4dbb-94cc-1b89451a5db5",   "RequestFor": "notifySounds" }</pre> |             |                                                                                                                                                                                                                                                     |

### 7.21.4.1 Subscription Status Notification

After subscribing for event sounds, the subscription status will be sent only once.

|          |                                                                                                                       |                       |                               |
|----------|-----------------------------------------------------------------------------------------------------------------------|-----------------------|-------------------------------|
| Type     | Name (case sensitive)                                                                                                 |                       |                               |
| Hub      | 2. norisHub<br>3. eventSoundsHub (lagacy)                                                                             |                       |                               |
| Function | notifySubscriptionStatus(status)                                                                                      |                       |                               |
|          | <b>Property</b>                                                                                                       | <b>Type</b>           | <b>Description</b>            |
|          | Status                                                                                                                | Subscription Response | See Subscription Status [201] |
| Example  | <pre>{   "ErrorCode":0,   "RequestId": "9565ca41-8556-4dbb-94cc-1b89451a5db5",   "RequestFor": "notifySounds" }</pre> |                       |                               |

## 7.22 Settings Service

Settings are stored on a “per user” basis.

| Resource              | Service Category | Requires Authentication | Requires SignalR (Push Notifications) |
|-----------------------|------------------|-------------------------|---------------------------------------|
| /settings/{settingId} | Internal         | x                       |                                       |

### 7.22.1 Read setting

Read a setting; the setting value is returned ad the body of the response.

|               |                                                                                                                                                        |             |                                                            |
|---------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|------------------------------------------------------------|
| Resource Path | /settings/:settingId                                                                                                                                   |             |                                                            |
| Category      | Internal                                                                                                                                               |             |                                                            |
| HTTP Method   | GET                                                                                                                                                    |             |                                                            |
| Parameters    | <b>Property</b>                                                                                                                                        | <b>Type</b> | <b>Description</b>                                         |
|               | settingId                                                                                                                                              | String      | Identifies the setting to be retrieved.<br><b>Required</b> |
| Response      | HTTP 200 OK<br>HTTP 204 No Content                                                                                                                     |             |                                                            |
| Examples      | <b>Request:</b><br>GET /api/settings/MySettingId<br><b>Response:</b><br>Returns status code 200 if the setting exists; otherwise code 204 is returned. |             |                                                            |

### 7.22.2 Write setting

Write a setting; the setting id is required as part of the URL while the content must be provided as part of the body of the request (a null or empty string is not valid and will result in a Bad Request response code). Note that a setting does not have to exist prior to writing it.

|               |                                                                                                                                  |             |                                                          |
|---------------|----------------------------------------------------------------------------------------------------------------------------------|-------------|----------------------------------------------------------|
| Resource Path | /settings/:settingId                                                                                                             |             |                                                          |
| Category      | Internal                                                                                                                         |             |                                                          |
| HTTP Method   | PUT                                                                                                                              |             |                                                          |
| Content Type  | application/x-www-form-urlencoded                                                                                                |             |                                                          |
| Parameters    | <b>Property</b>                                                                                                                  | <b>Type</b> | <b>Description</b>                                       |
|               | settingId                                                                                                                        | String      | Identifies the setting to be written.<br><b>Required</b> |
| Body          | The content of the setting to be stored for the current user.                                                                    |             |                                                          |
| Response      | HTTP 200 OK<br>HTTP 204 No Content<br>HTTP 400 Bad Request                                                                       |             |                                                          |
| Examples      | <b>Request:</b><br>PUT /api/settings/MySettingId<br><b>Post Data:</b><br>A string<br><b>Response:</b><br>Returns status code 200 |             |                                                          |

### 7.22.3 Delete setting

Delete a setting for the current user; note that deleting a non-existent setting returns success (200).

|               |                                                                                                    |             |                                                          |
|---------------|----------------------------------------------------------------------------------------------------|-------------|----------------------------------------------------------|
| Resource Path | /settings/:settingId                                                                               |             |                                                          |
| Category      | Internal                                                                                           |             |                                                          |
| HTTP Method   | DELETE                                                                                             |             |                                                          |
| Parameters    | <b>Property</b>                                                                                    | <b>Type</b> | <b>Description</b>                                       |
|               | settingId                                                                                          | String      | Identifies the setting to be deleted.<br><b>Required</b> |
| Response      | HTTP 200 OK                                                                                        |             |                                                          |
| Examples      | <b>Request:</b><br>DELETE /api/settings/MySettingId<br><b>Response:</b><br>Returns status code 200 |             |                                                          |

## 7.23 Users Service

| Resource        | Service Category | Requires Authentication | Requires SignalR (Push Notifications) |
|-----------------|------------------|-------------------------|---------------------------------------|
| /users/password | Internal         |                         |                                       |

### 7.23.1 Change Password

Change password by providing user details.



**NOTE:**

If user has a valid token and password is changed, token remains valid.

If the new password does not follow the password policy (configured in SMC), then password will not be allowed to be changed and HTTP 400 error will be returned.

|               |                                                                                                                                                                                                                                                                                    |             |                                     |
|---------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|-------------------------------------|
| Resource Path | api/users/password                                                                                                                                                                                                                                                                 |             |                                     |
| Category      | Internal                                                                                                                                                                                                                                                                           |             |                                     |
| HTTP Method   | PUT                                                                                                                                                                                                                                                                                |             |                                     |
| Body          | <b>Property</b>                                                                                                                                                                                                                                                                    | <b>Type</b> | <b>Description</b>                  |
|               | userName                                                                                                                                                                                                                                                                           | string      | Username<br><b>Required</b>         |
|               | currentPassword                                                                                                                                                                                                                                                                    | string      | Current password<br><b>Required</b> |
|               | newPassword                                                                                                                                                                                                                                                                        | string      | New password<br><b>Required</b>     |
| Response      | HTTP 200 OK<br>HTTP 400 Bad Request                                                                                                                                                                                                                                                |             |                                     |
| Examples      | <b>Request:</b><br>PUT /api/users/password<br><b>Body: (new password follows policy)</b><br><pre>{   "userName": "GmsUser",   "currentPassword": "XXXX",   "newPassword": "ZZZZ" }</pre> <b>Response:</b><br>HTTP 200 OK<br><br><b>Body: (new password does not follow policy)</b> |             |                                     |

|  |                                                                                                                                                                                                                                                                        |
|--|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | <pre>{   "userName": "GmsUser",   "currentPassword": "aaaa",   "newPassword": "bbbb" }</pre> <p><b>Response:</b><br/>HTTP 400 Bad Request</p> <pre>{   Details: "Password does not meet policy requirements."   Error: "password_not_compatible"   Id: 2400004 }</pre> |
|--|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

### 7.23.2 Get Login Information

Get login information for the requested user. This is open API, i.e. it does not requires any authentication. Provided a user, it will return account type of that user as integer. Based on this account type client of this API (especially Flex Client) can take decision on how to accept the password by showing correct UI to the end user. For any user which is not present in the DesigoCC, AccountType will be returned as DesigoCC (AccountType = 0)

Following are possible account type values:

0 = DesigoCC Account

1 = Windows Account

2 = Software Account (Not used for login from UI as of now)

3 = OpenID Account

Incase if the account type is OpenID (AccountType = 3), Additional information regarding OpenID LoginUri and LogoutUri will be sent in the same response as shown in the below example. In all the other cases this will be null.

When the OpenID configuration is missing in the system, fallback will happen for account type from OpenID to DesigoCC even though the user is correctly mapped to OpenID account in the system. Because if the necessary configuration is missing then we cannot prepare LoginUri and LogoutUri which are mandatory for the OpenID authentication.

|               |                                                                     |             |                                                                                                                               |
|---------------|---------------------------------------------------------------------|-------------|-------------------------------------------------------------------------------------------------------------------------------|
| Resource Path | api/users/logininfo/{username}                                      |             |                                                                                                                               |
| Category      | Public                                                              |             |                                                                                                                               |
| HTTP Method   | GET                                                                 |             |                                                                                                                               |
| Parameters    | <b>Property</b>                                                     | <b>Type</b> | <b>Description</b>                                                                                                            |
|               | username                                                            | string      | Username to login with.<br>(Recommendation: Double UriEncode parameter prior to making it to part of URI.)<br><b>Required</b> |
| Response      | HTTP 200 OK<br>HTTP 400 Bad Request<br>HTTP 503 Service Unavailable |             |                                                                                                                               |

|          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Example1 | <p><b>AccountType of the user is DesigoCC/Windows</b></p> <p><b>Request:</b><br/>GET /api/users/test</p> <p><b>Response:</b><br/>HTTP 200 OK<br/>{<br/>  "UserName": "test",<br/>  "AccountType": 0<br/>}</p>                                                                                                                                                                                                                                                                                                                                                                                            |
| Example2 | <p><b>AccountType of the user is OpenID, but the OpenID configuration is missing in DesigoCC system.</b></p> <p><b>Request:</b><br/>GET /api/users/test.test%40test.com</p> <p><b>Response:</b><br/>HTTP 200 OK<br/>{<br/>  "UserName": "test.test@test.com",<br/>  "AccountType": 0<br/>}</p>                                                                                                                                                                                                                                                                                                           |
| Example3 | <p><b>AccountType of the user is OpenID, OpenID configuration is present in the DesigoCC system.</b></p> <p><b>Request:</b><br/>GET /api/users/test.test%40test.com</p> <p><b>Response:</b><br/>HTTP 200 OK<br/>{<br/>  "UserName": "test.test@test.com",<br/>  "AccountType": 3,<br/>  "OpenIdUri": {<br/>    "Login": "https://MyAuth0Domain.com/authorize?client_id=MyTestClientId&amp;response_type=code&amp;scope=openid profile email&amp;connection=main-tenant-oidc",<br/>    "Logout": "https://MyIdServerDomain.com/v2/logout?returnTo=https:// MyAuth0Domain.com/v2/logout"<br/>  }<br/>}</p> |

As this API will be used while performing the login. There are some steps that needs to be additionally performed by client of this API. These steps are only applicable to users of type OpenID (AccountType = 3).

In case of such users, authentication will be performed by the ID server. For this client needs to be redirected to ID server. And after the authentication is done, it needs to be redirected back to the login page of the client to continue the authentication process.

On similar note, while performing the logout as well, client needs to logout from ID server as well, otherwise on next OpenID login, user will not be asked for credentials. Logout from ID server also takes advantage of redirection of the browser.

That's why in case of logout as well client needs to add redirect url as callback.

#### a. OpenIdUri.Login

Redirect URI of the client which is registered on Auth0 for the client application needs to be appended at the received login uri in following format:

Query string operator: &

Query string parameter Name: `redirect_uri`

Query string parameter Value: URI of the client

Final OpenIdUri.Login will look like following:

```
https://MyAuth0Domain.com/authorize?client_id=MyTestClientId&response_type=code&scope=openid profile email&connection=main-tenant-oidc&redirect_uri=https://MyMachineDomain:446/FlexApp
```

Text in the bold is the appended text.

#### b. OpenIdUri.Logout

Allowed Logout URI of the client which is register on the Auth0 for the client application needs to be appended at the received logout uri as following:

Query string operator: ?

Query string parameter Name: `returnTo`

Query string parameter Value: URI of the client

Final OpenIdUri.Logout will look like following:

```
https://MyIdServerDomain.com/v2/logout?returnTo=https://MyAuth0Domain.com/v2/logout?returnTo=https:// MyMachine-Domain:446/FlexApp
```

Text in the bold is the appended text.

## 7.23.3 Get Public Key

Gets public key in PEM format and a session key.

**SessionKey:** Session key is uniquely generated key, which is associated with and identifies corresponding public-private key pair.

**PublicKey:** Public key is RSA asymmetric algorithm key, which to be used for data encryption purposes.

|               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|---------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Resource Path | api/users/getpublickey                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| Category      | Internal                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| HTTP Method   | GET                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| Response      | HTTP 200 OK<br>HTTP 400 Bad Request                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| Examples      | <p><b>Request:</b><br/>GET /api/users/getpublickey</p> <p><b>Response:</b><br/>{ "SessionKey": "2023-02-27 05:10:36Z_3d7e2ff8-f731-47a3-8065-d5909ef4e228",<br/>"PublicKey":<br/>"-----BEGIN PUBLIC KEY-----\nMIIBIjANBgkqh-<br/>kiG9w0BAQEFAAOCAQ8AMIIB-<br/>CgKCAQEApz1NEQC3UFI0vZb2JT6f\nSk4cGwlhFnPrKM8x8FSVQRF-<br/>CbhdgxRtBKSjphb+G6J+n6FSX-<br/>oNK7/LbNGPK5dTJC\nQKdMI7H56P60tdWdjatxdAH4Ti2DEin2fFXE/<br/>hVXplFaA+sI33rERILhBGEXPITL\naoGgWQ+YciRYUNwkzcPm-<br/>FIQrOxADXH6xrShusoIh4gFCN17ZBBQexx1vhLoj/zVC\nnTpy-<br/>jRZpm/wZO2ywpI5FeYLCM1ZTq2Ui-<br/>UKdlp4m922Qi2sbFV8Dv+ld/oV2ST2b26\nnHyMwpMhH0f9Bi5RgcjuY<br/>CI3LMsw9kkO+Z6sE98qwJOY+mnAo2yN+aOBAKDyo8zte\n\nFQIDAQAB\n<br/>-----END PUBLIC KEY-----"}<br/>HTTP 200 OK</p> |

For encryption of data following steps are to be followed:

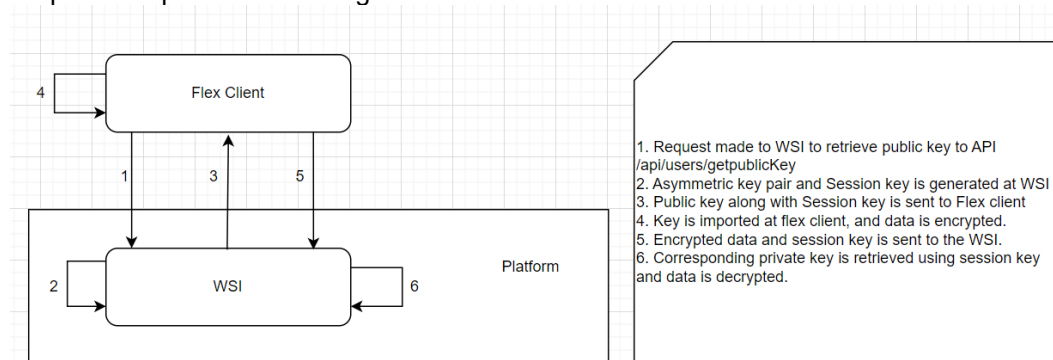
1. Import the RSA key with hash as SHA-256.

[https://de-](https://developer.mozilla.org/en-US/docs/Web/API/SubtleCrypto/importKey)

[vel-](https://developer.mozilla.org/en-US/docs/Web/API/SubtleCrypto/importKey)

2. Encrypt the data using the public key imported.  
<https://developer.mozilla.org/en-US/docs/Web/API/SubtleCrypto/encrypt>
3. Convert the resulting byte array into string.
4. Encode the resulting string in base64 format.  
<https://developer.mozilla.org/en-US/docs/Web/API/btoa>

Steps are depicted in the image below:



Send encrypted string (in base 64 format) and session key to the server for decryption.

Session key and public key gets expired in 5 minutes, so whole operation of encryption and decryption needs to be completed in 5 minutes, if key is expired you need to again make the request to get the public key and perform subsequent operations.

Also, this public key is one time use key, so you can perform encryption decryption using single key only once, if you need to perform these operations again you need to raise request for the new public key.

## 7.24 Scheduler Service

| Resource   | Service Category | Requires Authentication | Requires SignalR (Push Notifications) |
|------------|------------------|-------------------------|---------------------------------------|
| /scheduler | Internal         | x                       |                                       |

### 7.24.1 Gets BACnet schedule object based on the schedule object id

|               |                                                                                                                        |        |             |
|---------------|------------------------------------------------------------------------------------------------------------------------|--------|-------------|
| Resource Path | /scheduler/schedule/:objectId                                                                                          |        |             |
| Category      | Public                                                                                                                 |        |             |
| HTTP Method   | GET                                                                                                                    |        |             |
| Parameters    | Property                                                                                                               | Type   | Description |
|               | objectId                                                                                                               | string | Required    |
| Response      | HTTP 200 OK<br>Object of WeeklySchedule [->222]                                                                        |        |             |
| Examples      | <b>Request:</b><br>GET<br>/api/scheduler/schedule/System1%253AGmsDe-vice_1_1800_71303169<br><br><b>Response :</b><br>{ |        |             |



```

"DeviceId": "System1:GmsDevice_1_1800_33556232",
"Name": "System1:GmsDevice_1_1800_71303169",
"Setup": {
  "DefaultValue": "1.1",
  "StartDate": {
    "Date": "0001-01-01T00:00:00Z"
  },
  "EndDate": {
    "Date": "0001-01-01T00:00:00Z"
  },
  "Priority": 16,
  "DataType": "BACnetDataTypeReal",
  "ScheduleMin": -3.402823e+38,
  "ScheduleMax": 3.402823e+38,
  "TextGroupId": "TxG_EngineeringUnits",
  "Unit": 64
},
"Outputs": [
  "System1:GmsDevice_1_1800_8388609.Pre-
sent_Value"
],
"ScheduleType": "BACnet",
"Schedules": [
  {
    "Day": "monday",
    "Entries": [
      {
        "Time": "07:00:00",
        "Value": 18.5
      },
      {
        "Time": "21:00:00",
        "Value": 15.5
      }
    ]
  }
],
"Exceptions": [
  {
    "Priority": 6,
    "Mode": 3,
    "CalendarException": {
      "CalendarObjectId": "System1:GmsDe-
vice_1_1800_25165825",
      "CalendarEntries": [
        {
          "StartDate": {
            "Date": "0001-01-01T00:00:00Z",
            "Days": [
              6
            ],
            "Months": [
              1
            ],
            "DaysInMonth": [
              1
            ],
            "Years": [
              2005
            ]
          },
          "Mode": 0
        },
        {
          "StartDate": {
            "Date": "0001-01-01T00:00:00Z",
            "Days": [
              4
            ]
          }
        }
      ]
    }
  }
]

```

|  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|--|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | <pre> ],   "Months": [     6   ],   "DaysInMonth": [     27   ],   "Years": [     2019   ] }, "Mode": 0 } ], "CalendarName": "" }, "ExceptionTimes": [   {     "Time": "00:00:00",     "Value": 13.78   },   {     "Time": "08:32:32.3200000",     "Value": 66.6   },   {     "Time": "10:56:56.5600000",     "Value": 13.78   },   {     "Time": "12:11:11.1100000",     "Value": 66.6   },   {     "Time": "23:59:59.5900000",     "Value": 13.78   } ] } ] } </pre> |
|--|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

### 7.24.2 Save BACnet schedule values

|               |                                                                                                                                                                                                                     |                        |                    |
|---------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------|--------------------|
| Resource Path | /scheduler/schedule/save                                                                                                                                                                                            |                        |                    |
| Category      | Public                                                                                                                                                                                                              |                        |                    |
| HTTP Method   | POST                                                                                                                                                                                                                |                        |                    |
|               | <b>Property</b>                                                                                                                                                                                                     | <b>Type</b>            | <b>Description</b> |
| Body          | schedule                                                                                                                                                                                                            | WeeklySchedule [->222] |                    |
| Response      | HTTP 200 OK<br>true or false as body                                                                                                                                                                                |                        |                    |
| Examples      | <b>Request:</b><br>POST<br>/api/scheduler/schedule/save<br><pre> {   "DeviceId": "System1:GmsDevice_1_1800_33556232",   "Name": "System1:GmsDevice_1_1800_71303169",   "Setup": {     "DefaultValue": "1.1", </pre> |                        |                    |

|  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|--|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | <pre> "StartDate": {   "Date": "0001-01-01T00:00:00Z" }, "EndDate": {   "Date": "0001-01-01T00:00:00Z" }, "Priority": 16, "DataType": "BACnetDataTypeReal", "ScheduleMin": -3.402823e+38, "ScheduleMax": 3.402823e+38, "TextGroupId": "TxG_EngineeringUnits", "Unit": 64 }, "Outputs": [   "System1:GmsDevice_1_1800_8388609.Pre- sent_Value" ], "ScheduleType": "BACnet", "Schedules": [   {     "Day": "monday",     "Entries": [       {         "Time": "09:54:00",         "Value": 18.5       },       {         "Time": "21:00:00",         "Value": 15.5       }     ],     "OutputValuesActive": [],     "OutputValuesInactive": []   } ], "Exceptions": [   {     "Priority": 6,     "Mode": 3,     "CalendarException": {       "CalendarObjectId": "System1:GmsDe- vice_1_1800_25165825",       "CalendarEntries": [         {           "StartDate": {             "Date": "0001-01-01T00:00:00Z",             "Days": [               6             ],             "Months": [               1             ],             "DaysInMonth": [               1             ],             "Years": [               2005             ]           },           "Mode": 0         },         {           "StartDate": {             "Date": "0001-01-01T00:00:00Z",             "Days": [               4             ],             "Months": [ </pre> |
|--|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

|  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|--|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | <pre>         6       ],       "DaysInMonth": [         27       ],       "Years": [         2019       ]     },     "Mode": 0   } ], "CalendarName": "" }, "ExceptionTimes": [   {     "Time": "00:00:00",     "Value": 13.78   },   {     "Time": "08:32:32.3200000",     "Value": 66.6   },   {     "Time": "10:56:56.5600000",     "Value": 13.78   },   {     "Time": "12:11:11.1100000",     "Value": 66.6   },   {     "Time": "23:59:59.5900000",     "Value": 13.78   } ] ] } } </pre> <p><b>Response :</b><br/>200 OK<br/>true</p> |
|--|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

### 7.24.3 Save BACnet schedule exceptions

|               |                           |                           |             |
|---------------|---------------------------|---------------------------|-------------|
| Resource Path | /scheduler/exception/save |                           |             |
| Category      | Public                    |                           |             |
| HTTP Method   | POST                      |                           |             |
| Body          | Property                  | Type                      | Description |
|               | schedule                  | WeeklySchedule<br>[->222] |             |
| Response      | HTTP 200 OK               |                           |             |
| Examples      | <b>Request:</b>           |                           |             |

### 7.24.4 Save BACnet default value for schedule

|               |                                      |
|---------------|--------------------------------------|
| Resource Path | /scheduler/schedule/saveDefaultValue |
|---------------|--------------------------------------|

|             |                 |             |                    |
|-------------|-----------------|-------------|--------------------|
| Category    | Public          |             |                    |
| HTTP Method | GET             |             |                    |
| Query       | <b>Property</b> | <b>Type</b> | <b>Description</b> |
|             | defaultValue    | string      | <b>Required</b>    |
|             | objectId        | string      |                    |
| Response    | HTTP 200 OK     |             |                    |
| Examples    | <b>Request:</b> |             |                    |

### 7.24.5 Retrieve BACnet calendar list for schedule object id

|               |                                   |             |                    |
|---------------|-----------------------------------|-------------|--------------------|
| Resource Path | /scheduler/calendarList/:objectId |             |                    |
| Category      | Public                            |             |                    |
| HTTP Method   | GET                               |             |                    |
| Parameters    | <b>Property</b>                   | <b>Type</b> | <b>Description</b> |
|               | objectId                          | string      | <b>Required</b>    |
| Response      | HTTP 200 OK                       |             |                    |
| Examples      | <b>Request:</b>                   |             |                    |

### 7.24.6 Retrieve calendar object based on calendar object Id

|               |                                                                                                                                                                                                                                                                                                                                                                                                                               |             |                    |
|---------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|--------------------|
| Resource Path | /scheduler/calendar/:objectId                                                                                                                                                                                                                                                                                                                                                                                                 |             |                    |
| Category      | Public                                                                                                                                                                                                                                                                                                                                                                                                                        |             |                    |
| HTTP Method   | GET                                                                                                                                                                                                                                                                                                                                                                                                                           |             |                    |
| Parameters    | <b>Property</b>                                                                                                                                                                                                                                                                                                                                                                                                               | <b>Type</b> | <b>Description</b> |
|               | objectId                                                                                                                                                                                                                                                                                                                                                                                                                      | string      | <b>Required</b>    |
| Response      | HTTP 200 OK<br>Object of Calendar [->221]                                                                                                                                                                                                                                                                                                                                                                                     |             |                    |
| Examples      | <b>Request:</b><br>GET<br>/api/scheduler/calendar/System1%253AGmsDe-<br>vice_1_1800_25165825<br><b>Response:</b><br>200 OK<br><pre>{   "ScheduleCalendarType": 0,   "CalendarEntries": [     {       "StartDate": {         "Date": "0001-01-01T00:00:00Z",         "Months": [           1         ],         "DaysInMonth": [           1         ],         "Years": [           2005         ]       }     }   ], }</pre> |             |                    |

|  |                                                                                                                                                                                               |
|--|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | <pre>       "Mode": 0     }   ],   "AppliedSchedules": [     "System1:GmsDevice_1_1800_71303169"   ],   "ContainerObjectId": "System1:GmsDe- vice_1_1800_33556232",   "Device": 1800 } </pre> |
|--|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

### 7.24.7 Save BACnet calendar

|               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                  |             |
|---------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|-------------|
| Resource Path | /scheduler/calendar/save                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                  |             |
| Category      | Public                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                  |             |
| HTTP Method   | POST                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                  |             |
| Body          | Property                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Type             | Description |
|               | calendar                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Calendar [->221] |             |
| Response      | HTTP 200 OK<br>true or false in body                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                  |             |
| Examples      | <b>Request:</b><br>POST<br>/api/scheduler/calendar/save<br>Body:<br><pre> {   "ScheduleCalendarType": 0,   "CalendarEntries": [     {       "Mode": 0,       "StartDate": {         "Years": [           2005         ],         "Months": [           1         ],         "DaysInMonth": [           1         ]       }     },     {       "Mode": 0,       "StartDate": {         "Years": [           2019         ],         "Months": [           6         ],         "DaysInMonth": [           27         ]       }     }   ] } </pre> |                  |             |

|  |                                                                                                                                                                                                                              |
|--|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | <pre>"AppliedSchedules": [   "System1:GmsDevice_1_1800_71303169" ], "ContainerObjectId": "System1:GmsDevice_1_1800_33556232", "Device": 1800, "ObjectId": "System1:GmsDevice_1_1800_25165825" }  Response: 200 OK true</pre> |
|--|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

## 7.25 Objects Service

| Resource | Service Category | Requires Authentication | Requires SignalR (Push Notifications) |
|----------|------------------|-------------------------|---------------------------------------|
| /objects | Internal         | x                       |                                       |

### 7.25.1 Get available Objects types

|               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |             |                                                                                                                                                                    |
|---------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Resource Path | /objects/:designation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |             |                                                                                                                                                                    |
| Category      | Internal                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |             |                                                                                                                                                                    |
| HTTP Method   | GET                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |             |                                                                                                                                                                    |
| Parameters    | <b>Property</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | <b>Type</b> | <b>Description</b>                                                                                                                                                 |
|               | designation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | string      | Full view name;<br>Designation id of node in systembrowser tree<br>(Recommendation: Double UrlEncode parameter prior to making it to part of URI.) <b>Required</b> |
|               | includeChildInfo                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | bool        | If true, response will include child object type information<br>Default: false                                                                                     |
| Response      | HTTP 200 OK<br>An array of ChildObjectType objects See [→ 223]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |             |                                                                                                                                                                    |
| Examples      | <b>Request:</b><br>api/Objects/System1.ApplicationView%3AApplicationView.Trends.TrendViewDefinitions<br><br><b>Response:</b><br><pre>{   "Designation":   "System1.ApplicationView:ApplicationView.Trends.TrendViewDefinitions",   "Description": "Trends Folder",   "ObjectType": "_GmsTrendsFolder",   "IsGenericDeleteAllowed": false,   "ChildObjects": [     {       "IsGenericCreateAllowed": false,       "MaxChild": 4294967295,       "ExistingChildCount": 0,       "Description": "Trend View Definition",       "ObjectType": "GMS_TVD",       "IsGenericDeleteAllowed": true     }   ] }</pre> |             |                                                                                                                                                                    |

|  |                                                                                                                                                                                                                        |
|--|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | <pre> }, {   "IsGenericCreateAllowed": true,   "MaxChild": 4294967295,   "ExistingChildCount": 0,   "Description": "Trends Folder",   "ObjectType": "_GmsTrendsFolder",   "IsGenericDeleteAllowed": false } ] } </pre> |
|--|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

## 7.25.2 Create Objects

|               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                   |                                                |
|---------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------|------------------------------------------------|
| Resource Path | /objects                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                   |                                                |
| Category      | Internal                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                   |                                                |
| HTTP Method   | POST                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                   |                                                |
| Body          | <b>Property</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | <b>Type</b>                       | <b>Description</b>                             |
|               | childObject                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | ObjectRepresentation. See [→ 223] | Child node details which user needs to create. |
| Response      | HTTP 200 OK<br>BrowserNode objects See [→ 223 ]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                   |                                                |
| Examples      | <p><b>Request:</b><br/>api/Objects</p> <p>Body Parameter:</p> <pre> {   "Designation": "System1.ApplicationView:ApplicationView.Trends.TrendViewDefinitions",   "NameChildNode": "TestFolderTVD1",   "ObjectModelName": "_GmsTrendsFolder",   "Descriptor": {     "CommonText": "TestFolderTVDNode",     "MultiLangText": [       "TestFolderTVDNodeLang1",       "TestFolderTVDNodeLang1"     ]   } } </pre> <p><b>Response:</b></p> <pre> {   "HasChild": false,   "SystemId": 1,   "ViewId": 10,   "ViewType": 1,   "Name": "TestFolderTVD1",   "Descriptor": "TestFolderTVDNode",   "Designation":     "System1.ApplicationView:ApplicationView.Trends.TrendViewDefinitions.TestFolderTVD1",   "ObjectId": "System1:TestFolderTVD1",   "Attributes": {     "DefaultProperty": "StatusPropagation.AggregatedSummaryStatus",     "ObjectId": "System1:TestFolderTVD1",     "DisciplineDescriptor": "Management System",     "DisciplineId": 0,     "SubDisciplineDescriptor": "Unassigned",     "SubDisciplineId": 0,     "TypeDescriptor": "View Element", </pre> |                                   |                                                |



|  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|--|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | <pre> "TypeId": 8000, "SubTypeDescriptor": "User Defined Folder", "SubTypeId": 8016, "ManagedType": 59, "ManagedTypeName": "TrendsFolder", "ObjectModelName": "_GmsTrendsFolder", "ValidationRules": {   "CommentRule": "Optional",   "ReAuthentication": "NoNeed",   "Configuration": 0,   "IsFourEyesEnabled": false,   "_links": [] }, "Location": "System1.Application View:Applications.Trends.Trend View Definitions.string", "_links": [   {     "Rel": "properties",     "Href": "api/properties/System1%253ATestFolderTVD1?requestType=0",     "IsTemplated": false   } ] </pre> |
|--|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

### 7.25.3 Delete Objects

|               |                                                                                                                                                                            |        |             |
|---------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|-------------|
| Resource Path | /objects/:designation                                                                                                                                                      |        |             |
| Category      | Internal                                                                                                                                                                   |        |             |
| HTTP Method   | DELETE                                                                                                                                                                     |        |             |
| Parameters    | Property                                                                                                                                                                   | Type   | Description |
|               | designation                                                                                                                                                                | string |             |
| Response      |                                                                                                                                                                            |        |             |
| Examples      | <b>Request:</b><br>DELETE api/Objects/System1.ApplicationView%3AApplicationView.Trends.TrendViewDefinitions.TestFolder-TVD1<br><b>Response:</b><br>Returns status code 200 |        |             |

### 7.25.4 Get Information text

|               |                                                                                                       |        |                                                                                                 |
|---------------|-------------------------------------------------------------------------------------------------------|--------|-------------------------------------------------------------------------------------------------|
| Resource Path | /objects/servicetext/:objectId                                                                        |        |                                                                                                 |
| Category      | Internal                                                                                              |        |                                                                                                 |
| HTTP Method   | GET                                                                                                   |        |                                                                                                 |
| Parameters    | Property                                                                                              | Type   | Description                                                                                     |
|               | objectId                                                                                              | string | Object id<br>Recommendation: Double<br>UrlEncode parameter prior to<br>making it to part of URI |
| Response      | HTTP 200 OK<br>Array of ObjectInfoRepresentation[→224]                                                |        |                                                                                                 |
| Examples      | <b>Request:</b><br>GET api/Objects/servicetext/System1:ManagementView_ManagementSystem_Servers_Server |        |                                                                                                 |

|  |                                                                                                                                                                                                                                                                                                                                                                                               |
|--|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | <b>Response:</b><br>Returns status code 200<br><pre>[ {   "ObjectId": "System1:ManagementView_ManagementSystem_Servers_Server",   "ServiceText": {     "InformationText": "",     "Memo": ""   } }, {   "Error": {     "Id": 2402709,     "Error": "obj_invalid_object",     "Details": "Failed to resolve passed object id to a valid object"   },   "ObjectId": "InvalidObjectId" } ]</pre> |
|--|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

### 7.25.5 Put Memo text

|               |                                                                                                                                                                                                             |                                         |                                                                                                 |
|---------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------|-------------------------------------------------------------------------------------------------|
| Resource Path | /objects/servicetext/:objectid/memo                                                                                                                                                                         |                                         |                                                                                                 |
| Category      | Internal                                                                                                                                                                                                    |                                         |                                                                                                 |
| HTTP Method   | PUT                                                                                                                                                                                                         |                                         |                                                                                                 |
| Parameters    | <b>Property</b>                                                                                                                                                                                             | <b>Type</b>                             | <b>Description</b>                                                                              |
|               | objectId                                                                                                                                                                                                    | string                                  | Object id<br>Recommendation: Double<br>UrlEncode parameter prior<br>to making it to part of URI |
| Body          | memodetail                                                                                                                                                                                                  | ServiceTextMemoRepresentation<br>[→224] | Memo details                                                                                    |
| Response      | HTTP 200 OK<br>Array of ObjectInfoRepresentation[→224]                                                                                                                                                      |                                         |                                                                                                 |
| Examples      | <b>Request:</b><br>PUT api/Objects/servicetext/System1:ManagementView_ManagementSystem_Servers_Server/memo<br>BODY<br><pre>{   "Memo": "Sample Memo" }</pre><br><b>Response:</b><br>Returns status code 200 |                                         |                                                                                                 |

## 7.26 Related Items Service

The purpose of the related items API is to access items that are associated to another object. For example, an object in a graphic is said to be `related to` that graphic. In most situations the relationship between that object and the graphic is discovered by requesting the graphic – but it is often useful to know where that object is used (such as a graphic or re report) or if there is something external (such as a file or a website) that has a connection to the object.

| Resource      | Service Category | Requires Authentication | Requires SignalR (Push Notifications) |
|---------------|------------------|-------------------------|---------------------------------------|
| /relateditems | Public           |                         |                                       |

### 7.26.1 Read Related Items

Return related items for a list of objects. Please keep in mind that:

- The system information for each object is optional and if not provided, the WSI will assume that the object is on the “local” system. If the object does not exist in that system the list of related items for that object will be empty.
- The list of objects must all be from the same system. If they are not, the entire call will fail with an HTTP error code return of 400.
- A related item is just another object in the system and can appear in multiple locations within a hierarchy or within multiple hierarchies. Each related item will contain a BrowserNode for each location, where the BrowserNode has the object and view information.

|               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|---------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Resource Path | api/relateditems                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| Category      | Internal                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| HTTP Method   | POST                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Parameters    | None                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Body          | Array of Object Ids                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Response      | <p>HTTP 400 Bad Request</p> <p>HTTP 200 OK</p> <p>The top level object contains an array of results, one element for each incoming object id. Each element holds the object id, an error code, and an array of RelatedItem objects for that object id. For the full description of a RelatedItem go <a href="#">here</a>.</p>                                                                                                                                                                         |
| Example       | <p><b>Request:</b><br/>POST /api/relateditems</p> <p><b>Body:</b></p> <pre>[   "SystemName:Object1",   "SystemName:Object2" ]</pre> <p><b>Response:</b><br/>HTTP 200 OK</p> <pre>{   RelatedResults: [     {       ObjectId: "SystemName:Object1",       ErrorCode: 0,       RelatedItems: [         &lt;related item 1 for Object1&gt;         {           "ItemDescriptor": "",           "Reference": "",           "Mode": 1,           "SourceType": 1,           "Nodes": [             {</pre> |

```

        <BrowserNode: see [→ 55]>
      },
      {
        <BrowserNode: see [→ 55]>
      }
    ]
  },
  <related item 2 for Object1>
  {
    "ItemDescriptor": "",
    "Reference": "",
    "Mode": 2,
    "SourceType": 1,
    "Nodes": [
      {
        <BrowserNode: see [→ 55]>
      }
    ]
  },
  <... more related items for Object1>
]
},
{
  ObjectId: "SystemName:Object2",
  ErrorCode: 0,
  RelatedItems: [
    <related item 1 for Object2>
    {
      "ItemDescriptor": "",
      "Reference": "",
      "Mode": 1,
      "SourceType": 1,
      "Nodes": [
        {
          <BrowserNode: see [→ 55]>
        },
        {
          <BrowserNode: see [→ 55]>
        }
      ]
    },
    <related item 2 for Object2>
    {
      "ItemDescriptor": "",
      "Reference": "",
      "Mode": 2,
      "SourceType": 1,
      "Nodes": [
        {
          <BrowserNode: see [→ 55]>
        }
      ]
    },
    <... more related items for Object2>
  ]
}
]
}

```

## 7.27 Access rights Service

Application rights assigned to user group can be retrieved by this service.

| Resource      | Service Category | Requires Authentication | Requires SignalR (Push Notifications) |
|---------------|------------------|-------------------------|---------------------------------------|
| /accessrights | Internal         | x                       | x                                     |

## 7.27.1 Retrieve access rights

|               |                                                                                                                                                                                                                                                                                                                                                                                                                                |
|---------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Resource Path | /accessrights                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Category      | Internal                                                                                                                                                                                                                                                                                                                                                                                                                       |
| HTTP Method   | GET                                                                                                                                                                                                                                                                                                                                                                                                                            |
| Response      | HTTP 200 OK<br>Array of AccessRightResponseRepresentation object [ <b>→Error! Bookmark not defined.</b> ]                                                                                                                                                                                                                                                                                                                      |
| Examples      | <b>Request:</b><br>GET /api/accessrights<br><b>Response:</b><br>Returns status code 200 with applications and their respective access.<br><pre>{   "ApplicationRights": [     {       "Name": "BACnet Configuration",       "Id": 6,       "Operations": [         {           "Name": "Show",           "Id": 192         },         {           "Name": "Configure",           "Id": 194         }       ]     }   ] }</pre> |

## 7.27.2 Retrieve user roles

|               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|---------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Resource Path | /accessrights/roles                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Category      | Internal                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| HTTP Method   | GET                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Response      | HTTP 200 OK<br>Array of UserRoleRepresentation object                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Examples      | <b>Request:</b><br>GET /api/accessrights/roles<br><b>Response:</b><br>Returns status code 200 with applications and their respective access.<br><pre>{   "RolesMapping": [     {       "RoleId": 101,       "RoleState": false,       "RoleName": "DefaultAdmins",       "RoleDescription": "Default Administrator Group"     },     {       "RoleId": 100,       "RoleState": false,       "RoleName": "DefaultAdmins1",       "RoleDescription": "Default Administrator Group"     }   ] }</pre> |

### 7.27.3 Push Notification

Subscription for any changes in application rights configuration for logged in user will be notified.

| Type     | Name (case sensitive)       |                            |                                        |
|----------|-----------------------------|----------------------------|----------------------------------------|
| Hub      | norisHub                    |                            |                                        |
| Function | notifyAccessRights ( data ) |                            |                                        |
|          | Property                    | Type                       | Description                            |
|          | data                        | AccessRightsRepresentation | See AccessRightsRepresentation [→ 227] |

#### 7.27.3.1 Create channelize Access rights subscription

Subscription response will be pushed through notifySubscriptionStatus function available at client side. Application rights changes will be notified via notifyAccessRights.

| Resource Path | /api/sr/accessrightssubscriptions/channelize/:requestId/:connectionId?includedetails=:includedetails                                                                                                                                                                                                                                       |        |                                                                                                                                                                                                                                                                                                                                                |
|---------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Category      | Public                                                                                                                                                                                                                                                                                                                                     |        |                                                                                                                                                                                                                                                                                                                                                |
| HTTP Method   | POST                                                                                                                                                                                                                                                                                                                                       |        |                                                                                                                                                                                                                                                                                                                                                |
| Parameters    | Property                                                                                                                                                                                                                                                                                                                                   | Type   | Description                                                                                                                                                                                                                                                                                                                                    |
|               | requestId                                                                                                                                                                                                                                                                                                                                  | String | Request Identifier of subscription hub connection. RequestId is to be generated and provided by the client.<br>It is expected from client that client should use unique id for for each request.<br>Recommendation: Use GUID<br><b>Required</b>                                                                                                |
|               | connectionId                                                                                                                                                                                                                                                                                                                               | String | Identifier of subscription hub connection<br><b>Required</b>                                                                                                                                                                                                                                                                                   |
|               | includedetails                                                                                                                                                                                                                                                                                                                             | bool   | Default set to false, This flag will tell server to send all application rights as soon as something changes on server.Setting this flag to false, will result in empty object in push notification stating something is changed.<br><b>Optional</b><br><b>[Setting it to true could be resource heavy if changes are done too frequently]</b> |
| Response      | HTTP 200 OK                                                                                                                                                                                                                                                                                                                                |        |                                                                                                                                                                                                                                                                                                                                                |
| Examples      | <b>Request:</b><br>POST /api/sr/accessrightssubscriptions/channelize/9565ca41-8556-4dbb-94cc-1b89451a5db5/c6c95ece-133b-480b-af1f-067a173e0f5b?includedetails=false<br><br><b>notifySubscriptionStatus:</b><br><pre>{   "ErrorCode":0,   "RequestId": "9565ca41-8556-4dbb-94cc-1b89451a5db5",   "RequestFor":" notifyAccessRights" }</pre> |        |                                                                                                                                                                                                                                                                                                                                                |

### 7.27.3.2 Create User Roles Subscription

|               |                                                                                                                                                                                                                                                                                                                                            |             |                                                                                                                                                                                                                                                  |
|---------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Resource Path | /api/sr/accessrightssubscriptions/channelize/:requestid/:connectionId?includedetails=:includedetails/userroles                                                                                                                                                                                                                             |             |                                                                                                                                                                                                                                                  |
| Category      | Public                                                                                                                                                                                                                                                                                                                                     |             |                                                                                                                                                                                                                                                  |
| HTTP Method   | POST                                                                                                                                                                                                                                                                                                                                       |             |                                                                                                                                                                                                                                                  |
| Parameters    | <b>Property</b>                                                                                                                                                                                                                                                                                                                            | <b>Type</b> | <b>Description</b>                                                                                                                                                                                                                               |
|               | requestId                                                                                                                                                                                                                                                                                                                                  | String      | Request Identifier of subscription hub connection. RequestId is to be generated and provided by the client.<br>It is expected from client that client should use unique id for for each request.<br>Recommendation: Use GUID<br><b>Required</b>  |
|               | connectionId                                                                                                                                                                                                                                                                                                                               | String      | Identifier of subscription hub connection<br><b>Required</b>                                                                                                                                                                                     |
|               | includedetails                                                                                                                                                                                                                                                                                                                             | bool        | Default set to false, Setting this flag to false, will result in empty object in push notification stating something is changed.<br><b>[Currently setting it to true does not change the response. This might be implemented in the future.]</b> |
| Response      | HTTP 200 OK                                                                                                                                                                                                                                                                                                                                |             |                                                                                                                                                                                                                                                  |
| Examples      | <b>Request:</b><br>POST /api/sr/accessrightssubscriptions/channelize/9565ca41-8556-4dbb-94cc-1b89451a5db5/c6c95ece-133b-480b-af1f-067a173e0f5b?includedetails=false/userroles<br><br><b>notifySubscriptionStatus:</b><br><pre>{   "ErrorCode":0,   "RequestId": "9565ca41-8556-4dbb-94cc-1b89451a5db5",   "RequestFor":"userRoles" }</pre> |             |                                                                                                                                                                                                                                                  |

### 7.27.3.3 Delete Subscription for a Access Rights

|               |                                                                                                                                             |             |                                               |
|---------------|---------------------------------------------------------------------------------------------------------------------------------------------|-------------|-----------------------------------------------|
| Resource Path | /api/sr/ accessrightssubscriptions /{connectionId}                                                                                          |             |                                               |
| Category      | Public                                                                                                                                      |             |                                               |
| HTTP Method   | DELETE                                                                                                                                      |             |                                               |
| Parameters    | <b>Property</b>                                                                                                                             | <b>Type</b> | <b>Description</b>                            |
|               | connectionId                                                                                                                                | String      | Identifier of subscription<br><b>Required</b> |
| Response      | Object of UnsubscriptionStatus See Link [→228]                                                                                              |             |                                               |
| Examples      | <b>Request:</b><br>DELETE api/sr/ accessrightssubscriptions/aa82215b-b635-40a7-92a3-4fcc80a677d1<br><br><b>Response:</b><br>HTTP/1.1 200 OK |             |                                               |

#### 7.27.3.4 Delete Subscription for User Roles

|               |                                                                                                                                               |             |                                               |
|---------------|-----------------------------------------------------------------------------------------------------------------------------------------------|-------------|-----------------------------------------------|
| Resource Path | /api/sr/ accessrightssubscriptions /{connectionId}/userroles                                                                                  |             |                                               |
| Category      | Public                                                                                                                                        |             |                                               |
| HTTP Method   | DELETE                                                                                                                                        |             |                                               |
| Parameters    | <b>Property</b>                                                                                                                               | <b>Type</b> | <b>Description</b>                            |
|               | connectionId                                                                                                                                  | String      | Identifier of subscription<br><b>Required</b> |
| Response      | HTTP 200 OK                                                                                                                                   |             |                                               |
| Examples      | <b>Request:</b><br>DELETE api/sr/ accessrightssubscriptions/aa82215b-b635-40a7-92a3-4fcc80a677d1/userroles<br><b>Response:</b><br>HTTP 200 OK |             |                                               |



## 7.28 Operating Procedures Service

| Resource             | Service Category | Requires Authentication | Requires SignalR (Push Notifications) |
|----------------------|------------------|-------------------------|---------------------------------------|
| /operatingprocedures | Internal         | x                       | x                                     |

### 7.28.1 Retrieve Operating procedure

|               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|---------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Resource Path | /operatingprocedures                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Category      | Public                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| HTTP Method   | GET                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Response      | HTTP 200 OK<br>Object of Operating procedure [→226]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Examples      | <p><b>Request:</b><br/>GET /api/operatingprocedures?id=1.832765.2.23.2.131675.637082217374440000.0</p> <p><b>Response:</b><br/>Returns status code 200.</p> <pre>{   "AlertCount": 1,   "AlertSource": "System1",   "AlertTime": "2019-11-07T11:03:04.325Z",   "Id": "1.832765.2.23.2.131675.637082217374440000.0",   "InTreatmentBy": "#DefaultAdmin",   "IsClosed": true,   "ResetSteps": 0,   "Sequential": true,   "Steps": [     {       "HasConfirmedExecution": true,       "IsCompleted": true,       "Operator": "Admin",       "RuntimeStatus": "Runtime text",       "Status": "Failed",       "StepId": "System1:GraphicsStep",       "StepName": "GraphicsStep"     }   ],   "TemplateName": "GraphicsOperation" }</pre> |

### 7.28.2 Update Operating procedure step

|               |                                                          |                        |                                                                                                                                         |
|---------------|----------------------------------------------------------|------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| Resource Path | /operatingprocedures/{id}/step                           |                        |                                                                                                                                         |
| Category      | Internal                                                 |                        |                                                                                                                                         |
| HTTP Method   | PUT                                                      |                        |                                                                                                                                         |
| Parameters    | <b>Property</b>                                          | <b>Type</b>            | <b>Description</b>                                                                                                                      |
|               | id                                                       | string                 | <b>Required</b><br>Operating Procedure id                                                                                               |
| Body          | operatingStep                                            | OperatingProcedureStep | <b>Required</b><br>Object of Operating Procedure Step<br>All properties are not updatable, please refer Operating procedure step [→227] |
| Response      | HTTP 200 OK<br>Object of Operating procedure step [→227] |                        |                                                                                                                                         |

|          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Examples | <b>Request:</b><br>PUT api/operatingprocedures/1.832765.2.23.2.131675.637082217374440000.0/step<br><b>BODY:</b><br><pre>{   "HasConfirmedExecution": true,   "IsCompleted": true,   "Operator": "Admin",   "RuntimeStatus": "Runtime text",   "Status": "Failed",   "StepId": "System1:GraphicsStep" }</pre><br><b>Response:</b><br>Returns status code 200<br><pre>{   "HasConfirmedExecution": true,   "IsCompleted": true,   "Operator": "Admin",   "RuntimeStatus": "Runtime text",   "Status": "Failed",   "StepId": "System1:GraphicsStep",   "StepName": "GraphicsStep" }</pre> |
|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

### 7.28.3 Push Notification

After subscribing for Operating Procedures initial notification will be sent. Afterwards any change will be notified.

| Type     | Name (case sensitive)              |                                              |                                                          |
|----------|------------------------------------|----------------------------------------------|----------------------------------------------------------|
| Hub      | norisHub                           |                                              |                                                          |
| Function | notifyOperatingProcedures ( data ) |                                              |                                                          |
|          | Property                           | Type                                         | Description                                              |
|          | data                               | OperatingProcedureSubscriptionRepresentation | See OperatingProcedureSubscriptionRepresentation [→ 227] |

#### 7.28.3.1 Create channelize Operating procedures subscription

Subscription response will be pushed through notifySubscriptionStatus function available at client side. Operating procedure changes will be notified via notifyOperatingProcedures.

| Resource Path | /api/sr/operatingproceduressubscriptions/channelize/:requestId/:connectionId |        |                                                                                                                                                                                                                                                 |
|---------------|------------------------------------------------------------------------------|--------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Category      | Public                                                                       |        |                                                                                                                                                                                                                                                 |
| HTTP Method   | POST                                                                         |        |                                                                                                                                                                                                                                                 |
| Parameters    | Property                                                                     | Type   | Description                                                                                                                                                                                                                                     |
|               | requestId                                                                    | String | Request Identifier of subscription hub connection. RequestId is to be generated and provided by the client.<br>It is expected from client that client should use unique id for for each request.<br>Recommendation: Use GUID<br><b>Required</b> |
|               | connectionId                                                                 | String | Identifier of subscription hub connection<br><b>Required</b>                                                                                                                                                                                    |

|          |                                                                                                                                                                                                                                                                                                                                                                                                      |        |                                           |
|----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|-------------------------------------------|
|          | id                                                                                                                                                                                                                                                                                                                                                                                                   | string | Operating procedure id<br><b>Optional</b> |
| Response | HTTP 200 OK                                                                                                                                                                                                                                                                                                                                                                                          |        |                                           |
| Examples | <b>Request:</b><br>POST api/sr/operatingproceduresubscriptions /channel-ize/9565ca41-8556-4dbb-94cc-1b89451a5db5/c6c95ece-133b-480b-af1f-067a173e0f5b<br><br><b>notifySubscriptionStatus:</b><br><pre>{   "Id": "c6c95ece-133b-480b-af1f-067a173e0f5b",   "Key": "1a",   "ErrorCode": 0,   "RequestId": "9565ca41-8556-4dbb-94cc-1b89451a5db5",   "RequestFor": " notifyOperatingProcedures" }</pre> |        |                                           |

### 7.28.3.2 Subscription Status Notification

After subscribing for OperatingProcedure, for each key subscription status will be sent through notifySubscriptionStatus.

|          |                                                                                                                                                                                                                                                                                                                                                                                                           |                                            |                                      |
|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------|--------------------------------------|
| Type     | Name (case sensitive)                                                                                                                                                                                                                                                                                                                                                                                     |                                            |                                      |
| Hub      | norisHub                                                                                                                                                                                                                                                                                                                                                                                                  |                                            |                                      |
| Function | notifySubscriptionStatus(status)                                                                                                                                                                                                                                                                                                                                                                          |                                            |                                      |
|          | <b>Property</b>                                                                                                                                                                                                                                                                                                                                                                                           | <b>Type</b>                                | <b>Description</b>                   |
|          | Status                                                                                                                                                                                                                                                                                                                                                                                                    | OperatingProcedureSubscriptionKey Response | Subscription objects See Link [→228] |
| Example  | <pre>{   "Id": "c6c95ece-133b-480b-af1f-067a173e0f5b",   "Key": "1a",   "ErrorCode": 0,   "RequestId": "9565ca41-8556-4dbb-94cc-1b89451a5db5",   "RequestFor": " notifyOperatingProcedures",   "_links": [     {       "Rel": "unsubscribe",       "Href": "api/sr/operatingproceduresubscriptions/aa82215b-b635-40a7-92a3-4fcc80a677d1?subscriptionKey=1a",       "IsTemplated": false     }   ] }</pre> |                                            |                                      |

### 7.28.3.3 Delete Subscription for a Operating procedure

|               |                                                        |             |                                               |
|---------------|--------------------------------------------------------|-------------|-----------------------------------------------|
| Resource Path | /api/sr/operatingproceduresubscriptions/{connectionId} |             |                                               |
| Category      | Public                                                 |             |                                               |
| HTTP Method   | DELETE                                                 |             |                                               |
| Parameters    | <b>Property</b>                                        | <b>Type</b> | <b>Description</b>                            |
|               | connectionId                                           | String      | Identifier of subscription<br><b>Required</b> |

|          |                                                                                                                                                                                                                 |        |                                                          |
|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|----------------------------------------------------------|
|          | subscriptionKey                                                                                                                                                                                                 | String | Query Parameter with subscription key<br><b>Required</b> |
| Response | Object of UnsubscriptionStatus See Link [→228]                                                                                                                                                                  |        |                                                          |
| Examples | <b>Request:</b><br>DELETE api/sr/operatingproceduresubscriptions/aa82215b-b635-40a7-92a3-4fcc80a677d1?subscriptionKey=1a<br><br><b>Response:</b><br>HTTP/1.1 200 OK<br>{<br>"Key": "1a",<br>"ErrorCode": 0<br>} |        |                                                          |

## 7.29 Licenses Service

Settings are stored on a “per user” basis.

| Resource  | Service Category | Requires Authentication | Requires SignalR (Push Notifications) |
|-----------|------------------|-------------------------|---------------------------------------|
| /licenses | Public           | x                       |                                       |

### 7.29.1 Read License Mode

Read current Licences Mode; the License mode information is returned in the body of the response.

|               |                                                                                                                                                                                                        |
|---------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Resource Path | /licenses                                                                                                                                                                                              |
| Category      | Public                                                                                                                                                                                                 |
| HTTP Method   | GET                                                                                                                                                                                                    |
| Response      | Object of License Mode See Link [→228]                                                                                                                                                                 |
| Examples      | <b>Request:</b><br>GET /api/licenses<br><br><b>Response:</b><br>Returns status code 200.<br>{<br>"LicenseModeName": "Siemens Developer License",<br>"LicenseModeValue": 4,<br>"ExpirationTime": 0<br>} |

### 7.29.2 Read License Options

Read current Licences Mode; the License mode information is returned in the body of the response.

|               |                                                                             |
|---------------|-----------------------------------------------------------------------------|
| Resource Path | /licenses/ <b>licenseoptions</b>                                            |
| Category      | Public                                                                      |
| HTTP Method   | GET                                                                         |
| Response      | Object of License Option See Link [→8.24]                                   |
| Examples      | <b>Request:</b><br>GET /api/licenses/licenseoptions<br><br><b>Response:</b> |

|  |                                                                                                                                                                                                                                                                                                                                                                           |
|--|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | <p>Returns status code 200.</p> <pre>[   {     "Id": "sbt_gms_intOnly_allfeature",     "Required": 0,     "Remaining": 4,     "Available": 1,     "Description": "Unlimited use for Development Pur- poses"   },   {     "Id": "sbt_gms_ext_sm",     "Required": 0,     "Remaining": -1,     "Available": -1,     "Description": "System Manager Application"   } ]</pre> |
|--|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

## 7.30 Multimonitor Service

Multimonitor Configuration assigned to user group can be retrieved by this service.

| Resource       | Service Category | Requires Authentication | Requires SignalR (Push Notifications) |
|----------------|------------------|-------------------------|---------------------------------------|
| /multimonitors | Internal         | x                       |                                       |

### 7.30.1 Retrieve Multi Monitor Configuration

| Description   | <ul style="list-style-type: none"> <li>Returns the default multi-monitor configuration for the specified client.</li> <li>The client is identified by client certificate or optionally by the StationIdentifier parameter</li> </ul> |        |                                                                                                                                                                                                                                                                                                |
|---------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Resource Path | /multimonitors                                                                                                                                                                                                                       |        |                                                                                                                                                                                                                                                                                                |
| Category      | Internal                                                                                                                                                                                                                             |        |                                                                                                                                                                                                                                                                                                |
| HTTP Method   | GET                                                                                                                                                                                                                                  |        |                                                                                                                                                                                                                                                                                                |
| Parameters    | Property                                                                                                                                                                                                                             | Type   | Description                                                                                                                                                                                                                                                                                    |
|               | stationIdentifier                                                                                                                                                                                                                    | String | <ul style="list-style-type: none"> <li>Identifier for the particular station preferably hostname of any particular machine</li> <li>or it can unique id. If valid Client Certificate is present then it is given higher precedence than stationIdentifier.</li> <li><b>Optional</b></li> </ul> |
| Response      | Object of <a href="#">Multimonitor Service</a>                                                                                                                                                                                       |        |                                                                                                                                                                                                                                                                                                |
| Examples      | <b>Request Url</b>                                                                                                                                                                                                                   |        |                                                                                                                                                                                                                                                                                                |

|  |                                                                                                                       |
|--|-----------------------------------------------------------------------------------------------------------------------|
|  | /api/multimonitors/user?stationIdentifier=stat1&api_key=defaultadmin%3AGau-rav%4026                                   |
|  | <b>Response Body</b><br><pre>{   "Configuration": "conf1",   "OverrideSettings": true,   "HostName": "host19" }</pre> |

### 7.30.2 Retrieve MultiMonitor Configuration For User

|               |                                                                                                                                                                                                                                        |             |                                                                                                                                                                                                                                                                                                         |
|---------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Description   | <ul style="list-style-type: none"> <li>Returns the multi-monitor configuration for the specified client and logged in user.</li> <li>The client is identified by client certificate or optionally by the clientid parameter</li> </ul> |             |                                                                                                                                                                                                                                                                                                         |
| Resource Path | /multimonitors                                                                                                                                                                                                                         |             |                                                                                                                                                                                                                                                                                                         |
| Category      | Internal                                                                                                                                                                                                                               |             |                                                                                                                                                                                                                                                                                                         |
| HTTP Method   | GET                                                                                                                                                                                                                                    |             |                                                                                                                                                                                                                                                                                                         |
| Parameters    | <b>Property</b>                                                                                                                                                                                                                        | <b>Type</b> | <b>Description</b>                                                                                                                                                                                                                                                                                      |
|               | stationIdentifier                                                                                                                                                                                                                      | String      | <ul style="list-style-type: none"> <li>Identifier for the particular station preferably hostname of any particular machine</li> <li>or it can unique id.</li> <li>If valid Client Certificate is present then it is given higher precedence than stationIdentifier.</li> <li><b>Optional</b></li> </ul> |
| Response      | Object of <a href="#">Multimonitor Service</a>                                                                                                                                                                                         |             |                                                                                                                                                                                                                                                                                                         |
| Examples      | <b>Request Url</b>                                                                                                                                                                                                                     |             |                                                                                                                                                                                                                                                                                                         |
|               | /api/multimonitors/user?stationIdentifier=z&api_key=defaultadmin%3A9                                                                                                                                                                   |             |                                                                                                                                                                                                                                                                                                         |
| Examples      | <b>Response Body</b>                                                                                                                                                                                                                   |             |                                                                                                                                                                                                                                                                                                         |
|               | <pre>{   "Configuration": "y",   "OverrideSettings": false,   "HostName": "x" }</pre>                                                                                                                                                  |             |                                                                                                                                                                                                                                                                                                         |

### 7.30.3 Add/Update Multi Monitor Configuration

|             |                                                                                                                                                                                                                                                                                                                                        |             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|-------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Description | <ul style="list-style-type: none"> <li>Stores the default multi-monitor configuration for the specified client.</li> <li>The client is identified by client certificate or optionally by the clientid parameter</li> <li>Saving checks, if the application right 'Multi-Monitor – Configure' is set for the logged in user.</li> </ul> |             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| Category    | Internal                                                                                                                                                                                                                                                                                                                               |             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| HTTP Method | POST                                                                                                                                                                                                                                                                                                                                   |             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| Parameters  | <b>Property</b>                                                                                                                                                                                                                                                                                                                        | <b>Type</b> | <b>Description</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|             | Configuration                                                                                                                                                                                                                                                                                                                          | string      | Json object for configuration of MultiMonitor Settings                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|             | OverrideSettings                                                                                                                                                                                                                                                                                                                       | bool        | <p>Any user can overrule the default multi monitor configuration with its own setting. This setting is used by the client to enable/disable the respective UI controls which support the multi monitor configuration. I.e. if it is set to false, the user (except the user which has the application right 'Configure multi monitor') <b>cannot</b>:</p> <ul style="list-style-type: none"> <li>Open an additional system manager.</li> <li>Undock a dedicated event manager.</li> <li>Close an additional system or event manager.</li> </ul> <p>Move windows.</p> |
|             | HostName                                                                                                                                                                                                                                                                                                                               | string      | HostName of the client where multimonitor configuration being used.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|             | stationIdentifier                                                                                                                                                                                                                                                                                                                      | String      | <ul style="list-style-type: none"> <li>Identifier for the particular station preferably hostname of any particular machine</li> <li>or it can unique id.</li> </ul> <p>If valid Client Certificate is present then it is given higher precedence than stationIdentifier.</p> <ul style="list-style-type: none"> <li><b>Optional</b></li> </ul>                                                                                                                                                                                                                       |
| Response    | Object of <a href="#">Multimonitor Service</a>                                                                                                                                                                                                                                                                                         |             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| Examples    | <p><b>Request Url</b></p> <p>/api/multimonitors/user?stationIdentifier=stat1&amp;api_key=defaultadmin%3AGurav%4026</p> <p><b>Response Body</b></p> <pre>{   "StationIdentifier": "stat1",   "ErrorCode": 0 }</pre>                                                                                                                     |             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |

#### 7.30.4 Add/Update MultiMonitor Configuration For User

| Description       | <ul style="list-style-type: none"><li>• Stores the multi-monitor configuration for the specified client and logged in user.</li><li>• The client is identified by client certificate or optionally by the clientid parameter</li><li>• Saving checks if <b>either one</b> of the following options is met<ul style="list-style-type: none"><li>- No default multi-monitor configuration is available for the specified client.</li><li>- A default multi-monitor configuration is available for the specified client and the flag 'OverrideSettings' is set.</li></ul></li><li>• Note, that the 'Multi-Monitor – Configure' application right is <b>not</b> applied/needed!</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |      |             |               |        |                                                        |                  |      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |          |        |                                                                     |                   |        |                                                                                                                                                                                                                                                                                                  |  |  |
|-------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|-------------|---------------|--------|--------------------------------------------------------|------------------|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|--------|---------------------------------------------------------------------|-------------------|--------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|--|
| Category          | Internal                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |      |             |               |        |                                                        |                  |      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |          |        |                                                                     |                   |        |                                                                                                                                                                                                                                                                                                  |  |  |
| HTTP Method       | POST                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |      |             |               |        |                                                        |                  |      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |          |        |                                                                     |                   |        |                                                                                                                                                                                                                                                                                                  |  |  |
| Parameters        | <table><tr><th>Property</th><th>Type</th><th>Description</th></tr><tr><td>Configuration</td><td>string</td><td>Json object for configuration of MultiMonitor Settings</td></tr><tr><td>OverrideSettings</td><td>bool</td><td>Any user can override the default multi monitor configuration with its own setting. This setting is used by the client to enable/disable the respective UI controls which support the multi monitor configuration. I.e. if it is set to false, the user (except the user which has the application right 'Configure multi monitor') <b>cannot</b>:<ul style="list-style-type: none"><li>• Open an additional system manager.</li><li>• Undo a dedicated event manager.</li><li>• Close an additional system or event manger.</li></ul>Move windows.</td></tr><tr><td>HostName</td><td>string</td><td>HostName of the client where multimonitor configuration being used.</td></tr><tr><td>stationIdentifier</td><td>string</td><td><ul style="list-style-type: none"><li>• Identifier for the particular station preferably hostname of any particular machine</li><li>• or it can unique id. If valid Client Certificate is present then it is given higher precedence than stationIdentifier.</li><li>• <b>Optional</b></li></ul></td></tr></table> | Property                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Type | Description | Configuration | string | Json object for configuration of MultiMonitor Settings | OverrideSettings | bool | Any user can override the default multi monitor configuration with its own setting. This setting is used by the client to enable/disable the respective UI controls which support the multi monitor configuration. I.e. if it is set to false, the user (except the user which has the application right 'Configure multi monitor') <b>cannot</b> : <ul style="list-style-type: none"><li>• Open an additional system manager.</li><li>• Undo a dedicated event manager.</li><li>• Close an additional system or event manger.</li></ul> Move windows. | HostName | string | HostName of the client where multimonitor configuration being used. | stationIdentifier | string | <ul style="list-style-type: none"><li>• Identifier for the particular station preferably hostname of any particular machine</li><li>• or it can unique id. If valid Client Certificate is present then it is given higher precedence than stationIdentifier.</li><li>• <b>Optional</b></li></ul> |  |  |
| Property          | Type                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |      |             |               |        |                                                        |                  |      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |          |        |                                                                     |                   |        |                                                                                                                                                                                                                                                                                                  |  |  |
| Configuration     | string                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Json object for configuration of MultiMonitor Settings                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |      |             |               |        |                                                        |                  |      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |          |        |                                                                     |                   |        |                                                                                                                                                                                                                                                                                                  |  |  |
| OverrideSettings  | bool                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Any user can override the default multi monitor configuration with its own setting. This setting is used by the client to enable/disable the respective UI controls which support the multi monitor configuration. I.e. if it is set to false, the user (except the user which has the application right 'Configure multi monitor') <b>cannot</b> : <ul style="list-style-type: none"><li>• Open an additional system manager.</li><li>• Undo a dedicated event manager.</li><li>• Close an additional system or event manger.</li></ul> Move windows. |      |             |               |        |                                                        |                  |      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |          |        |                                                                     |                   |        |                                                                                                                                                                                                                                                                                                  |  |  |
| HostName          | string                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | HostName of the client where multimonitor configuration being used.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |      |             |               |        |                                                        |                  |      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |          |        |                                                                     |                   |        |                                                                                                                                                                                                                                                                                                  |  |  |
| stationIdentifier | string                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | <ul style="list-style-type: none"><li>• Identifier for the particular station preferably hostname of any particular machine</li><li>• or it can unique id. If valid Client Certificate is present then it is given higher precedence than stationIdentifier.</li><li>• <b>Optional</b></li></ul>                                                                                                                                                                                                                                                       |      |             |               |        |                                                        |                  |      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |          |        |                                                                     |                   |        |                                                                                                                                                                                                                                                                                                  |  |  |
| Response          | Object of <a href="#">Multimonitor Service</a>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |      |             |               |        |                                                        |                  |      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |          |        |                                                                     |                   |        |                                                                                                                                                                                                                                                                                                  |  |  |
| Examples          | <b>Request Url</b><br>/api/multimonitors/user?stationIdentifier=z&api_key=defaultadmin%3A9<br><b>Response Body</b><br>{<br>"UserName": "DefaultAdmin",<br>"StationIdenfier": "z",<br>"ErrorCode": 0<br>}                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |      |             |               |        |                                                        |                  |      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |          |        |                                                                     |                   |        |                                                                                                                                                                                                                                                                                                  |  |  |



### 7.30.5 Delete MultiMonitor Configuration

|               |                                                                                                                                                                                                     |             |                                                                                                                                                                                                                                                                                                |
|---------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Description   | <ul style="list-style-type: none"> <li>Delete the default multi-monitor configuration for the specified client.</li> </ul>                                                                          |             |                                                                                                                                                                                                                                                                                                |
| Resource Path | /multimonitors                                                                                                                                                                                      |             |                                                                                                                                                                                                                                                                                                |
| Category      | Internal                                                                                                                                                                                            |             |                                                                                                                                                                                                                                                                                                |
| HTTP Method   | DELETE                                                                                                                                                                                              |             |                                                                                                                                                                                                                                                                                                |
| Parameters    | <b>Property</b>                                                                                                                                                                                     | <b>Type</b> | <b>Description</b>                                                                                                                                                                                                                                                                             |
|               | stationIdentifier                                                                                                                                                                                   | String      | <ul style="list-style-type: none"> <li>Identifier for the particular station preferably hostname of any particular machine</li> <li>or it can unique id. If valid Client Certificate is present then it is given higher precedence than stationIdentifier.</li> <li><b>Optional</b></li> </ul> |
| Response      | Object of <a href="#">Multimonitor Service</a>                                                                                                                                                      |             |                                                                                                                                                                                                                                                                                                |
| Examples      | <b>Request Url</b><br>/api/multimonitors/user?stationIdentifier=stat1&api_key=defaultadmin%3AGaurav%4026<br><b>Response Body</b><br><pre>{   "StationIdentifier": "stat1",   "ErrorCode": 0 }</pre> |             |                                                                                                                                                                                                                                                                                                |

### 7.30.6 Delete MultiMonitorConfiguration For User

|               |                                                                                                                                                                                                                 |             |                                                                                                                                                                                                                                                                                                |
|---------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Description   | Deletes the multi-monitor configuration for the specified client and logged in user.                                                                                                                            |             |                                                                                                                                                                                                                                                                                                |
| Resource Path | /api/sr/ accessrightssubscriptions /{connectionId}/userroles                                                                                                                                                    |             |                                                                                                                                                                                                                                                                                                |
| Category      | Public                                                                                                                                                                                                          |             |                                                                                                                                                                                                                                                                                                |
| HTTP Method   | DELETE                                                                                                                                                                                                          |             |                                                                                                                                                                                                                                                                                                |
| Parameters    | <b>Property</b>                                                                                                                                                                                                 | <b>Type</b> | <b>Description</b>                                                                                                                                                                                                                                                                             |
|               | stationIdentifier                                                                                                                                                                                               | String      | <ul style="list-style-type: none"> <li>Identifier for the particular station preferably hostname of any particular machine</li> <li>or it can unique id. If valid Client Certificate is present then it is given higher precedence than stationIdentifier.</li> <li><b>Optional</b></li> </ul> |
| Response      | Object of <a href="#">Multimonitor Service</a>                                                                                                                                                                  |             |                                                                                                                                                                                                                                                                                                |
| Examples      | <b>Request Url</b><br>/api/multimonitors/user?stationIdentifier=a&api_key=defaultadmin%3A9<br><b>Response Body</b><br><pre>{   "UserName": "DefaultAdmin",   "StationIdentifier": "a",   "ErrorCode": 0 }</pre> |             |                                                                                                                                                                                                                                                                                                |

## 7.31 Validation Service

| Resource    | Service Category | Requires Authentication | Requires SignalR (Push Notifications) |
|-------------|------------------|-------------------------|---------------------------------------|
| /validation | Public           | x                       |                                       |

### 7.31.1 Command Validation Operation

This method checks the incoming properties (and the command group) for the most restrictive validation requirements.

The return will also include the contents of a text group if all objects have been configured for "pre-defined" comments

if not all are configured that way, then the text group is not provided.

|               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |             |                                                                     |
|---------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|---------------------------------------------------------------------|
| Resource Path | /api/validation/getCommandValidationOperation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |             |                                                                     |
| Category      | Public                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |             |                                                                     |
| HTTP Method   | POST                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |             |                                                                     |
| Parameters    | <b>Property</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | <b>Type</b> | <b>Description</b>                                                  |
|               | PropertyIds                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | String[]    | List of Property Ids to validate for commanding.<br><b>Required</b> |
|               | Group Number                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Int         | Group Number of Command to execute.<br><b>Required</b>              |
| Response      | Object of ValidateOpResponse See Link <a href="#">8.1.4</a>                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |             |                                                                     |
| Examples      | <b>Request:</b><br>POST /api/validation/getCommandValidationOperation<br><br><b>Response:</b><br>Returns status code 200<br><pre>{   "IncrVersion": true,   "Log": true,   "PredefinedComments": [     {       "Value": 0,       "Text": "string",       "Color": 0,       "SubText": [         {}       ]     }   ],   "CommentRule": "string",   "ReAuthentication": "string",   "IsFourEyesEnabled": true,   "_links": [     {       "Rel": "string",       "Href": "string",       "IsTemplated": true     }   ] }</pre> |             |                                                                     |



### 7.31.2 Edit Validation Operation

This method checks the incoming collection of objects (DplIdentifiers) for the most restrictive validation requirements.

The return will also include the contents of a text group if all objects have been configured for "pre-defined" comments

if not all are configured that way, then the text group is not provided.

|               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |             |                                                                   |
|---------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|-------------------------------------------------------------------|
| Resource Path | /api/validation/getEditValidationOperation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |             |                                                                   |
| Category      | Public                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |             |                                                                   |
| HTTP Method   | POST                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |             |                                                                   |
| Parameters    | <b>Property</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | <b>Type</b> | <b>Description</b>                                                |
|               | objectIds                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | String[]    | List of object Ids to validate to edit object.<br><b>Required</b> |
| Response      | Object of ValidateOpResponse See link <a href="#">8.1.4</a>                                                                                                                                                                                                                                                                                                                                                                                                                                                               |             |                                                                   |
| Examples      | <b>Request:</b><br>POST /api/validation/getEditValidationOperation<br><br><b>Response:</b><br>Returns status code 200<br><pre>{   "IncrVersion": true,   "Log": true,   "PredefinedComments": [     {       "Value": 0,       "Text": "string",       "Color": 0,       "SubText": [         {}       ]     }   ],   "CommentRule": "string",   "ReAuthentication": "string",   "IsFourEyesEnabled": true,   "_links": [     {       "Rel": "string",       "Href": "string",       "IsTemplated": true     }   ] }</pre> |             |                                                                   |

### 7.31.3 Validate Credential

Validates the User and Supervisor Credentials.

|               |                                    |             |                    |
|---------------|------------------------------------|-------------|--------------------|
| Resource Path | /api/validation/validateCredential |             |                    |
| Category      | Public                             |             |                    |
| HTTP Method   | POST                               |             |                    |
| Parameters    | <b>Property</b>                    | <b>Type</b> | <b>Description</b> |
|               |                                    |             |                    |

|          |                                                                                                               |          |                                                                                                               |
|----------|---------------------------------------------------------------------------------------------------------------|----------|---------------------------------------------------------------------------------------------------------------|
|          | objectIds                                                                                                     | String[] | List of Object Ids to validate for provided credentials.<br><b>Required</b>                                   |
|          | CheckCredentials                                                                                              | int      | 0 validate both user and supervisor<br>1 validate only user<br>2 validate supervisor only.<br><b>Required</b> |
|          | UserPassword                                                                                                  | string   | Password of logged in user.                                                                                   |
|          | SupervisorName                                                                                                | string   | Name of Supervisor.                                                                                           |
|          | SupervisorPassword                                                                                            | string   | Password of Supervisor                                                                                        |
| Response | HTTP 200                                                                                                      |          |                                                                                                               |
| Examples | <b>Request:</b><br>POST /api/validation/validateCredential<br><br><b>Response:</b><br>Returns status code 200 |          |                                                                                                               |

## 8 Objects and Data Types


**NOTE:**

In case a property has an empty or null value it may be omitted.

### 8.1 Common

#### 8.1.1 Link

Models a hyperlink to another resource.

| Property    | Type   | Description                                                       |
|-------------|--------|-------------------------------------------------------------------|
| Href        | string | Hyperlink to a resource (URI [RFC3986] or URI template [RFC6570]) |
| IsTemplated | bool   | If true, Href contains a placeholder in its link.                 |
| Rel         | string | Link relation type (RFC5988)                                      |

#### 8.1.2 Page<Name, Type>

| Property         | Type            | Description                             |
|------------------|-----------------|-----------------------------------------|
| _links           | Array of Link   | See Link [→ 50]                         |
| Events           | Array of <Type> | List of events See Link [→ 195 ]        |
| Page             | int             | Number of current page (first page = 1) |
| Size             | int             | Number of items per page                |
| Total            | Int             | Total number of items                   |
| CreationTimeFrom | DateTime        |                                         |
| CreationTimeTo   | DateTime        |                                         |

#### 8.1.3 KeyValue<Name, Type>

| Property | Type   | Description                              |
|----------|--------|------------------------------------------|
| <Name>   | <Type> | Name and type depend on current instance |

#### 8.1.4 NameValue

| Property | Type        | Description               |
|----------|-------------|---------------------------|
| DataType | ApiDataType | Data type of the property |
| Name     | string      | Name of the property      |
| Value    | string      | Value of the property     |

#### 8.1.5 Subscription

| Property   | Type          | Description                                       |
|------------|---------------|---------------------------------------------------|
| _links     | Array of Link | See Link [→ 190]                                  |
| Key        | int           | Subscription key is specific for the requested ID |
| ErrorCode  | int           | If value is >0, an error occurred.                |
| PropertyId | string        | Address of a property                             |

## 8.1.6 UnSubscription

| Property  | Type | Description                                       |
|-----------|------|---------------------------------------------------|
| Key       | int  | Subscription key is specific for the requested ID |
| ErrorCode | int  | If value is >0, an error occurred.                |

## 8.1.7 Value

| Property         | Type     | Description                                                                                                |
|------------------|----------|------------------------------------------------------------------------------------------------------------|
| Value            | string   | Raw value                                                                                                  |
| DisplayValue     | string   | Value for display purpose (textual representation of the raw value)                                        |
| Timestamp        | DateTime | Timestamp of this value                                                                                    |
| QualityGood      | bool     |                                                                                                            |
| Quality          | string   | See Quality Flags [→ 191]                                                                                  |
| IsPropertyAbsent | bool     | Driver reported that property is not present<br><b>Note:</b> Only set if true and not available for trends |

### 8.1.7.1 Quality Flags

**Quality** is a string representation of a 64-bit bit-set. The string contains a ulong numeric value. Bits not listed are not in use.

| Bit | Description                                                                   | QualityGood | Effect on Value |
|-----|-------------------------------------------------------------------------------|-------------|-----------------|
| 0   | variable active/inactive                                                      | TRUE        | value ok        |
| 1   | default value set explicitly                                                  | TRUE        | value ok        |
| 2   | default value set automatically                                               | TRUE        | value ok        |
| 3   | WinCC OA value range violated                                                 | FALSE       | value not ok    |
| 4   | value of the variable "out of range"                                          | FALSE       | value not ok    |
| 5   | explicit invalid                                                              | FALSE       | value not ok    |
| 6   | invalid set by an interface driver                                            | FALSE       | value not ok    |
| 8   | default value invalid                                                         | TRUE        | value ok        |
| 9   | set during general query                                                      | TRUE        | value ok        |
| 10  | set during an individual query                                                | TRUE        | value ok        |
| 11  | interface driver active                                                       | TRUE        | value ok        |
| 12  | value corrected                                                               | TRUE        | value ok        |
| 13  | value condensed or compressed, used (HDB & RAIMA) with archiving/compression. | TRUE        | value ok        |
| 14  | corrected value condensed                                                     | TRUE        | value ok        |
| 15  | additional correction value                                                   | TRUE        | value ok        |
| 16  | compressed value invalid                                                      | TRUE        | value ok        |
| 17  | source time invalid (corrected by the Event Manager)                          | TRUE        | value ok        |
| 19  | disable last value storage                                                    | TRUE        | value ok        |
| 20  | value changed                                                                 | TRUE        | value ok        |
| 21  | value up (or set to same value)                                               | TRUE        | value ok        |
| 24  | for internal use only                                                         | TRUE        | value ok        |
| 32  | driver out of service                                                         | TRUE        | value ok        |

|                 |                                     |       |                       |
|-----------------|-------------------------------------|-------|-----------------------|
| 33              | driver alarm                        | TRUE  | value ok              |
| 34              | driver fault                        | TRUE  | unreliable            |
| 35              | driver overridden                   | TRUE  | value ok              |
| 36              | driver subscribed                   | TRUE  | value ok              |
| 37              | driver property absent              | FALSE | value not ok          |
| 40              | trend: time shift                   | FALSE | no value, is event    |
| 41              | trend: log enabled                  | FALSE | no value, is event    |
| 42              | trend: error                        | FALSE | no value, is event    |
| 43              | trend: purge                        | FALSE | no value, is event    |
| 44              | trend: rollover                     | FALSE | no value, is event    |
| 45              | trend: value is status              | FALSE | value is status/error |
| 46              | trend: log interrupted              | FALSE | no value, is event    |
| 48              | trend: start logging                | TRUE  | value ok              |
| 49              | trend: value reduced                | TRUE  | value ok              |
| 50 <sup>1</sup> | priority 1 (0 bit; 2 <sup>0</sup> ) | TRUE  | value ok              |
| 51 <sup>1</sup> | priority 2 (1 bit; 2 <sup>1</sup> ) | TRUE  | value ok              |
| 52 <sup>1</sup> | priority 4 (2 bit; 2 <sup>2</sup> ) | TRUE  | value ok              |

<sup>1</sup> Bit 50 / Bit 51 / Bit 52 are to be combined into a single value (priority 0..7)

## 8.1.8 ColorRepresentation

| Property | Type | Description      |
|----------|------|------------------|
| A        | byte | Color code       |
| R        | byte | Red color code   |
| G        | byte | Green color code |
| B        | byte | Blue color code  |

## 8.1.1 CommandsSubscription

| Property       | Type          | Description                                                                                                                 |
|----------------|---------------|-----------------------------------------------------------------------------------------------------------------------------|
| _links         | Array of Link | See Link [→ 190]                                                                                                            |
| Key            | int           | Subscription key is specific for the requested ID                                                                           |
| ErrorCode      | int           | If value is >0, an error occurred.                                                                                          |
| PropertyId     | string        | Address of a property                                                                                                       |
| RequestId      | string        | Channelized subscription request identifier<br><b>Note:</b> Only available for channelize requests                          |
| RequestFor     | string        | Notification function name where client will receive further updates<br><b>Note:</b> Only available for channelize requests |
| SubscriptionId | string        | Original requested string for subscription                                                                                  |



## 8.1.2 ValuesSubscription

| Property                   | Type          | Description                                                                                                                 |
|----------------------------|---------------|-----------------------------------------------------------------------------------------------------------------------------|
| _links                     | Array of Link | See Link [→ 190]                                                                                                            |
| Key                        | int           | Subscription key is specific for the requested ID                                                                           |
| ErrorCode                  | int           | If value is >0, an error occurred.                                                                                          |
| PropertyId                 | string        | Address of a property                                                                                                       |
| RequestId                  | string        | Channelized subscription request identifier<br><b>Note:</b> Only available for channelize requests                          |
| RequestFor                 | string        | Notification function name where client will receive further updates<br><b>Note:</b> Only available for channelize requests |
| OriginalObjectOrPropertyId | string        | Original requested string for subscription                                                                                  |
| ObjectOrPropertyId         | string        | Original requested string for subscription                                                                                  |
| AttributeId                | string        | Object Name                                                                                                                 |
| ObjectId                   | string        | Object Id only of requested ObjectOrPropertyId                                                                              |
| PropertyName               | string        | PropertyName of requested ObjectOrPropertyId                                                                                |

## 8.1.3 Attributes

| Property                | Type   | Description                           |
|-------------------------|--------|---------------------------------------|
| DefaultProperty         | string | Default property of object-model      |
| DisciplineDescriptor    | string | Localized name of discipline          |
| DisciplineId            | int    | ID of discipline                      |
| FunctionDefaultProperty | string | Default property of assigned function |
| FunctionName            | string |                                       |
| ManagedType             | uint   |                                       |
| ManagedTypeName         | string |                                       |
| ObjectModelName         | string |                                       |
| ObjectId                | string |                                       |
| SubDisciplineDescriptor | string | Localized name of subdiscipline       |
| SubDisciplineId         | int    | ID of subdiscipline                   |
| SubTypeDescriptor       | string | Localized name of subtype             |
| SubTypeId               | int    | ID of subtype                         |
| TypeDescriptor          | string | Localized name of type                |
| TypeId                  | int    | ID of type                            |

## 8.1.4 ApiDataType

| Name        | Description | BaseType |
|-------------|-------------|----------|
| None        |             |          |
| BasicChar   |             | char     |
| BasicUInt   |             | uint     |
| BasicInt    |             | int      |
| BasicFloat  |             | double   |
| BasicBool   |             | boolean  |
| BasicBit32  |             | uint     |
| BasicString |             | string   |
| BasicTime   |             | DateTime |

|                         |                                                                                                                                                 |        |
|-------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|--------|
| BasicObjectOrPropertyId |                                                                                                                                                 | string |
| BasicLangText           |                                                                                                                                                 | string |
| BasicBlob               |                                                                                                                                                 | string |
| ExtendedBool            | Extends BasicBool with a language text for true/false.                                                                                          |        |
| ExtendedInt             | Extends BasicInt with Min, Max, Default, Unit.                                                                                                  |        |
| ExtendedUInt            | Extends BasicUInt with Min, Max, Default, Unit.                                                                                                 |        |
| ExtendedReal            | Extends BasicFloat with Min, Max, Default, Unit.                                                                                                |        |
| ExtendedEnum            | Extends BasicUInt with enumeration texts.                                                                                                       |        |
| ExtendedBitString       | Extends BasicUInt with bit set texts.                                                                                                           |        |
| ExtendedDateTime        | Extends BasicString with semantics for a DateTime string format.                                                                                |        |
| ExtendedAppSpecific     | Extends BasicString with the knowledge that the string has a dedicated meaning for some application. Typically, the string will be XML-encoded. |        |
| ExtendedAny             | Extends BasicFloat with the permission to switch its type.                                                                                      |        |
| ExtendedComplex         | Extends BasicBlob.                                                                                                                              |        |
| ExtendedDuration        | Extends BasicUInt with Min, Max, Default and time duration specific configurations.                                                             |        |

### 8.1.5 Subgroups

| Property   | Type   | Description                |
|------------|--------|----------------------------|
| Id         | UInt   | Id of the SubGroup         |
| Descriptor | string | Descriptor of the SubGroup |

### 8.1.6 StatusNotification

| Property   | Type   | Description                                                |
|------------|--------|------------------------------------------------------------|
| ErrorCode  | int    | Error code of operation:<br>0 – Success<br>1 - Error       |
| RequestId  | string | Unique Identifier passed by client in subscription request |
| RequestFor | string | Client Method where push notification will be served       |

## 8.2 Token Service

### 8.2.1 Login

| Property                | Type   | Description                                                                                                                          |
|-------------------------|--------|--------------------------------------------------------------------------------------------------------------------------------------|
| access_token            | string | Token to be provided for requesting protected resources.                                                                             |
| expires_in              | int    | Seconds until token expires at the latest<br><b>NOTE:</b> Token will expire sooner in case of inactivity; see Authentication [→ 26]. |
| token_type              | string | Only <b>Bearer</b> is supported at this time.                                                                                        |
| user_name               | string | Logged in user name                                                                                                                  |
| user_descriptor         | string | Localized user descriptor in user language                                                                                           |
| user_profile            | string | name of currently logged in user's profile                                                                                           |
| flex_user_profile       | string | name of currently logged in flex-user profile                                                                                        |
| user_inactivity_timeout | string | Inactivity timeout of user's group (uint) in minutes.                                                                                |

|                           |        |                                                                                                                 |
|---------------------------|--------|-----------------------------------------------------------------------------------------------------------------|
| password_reminder_counter | string | Days remaining for password expiry                                                                              |
| personally_used           | string | “true” : machine is personally used<br>“false” : machine is not personally used for provided client certificate |

**password\_reminder\_counter:** this value gives counter i.e., number of days. Non-zero value of this counter gives number of days left before password expires. If the value is Zero then, the number of days left before password expires, are greater than the set number of days in “Reminder for password expiration” property of passwordpolicy of that System. The values of this counter in login response can be interpreted as shown in following table.

| Response                                                                                                                                            | Interpretation                                                                        |
|-----------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
| password_reminder_counter = “0”                                                                                                                     | Password is active, no need to change it.                                             |
| password_reminder_counter= “10”                                                                                                                     | Password will be expired in 10 days; it can be changed proactively before it expires. |
| { "Id":2400003, "Error":“password_expired”,<br>"Details":“Logon rejected. Password of this account is expired and needs to get changed first.”<br>} | Password of the account is already expired and need to be changed.                    |

## 8.3 Event Service

### 8.3.1 Event

| Property                | Type                   | Description                                                                                 |
|-------------------------|------------------------|---------------------------------------------------------------------------------------------|
| _links                  | Array of Link          | See Link [→ 190]                                                                            |
| CategoryDescriptor      | string                 | Localized textual representation of CategoryId                                              |
| CategoryId              | int                    | Alarm category                                                                              |
| Cause                   | string                 | Cause came with alarm (localized)                                                           |
| Commands                | Array of EventComm and | List of available commands. See Event Command [→ 199]                                       |
| CreationTime            | DateTime               | Time when alarm created.                                                                    |
| Deleted                 | bool                   | Flag whether an event was deleted. This is for server sent notifications only.              |
| Direction               | EventDirection         | See EventDirection [→ 197]                                                                  |
| EventId                 | uint                   | Event counter (non-unique)                                                                  |
| EventText               | string                 | Service text attached to an event.                                                          |
| Id                      | string                 | Desigo CC alarm identifier<br>→ Invariant across sessions                                   |
| InfoDescriptor          | string                 | Additional information to an event                                                          |
| SrcDescriptor           | string                 | Tree node descriptor<br><b>OBSOLETE</b> : use DescriptionList instead                       |
| SrcDesignation          | string                 | Concatenated tree node names<br><b>OBSOLETE</b> : use DesignationList instead               |
| SrcDisciplineDescriptor | string                 | Localized textual representation of DisciplineId                                            |
| SrcDisciplineId         | int                    | Alarm discipline                                                                            |
| SrcLocation             | string                 | Concatenated tree-node descriptors<br><b>OBSOLETE</b> :use DescriptionLocationsList instead |

|                          |                                 |                                                                           |
|--------------------------|---------------------------------|---------------------------------------------------------------------------|
| SrcName                  | string                          | Tree node name                                                            |
| SrcObservedPropertyId    | string                          | Observed property which lets SrcPropertyId trigger an event               |
| SrcPropertyId            | string                          | Source of an event                                                        |
| SrcState                 | SourceState                     | See SourceState [→ 196]                                                   |
| SrcSystemId              | uint                            | Unique system identifier                                                  |
| SrcViewDescriptor        | string                          | View descriptor                                                           |
| SrcViewName              | string                          | View name                                                                 |
| State                    | EventState                      | See EventState [→ 196]                                                    |
| SuggestedAction          | SuggestedAction                 | See SuggestedAction[→198 ]                                                |
| NextCommand              | NextCommand                     | See NextCommand [→198 ]                                                   |
| SourceDesignationList    | Array of Event Designation List | List of node name.<br>See EventDetailsList [→200]                         |
| DescriptionLocationsList | Array of EventsLocations        | List of Concatenated tree-node descriptors<br>See EventDetailsList [→200] |
| DesignationList          | Array of EventsDesignation      | List of Concatenated tree node names<br>See EventDetailsList [→200]       |
| DescriptionList          | Array of EventsDescription      | List of Tree node descriptor<br>See EventDetailsList [→200]               |
| MessageText              | Array of string                 | Information Texts of events                                               |
| SrcAlias                 | string                          | Event Source Alias                                                        |
| OperatingProcedureId     | string                          | Operating procedure id associated with event                              |
| Sound                    | string                          | Path of sound file associated with event                                  |
| AutomaticTreatmentData   | AutomaticTreatmentData          | See Automatic Treatment Data[→199]                                        |

### 8.3.1.1 SourceState

**SrcState** must have one of the following values.

| SrcState | Description      |
|----------|------------------|
| Active   | Object is active |
| Quiet    | Object is quiet  |

### 8.3.1.2 EventState

**State** must have one of the following values.

| State                   | Description                                  |
|-------------------------|----------------------------------------------|
| UnprocessedWithTimer    | Unprocessed with Timer for user action       |
| Unprocessed             | Unprocessed                                  |
| ReadyToBeResetWithTimer | Ready to be reset with timer for user action |
| ReadyToBeReset          | Ready to be reset                            |
| ReadyToBeClosed         | Ready to be closed                           |

|                     |                        |
|---------------------|------------------------|
| WaitingOPCompletion | Waiting for completion |
| Acked               | Acknowledged           |
| Closed              | Closed                 |

### 8.3.1.3 EventDirection

**Direction** must have one of the following values:

| Direction | Description              |
|-----------|--------------------------|
| In        | IN alarm                 |
| Out       | OUT alarm                |
| None      | Neither IN nor OUT alarm |

### 8.3.1.4 EventTimeFilter

**creationTime** must have one of the following values:

| creationTime | Description                             |
|--------------|-----------------------------------------|
| 0            | All events (default, can be omitted)    |
| 1            | Events generated in the last 15 minutes |
| 2            | Events generated in the last 30 minutes |
| 3            | Events generated in the last hour       |
| 4            | Events generated last night             |
| 5            | Events generated yesterday              |
| 6            | Events generated today                  |

### 8.3.1.5 EventSorting

**sorting** must have one of the following values.

| sorting | Description                                                         |
|---------|---------------------------------------------------------------------|
| 0       | Sort by StatId, CategoryId, and descending CreationTime             |
| 1       | Sort by StatId, CategoryId, DirectionId, and ascending CreationTime |

### 8.3.1.6 EventFilter

| Property         | Type             | Description                  |
|------------------|------------------|------------------------------|
| ids              | Array of strings | Event identifier             |
| caseSensitive    | bool             |                              |
| categories       | Array of int     |                              |
| cause            | string           |                              |
| creationTime     | int              | See EventTimeFilter [→ 197 ] |
| creationTimeFrom | DateTime         |                              |
| creationTimeTo   | DateTime         |                              |
| disciplines      | Array of int     |                              |
| systemId         | UInt             | System identifier            |
| states           | Array of int     |                              |

### 8.3.1.7 EventSubscriptionStatus

| Property     | Type               | Description                                                                                                                  |
|--------------|--------------------|------------------------------------------------------------------------------------------------------------------------------|
| ErrorCode    | int                | 0: success<br>1: Failure                                                                                                     |
| Sorting      | int                | See Event Sorting [→ 197]                                                                                                    |
| EventsFilter | EventFilter object | See EventFilter [→ 197]                                                                                                      |
| RequestId    | string             | Channelized subscription request identifier<br><b>Note:</b> Only available for channelize requests                           |
| RequestFor   | string             | Notification function name where client will receive further updates.<br><b>Note:</b> Only available for channelize requests |

### 8.3.1.8 Maintenance State

“Maintenance” must have one of the following values.

| SrcState        | Description                     |
|-----------------|---------------------------------|
| All             | Maintenance is All              |
| MaintenanceOnly | MaintenanceState is Maintenance |
| GenuineOnly     | MaintenanceState is Genuine     |

### 8.3.1.9 SuggestedAction

“SuggestedAction” must have one of the following values.

| SuggestedAction  | Description        |
|------------------|--------------------|
| Acknowledge      |                    |
| Silence          |                    |
| Suspend          |                    |
| Reset            |                    |
| Close            |                    |
| CompleteOP       | Complete operation |
| WaitforCondition |                    |
| None             |                    |

### 8.3.1.10 Next Command

“NextCommand” must have one of the following values.

| NextCommand      | Description |
|------------------|-------------|
| None             |             |
| Acknowledge      |             |
| Reset            |             |
| NoActionPossible |             |
| StepMandatory    |             |
| Silence          |             |
| Close            |             |

### 8.3.1.11 Automatic Treatment data

| Property                 | Type     | Description                                                                                                                                                                                        |
|--------------------------|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| HasFilter                | boolean  |                                                                                                                                                                                                    |
| Stations                 | string[] |                                                                                                                                                                                                    |
| Timeout                  | uint     |                                                                                                                                                                                                    |
| Users                    | uint[]   |                                                                                                                                                                                                    |
| CloseTreatmentWhen       | string   | Configuration value specifying condition for closing the treatment.<br>Valid values:<br>1.SourceToNormal<br>2.UponUserRequest<br>3.EventAcknowledged<br>4.EventReset<br>5.EventClosed<br>6.Timeout |
| OnNewEvent               | string   | Configuration value specifying action on new event.<br>Valid values:<br>1.DoNothing<br>2.OpenEventList<br>3.StartFastTreatment<br>4.StartInvestigativeTreatment<br>5.StartAssistedTreatment        |
| OnNewHigherPriorityEvent | string   | Configuration value specifying action on new higher priority event.<br>Valid values:<br>1. DoNothing<br>2. SwitchToNew<br>3. AskUser                                                               |

### 8.3.2 EventCommand

| Property      | Type           | Description                                                                                                                                                                                                                                                                                                                                                                                                                              |
|---------------|----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| _links        | Array of Link  | See Link [→ 190]                                                                                                                                                                                                                                                                                                                                                                                                                         |
| EventId       | string         | Unique identifier of an event                                                                                                                                                                                                                                                                                                                                                                                                            |
| Id            | EventCommandId | See Event Command ID [→ 49]                                                                                                                                                                                                                                                                                                                                                                                                              |
| Configuration | Int            | ValidationId of a propertyid<br>If Configuration :0 ->None is required<br>If Configuration :1 -> Only comment is required<br>If Configuration : 2->Both Comment and Password(to re-authenticate) are required.<br>if Configuration =3 : Comment , Current logged in user's credentials, Supervisor's Username and Supervisor's Password is expected<br>if Configuration =4 : Supervisor's Username and Supervisor's Password is expected |

### 8.3.3 ValidationData

| Property        | Type            | Description                                         |
|-----------------|-----------------|-----------------------------------------------------|
| Password        | String          | Password required to re-authenticate event-commands |
| LangTextContent | LangTextContent | Input to provide comments<br>Refer LangTextContent  |

|               |        |                                                               |
|---------------|--------|---------------------------------------------------------------|
| SuperName     | String | Supervisor's username required to authenticate event-commands |
| SuperPassword | String | Supervisor's password required to authenticate event-commands |

### 8.3.4 LangTextContent

| Property        | Type     | Description                                                                                                                                                                                                                           |
|-----------------|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| CommonText      | string   | This property takes priority over MultiLangText. If value is assigned to CommonText then property MultiLangText will not be considered. Content of CommonText will be applied to all languages. Whitespace is treated as valid input. |
| LangTextContent | string[] | This property will accept input text for each language in system. Error is returned if number of entries does not match the number of languages in system. Whitespace is treated as valid input.                                      |

### 8.3.5 BatchEventsCommandExecution

| Property        | Type           | Description                                                  |
|-----------------|----------------|--------------------------------------------------------------|
| ValidationInput | ValidationData | Validation information of command execution See Link [→ 199] |
| EventIds        | String[]       | List of Event Id's for execution                             |

### 8.3.6 EventDetailsList

| Property   | Type   | Description                              |
|------------|--------|------------------------------------------|
| ViewId     | uint   | Unique identifier of a view              |
| Descriptor | string | View details (Tree node or node details) |

## 8.4 Event Counter Service

### 8.4.1 EventCounterList

| Property                 | Type                  | Description                                                 |
|--------------------------|-----------------------|-------------------------------------------------------------|
| EventCategoryCounters    | Array of EventCounter | Categorized list of event counters                          |
| TotalCounters            | int                   | Total counters in the system for logged-in user             |
| TotalUnprocessedCounters | int                   | Total unprocessed counters in the system for logged-in user |

### 8.4.2 EventCounter

| Property   | Type          | Description      |
|------------|---------------|------------------|
| _links     | Array of Link | See Link [→ 190] |
| CategoryId | int           | Category key     |



|                               |        |                                                              |
|-------------------------------|--------|--------------------------------------------------------------|
| CategoryDescriptor            | string | Localized category name                                      |
| TotalCount                    | int    | Total events of this category                                |
| UnprocessedCount              | int    | Total unprocessed events of this category                    |
| TotalSubsequentGrouping       | int    | Total number of grouped for subsequent Events for a category |
| UnprocessedSubsequentGrouping | int    | Total number of unprocessed subsequent Events for a category |

### 8.4.1 Category color details

| EColorPalette Type(int) | EColorPalette description   |
|-------------------------|-----------------------------|
| 1                       | TextButtonNormal            |
| 2                       | TextButtonPressed           |
| 3                       | TextEventSelected           |
| 4                       | TextEventNormal             |
| 5                       | TextEventHover              |
| 6                       | ButtonGradientBright        |
| 7                       | ButtonGradientDark          |
| 8                       | ButtonPressedGradientBright |
| 9                       | ButtonPressedGradientDark   |
| 10                      | ButtonBlinkingBright        |
| 11                      | ButtonBlinkingDark          |
| 12                      | EventDescriptorSelected     |
| 13                      | EventDescriptorNormal       |

### 8.4.2 Subscription Status

| Key        | Type   | description                                                                                                                 |
|------------|--------|-----------------------------------------------------------------------------------------------------------------------------|
| ErrorCode  | int    | 0: Success<br>1: Failure                                                                                                    |
| RequestId  | string | Channelized subscription request identifier<br><b>Note:</b> Only available for channelize requests                          |
| RequestFor | string | Notification function name where client will receive further updates<br><b>Note:</b> Only available for channelize requests |

## 8.5 System Browser Service

### 8.5.1 View

| Property    | Type          | Description                                                                                                                                                                                                   |
|-------------|---------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| _links      | Array of Link | See Link [→ 50]                                                                                                                                                                                               |
| Designation | string        | Full view name                                                                                                                                                                                                |
| Descriptor  | string        | Localized view name                                                                                                                                                                                           |
| Name        | string        | View name                                                                                                                                                                                                     |
| SystemId    | uint          | Unique ID of a system                                                                                                                                                                                         |
| SystemName  | String        | Name of system                                                                                                                                                                                                |
| ViewId      | uint          | Unique ID of a view within a system                                                                                                                                                                           |
| ViewType    | int           | The type of view in Desigo CC:<br>-3: Reserved (used internally)<br>-2: Isolated (used for isolated nodes)<br>-1: Undefined<br>0: Management<br>1: Application<br>2: Logical<br>3: Physical<br>4: UserDefined |

## 8.5.2 BrowserNode

| Property    | Type          | Description                                                                                               |
|-------------|---------------|-----------------------------------------------------------------------------------------------------------|
| _links      | Array of Link | See Link [→ 50]                                                                                           |
| Attributes  | Attributes    | See Attributes [→ 51]                                                                                     |
| Descriptor  | string        | Localized browser object name                                                                             |
| Designation | string        | Full view name                                                                                            |
| HasChild    | bool          | Flag whether child nodes are available<br><b>Note:</b> Property omitted for searches and in notifications |
| Name        | string        | View name                                                                                                 |
| Location    | string        | Localized full browser object name                                                                        |
| ObjectId    | string        | Node address                                                                                              |
| SystemId    | uint          | Unique ID of a system                                                                                     |
| ViewId      | uint          | Unique ID of a view within a system                                                                       |

## 8.5.3 SystemBrowserSearchList

| Property  | Type                 | Description                                                              |
|-----------|----------------------|--------------------------------------------------------------------------|
| ErrorCode | int                  | If value is >0, an error occurred                                        |
| ObjectId  | string               | Searched Object Id                                                       |
| Nodes     | Array of BrowserNode | Array of Browser nodes returned as search result for ObjectId see [→ 77] |

## 8.5.4 SystemBrowserSubscription

| Property        | Type              | Description                                                                                           |
|-----------------|-------------------|-------------------------------------------------------------------------------------------------------|
| Action          | ObservationAction | Enumeration to convey additional information about the action. See Enum for Action [→ 78]             |
| Change          | ObservationChange | Enumeration to convey the possible reasons for the event to have occurred. See Enum for Change [→ 78] |
| Node            | BrowserObject     | See BrowserObject [→ 77]                                                                              |
| View            | View              | See View [→ 77]                                                                                       |
| SubscriptionKey | int               | Subscription Key of View                                                                              |
| RequestId       | string            | Channelized subscription request identifier<br><b>Note:</b> Only available for channelize requests    |

### 8.5.4.1 ObservationChange

| Change type | Description                                                                                                                                                         |
|-------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0           | Reserved for 'no reason known'                                                                                                                                      |
| 1           | System name changed                                                                                                                                                 |
| 2           | View separator changed                                                                                                                                              |
| 3           | Change in the Nodes display name OR Views display name                                                                                                              |
| 4           | Tree data changed                                                                                                                                                   |
| 5           | Structural change in the specified View, Tree OR Node. This also handles creation/deletion of System Browser-Views, a System Browser-Node OR System Browser-Subtree |
| 100         | Provider Reconnect                                                                                                                                                  |

### 8.5.4.2 ObservationAction

| Action type | Description                                                           |
|-------------|-----------------------------------------------------------------------|
| 0           | Reserver for 'no action known'                                        |
| 1           | Changes in the System name                                            |
| 2           | Creation of System Browser-View                                       |
| 3           | Deletion of System Browser-View                                       |
| 5           | Editing of System Browser-View                                        |
| 6           | Editing of System Browser-View separator                              |
| 7           | Creation of System Browser-Tree                                       |
| 8           | Deletion of System Browser-Tree                                       |
| 9           | Editing of System Browser-Tree                                        |
| 10          | Adding of a System Browser-Tree i.e. a new node somewhere in the Tree |
| 11          | Changes in the Node name                                              |
| 12          | Changes in the Node data                                              |
| 100         | All nodes in the client-hosted tree have been collapsed.              |

### 8.5.5 SystemBrowserSubscriptionKey

| Property     | Type          | Description                                                                                                                 |
|--------------|---------------|-----------------------------------------------------------------------------------------------------------------------------|
| _links       | Array of Link | See Link [→ 190]                                                                                                            |
| Key          | int           | Subscription key is specific for the requested ID                                                                           |
| ErrorCode    | int           | If value is >0, an error occurred.                                                                                          |
| Designations | String[]      | String array of designations for which client requested subscription                                                        |
| RequestId    | string        | Channelized subscription request identifier<br><b>Note:</b> Only available for channelize requests                          |
| RequestFor   | string        | Notification function name where client will receive further updates<br><b>Note:</b> Only available for channelize requests |

## 8.6 Value Service

### 8.6.1 ValueDetails

| Property                   | Type          | Description                                                                                                                            |
|----------------------------|---------------|----------------------------------------------------------------------------------------------------------------------------------------|
| _links                     | Array of Link | See [→ 42]                                                                                                                             |
| Attributeld                | string        | Address of attribute to which value belongs to<br><br><b>Note:</b> In channelized subscription response data, this property is not set |
| DataType                   | ApiDataType   |                                                                                                                                        |
| ErrorCode                  | int           | If value is >0, an error occurred                                                                                                      |
| ObjectId                   | string        | Address of object to which value belongs to<br><br><b>Note:</b> In channelized subscription response data, this property is not set    |
| OriginalObjectOrPropertyId | string        | Address of requested object or property<br><br><b>Note:</b> In channelized subscription response data, this property is not set        |
| PropertyName               | string        | Name of property                                                                                                                       |

|                 |                               |                                                                                  |
|-----------------|-------------------------------|----------------------------------------------------------------------------------|
|                 |                               | <b>Note:</b> In channelized subscription response data, this property is not set |
| Value           | Value                         | See Value [→ 191]                                                                |
| IsArray         | bool                          | Flag whether value is array or not                                               |
| BackgroundColor | ColorRepresentation[See→ 192] | Background color of the icon, which denotes alarm state.                         |
| SubscriptionKey | Int                           | Subscription Key of OriginalObjectOrPropertyId                                   |

## 8.7 Property Value Service

### 8.7.1 Object<Type>

| Property           | Type            | Description                                 |
|--------------------|-----------------|---------------------------------------------|
| _links             | Array of Link   | See Link [→ 190]                            |
| Attributes         | Attributes      | See Attributes [→ 193]                      |
| ErrorCode          | int             | If value is >0, an error occurred.          |
| ObjectId           | string          | Address of object to which value belongs to |
| Properties         | Array of <Type> |                                             |
| FunctionProperties | Array of <Type> |                                             |

### 8.7.2 PropertyDetails

| Property                 | Type             | Description                                                                                                                                                                                      |
|--------------------------|------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| _links                   | Array of Link    | See Link [→ 190]                                                                                                                                                                                 |
| AllowWildcards           | bool             | Boolean flag for BACnet date/time values; indicates if the user can provide wildcards for any of the fields.                                                                                     |
| AllowDayOfWeek           | bool             | Boolean flag for BACnet date/time values; indicates if the day-of-week field is supported.                                                                                                       |
| BackgroundColor          | Color            |                                                                                                                                                                                                  |
| BACnetDateTimeDetail     | uint             | Indicates which parts of the BACnet date/time value to use:<br>1: date<br>2: time<br>3: date and time                                                                                            |
| BACnetDateTimeResolution | uint             | When the time part of the BACnet date/time value is to be used, what is the resolution of the seconds?<br>0: seconds<br>1: tenths of seconds<br>2: hundredths of seconds                         |
| Descriptor               | string           |                                                                                                                                                                                                  |
| DisplayOffNormalOnly     | Boolean          | When true, the user interface should only display the property when the value is off normal.                                                                                                     |
| DisplayType              | integer          | Provides a 'hint' as to how the property value should be displayed. See DisplayType [→ 206]                                                                                                      |
| DurationDisplayFormat    | unsigned integer | If the property is of type "duration" this field will indicate how the duration is to be displayed.                                                                                              |
| DurationValueUnits       | unsigned integer | If the property is of type "duration" this field will indicate the units of the raw property value (the raw value is an unsigned integer that can represent anything from milliseconds to days). |

| Property          | Type                        | Description                                                                                                                                                                                                                                                                    |
|-------------------|-----------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ElementLabels     | Array of strings            | [Obsolete]<br>Will be non-null (and contain string(s)) when a secondary text group has been assigned to the property. Normally used when the property value contains either an array of values or the value is a bit string. Strings will be in sorted order of Array indexes. |
| Elements          | Array of Elements           | Will be non-null when a secondary text group has been assigned to the property. Normally used when the property value contains either an array of values or the value is a bit string. Elements will be in sorted order of indexes. [→ 206]                                    |
| MappedPropertId   | string                      | Identifier of property to which a function property maps to.                                                                                                                                                                                                                   |
| Max               | string                      | Allowed maximum value; empty if maximum value of datatype                                                                                                                                                                                                                      |
| Min               | string                      | Allowed minimum value; empty if minimum value of data type                                                                                                                                                                                                                     |
| Normal            | String                      | The „normal“ value for the property. Used primarily to determine when the property is in the „off-normal“ state.                                                                                                                                                               |
| Order             | int                         | Order of the properties                                                                                                                                                                                                                                                        |
| PropertyAbsent    | Boolean                     | True if the property is defined in the model but does not exist in the actual object.                                                                                                                                                                                          |
| PropertyName      | string                      | The property's name. Name is invariant and serves as property identifier.                                                                                                                                                                                                      |
| Resolution        | int                         | If the property is floating point, this will be the desired resolution for display of values.                                                                                                                                                                                  |
| TextTable         | string                      | If the property is of type enumeration, boolean, or bit string: this field will contain the name of the text table used to represent the property value.                                                                                                                       |
| Type              | ApiDataType                 |                                                                                                                                                                                                                                                                                |
| UnitDescriptor    | string                      | Localized representation of unit                                                                                                                                                                                                                                               |
| UnitId            | uint                        | Unit of value                                                                                                                                                                                                                                                                  |
| Usage             | uint                        |                                                                                                                                                                                                                                                                                |
| Value             | Value                       |                                                                                                                                                                                                                                                                                |
| ScaledInformation | ScaledFactor Representation | Scaled details of the property if applied. [→ 206]                                                                                                                                                                                                                             |

### 8.7.3 Elements

| Property   | Type   | Description                                  |
|------------|--------|----------------------------------------------|
| Id         | uint   | Id assigned to the                           |
| Descriptor | string | raw unit (index) of a property if configured |

### 8.7.4 Scaled Property Details

| Property | Description                                  |
|----------|----------------------------------------------|
| Factor   | scaling factor of a property if configured   |
| Index    | raw unit (index) of a property if configured |
| Offset   | scaling offset of a property if configured   |

### 8.7.5 DisplayType

| Change type | Description                                                                                                       |
|-------------|-------------------------------------------------------------------------------------------------------------------|
| 0           | Reserved for 'use default display'                                                                                |
| 1           | Property is of type "command priority array" (normally 16 values, where any or all of those values can be "null") |
| 2           | Property is of type "event timestamps" (an array of three timestamps)                                             |

## 8.8 Properties Service

### 8.8.1 PropertyNames

| Property           | Type            | Description                                                    |
|--------------------|-----------------|----------------------------------------------------------------|
| ObjectId           | string          | Address of object to which value belongs to                    |
| Properties         | Array of string | Array of property names corresponding to the ObjectId          |
| FunctionProperties | Array of string | Array of function property names corresponding to the ObjectId |

## 8.9 Command Service

### 8.9.1 PropertyCommand

| Property        | Type             | Description                                                   |
|-----------------|------------------|---------------------------------------------------------------|
| Commands        | Array of Command | See Command [→ 207]                                           |
| PropertyId      | int              |                                                               |
| ErrorCode       | int              | -ErrorCode >0, an error occurred.<br>-ErrorCode = 0 , Success |
| SubscriptionKey | int              | -subscription key of propertyId                               |

### 8.9.2 Command

| Property      | Type                      | Description                                                                                                                                                                                                |
|---------------|---------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| _links        | Array of Link             | See Link [→ 190]                                                                                                                                                                                           |
| Descriptor    | string                    | Localized command name                                                                                                                                                                                     |
| GroupNumber   | int                       | If provided, commands with same GroupNumber belong together                                                                                                                                                |
| Parameters    | Array of CommandParameter | See CommandParameter [→ 208]s                                                                                                                                                                              |
| PropertyId    | string                    | Address of property to which command belongs to                                                                                                                                                            |
| Id            | string                    | Unique identifier of command                                                                                                                                                                               |
| IsDefault     | bool                      | True if this is the default command for the property.                                                                                                                                                      |
| Configuration | int                       | ValidationId of a propertyid<br>If Configuration :0 ->None is required<br>If Configuration :1 -> Only comment is required<br>If Configuration : 2-> Comment and Password(to re-authenticate) are required. |

### 8.9.3 CommandParameters

| Property                 | Type              | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|--------------------------|-------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| AllowWildcards           | bool              | Boolean flag for BACnet date/time command parameters; indicates if the user can provide wildcards for any of the fields.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| AllowDayOfWeek           | bool              | Boolean flag for BACnet date/time command parameters; indicates if the day-of-week field is supported.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Application              | int               | Integer indicating the application use of the parameter<br>Application: 0 -> Normal<br>Application: 1 -> The parameter is an index                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| BACnetDateTimeDetail     | uint              | Indicates which parts of the BACnet date/time value to use:<br>1: date<br>2: time<br>3: date and time                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| BACnetDateTimeResolution | uint              | When the time part of the BACnet date/time value is to be used, what is the resolution of the seconds?<br>0: seconds<br>1: tenths of seconds<br>2: hundredths of seconds                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| DataType                 | ApiDataType       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| DefaultValue             | string            | Default value of parameter                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Descriptor               | string            | Localized command parameter name                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| DurationDisplayFormat    | uint              | How the duration is to be displayed. The following abbreviations are used: d: days, h: hours, m: minutes, s: seconds, ms: milliseconds<br>0: display raw value as-is<br>1: display is to be in units of d<br>2: display is to be in d + h<br>3: display is to be in d + h + m<br>4: display is to be in d + h + m + s<br>5: display is to be in d + h + m + s + ms<br>6: display is to be in h<br>7: display is to be in h + m<br>8: display is to be in h + m + s<br>9: display is to be in h + m + s + ms<br>10: display is to be in m<br>11: display is to be in m + s<br>12: display is to be in m + s + ms<br>13: display is to be in s<br>14: display is to be in s + ms |
| DurationValueUnits       | uint              | The content of the unsigned integer representing a duration value:<br>1: number of days<br>2: number of hours<br>3: number of minutes<br>4: number of seconds<br>5: number of deciseconds<br>6: number of centiseconds<br>7: number of milliseconds                                                                                                                                                                                                                                                                                                                                                                                                                            |
| EnumerationTexts         | Array of EnumItem | See EnumItem [→ 209]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| IsPassword               | Bool              | Boolean indicating if the parameter represents a password. This flag will only exist when the parameter is of type "BasicString"                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |



|                |        |                                                                                                                                                                   |
|----------------|--------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Max            | string | Maximum value if applicable                                                                                                                                       |
| Min            | string | Minimum value if applicable                                                                                                                                       |
| Name           | string | Unique identifier of command parameter within command                                                                                                             |
| Order          | int    | Order of parameters within command                                                                                                                                |
| Resolution     | int    | For floating point parameters, the resolution                                                                                                                     |
| UnitDescriptor | string | For parameters of type real / integer / unsigned integer, if an engineering unit has been assigned (and can be found) this field will contain the localized text. |

## 8.9.4 EnumItem

| Property   | Type   | Description                               |
|------------|--------|-------------------------------------------|
| Descriptor | string | Localized value                           |
| Value      | int    | Numeric value (unique within enumeration) |

## 8.9.5 BatchExecution

| Property     | Type   | Description                                                                                                     |
|--------------|--------|-----------------------------------------------------------------------------------------------------------------|
| PropertyId   | string | Address of property to which command belongs to                                                                 |
| ErrorCode    | int    | -ErrorCode >0, an error occurred.<br>-ErrorCode = 0 , Success                                                   |
| ErrorMessage | string | Error message containing reason for failure – comments not passed or missing validation information in request. |

## 8.9.6 CommandInputForExecutionValidateParam

Inherits ValidationData See[→ 199]

| Property          | Type        | Description                                                                                                                                               |
|-------------------|-------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|
| Name              | string      | Parameter name                                                                                                                                            |
| Value             | int         | Value of the parameter                                                                                                                                    |
| DataType          | ApiDataType |                                                                                                                                                           |
| MultiLangComments | string[]    | Validation Profile : String array of comments for validation profile. Number of items in array should match the number of languages installed in project. |
| Password          | string      | Validation Profile : Logged in user's password for reauthentication                                                                                       |
| SuperName         | string      | Validation Profile : Supervisor's username                                                                                                                |
| SuperPassword     | string      | Validation Profile : Supervisor's password                                                                                                                |

## 8.9.7 BatchCommandExecution

| Property                 | Type                                    | Description                                               |
|--------------------------|-----------------------------------------|-----------------------------------------------------------|
| CommandInputForExecution | CommandInputForExecutionValidateParam[] | Command parameter required for execution of commands      |
| PropertyIds              | string[]                                | String array of propertyids required for batch commanding |

### 8.9.8 CommandInputForExecution

| Property | Type        | Description            |
|----------|-------------|------------------------|
| Name     | string      | Parameter name         |
| Value    | int         | Value of the parameter |
| DataType | ApiDataType |                        |

### 8.9.9 BatchCommandExecutionValidateProperty

Inherits ValidationData See[→ 162]

| Property                 | Type                       | Description                                                         |
|--------------------------|----------------------------|---------------------------------------------------------------------|
| CommandInputForExecution | CommandInputForExecution[] | Command parameter required for execution of commands                |
| PropertyIds              | string[]                   | String array of propertyids required for batch commanding           |
| LangTextContent          | LangTextContent            | Comments for logging                                                |
| Password                 | String                     | Validation Profile : Logged in user's password for reauthentication |
| SuperName                | String                     | Validation Profile : Supervisor's username                          |
| SuperPassword            | String                     | Validation Profile : Supervisor's password                          |

### 8.9.10 PropertyCommandInput

| Property                 | Type                                    | Description                                          |
|--------------------------|-----------------------------------------|------------------------------------------------------|
| CommandInputForExecution | CommandInputForExecutionValidateParam[] | Command parameter required for execution of commands |
| PropertyId               | String                                  | Property Id for execution                            |
| CommandId                | string                                  | Command Id for execution                             |

### 8.9.1 PropertyCommandInputValidateProperty

Inherits ValidationData See[→ 162]

| Property                 | Type                       | Description                                                         |
|--------------------------|----------------------------|---------------------------------------------------------------------|
| CommandInputForExecution | CommandInputForExecution[] | Command parameter required for execution of commands                |
| PropertyId               | string                     | Property Id for execution                                           |
| CommandId                | string                     | Command Id for execution                                            |
| LangTextContent          | LangTextContent            | Comments for logging                                                |
| Password                 | string                     | Validation Profile : Logged in user's password for reauthentication |
| SuperName                | string                     | Validation Profile : Supervisor's username                          |
| SuperPassword            | string                     | Validation Profile : Supervisor's password                          |

## 8.10 Trend Service

### 8.10.1 TrendCollector

| Property | Type | Description |
|----------|------|-------------|
|----------|------|-------------|

|                             |        |                                                                    |
|-----------------------------|--------|--------------------------------------------------------------------|
| CollectorObjectOrPropertyId | string | Address of collector object                                        |
| PropertyIndex               | uint   | Index of Trended Property in the Trended Object.<br>Start index: 1 |
| ObjectId                    | string | Object ID of trended object                                        |
| PropertyName                | string | Trended property name                                              |
| TrendseriesId               | string | Identifier of corresponding trend series                           |

### 8.10.2 TrendBorder

| Property | Type     | Description                                                           |
|----------|----------|-----------------------------------------------------------------------|
| From     | DateTime | Minimum date/time from available values for requested trend series ID |
| To       | DateTime | Maximum date/time from available values for requested trend series ID |

### 8.10.3 TrendSeries

| Property         | Type           | Description        |
|------------------|----------------|--------------------|
| Id               | string         | ID of trend series |
| Series           | Array of Value | See Value [→ 191]  |
| SeriesPropertyId | string         | Trended object     |

### 8.10.4 TrendAggregateSeries

| Property         | Type                    | Description            |
|------------------|-------------------------|------------------------|
| Id               | string                  | ID of trend series     |
| Series           | Array of TrendAggregate | See Value [→ 211]      |
| SeriesPropertyId | string                  | Trend Aggregate object |

### 8.10.5 TrendAggregate

Statistical analysis will be done for the samples of Values. All details are below are in reference of a Sample of Values

| Property            | Type     | Description                                   |
|---------------------|----------|-----------------------------------------------|
| AvgX                | double   | $\text{mean } x = (\sum x_i) / n$             |
| AvgY                | double   | $\text{mean } y = (\sum y_i) / n$             |
| Count               | double   | Total numbers of values taken for calculation |
| FromValue           | double   | First value in the Sample                     |
| FromValueDescriptor | string   | Localized value to display FromValue          |
| FromTime            | DateTime | Time of the first value in the sample         |
| Min                 | double   | Minimum value in the Sample                   |
| Max                 | double   | Maximum Value in the Sample                   |
| StDevX              | double   | Standard Deviation X                          |
| StDevY              | double   | Standard Deviation Y                          |
| SumX                | double   | Sum of all X Values                           |
| SumXX               | double   | Sum of all $X^2$                              |

|                   |          |                                      |
|-------------------|----------|--------------------------------------|
| SumXY             | double   | Sum of all X*Y                       |
| SumY              | double   | Sum of all Y                         |
| SumYY             | double   | Sum of all Y <sup>2</sup>            |
| ToTime            | DateTime | Last value in the Sample             |
| ToValueDescriptor | double   | Localized value to display ToValue   |
| ToValue           | double   | Time of the last value in the sample |
| VarX              | double   | Variance of X                        |
| VarY              | double   | Variance of Y                        |

### 8.10.6 TrendViewDefinitionRepresentation

| Property                                 | Type                                                                  | Description |
|------------------------------------------|-----------------------------------------------------------------------|-------------|
| AxisX                                    | AxisDefinitionRepresentation <a href="#">[→ 212]</a>                  |             |
| AxisYLeft                                | AxisDefinitionRepresentation <a href="#">[→ 212]</a>                  |             |
| AxisYRight                               | AxisDefinitionRepresentation <a href="#">[→ 212]</a>                  |             |
| DescriptionTvd                           | string                                                                |             |
| ObjectIdOfTvd                            | string                                                                |             |
| IdTvd                                    | string                                                                |             |
| NameTvd                                  | string                                                                |             |
| NumberDisplayedSamplesPerTrendSeries     | integer                                                               |             |
| RemoveOnlineTrendLogOfDeletedTrendSeries | boolean                                                               |             |
| TimeRange                                | TimeRange <a href="#">[→ 212]</a>                                     |             |
| TitleLeft                                | string                                                                |             |
| TitleRight                               | string                                                                |             |
| TitleTop                                 | string                                                                |             |
| TsdCollectionInfo                        | TrendSeriesDefinitionCollectionRepresentation <a href="#">[→ 214]</a> |             |
| TvdType                                  | integer                                                               |             |
| TvCovType                                | string                                                                |             |
| _links                                   | Array of Link <a href="#">[→ 190]</a>                                 |             |

### 8.10.7 AxisDefinitionRepresentation

| Property     | Type    | Description |
|--------------|---------|-------------|
| AutoScale    | boolean |             |
| AxisType     | integer |             |
| ScaleMax     | number  |             |
| ScaleMin     | number  |             |
| ShowTitle    | boolean |             |
| ShowZeroLine | boolean |             |
| Title        | string  |             |

### 8.10.8 TimeRange

| Property | Type | Description |
|----------|------|-------------|
|----------|------|-------------|

|                   |                                          |  |
|-------------------|------------------------------------------|--|
| ValidTimeRange    | integer                                  |  |
| AbsoluteTimeRange | AbsoluteTimeRange <a href="#">[→213]</a> |  |
| RelativeTimeRange | RelativeTimeRange <a href="#">[→213]</a> |  |

### 8.10.9 TrendSeriesDefinitionRepresentation

| Property                 | Type    | Description |
|--------------------------|---------|-------------|
| ReservedDateTime         | string  |             |
| AccessDenied             | boolean |             |
| Alias                    | string  |             |
| ArchiveOn                | boolean |             |
| AxisAttachment           | string  |             |
| ChartLineStyle           | string  |             |
| ChartLineType            | string  |             |
| Color                    | string  |             |
| DescriptionTIField       | string  |             |
| DescriptionTIGms         | string  |             |
| DescriptionTo            | string  |             |
| DescriptionToProperty    | string  |             |
| ObjectIdOfTrendLog       | string  |             |
| ObjectIdOfTrendLogOnline | string  |             |
| ObjectIdOfTrendedObject  | string  |             |
| Index                    | integer |             |
| IsLineVisible            | boolean |             |
| LineWidth                | integer |             |
| ManagedType              | string  |             |
| MaxTimeStamp             | string  |             |
| MinTimeStamp             | string  |             |
| ShowDataLabels           | boolean |             |
| ShowMarkers              | boolean |             |
| ShowQualityIcons         | boolean |             |
| TimeZoneServer           | string  |             |
| Type                     | string  |             |
| UnitText                 | string  |             |
| ValueType                | string  |             |

### 8.10.10 RelativeTimeRange

| Property          | Type    | Description |
|-------------------|---------|-------------|
| AnchorMode        | integer |             |
| AnchorTime        | string  |             |
| NumberOfTimeUnits | integer |             |
| TimeUnit          | integer |             |

### 8.10.11 AbsoluteTimeRange

| Property | Type | Description |
|----------|------|-------------|
|----------|------|-------------|

|      |        |  |
|------|--------|--|
| From | string |  |
| To   | string |  |

### 8.10.12 TrendSeriesDefinitionCollectionRepresentation

| Property               | Type                                                              | Description |
|------------------------|-------------------------------------------------------------------|-------------|
| TrendSeriesDefinitions | Array of TrendSeriesDefinitionRepresentation <a href="#">↗190</a> |             |

### 8.10.13 TrendViewDefinitionUpdateRepresentation

| Property                                | Type                                                                     | Description |
|-----------------------------------------|--------------------------------------------------------------------------|-------------|
| AxisX                                   | AxisDefinitionUpdateRepresentation <a href="#">↗212</a>                  |             |
| AxisYLeft                               | AxisDefinitionUpdateRepresentation <a href="#">↗212</a>                  |             |
| AxisYRight                              | AxisDefinitionUpdateRepresentation <a href="#">↗212</a>                  |             |
| CNSDescription                          | string                                                                   |             |
| Location                                | string                                                                   |             |
| TvdObjectId                             | string                                                                   |             |
| NumberDisplayedSamplesPerTrendSerie     | integer                                                                  |             |
| RemoveOnlineTrendLogOfDeletedTrendSerie | boolean                                                                  |             |
| TimeRange                               | TimeRange <a href="#">↗212</a>                                           |             |
| TitleLeft                               | string                                                                   |             |
| TitleRight                              | string                                                                   |             |
| TitleTop                                | string                                                                   |             |
| TsdCollectionInfo                       | TrendSeriesDefinitionUpdateCollectionRepresentation <a href="#">↗214</a> |             |
| TvCovType                               | string                                                                   |             |

### 8.10.14 TrendSeriesDefinitionUpdateRepresentation

| Property               | Type    | Description |
|------------------------|---------|-------------|
| ArchiveOn              | boolean |             |
| AxisAttachment         | string  |             |
| ChartLineStyle         | string  |             |
| ChartLineType          | string  |             |
| Color                  | string  |             |
| TrendLogObjectId       | string  |             |
| TrendLogOnlineObjectId | string  |             |
| ObjectPropertyId       | string  |             |
| TrendedObjectId        | string  |             |
| IsLineVisible          | boolean |             |
| LineWidth              | integer |             |
| ShowDataLabels         | boolean |             |

|                   |         |  |
|-------------------|---------|--|
| ShowMarkers       | boolean |  |
| ShowQualityIcons  | boolean |  |
| Type              | string  |  |
| Smoothing         | boolean |  |
| CustomDescription | string  |  |

### 8.10.15 TrendSeriesDefinitionUpdateCollectionRepresentation

| Property               | Type                                               | Description |
|------------------------|----------------------------------------------------|-------------|
| TrendSeriesDefinitions | Array of TrendSeriesDefinitionUpdateRepresentation |             |

### 8.10.16 AxisDefinitionUpdateRepresentation

| Property     | Type    | Description |
|--------------|---------|-------------|
| AutoScale    | boolean |             |
| ScaleMax     | number  |             |
| ScaleMin     | number  |             |
| ShowTitle    | boolean |             |
| ShowZeroLine | boolean |             |
| Title        | string  |             |

## 8.11 History Logs Service

### 8.11.1 LogTable

| Property                     | Type            | Description                                                                                                                                                                                                                                                    |
|------------------------------|-----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Name                         | string          | Name of table which is unique and will be used for further operations in subsequent requests                                                                                                                                                                   |
| Descriptor                   | string          | Display name of table                                                                                                                                                                                                                                          |
| Description                  | string          | Description of table                                                                                                                                                                                                                                           |
| HasChildColumns              | bool            | Flag whether table supports child columns or not                                                                                                                                                                                                               |
| HasConditionFilter           | bool            | Flag whether table supports Condition Filter or not                                                                                                                                                                                                            |
| HasNameFilter                | bool            | Flag whether table supports Name Filter or not                                                                                                                                                                                                                 |
| HasTimeFilter                | bool            | Flag whether table supports Time Filter or not                                                                                                                                                                                                                 |
| HasTimeFilterDefinedBySource | bool            | Flag whether table supports Time Filter defined by source or not. Used in case of trend data retrieval.<br>If set to TRUE then Time Filter provided in the request will be ignored and instead Time Filter defined by source will be considered to filter data |
| HasAndOperatorOnly           | bool            | Flag whether table only supports And-operator or not                                                                                                                                                                                                           |
| HasMaxAge                    | bool            | Flag whether table supports Max Age or not                                                                                                                                                                                                                     |
| HasSingleNameFilter          | bool            | Flag whether table supports Single Name Filter or not                                                                                                                                                                                                          |
| SupportedTypes               | Array of string | This is array of Managed Types which are supported by the table                                                                                                                                                                                                |

### 8.11.2 LogColumn

| Property     | Type   | Description                                                                                                                 |
|--------------|--------|-----------------------------------------------------------------------------------------------------------------------------|
| Name         | string | Name of column which is unique and will be used for further operations in subsequent requests                               |
| Descriptor   | string | Display name of column                                                                                                      |
| DataType     | string | Data type of the data which column supports                                                                                 |
| ErrorSupport | bool   | Flag whether column supports display of error data (e.g., #COM, #ERR) or not                                                |
| IsArray      | bool   | Flag whether column contains a data array or not                                                                            |
| IsDefault    | bool   | Flag whether column is displayed by default or not. This will be useful while developing a user interface to create tables. |
| IsHidden     | bool   | Flag whether column is hidden or not                                                                                        |
| IsSortable   | bool   | Flag whether column is sortable or not                                                                                      |
| IsEnum       | bool   | Flag whether column contains enumerated data or not                                                                         |
| IsFilterable | bool   | Flag whether data from the column can be filtered using condition filter or not                                             |

### 8.11.3 LogConditionFilter

| Property  | Type             | Description                                                             |
|-----------|------------------|-------------------------------------------------------------------------|
| Operators | Array of strings | Supported condition filter operators for column                         |
| ErrorInfo | Array of strings | Localized list of error information. It describes specific error cases. |

### 8.11.4 LogEnumValues

| Property   | Type             | Description                                                             |
|------------|------------------|-------------------------------------------------------------------------|
| EnumValues | Array of strings | Supported enum values for column                                        |
| ErrorInfo  | Array of strings | Localized list of error information. It describes specific error cases. |

### 8.11.5 LogSortInfo

| Property | Type        | Description                                                 |
|----------|-------------|-------------------------------------------------------------|
| Name     | string      | Name of the Column which is to be sorted<br><b>Required</b> |
| SortType | LogSortType | See LogSortType [→ 216]<br><b>Required</b>                  |

#### 8.11.5.1 LogSortType

**LogSortType** must have one of the following values.

| LogSortType | Description                    |
|-------------|--------------------------------|
| Ascending   | Sorts data in ascending order  |
| Descending  | Sorts data in descending order |



### 8.11.6 LogColumnList

| Property | Type            | Description                  |
|----------|-----------------|------------------------------|
| Parent   | Array of string | Column list for parent table |
| Child    | Array of string | Column list for child table  |

### 8.11.7 LogTableData

| Property   | Type             | Description                                                                                 |
|------------|------------------|---------------------------------------------------------------------------------------------|
| TableName  | string           | Name of the table                                                                           |
| Page       | int              | Current page number                                                                         |
| Size       | int              | Number of records per page                                                                  |
| Result     | Array of objects | At runtime each object is composed using selected columns to retrieve data and there value. |
| SnapshotId | string           | Snapshot id of the snapshot created                                                         |
| Total      | long             | Total number of records found for the requested configuration                               |
| ErrorInfo  | Array of strings | Localized list of error information. It describes specific error cases.                     |
| _links     | Array of Link    | See [→190]                                                                                  |

### 8.11.8 LogDiscardedSnapshot

| Property            | Type   | Description                                                        |
|---------------------|--------|--------------------------------------------------------------------|
| TableName           | string | Name of the table                                                  |
| SnapshotId          | string | Snapshot Id which is discarded                                     |
| IsSnapshotDiscarded | bool   | Status of the discard request. If TRUE then discard is successful. |

### 8.11.9 LogConditionFilterValidation

| Property               | Type   | Description                                                                                                                      |
|------------------------|--------|----------------------------------------------------------------------------------------------------------------------------------|
| ErrorPosition          | int    | In case of invalid string, Error position in the invalid string                                                                  |
| ErrorMessageDescriptor | string | In case of invalid string, Localized Error message if string is not valid as per the standards                                   |
| ToDoMessageDescriptor  | string | In case of invalid string, Localized to-do message using which client can correct the Condition filter string if it is incorrect |
| IsValidConditionFilter | bool   | Status of Condition filter string validation. If TRUE then string as per the standards.                                          |

### 8.11.10 LogChildData

| Property | Type             | Description                                                                                 |
|----------|------------------|---------------------------------------------------------------------------------------------|
| Page     | int              | Current page number                                                                         |
| Size     | int              | Number of records per page                                                                  |
| Total    | long             | Total number of records found for the requested configuration                               |
| Result   | Array of objects | At runtime each object is composed using selected columns to retrieve data and there value. |

|           |                  |                                                                         |
|-----------|------------------|-------------------------------------------------------------------------|
| ErrorInfo | Array of strings | Localized list of error information. It describes specific error cases. |
| _links    | Array of Link    | See [→190]                                                              |

## 8.12 Language Service

### 8.12.1 Language

| Property   | Type   | Description                                       |
|------------|--------|---------------------------------------------------|
| Descriptor | string | Full localized language name                      |
| Code       | string | IETF language tag according to RFC5646 [5] [→ 16] |

## 8.13 Tables Service

### 8.13.1 SubDisciplines

| Property             | Type               | Description                                                                       |
|----------------------|--------------------|-----------------------------------------------------------------------------------|
| DisciplineId         | Uint               | Id of the Discipline                                                              |
| DisciplineDescriptor | string             | Descriptor of the Discipline                                                      |
| SubDisciplines       | Array of SubGroups | Array of SubDisciplines in SubGroups form for which filter is applied. See [→ 55] |

### 8.13.2 SubObjectTypes

| Property             | Type               | Description                                                                       |
|----------------------|--------------------|-----------------------------------------------------------------------------------|
| ObjectTypeId         | Uint               | Id of the ObjectType                                                              |
| ObjectTypeDescriptor | string             | Descriptor of the ObjectType                                                      |
| SubObjectTypes       | Array of SubGroups | Array of SubObjectTypes in SubGroups form for which filter is applied. See [→ 55] |

### 8.13.3 TextTableEntry

| Property | Type                    | Description                                                                                                                                     |
|----------|-------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| Value    | uint                    | Numeric value of the entry                                                                                                                      |
| Text     | string                  | Text of the entry.                                                                                                                              |
| Color    | uint?                   | The color assigned to the entry; this field is optional and if it does not exist, the property will not be included in the JSON representation. |
| Subtext  | Array of TextTableEntry | Can be nullCollection of subtext.                                                                                                               |

### 8.13.4 LangTextTableEntry

| Property | Type  | Description                                                                                                                                     |
|----------|-------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| Value    | uint  | Numeric value of the entry                                                                                                                      |
| Color    | uint? | The color assigned to the entry; this field is optional and if it does not exist, the property will not be included in the JSON representation. |

|          |          |                                                    |
|----------|----------|----------------------------------------------------|
| LangText | String[] | Array of strings, one for each installed language. |
|----------|----------|----------------------------------------------------|

### 8.13.5 TextTableEntry Color Calculation

Note that Color is an optional property of a text table entry and will only appear in the JSON string if a color has been assigned to that entry. If a color property is provided, it is in the form of an unsigned 4 byte integer containing the following information:

| Channel | Calculation            |
|---------|------------------------|
| Alpha   | ( Color >> 24 ) & 0xFF |
| Red     | ( Color >> 16 ) & 0xFF |
| Green   | ( Color >> 8 ) & 0xFF  |
| Blue    | Color & 0xFF           |

## 8.14 Systems Service

### 8.14.1 System Languages

| Property   | Type   | Description                                       |
|------------|--------|---------------------------------------------------|
| Descriptor | string | Full localized language name                      |
| Code       | string | IETF language tag according to RFC5646 [5] [→ 16] |
| ArrayIndex | Int    | Index of the installed language                   |

### 8.14.2 Systems

| Property      | Type               | Description                                     |
|---------------|--------------------|-------------------------------------------------|
| Systems       | SystemsInformation | Array of systems information. See[→219 ]        |
| Languages     | SystemsLanguages   | Array of systems languages See[→219 ]           |
| IsDistributed | bool               | System is in distribution environment or not    |
| IdLocal       | uint               | Id of system that is currently serving requests |

### 8.14.3 SystemsInformation

| Property | Type   | Description                                                                                           |
|----------|--------|-------------------------------------------------------------------------------------------------------|
| Name     | string | System name                                                                                           |
| Id       | uint   | System id                                                                                             |
| IsOnline | bool   | Obsolete: If system is available or not.<br>MP4.0 onward: This field is not available in notification |

### 8.14.4 ServicesSubscriptionsRepresentation

| Property  | Type | Description                                                                                                    |
|-----------|------|----------------------------------------------------------------------------------------------------------------|
| SystemId  | int  | System id                                                                                                      |
| Serviceid | int  | Service id of the respective system id                                                                         |
| ErrorCode | int  | If that particular subscription for the respective system AND service is successful or not. 1 for error, 0 for |

|  |  |          |
|--|--|----------|
|  |  | success. |
|--|--|----------|

### 8.14.5 ServicesRequestSubscriptionsRepresentation

| Property | Type      | Description                                                             |
|----------|-----------|-------------------------------------------------------------------------|
| SystemId | int       | System id                                                               |
| Ids      | List<int> | List of Service ids of the respective system which are to be subscribed |

### 8.14.6 ServicesNotificationRepresentation

| Property    | Type   | Description                                                                                                                                          |
|-------------|--------|------------------------------------------------------------------------------------------------------------------------------------------------------|
| SystemId    | uint   | System id                                                                                                                                            |
| Serviceid   | int    | Service id of the respective system id                                                                                                               |
| Description | string | Name/description of the particular service for which notification is being received. NOTE: this will only be non-null when the service is connected. |
| IsConnected | bool   | The current status of the service                                                                                                                    |

## 8.15 Products Service

### 8.15.1 ProductInfo

| Property          | Type           | Description                                                                                                                    |
|-------------------|----------------|--------------------------------------------------------------------------------------------------------------------------------|
| ProductNameShort  | string         | The name of the product; not localizable<br>e.g. "Desigo CC" or "Cerberus DMS"                                                 |
| ProductName       | string         | The long name of the product;<br>not localizable<br>e.g. "Desigo CC ™"                                                         |
| VendorName        | string         | The name of the vendor of the product;<br>not localizable<br>e.g. "Siemens Switzerland Ltd."<br>or "Daikin Ltd."               |
| ProductCopyRights | string         | Any statement;<br>not localizable:<br>© Siemens Switzerland Ltd, 2016                                                          |
| ProductVersion    | string         | Desigo CC project version                                                                                                      |
| SupportLink       | String         | Technical support URL                                                                                                          |
| Links             | Array of Links | Direct URLs to access <ul style="list-style-type: none"> <li>- Product picture wide</li> <li>- Product picture tall</li> </ul> |

## 8.16 Files Service

### 8.16.1 BasicFiles (Enum type)

Contain different file types.

| Property        | Type | Description |
|-----------------|------|-------------|
| ProductPictures | int  | a file type |

## 8.16.2 File Details.

| Property | Type   | Description        |
|----------|--------|--------------------|
| FileName | string | contains file name |
| _links   | Link   | See Link [→ 190]   |

## 8.17 Graphics Service

### 8.17.1 GraphicalItem

Objects that are graphical in nature will be represented by this object type.

| Property    | Type   | Description                                                                                                                                       |
|-------------|--------|---------------------------------------------------------------------------------------------------------------------------------------------------|
| Context     | String | The designation of the object that 'owns' the graphical item.                                                                                     |
| ObjectId    | String | The id of the object within DCC. Note that this id must include the system information to distinguish the object in a multi-system configuration. |
| ManagedType | String | The DCC managed type of the object.                                                                                                               |
| DisplayName | String | A string that can be displayed to the user.                                                                                                       |

## 8.18 Scheduler Service

### 8.18.1 Calendar

| Property             | Type                           | Description |
|----------------------|--------------------------------|-------------|
| ScheduleCalendarType | integer                        |             |
| CalendarEntries      | Array of CalendarEntry[->221 ] |             |
| AppliedSchedules     | Array of string                |             |
| ContainerObjectId    | string                         |             |
| ObjectId             | string                         |             |
| Device               | integer                        |             |

### 8.18.2 CalendarEntry

| Property     | Type                 | Description |
|--------------|----------------------|-------------|
| StartDate    | CalendarDate [->221] |             |
| EndDate      | CalendarDate [->221] |             |
| Restrictions | CalendarDate [->221] |             |
| Mode         | integer              |             |

### 8.18.3 CalendarDate

| Property       | Type             | Description |
|----------------|------------------|-------------|
| Date           | string           |             |
| IsAny          | boolean          |             |
| Days           | Array of integer |             |
| Months         | Array of integer |             |
| DaysInMonth    | Array of integer |             |
| LastDayInMonth | boolean          |             |

|                 |                  |  |
|-----------------|------------------|--|
| Years           | Array of integer |  |
| Weeks           | Array of integer |  |
| LastWeekInMonth | boolean          |  |

#### 8.18.4 WeeklySchedule

| Property                | Type                                     | Description |
|-------------------------|------------------------------------------|-------------|
| DeviceId                | string                                   |             |
| Name                    | string                                   |             |
| Setup                   | ScheduleSetup [->221]                    |             |
| Outputs                 | Array of string                          |             |
| ScheduleType            | string                                   |             |
| Schedules               | Array of Schedule [->221]                |             |
| ReferencedOutputDetails | Array of ReferencedOutputDetails [->222] |             |
| Exceptions              | Array of ScheduleException [->222]       |             |

#### 8.18.5 ScheduleSetup

| Property     | Type                 | Description |
|--------------|----------------------|-------------|
| DefaultValue | string               |             |
| StartDate    | CalendarDate [->221] |             |
| EndDate      | CalendarDate [->221] |             |
| Priority     | integer              |             |
| DataType     | string               |             |
| ScheduleMin  | number               |             |
| ScheduleMax  | number               |             |
| TextGroupId  | string               |             |
| Unit         | integer              |             |

#### 8.18.6 Schedule

| Property             | Type                         | Description |
|----------------------|------------------------------|-------------|
| Day                  | String                       |             |
| Entries              | Array of TimeTrigger [->223] |             |
| OutputValuesActive   | Array of string              |             |
| OutputValuesInactive | Array of string              |             |

#### 8.18.7 ReferencedOutputDetails

| Property    | Type    | Description |
|-------------|---------|-------------|
| Output      | string  |             |
| TextGroupId | string  |             |
| Unit        | integer |             |

#### 8.18.8 ScheduleException

| Property | Type | Description |
|----------|------|-------------|
|----------|------|-------------|

|                         |                              |  |
|-------------------------|------------------------------|--|
| Priority                | integer                      |  |
| Mode                    | integer                      |  |
| CalendarException       | CalendarException [->223]    |  |
| CalendarEntry           | CalendarEntry[->221 ]        |  |
| ExceptionTimes          | Array of TimeTrigger [->223] |  |
| ExceptionValuesActive   | Array of string              |  |
| ExceptionValuesInactive | Array of string              |  |

### 8.18.9 TimeTrigger

| Property | Type   | Description |
|----------|--------|-------------|
| Time     | string |             |
| Value    | number |             |

### 8.18.10 CalendarException

| Property         | Type                            | Description |
|------------------|---------------------------------|-------------|
| CalendarObjectId | string                          |             |
| CalendarEntries  | Array of CalendarEntry [->221 ] |             |
| CalendarName     | string                          |             |

## 8.19 Objects Service

### 8.19.1 Object

Objects will be represented by this type.

| Property      | Type          | Description                                              |
|---------------|---------------|----------------------------------------------------------|
| Designation   | string        | Full node name under which child is supposed to be added |
| NameChildNode | string        | Name of the newly created system browser node.           |
| ObjectType    | string        | The type of the object to create.                        |
| Descriptor    | Descriptor or | Display name of the browser node. Multilingual text      |

### 8.19.2 Descriptor

Priority based text value definition.

| Property      | Type     | Description                                                                                                                                                                                                                                                                                       |
|---------------|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| CommonText    | string   | Text to be initiated for all the languages<br>Prio 1: If value is assigned to CommonText then attribute MultiLangText will not be considered.<br>To assign value in MultiLangText set Descriptor to null.<br>To be created node display name. Common descriptor will be applied in all languages. |
| MultiLangText | string[] | Language specific text for object<br>Prio 2: MultiLangText has lower priority to Descriptor. It will not be considered if any value is assigned to attribute Descriptor<br>Set Descriptor to null to assign MultiLangText. Language specific text will be updated for the node.                   |

### 8.19.3 ObjectModel

Objects type will be represented by this type.

| Property                | Type                              | Description                                 |
|-------------------------|-----------------------------------|---------------------------------------------|
| Designation             | string                            | Full node name                              |
| Description             | string                            | Description of the object type.             |
| ObjectType              | string                            | The type of the object to create.           |
| IsGenericDelete Allowed | bool                              | Generic delete configuration for the object |
| ChildObjects            | ChildObjectModel Representation[] | List of available child types.              |

### 8.19.1 ChildObjectModel

Objects type will be represented by this type.

| Property                | Type   | Description                                                                                                                                                                                   |
|-------------------------|--------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Description             | string | Description of the object type.                                                                                                                                                               |
| ObjectType              | string | The type of the object to create.                                                                                                                                                             |
| IsGenericDelete Allowed | bool   | Generic delete configuration for the object                                                                                                                                                   |
| IsGenericCreate Allowed | bool   | Generic create configuration for the object                                                                                                                                                   |
| MaxChild                | uint   | Max number of Childs allowed for particular type                                                                                                                                              |
| ExistingChildCount      | int    | Count of the Childs already added under the parent for current object types.<br>Difference of ExistingChildCount and MaxChild can be used to identify that how many more childs can be added. |

### 8.19.2 ServiceTextRepresentation

Information Text and Memo information will be represented by this type.

| Property        | Type   | Description                                                       |
|-----------------|--------|-------------------------------------------------------------------|
| InformationText | string | Information Text saved for an object in loggedin user's language. |
| Memo            | string | Memo content for an object.                                       |

### 8.19.3 ObjectInfo

Response type for Information Text and Memo content.

| Property    | Type                              | Description                                                                                                            |
|-------------|-----------------------------------|------------------------------------------------------------------------------------------------------------------------|
| ObjectId    | string                            | Object Id.                                                                                                             |
| ServiceText | ServiceTextRepresentation [->224] | Information text and Memo details for an object.<br>In case of failed request, this field is missing in Json response. |

### 8.19.4 ServiceTextMemoRepresentation

Request type for accepting Memo text.

| Property | Type   | Description                |
|----------|--------|----------------------------|
| Memo     | string | Memo content for an object |



## 8.20 Related Items Service

### 8.20.1 RelatedItemResult

This is the top level object returned from the “get related items” API call. It contains a single property at this time.

| Property       | Type                | Description                                                                                                                                                   |
|----------------|---------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| RelatedResults | ObjectRelatedItem[] | An array: each element corresponds to an object id passed into the API call, where each element contains information about the related items for that object. |

### 8.20.2 ObjectRelatedItem

This object contains the related item information for each object id passed to the “get related items” API.

| Property     | Type          | Description                                                                     |
|--------------|---------------|---------------------------------------------------------------------------------|
| ErrorCode    | integer       | 0: No error<br>1: Error                                                         |
| ObjectId     | string        | The object id passed into the API.                                              |
| RelatedItems | RelatedItem[] | A collection of the related items for the object; this collection can be empty. |

### 8.20.3 RelatedItem

As described elsewhere, two objects can be associated with each other – for example, an object in a report is ‘related to’ that report. This table defines the information that returned for each item that the object is related to.

| Property        | Type          | Description                                                                                                                                                                                                                                                                                                                                                                                                                   |
|-----------------|---------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| GroupDescriptor | string        | If provided and non-blank, can be used to group items.                                                                                                                                                                                                                                                                                                                                                                        |
| GroupOrder      | ushort?       | If a group descriptor is provided, this property can be used to sort the items within a group                                                                                                                                                                                                                                                                                                                                 |
| ItemDescriptor  | string        | If provided and non-blank, should be used for the item's display name                                                                                                                                                                                                                                                                                                                                                         |
| Mode            | ushort        | The Mode in which the item should be visible:<br>0: All<br>1: Operation<br>2: Engineering                                                                                                                                                                                                                                                                                                                                     |
| Nodes           | BrowserNode[] | The array will contain one BrowserNode [→ 77] for each reference in a hierarchy to the object.                                                                                                                                                                                                                                                                                                                                |
| Parameter       | string        | An optional parameter added by a dynamic link provider.                                                                                                                                                                                                                                                                                                                                                                       |
| Reference       | string        | If the related item is an external entity, this property will contain a URL – where the URL is of the form {keyword}{reference}. The list of valid keywords include: <ul style="list-style-type: none"><li>- file:// (reference is to a file)</li><li>- file:/// (alternate form of file://)</li><li>- http:// (HTTP location)</li><li>- https:// (Secured HTTP location)</li><li>- ftp:// (file transfer location)</li></ul> |
| SourceType      | ushort        | The source of the link:<br>0: NotSpecified<br>1: System<br>2: User<br>3: Dynamic                                                                                                                                                                                                                                                                                                                                              |

## 8.21 Access rights Service

### 8.21.1 AccessRightsRepresentation

| Property           | Type                                                                              | Description                                                 |
|--------------------|-----------------------------------------------------------------------------------|-------------------------------------------------------------|
| Application Rights | Array of ApplicationRightRepresentation [-> <b>Error! Bookmark not defined.</b> ] | Collection of application rights assigned to logged in user |

### 8.21.2 ApplicationRightRepresentation

| Property   | Type                                      | Description                                             |
|------------|-------------------------------------------|---------------------------------------------------------|
| Name       | string                                    | Name of AccessRight in loggedin user's current language |
| Id         | uint                                      | Unique identifier assigned to access right              |
| Operations | Array of OperationsRepresentation [->226] | Operations applicable within current access right       |

### 8.21.3 UserRoleRepresentation

| Property     | Type                  | Description                                    |
|--------------|-----------------------|------------------------------------------------|
| RolesMapping | List of type RoleInfo | Collection of roles assigned to logged in user |

### 8.21.4 RoleInfo

| Property        | Type   | Description                                      |
|-----------------|--------|--------------------------------------------------|
| RoleName        | string | Name of Role in loggedin user's current language |
| RoleId          | uint   | Unique identifier assigned to role               |
| RoleDescription | string | Description of role functionality                |
| RoleState       | bool   | Active or not.                                   |

### 8.21.5 OperationsRepresentation

| Property | Type   | Description                                                 |
|----------|--------|-------------------------------------------------------------|
| Name     | string | Name of AccessRight in loggedin user's current language     |
| Id       | uint   | Unique identifier assigned to operation within access right |

## 8.22 Operating Procedures Service

### 8.22.1 Operating Procedure

| Property    | Type     | Description               |
|-------------|----------|---------------------------|
| AlertCount  | uint     |                           |
| AlertSource | string   |                           |
| AlertTime   | datetime |                           |
| Id          | string   | Id of Operating procedure |

|               |                                           |                                        |
|---------------|-------------------------------------------|----------------------------------------|
| InTreatmentBy | string                                    | User performing the treatment          |
| IsClosed      | boolean                                   |                                        |
| ResetSteps    | uint                                      |                                        |
| Sequential    | boolean                                   | Steps need to be sequential            |
| Steps         | Array of Operating Procedure Step [->227] | Steps available in Operating procedure |
| Subsequent    | uint                                      |                                        |
| TemplateName  | string                                    | Name of Operating procedure template   |

## 8.22.2 Operating Procedure Step

| Property              | Type   | Description                                                                                                                          |
|-----------------------|--------|--------------------------------------------------------------------------------------------------------------------------------------|
| Attachments           | string | [Property can be updated]                                                                                                            |
| Attributes            | string |                                                                                                                                      |
| AutomaticDPE          | string |                                                                                                                                      |
| Configuration         | string |                                                                                                                                      |
| ErrorText             | string | [Property can be updated]                                                                                                            |
| FixedLink             | string |                                                                                                                                      |
| HasConfirmedExecution | bool   | [Property can be updated]                                                                                                            |
| IsCompleted           | bool   | [Property can be updated]                                                                                                            |
| ManagedType           | string | Internal managed type of Operating procedure step                                                                                    |
| Notes                 | string | Notes configured during Operating procedure step creation                                                                            |
| Operator              | string | [Property can be updated]<br>Operator name                                                                                           |
| RuntimeStatus         | string | [Property can be updated]                                                                                                            |
| Status                | string | [Property can be updated]<br>Status of Operating procedure step<br>Valid values are:<br>Unknown, PartiallyFailed, Failed, Succesfull |
| StepId                | string | [Required]<br>Id of Operating procedure step                                                                                         |
| StepName              | string | Name of Operating procedure step                                                                                                     |

## 8.22.3 Operating Procedure Subscription Representation

| Property      | Type     | Description                   |
|---------------|----------|-------------------------------|
| AlertCount    | uint     |                               |
| AlertSource   | string   |                               |
| AlertTime     | datetime |                               |
| Id            | string   | Id of Operating procedure     |
| InTreatmentBy | string   | User performing the treatment |
| IsClosed      | boolean  |                               |

|                 |                                           |                                                     |
|-----------------|-------------------------------------------|-----------------------------------------------------|
| ResetSteps      | uint                                      |                                                     |
| Sequential      | boolean                                   | Steps need to be sequential                         |
| Steps           | Array of Operating Procedure Step [->227] | Steps available in Operating procedure              |
| Subsequent      | uint                                      |                                                     |
| TemplateName    | string                                    | Name of Operating procedure template                |
| SubscriptionKey | string                                    | Subscription key for which Notification is recieved |

### 8.22.4 Operating Procedure Subscription Key Representation

| Property   | Type   | Description                                            |
|------------|--------|--------------------------------------------------------|
| Id         | string | Id of Operating procedure                              |
| Key        | string | Subscription Key                                       |
| ErrorCode  | Int    | Success = 0, Failure= -1                               |
| RequestId  | string | Unique Id passed by client during subscription request |
| RequestFor | string | Hub method where notification will arrive              |

### 8.22.5 UnsubscriptionStatus

| Property  | Type   | Description              |
|-----------|--------|--------------------------|
| Key       | string | Subscription Key         |
| ErrorCode | Int    | Success = 0, Failure= -1 |

## 8.23 Diagnostics Service

### 8.23.1 DiagnosticsRepresentation

| Property       | Type     | Description                                                                                                                                                                                  |
|----------------|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ServerDateTime | DateTime | Server's UTC Datetime                                                                                                                                                                        |
| ClientDateTime | DateTime | [optional] Client's UTC Datetime passed as input                                                                                                                                             |
| NTPt1t0        | string   | [optional] Total milliseconds difference between server time and client time. Server's datetime – Client's datetime<br><b>NOTE:</b> This value would be double value and could be negative . |

## 8.24 Licenses Service

### 8.24.1 LicenseMode

| Property        | Type   | Description                      |
|-----------------|--------|----------------------------------|
| LicenseModeName | string | The mode name of License Service |

|                  |      |                                                      |
|------------------|------|------------------------------------------------------|
| LicenseModeValue | uint | License mode value                                   |
| ExpirationTime   | uint | ExpirationTime of relative mode expression in second |

## 8.24.2 LicenseOptions

| Property    | Type   | Description                     |
|-------------|--------|---------------------------------|
| Id          | string | The License option Id           |
| Required    | Int    | Required property as int        |
| Remaining   | Int    | Remaining property as int       |
| Description | string | Description of licensing option |
| Available   | int    | Available property as int       |

## 8.25 Multi Monitors Service

### 8.25.1 MultiMonitorRepresentation

| Property         | Type   | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|------------------|--------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Configuration    | string | Json object for configuration of MultiMonitor Settings                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| OverruleSettings | bool   | Any user can overrule the default multi monitor configuration with its own setting.<br>This setting is used by the client to enable/disable the respective UI controls which support the multi monitor configuration. I.e. if it is set to false, the user (except the user which has the application right 'Configure multi monitor') <b>cannot</b> : <ul style="list-style-type: none"> <li>• Open an additional system manager.</li> <li>• Undock a dedicated event manager.</li> <li>• Close an additional system or event manger.</li> </ul> Move windows. |
| HostName         | string | HostName of the client where multimonitor configuration being used.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |

### 8.25.2 MultiMonitorStationStatus

| Property          | Type   | Description                                                                                                                                                                                                                                                |
|-------------------|--------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| stationIdentifier | string | <ul style="list-style-type: none"> <li>• Identifier for the particular station preferably hostname of any particular machine</li> <li>• or it can unique id.<br/>If valid Client Certificate is present then it is given higher precedence than</li> </ul> |

|                          |  |                                                                                                                          |
|--------------------------|--|--------------------------------------------------------------------------------------------------------------------------|
|                          |  | stationIdentifier.<br><ul style="list-style-type: none"> <li>• <b>Optional</b></li> </ul>                                |
| DistinguishedSubjectName |  | <ul style="list-style-type: none"> <li>• Subject name for client certificate</li> </ul>                                  |
| ErrorCode                |  | <ul style="list-style-type: none"> <li>• Status of response</li> <li>• 0 for success</li> <li>• 1 for failure</li> </ul> |

### 8.25.3 MultiMonitorUserStatus

| Property | Type   | Description                                                                                                                                                              |
|----------|--------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| UserName | string | <ul style="list-style-type: none"> <li>• LoggedIn UserId</li> <li>• an identification used by a person with access to a computer, network, or online service.</li> </ul> |

## 8.26 Validation Service

### 8.26.1 ValidationCommandOpRepresentation

| Property    | Type     | Description                              |
|-------------|----------|------------------------------------------|
| PropertyIds | String[] | The properties that are to be commanded. |
| GroupNumber | uint     | Group Number of Command to execute.      |

### 8.26.2 ValidationEditOpRepresentation

| Property  | Type     | Description                    |
|-----------|----------|--------------------------------|
| ObjectIds | String[] | Objects that are being edited. |

### 8.26.3 ValidationCredentialRepresentation

| Property           | Type     | Description                                                                                 |
|--------------------|----------|---------------------------------------------------------------------------------------------|
| ObjectIds          | String[] | The objects the supervisor should have access to.                                           |
| CheckCredentials   | int      | 0 validate both user and supervisor<br>1 validate user only,<br>2 validate supervisor only. |
| UserPassword       | string   | Password of user.                                                                           |
| SupervisorName     | string   | Name of supervisor.                                                                         |
| SupervisorPassword | string   | Password of supervisor.                                                                     |

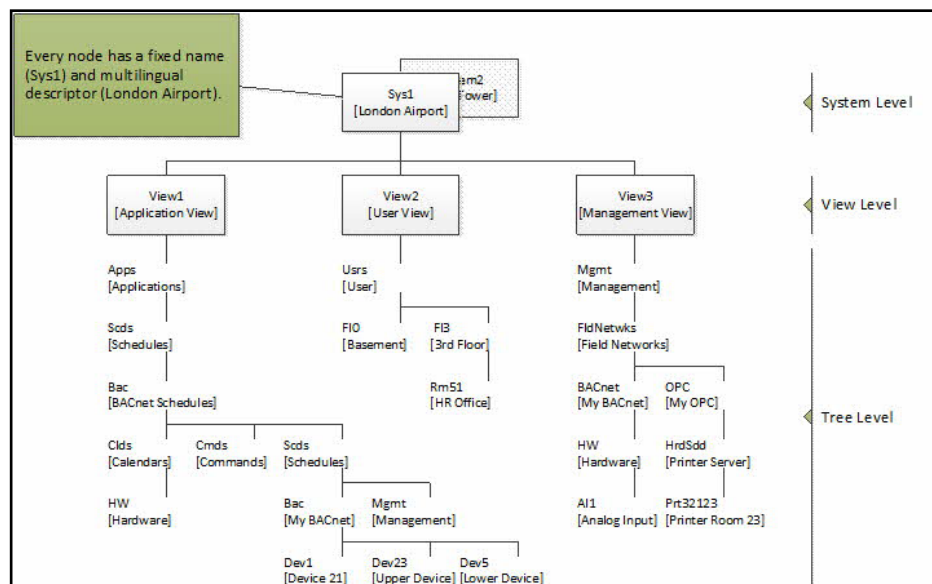
## 8.26.4 ValidateOpResponse

| Property            | Type           | Description                                                                                                                                                                                 |
|---------------------|----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| IncrVersion         | boolean        | Should the Version number be incremented?                                                                                                                                                   |
| Log                 | boolean        | Should a log entry be created?                                                                                                                                                              |
| Predefined Comments | TextGroupEntry | Pre-defined comments assigned to the object(s). If there are no pre-defined comments this will be an empty array.                                                                           |
| CommentRule         | string         | Is a comment required? The value is an enumeration with values of "Never", "Optional", and "Mandatory". In practice the value is never or mandatory.                                        |
| ReAuthentication    | string         | What kind of re-authentication is required? The value is an enumeration with values of "NoNeed" (no re-authentication is required) and "ReenterPw" (the user must re-enter their password). |
| IsFourEyes Enabled  | boolean        | Are a second user's credentials required?                                                                                                                                                   |

# 9 Concepts

## 9.1 Naming

The following is an overview about data hierarchies and naming conventions in the system.



**Name:** The name is a fixed (untranslatable) literal of a node. In order to uniquely identify nodes deeper in the hierarchy, nodes can be concatenated. Further, to ensure uniqueness sibling nodes can never have the same name. Concatenated Names are called **Designation**.

`<Designation> ::= <SystemName>.<ViewName>:{<TreeNodeName>}`

Example of Designation (node **HW [Hardware]**):

`Sys1.View1:Apps.Scds.Bac.Clds.HW`

**Descriptor:** The descriptor is a multilingual text of a node. Although nodes deeper in the hierarchy can be concatenated, they DO NOT have to be unique! Concatenated Descriptors are called **Location**.

```
<Location> ::=
<SystemDescriptor>.<ViewDescriptor>:{<TreeNodeDescriptor>}
```

Example of Location (node **HW** [Hardware]):

```
London Airport.Application View:Applications.Schedules.BACnet
Schedules.Calendar.Hardware
```

## 9.2 Extensibility

In order to deploy its own Web Service Interface extension one can create and deploy its own DLL. The DLL has to meet the following prerequisites:

- DLL needs to have literal “WsiExtension” as part of its name
- DLL needs to be deployed into the GMSMainProject\bin folder
- Class Attribute [Authorize]: It is an optional attribute and that is required if GMS user authorization is required.
- Class Attribute [ControllerLicense(<Put License Name>)]: It is a mandatory attribute. Provide here a valid GMS License information against which extension module feature should be available.
- For 3<sup>rd</sup> party license key “sbt\_gms\_opt\_ws\_3rd” can be used.
- If WSI extension wants license verification feature to be implemented then extension team should inform WSI team to incorporate License key along with it's Service ApiController name.

Each service comes as a class named with the service-name and a “Controller”-suffix. It derives from the ApiController base class. The following service could be reached at `http://[Your Host]/api/extensions/` :

**Example with controller mapping:**

```
namespace Siemens.Gms.WsiExtensionTest.Controller
{
    [Authorize]
    [ControllerLicense(ControllerNames.ServiceApiController, Licensin-
gOptions.sbt_gms_opt_ws_base)]
    [RoutePrefix("api/MyService")]
    public class ExtensionsApiController : ControllerBase
    {
        private const string TraceChannel = "MyService";
        public MyServiceController()
            : base(TraceChannel){}

        public MyServiceController(IStaticManager staticManager)
            : base(staticManager, TraceChannel){ }

        public HttpResponseMessage Get()
        {
            return Request.CreateResponse(HttpStatusCode.OK);
        }
    }
}
```

**Example without controller mapping:**

```
namespace Siemens.Gms.WsiExtensionTest.Controller
{
    [Authorize]
```



```
[ControllerLicense(<Put License Name>)]
[RoutePrefix("api/MyService")]
public class MyServiceController : ControllerBase
{
    private const string TraceChannel = "MyService";
    public MyServiceController()
        : base(TraceChannel){}

    public MyServiceController(IStaticManager staticManager)
        : base(staticManager, TraceChannel){ }
}

public HttpResponseMessage Get()
{
    return Request.CreateResponse(HttpStatusCode.OK);
}
}
```

For using push-notifications one can take advantage of the existing push infrastructure.

```
[System.Web.Http.Authorize]
[ControllerLicense("sbt_gms_opt_ws_base")]
[RoutePrefix("api/sr/SubExtensions")]
public class SubExtensionsController : ControllerBase
{
    #region SignalR Hub
    private const string TraceChannel = "SubExtensions";
    readonly Lazy<IHubContext> _hub = new Lazy<IHubContext>(() =>
        GlobalHost.ConnectionManager.GetHubContext<ExtensionsHub>());

    protected IHubContext Hub { get { return _hub.Value; } }
    #endregion

    public SubExtensionsController(): base(TraceChannel){}

    public SubExtensionsController(IStaticManager staticManager)
        : base(staticManager, TraceChannel)

    public HttpResponseMessage Post(string id)
    {
        //send message to all clients
        Hub.Clients.All.notifyMessage(id);
        return Request.CreateResponse(HttpStatusCode.OK);
    }
}
```

For details on how push notifications work check out SignalR documentation [17].

## 9.3 Support for Web Service Interface extensions

### 9.3.1 Initialization support

- WSI now provides support for an extension module to initialize.

```
namespace Siemens.Gms.WsiExtensionTest.Init
{
```

```

public class ServiceInitialization : ExtensionBase
{
    public IDpProxy DpProxy { get; private set; }
    public override void CleanUp()
    {
    }
    public override void Initialize()
    {
        WsiTracingManager.Instance.TraceDebug("Client1");
        DpProxy = new DpProxy(StaticManager.OperationContext,
GlobalDefinitions.ClientName, null, GlobalDefinitions.ClientName);
        var dpConn = DpProxy.Connect(null, null);
        dpConn.WaitCompleted();
    }
}

```

### 9.3.2 Caching support

WSI now provides support for an extension module to cache.

```

namespace Siemens.Gms.WsiExtensionTest.Init
{
    public class CacheInitialization : InitializeCacheExtensionBase
    {
        public override void Initialize()
        {
        }
    }
}

```

## 9.4 Support for Browser Caching

WSI now support browser caching support via Cache-Control and Etag headers.

Browser handles caching based on response headers thus client application does not need to make any explicit change to take advantage of caching.

Following services now returns Cache-Control headers in response.

- Files service [→147]
- Graphics service [→148]
- Image service [→127]
- Properties service [→84]
- Tables service [→128]

Following services returns Etag headers in response.

- Graphics service [→148]

WSI also returns Access-Control-Max-Age for preflight requests allowing browsers to cache preflight response.

In default configuration, Caching max time is set to 30 minutes and Vary header is set with value *Authorization* to refresh cache based on user login session.

Caching configuration can be modified in config file [→Error! Bookmark not defined.] .

**Developer Resources:**

- **Cache-Control** : <https://developer.mozilla.org/en-US/docs/Web/HTTP/Headers/Cache-Control>
- **Access-Control-Max-Age** : <https://developer.mozilla.org/en-US/docs/Web/HTTP/Headers/Access-Control-Max-Age>
- **Etag** : <https://developer.mozilla.org/en-US/docs/Web/HTTP/Headers/ETag>
- **Vary** : <https://developer.mozilla.org/en-US/docs/Web/HTTP/Headers/Vary>



**NOTE:**

Browser supports only GET response to be cached. POST ,PUT and DELETE response is not cached by browsers.

In case of HTTPS connection , Chrome supports caching only on valid certificate.

## 9.5 Support for CORS

WSI provides support to apply CORS policy for list of known domains.

In default configuration , CORS policy allows all domains.

White list of domains can be provided in config file to modify default configuration and apply CORS policy for specific domains. See [→Error! Bookmark not defined. ].

## 10 Appendix

### 10.1 Error Response

To identify errors in the responses WSI follows standards so that error location and service can be easily identified. Any error response shall contain below information:

| Property | Type   | Description                                                                                    |
|----------|--------|------------------------------------------------------------------------------------------------|
| Id       | Int    | Error codes. It will be return code from service specific range mentioned in Error Codes [236] |
| Error    | String | Hint text to identify error and service. For details refer Errors [237]                        |
| Details  | String | One liner information for short description of error                                           |

#### 10.1.1 Errors Codes

| S.No. | Service                          | Max     | Min     |
|-------|----------------------------------|---------|---------|
| 1     | Authentication/ Token            | 2400000 | 2400099 |
| 2     | Licensing                        | 2400100 | 2400199 |
| 3     | General                          | 2400200 | 2400299 |
| 4     | Command                          | 2400300 | 2400399 |
| 5     | commandSubscriptions             | 2400400 | 2400499 |
| 6     | EventCounters                    | 2400500 | 2400599 |
| 7     | EventCountersSubscriptions       | 2400600 | 2400699 |
| 8     | EventsCommands                   | 2400700 | 2400799 |
| 9     | Events                           | 2400800 | 2400899 |
| 10    | EventsSubscriptions              | 2400900 | 2400999 |
| 11    | Images                           | 2401000 | 2401099 |
| 12    | Properties                       | 2401100 | 2401199 |
| 13    | PropertyValues                   | 2401200 | 2401299 |
| 14    | SystemBrowser                    | 2401300 | 2401399 |
| 15    | SystemBrowserSubscriptions       | 2401400 | 2401499 |
| 16    | Tables                           | 2401500 | 2401599 |
| 17    | TrendSeries                      | 2401600 | 2401699 |
| 18    | TrendSeriesInfo                  | 2401700 | 2401799 |
| 19    | Values                           | 2401800 | 2401899 |
| 20    | ValuesSubscriptions              | 2401900 | 2401999 |
| 21    | HistoryLog                       | 2402000 | 2402099 |
| 22    | Files                            | 2402100 | 2402199 |
| 23    | Graphics                         | 2402200 | 2402300 |
| 24    | EventCategorySounds              | 2402400 | 2402499 |
| 25    | EventCategorySoundsSubscriptions | 2402301 | 2402399 |
| 26    | SystemsSubscriptions             | 2402500 | 2402599 |
| 27    | Schedule                         | 2402600 | 2402699 |
| 28    | Licenses                         | 2403200 | 2403299 |
| 29    | Validation                       | 2404000 | 2404099 |
| 30    | MultiMonitors                    | 2403800 | 2403899 |

## 10.1.2 Errors

| S.No. | Resource Path                | Error                        | Id      | Details                                                           | Remarks                                                                                              |
|-------|------------------------------|------------------------------|---------|-------------------------------------------------------------------|------------------------------------------------------------------------------------------------------|
| 1     | POST<br>/token               | invalid_grant                | 2400000 | The user name or password is incorrect.                           | Details not displayed because of some design constraint of OAuthGrantResourceOwnerCredentialsContext |
| 2     |                              | invalid_token_grant          | 2400001 | The SSO Token is incorrect or Expired                             | Provided SSO Token is either expired or invalid, continue with new SSO Token.                        |
| 3     |                              | password_expired             | 2400003 | Credentials of account are expired and need to get changed first. | Credentials of account are expired and should be changed from DesigoCC first.                        |
| 4     |                              | license_not_available        | 2400100 | A needed license is missing or expired                            | Required license is consumed or not available                                                        |
| 5     |                              | license_option_not_available | 2400102 | A required license option is missing                              | This resource requires a license option which is not available.                                      |
| 6     | PUT<br>/users/password       | invalid_details              | 2400002 | The user name or password is incorrect or missing.                | Provided user details are invalid or missing.                                                        |
|       |                              | server_error                 | 2400008 | Server is not able to process the request                         | Problem with reading from the server.                                                                |
|       |                              | password_not_compatible      | 2400004 | The password does not meet policy requirements.                   | The password does not meet policy requirements.                                                      |
| 7     | GET<br>/commands/:propertyId | cmd_missing_input            | 2400300 | propertyIds are missing                                           | propertyIds is blank or Null                                                                         |
| 8     |                              | cmd_invalid_input            | 2400301 | Unknown clientType                                                | Invalid Client Type                                                                                  |
| 9     |                              | cmd_failure                  | 2400399 | Couldn't retrieve a list of commands                              | Exception occurred                                                                                   |
| 10    | GET<br>/commands             | cmd_missing_input            | 2400300 | propertyIds are missing                                           | propertyIds is blank or Null                                                                         |
| 11    |                              | cmd_invalid_input            | 2400301 | Unknown clientType                                                | Invalid Client Type                                                                                  |

| S.No. | Resource Path                                                                                                                      | Error                      | Id      | Details                                                                               | Remarks                                            |
|-------|------------------------------------------------------------------------------------------------------------------------------------|----------------------------|---------|---------------------------------------------------------------------------------------|----------------------------------------------------|
| 12    |                                                                                                                                    | cmd_failure                | 2400399 | Couldn't retrieve a list of commands                                                  | Exception occurred                                 |
| 13    | POST<br>/commands/:propertyId:commandId                                                                                            | cmd_missing_input          | 2400300 | propertyIds are missing                                                               | propertyIds is blank or Null                       |
| 14    |                                                                                                                                    | cmd_invalid_input          | 2400301 | propertyId is invalid                                                                 | Invalid Property Id - DpIdentifier instantiation   |
| 15    |                                                                                                                                    | cmd_execution_failed       | 2400398 | command execution failed!                                                             | Command execution could not succeed                |
| 16    |                                                                                                                                    | cmd_failure                | 2400399 | Command execution failed                                                              | Command execution could not succeed                |
| 17    | POST<br>/commands/:commandId                                                                                                       | cmd_missing_input          | 2400300 | propertyIds are missing                                                               | propertyIds is blank or Null                       |
| 18    |                                                                                                                                    | cmd_execution_failed       | 2400398 | command execution failed!                                                             | Command execution could not succeed                |
| 19    |                                                                                                                                    | cmd_failure                | 2400399 | Command could not execute                                                             | Exception occurred                                 |
| 20    | POST<br>/sr/commandssubscriptions/:connectionId<br>or<br>POST<br>/sr/commandssubscriptions<br>/channelize/:requestId/:connectionId | cmdsub_invalid_input       | 2400400 | Specify a valid propertyIdList                                                        |                                                    |
| 21    |                                                                                                                                    | cmdsub_missing_input       | 2400401 | connectionId is missing                                                               |                                                    |
| 22    |                                                                                                                                    | cmdsub_subscription_failed | 2400402 | Error occurred while creating command subscription<br>--> Command subscription failed |                                                    |
| 23    |                                                                                                                                    | cmdsub_failure             | 2400499 | Could not retrieve command(s)                                                         | Exception occurred                                 |
| 24    | DELETE<br>/sr/commandssubscriptions/:connectionId                                                                                  | cmdsub_missing_input       | 2400399 | connectionId or subscriptionkey not available                                         | subscriptionkey or connectionId null or whitespace |
| 25    |                                                                                                                                    | cmdsub_unsubscribe_failed  | 2400498 | Command unsubscription failed for commands with subscriptionkey                       |                                                    |
| 26    |                                                                                                                                    | cmdsub_failure             | 2400499 | Command unsubscription failed for commands with connectionId                          | Exception occurred                                 |
| 27    | GET<br>/eventcounters/:categoryId                                                                                                  | evtcnt_failure             | 2400599 | Could not retrieve Event Counters                                                     |                                                    |
| 28    | POST<br>/sr/eventcounterssubscriptions/:connectionId                                                                               | evtcntsub_missing_input    | 2400600 | connectionId is missing                                                               |                                                    |

| S.No. | Resource Path                                                                        | Error                     | Id      | Details                                                                      | Remarks            |
|-------|--------------------------------------------------------------------------------------|---------------------------|---------|------------------------------------------------------------------------------|--------------------|
| 29    | or<br>POST<br>/sr/eventcounterssubscriptions<br>/channelize/:requestId/:connectionId | evtcntsub_failure         | 2400699 | Event Counter subscription unsuccessful for connection-id                    | Exception occurred |
| 30    | DELETE<br>/sr/eventcounterssubscriptions/:connectionId                               | evtcntsub_missing_input   | 2400600 | connectionId is missing                                                      |                    |
| 31    |                                                                                      | evtcntsub_failure         | 2400699 | Event-counters unsubscription unsuccessful for connection-id                 | Exception occurred |
| 32    | POST<br>/eventscommands/:eventId/:commandId                                          | evtcmd_unknown_command    | 2400700 | Unknown command                                                              |                    |
| 33    |                                                                                      | evtcmd_missing_supervisor | 2400794 | Missing supervisor credentials for validation input                          |                    |
| 34    |                                                                                      | evtcmd_invalid_supervisor | 2400795 | Invalid supervisor's credential passed                                       |                    |
| 35    |                                                                                      | evtcmd_invalid_comment    | 2400796 | Mandatory comments not passed in case validation profile is applicable       |                    |
| 36    |                                                                                      | evtcmd_invalid_password   | 2400797 | Invalid user credentials are passed in case validation profile is applicable |                    |
| 37    |                                                                                      | evtcmd_execution_failure  | 2400798 | Command execution failed                                                     |                    |
| 38    |                                                                                      | evtcmd_failed             | 2400799 | Command execution failed                                                     |                    |
| 39    | GET<br>/events                                                                       | evt_failure               | 2400899 | Could not retrieve Events List                                               | Exception occurred |
| 40    | POST<br>/sr/eventssubscriptions/:connectionId                                        | evtsub_missing_input      | 2400900 | connectionId is missing                                                      |                    |
| 41    | or<br>POST<br>/sr/eventssubscriptions /channelize/:requestId/:connectionId           | evtsub_failure            | 2400999 | Event Subscription unsuccessful for connection-id                            | Exception occurred |
| 42    | PUT<br>/sr/eventssubscriptions/:connectionId                                         | evtsub_missing_input      | 2400900 | connectionId is missing                                                      |                    |

| S.No. | Resource Path                                   | Error                  | Id      | Details                                                           | Remarks            |
|-------|-------------------------------------------------|------------------------|---------|-------------------------------------------------------------------|--------------------|
| 43    |                                                 | evtcntsub_failure      | 2400699 | Error occurred while changing an event subscription               | Exception occurred |
| 44    | DELETE<br>/sr/eventssubscriptions/:connectionId | eventsub_missing_input | 2400900 | connectionId is missing                                           |                    |
| 45    | GET<br>/images/:imageId                         | img_format_unknown     | 2401000 | Image format unknown!, format                                     |                    |
| 46    |                                                 | img_invalid_path       | 2401001 | Image path not allowed!                                           |                    |
| 47    |                                                 | img_failure            | 2401099 | Could not retrieve image                                          | Exception occurred |
| 48    | GET<br>/properties/:objectOrPropertyId          | prop_missing_input     | 2401100 | ObjectOrPropertyId is missing                                     |                    |
| 49    |                                                 | prop_invalid_input     | 2401101 | Invalid Request Type requestType                                  |                    |
| 50    |                                                 | prop_invalid_input     | 2401101 | Specify a valid Object OR Property Id                             |                    |
| 51    |                                                 | prop_failure           | 2401199 | Failed to read properties for ObjectOrPropertyId                  | Exception occurred |
| 52    | POST<br>/properties                             | prop_missing_input     | 2401100 | Object OR Property Ids are missing!                               |                    |
| 53    |                                                 | prop_invalid_input     | 2401101 | Invalid Request Type                                              |                    |
| 54    |                                                 | prop_failure           | 2401199 | Failed to read properties                                         | Exception occurred |
| 55    | GET<br>/propertyvalues/:objectOrPropertyId      | propVal_missing_input  | 2401200 | Object OR Property Ids are missing!                               |                    |
| 56    |                                                 | propVal_invalid_input  | 2401201 | Specify a valid Object OR Property Id                             |                    |
| 57    |                                                 | propVal_failure        | 2401299 | Failed to get values for properties                               | Exception occurred |
| 58    | GET<br>/systembrowser                           | sysbrow_failure        | 2401399 | Failed to retrieve Views                                          | Exception occurred |
| 59    | GET<br>/systembrowser/:systemId/:viewId/:node   | sysbrow_missing_input  | 2401300 | View OR Node details are missing while retrieving Browser Objects |                    |
| 60    |                                                 | sysbrow_failure        | 2401399 | Failed to retrieve Nodes                                          | Exception occurred |



| S.No. | Resource Path                                                                                                                                         | Error                              | Id      | Details                            | Remarks            |
|-------|-------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------|---------|------------------------------------|--------------------|
| 61    | POST<br>/systembrowser/:systemId/:viewId                                                                                                              | sysbrow_miss-<br>ing_input         | 2401300 | Object Ids are missing             |                    |
| 62    |                                                                                                                                                       | sysbrow_failure                    | 2401399 | Failed to search multiple object   | Exception occurred |
| 63    | POST<br>/sr/SystemBrowserSubscrip-<br>tions/:connectionId<br>or<br>POST /sr/SystemBrowserSub-<br>scriptions /channelize/:reques-<br>tId/:connectionId | sys-<br>browsub_miss-<br>ing_input | 2401400 | connectionId is missing            |                    |
| 64    |                                                                                                                                                       | sysbrowsub_fail-<br>ure            | 2401499 | SystemBrowser Subscription failed  | Exception occurred |
| 65    | DELETE<br>/sr/SystemBrowserSubscrip-<br>tions/:connectionId                                                                                           | sys-<br>browsub_miss-<br>ing_input | 2401400 | connectionId is missing            |                    |
| 66    |                                                                                                                                                       | sysbrowsub_fail-<br>ure            | 2401499 | Failed to unsubscribe Cns          | Exception occurred |
| 67    | GET<br>/tables/:tableId                                                                                                                               | tbl_missing_in-<br>put             | 2401500 | tableId is missing                 |                    |
| 68    |                                                                                                                                                       | tbl_un-<br>known_tableId           | 2401500 | Unknown table                      |                    |
| 69    |                                                                                                                                                       | tbl_failure                        | 2401599 | Failed to Get Tables using TableId | Exception occurred |
| 70    | GET<br>/tables/:tableId/:tableEntryId                                                                                                                 | tbl_missing_in-<br>put             | 2401500 | tableId is missing                 |                    |
| 71    |                                                                                                                                                       | tbl_failure                        | 2401599 | Failed to get table                | Exception occurred |
| 72    |                                                                                                                                                       | tbl_unknown_id                     | 2401501 | Unknown id                         |                    |
| 73    |                                                                                                                                                       | tbl_un-<br>known_tableId           | 2401502 | Unknown table                      |                    |
| 74    | GET<br>/tables/:tableId/subtables/:subta-<br>bleId                                                                                                    | tbl_missing_in-<br>put             | 2401500 | subtableId is missing!             |                    |
| 75    |                                                                                                                                                       | tbl_un-<br>known_subtable          | 2401503 | Unknown subtable                   |                    |
| 76    |                                                                                                                                                       | tbl_failure                        | 2401599 | Failed to Get Sub Tables           | Exception occurred |
| 77    | GET<br>/tables/:tableId/subgroups                                                                                                                     | tbl_missing_in-<br>put             | 2401500 | tableId is missing!                |                    |
| 78    |                                                                                                                                                       | tbl_un-<br>known_subgroup          | 2401504 | Unknown subGroups                  |                    |

| S.No. | Resource Path                                                                                                                | Error                            | Id      | Details                                           | Remarks                                    |
|-------|------------------------------------------------------------------------------------------------------------------------------|----------------------------------|---------|---------------------------------------------------|--------------------------------------------|
| 79    |                                                                                                                              | tbl_failure                      | 2401599 | Failed to Get Sub Groups                          | Exception occurred                         |
| 80    | GET<br>/trendseries/:trendseriesId/borders                                                                                   | tnd_missing_input                | 2401600 | TrendSeriesId is missing                          |                                            |
| 81    |                                                                                                                              | tnd_failure                      | 2401699 | Could not retrieve Borders                        | Exception occurred                         |
| 82    | GET<br>/trendseries/:trendseriesId                                                                                           | tnd_missing_input                | 2401600 | one or some of the input fields are missing       |                                            |
| 83    |                                                                                                                              | tnd_notfound                     | 2401601 | Timeseries data not available                     |                                            |
| 84    |                                                                                                                              | tnd_failure                      | 2401699 | Could not retrieve Timeseries data                | Exception occurred                         |
| 85    | GET<br>/trendseriesinfo/:systemId                                                                                            | tndinfo_notfound                 | 2401700 | Get All TrendSeriesInfo Trendseries not available |                                            |
| 86    | GET<br>/trendseriesinfo/:objectOrPropertyId                                                                                  | tndinfo_notfound                 | 2401702 | Trendseries not available.                        |                                            |
| 87    |                                                                                                                              | tndinfo_failure                  | 2401799 | Could not retrieve Trendseries Information        | Exception occurred                         |
| 88    | GET<br>/trendseriesinfo/tvd/:tvdDpld-Name                                                                                    | tnd_missing_input                | 2401600 | one or some of the input fields are missing       |                                            |
| 89    |                                                                                                                              | tndinfo_failure                  | 2401799 | Could not retrieve Trendseries Information        | Not a valid objectid or Exception occurred |
| 90    | GET<br>/values/:objectOrPropertyId                                                                                           | val_missing_input                | 2401800 | Object OR Property Ids are missing!               |                                            |
| 91    |                                                                                                                              | val_failure                      | 2401899 | Could not retrieve Value                          | Exception occurred                         |
| 92    | POST<br>/values                                                                                                              | val_missing_input                | 2401800 | objectOrPropertyIds not provided                  |                                            |
| 93    |                                                                                                                              | val_failure                      | 2401899 | Could not retrieve Value                          | Exception occurred                         |
| 94    | POST<br>/sr/valuessubscriptions/:connectionId<br>or<br>POST<br>/sr/valuessubscriptions /channel-ize/:requestId/:connectionId | valsub_missing_input             | 2401900 | connectionId or objectOrPropertyIds not available |                                            |
| 95    |                                                                                                                              | valsub_subscription_unsuccessful | 2401901 | Subscription unsuccessful                         |                                            |
| 96    |                                                                                                                              | valsub_failure                   | 2401999 | Could not subscribe                               | Exception occurred                         |
| 97    | DELETE<br>/sr/valuessubscriptions/:connectionId                                                                              | valsub_missing_input             | 2401900 | connectionId or subscriptionkey not available     |                                            |

| S.No. | Resource Path                                                   | Error               | Id      | Details                                                                           | Remarks                                      |
|-------|-----------------------------------------------------------------|---------------------|---------|-----------------------------------------------------------------------------------|----------------------------------------------|
| 98    |                                                                 | valsub_notfound     | 2401902 | subscription not available for provided subscription key                          | unsubscription not done for any of the input |
| 99    |                                                                 | valsub_failure      | 2401999 | Could not remove subscription                                                     | Exception occurred                           |
| 100   | GET<br>/historylogs/:systemId                                   | htlog_failure       | 2402099 | Could not retrieve list of available Tables.                                      | Exception occurred                           |
| 101   | GET<br>/historylogs/:systemId/:tableName                        | htlog_missing_input | 2402000 | Table name is missing while retrieving column names for the table.                | tableName is blank or Null                   |
| 102   |                                                                 | htlog_failure       | 2402099 | Could not retrieve list of column details for Table                               | Exception occurred                           |
| 103   | GET<br>/historylogs/:systemId/:tableName/operators/:columnName  | htlog_missing_input | 2402000 | Table name is missing while retrieving condition filter operators for the column. | tableName is blank or Null                   |
| 104   |                                                                 | htlog_missing_input | 2402000 | Column name is missing while retrieving condition filter operators for the Table  | columnName is blank or Null                  |
| 105   |                                                                 | htlog_failure       | 2402099 | Could not retrieve Condition filter operators for Column                          | Exception occurred                           |
| 106   | GET<br>/historylogs/:systemId/:tableName/enumvalues/:columnName | htlog_missing_input | 2402000 | Table name is missing while retrieving Enum values for the column.                | tableName is blank or Null                   |
| 107   |                                                                 | htlog_missing_input | 2402000 | Column name is missing while retrieving Enum values for Table                     | columnName is blank or Null                  |
| 108   |                                                                 | htlog_failure       | 2402099 | Could not retrieve Enum values for Column                                         | Exception occurred                           |
| 109   | GET<br>/historylogs/:systemId/:tableName/:snapshotId            | htlog_missing_input | 2402000 | Table name is missing while retrieving Table Data.                                | tableName is blank or Null                   |
| 110   |                                                                 | htlog_missing_input | 2402000 | Column names not available while retrieving Table Data from Table                 | No valid list of columns                     |
| 111   |                                                                 | htlog_failure       | 2402099 | Could not retrieve page wise data for Table                                       | Exception occurred                           |
| 112   | DELETE<br>/historylogs/:systemId/:tableName/:snapshotId         | htlog_missing_input | 2402000 | Table name is missing in the request to discard the snapshot.                     | tableName is blank or Null                   |
| 113   |                                                                 | htlog_missing_input | 2402000 | Snapshot id is missing in the request to discard the snapshot for Table           | snapshotId is blank or Null                  |

| S.No. | Resource Path                                                                     | Error                  | Id      | Details                                                                                         | Remarks                                               |
|-------|-----------------------------------------------------------------------------------|------------------------|---------|-------------------------------------------------------------------------------------------------|-------------------------------------------------------|
| 114   |                                                                                   | htlog_failure          | 2402099 | Could not discard snapshot with SnapshotId                                                      | Exception occurred                                    |
| 115   | GET<br>/historylogs/:systemId/:tableName/validateconditionfilter/:conditionFilter | htlog_missing_input    | 2402000 | Table name is missing in the request to validate condition filter string.                       | tableName is blank or Null                            |
| 116   |                                                                                   | htlog_missing_input    | 2402000 | Condition filter string is missing in the request to validate condition filter string for Table | conditionFilter is blank or Null                      |
| 117   |                                                                                   | htlog_failure          | 2402099 | Could not validate condition filter string                                                      | Exception occurred                                    |
| 118   | GET<br>/historylogs/:systemId/:tableName/:snapshotId/:parentId                    | htlog_missing_input    | 2402000 | Table name is missing in the request to retrieve child data.                                    | tableName is blank or Null                            |
| 119   |                                                                                   | htlog_missing_input    | 2402000 | Snapshot Id is missing in the request to retrieve child data for Table                          | snapshotId is blank or Null                           |
| 120   |                                                                                   | htlog_missing_input    | 2402000 | Parent information is missing in the request to retrieve child data                             | parentId is blank or Null                             |
| 121   |                                                                                   | htlog_missing_input    | 2402000 | Column names not available while retrieving child Data for Table                                | No valid list of columns                              |
| 122   |                                                                                   | htlog_cancelled        | 2402098 | Request preliminary canceled by the user                                                        | Report cancellation request aborted report generation |
| 123   |                                                                                   | htlog_failure          | 2402099 | Could not retrieve Child data for Table                                                         | Exception occurred                                    |
| 124   | GET<br>/files/:filetype                                                           | file_retrieval_failed  | 2402199 |                                                                                                 |                                                       |
| 125   | GET<br>/files/:systemId/:relativeFilePath                                         | file_directorynotfound | 2402100 |                                                                                                 |                                                       |
| 126   |                                                                                   | file_filenotfound      | 2402101 |                                                                                                 |                                                       |
| 127   |                                                                                   | file_inputdatainvalid  | 2402102 |                                                                                                 |                                                       |

| S.No. | Resource Path                        | Error                          | Id      | Details                                                                                             | Remarks                                      |
|-------|--------------------------------------|--------------------------------|---------|-----------------------------------------------------------------------------------------------------|----------------------------------------------|
| 128   |                                      | file_inputdatano-<br>texpected | 2402103 |                                                                                                     |                                              |
| 129   |                                      | file_pathinvalid               | 2402104 |                                                                                                     |                                              |
| 130   |                                      | file_pathnull                  | 2402105 |                                                                                                     |                                              |
| 131   |                                      | file_pathtoolong               | 2402106 |                                                                                                     |                                              |
| 132   |                                      | file_unauthoir-<br>zedaccess   | 2402107 |                                                                                                     |                                              |
| 133   |                                      | file_re-<br>trieval_failed     | 2402199 |                                                                                                     |                                              |
| 134   | GET<br>/files/documents/:designation | file_inputdatain-<br>valid     | 2402102 |                                                                                                     |                                              |
| 135   |                                      | file_retri-<br>val_failed      | 2402199 |                                                                                                     |                                              |
| 136   | GET<br>/graphics/:objectId           | ob-<br>ject_not_known          | 2402200 | The object id is invalid.                                                                           |                                              |
| 137   |                                      | failure                        | 2402201 | Could not identify the object or could not determine if it is graphical.                            |                                              |
| 138   | GET<br>/graphics/itemIds/:objectId   | ob-<br>ject_not_known          | 2402200 | The object id is invalid.                                                                           |                                              |
| 139   |                                      | failure                        | 2402201 | Could not identify the object or there was a problem finding graphical items related to the object. |                                              |
| 140   | GET<br>/graphics/items/:objectId     | ob-<br>ject_not_known          | 2402200 | The object id is invalid.                                                                           |                                              |
| 141   |                                      | failure                        | 2402201 | Could not identify the object or there was a problem finding (or loading) the specified graphic.    | The object id must specify a graphical item. |

| S.No. | Resource Path                                                                                                                                                              | Error                                          | Id      | Details               | Remarks |
|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------|---------|-----------------------|---------|
| 142   | GET<br>api/eventcategorysounds                                                                                                                                             | evtsound_re-<br>trieval_failure                | 2402499 |                       |         |
| 143   | POST<br>api/sr/eventcategorysoundssub-<br>scriptions/:connectionId<br>or<br>POST<br>api/sr/eventcategorysoundssub-<br>scriptions /channelize/:request-<br>Id/:connectionId | evtsound-<br>sub_missing_in-<br>put            | 2402301 | Connection id missing |         |
| 144   |                                                                                                                                                                            | evtsoundsub_cli-<br>ent_alreadysub-<br>scribed | 2402302 |                       |         |
| 145   |                                                                                                                                                                            | evtsound-<br>sub_failure                       | 2402399 |                       |         |
| 146   | DELETE<br>api/sr/eventcategorysoundssub-<br>scriptions/:connectionId                                                                                                       | evtsoundUn-<br>sub_failure                     | 240398  |                       |         |
| 147   | POST<br>/api/sr/systemssubscrip-<br>tions/channelize/:requestId/:con-<br>nectionId                                                                                         | systems-<br>sub_missing_in-<br>put             | 2402500 |                       |         |
| 148   | and<br>DELETE<br>/api/sr/systemssubscrip-<br>tions/channelize/:connectionId                                                                                                | systemssub_in-<br>valid_connek-<br>tionId      | 2402501 |                       |         |
| 149   | POST<br>/api/sr/systemssubscrip-<br>tions/channelize/:requestId/:con-<br>nectionId                                                                                         | systemsun-<br>sub_failure                      | 2402598 |                       |         |
| 150   | POST<br>/api/sr/systemssubscrip-<br>tions/channelize/:requestId/:con-<br>nectionId                                                                                         | systemssub_fail-<br>ure                        | 2402599 |                       |         |
| 151   | GET<br>/api/scheduler/schedule/:objectId                                                                                                                                   | sch_Inva-<br>lid_schedule                      | 2402600 |                       |         |
| 152   |                                                                                                                                                                            | sch_un-<br>kown_schedule                       | 2402603 |                       |         |
| 153   |                                                                                                                                                                            | sch_failure                                    | 2402602 |                       |         |
| 154   | POST /api/scheduler/sched-<br>ule/save                                                                                                                                     | sch_Inva-<br>lid_schedule                      | 2402600 |                       |         |

| S.No. | Resource Path                                | Error                     | Id      | Details | Remarks |
|-------|----------------------------------------------|---------------------------|---------|---------|---------|
| 155   |                                              | sch_un-<br>kown_schedule  | 2402603 |         |         |
| 156   |                                              | sch_failure               | 2402602 |         |         |
| 157   | POST /api/scheduler/exception/save           | sch_Inva-<br>lid_schedule | 2402600 |         |         |
| 158   |                                              | sch_un-<br>kown_schedule  | 2402603 |         |         |
| 159   |                                              | sch_failure               | 2402602 |         |         |
| 160   | GET /api/scheduler/schedule/saveDefaultValue | sch_Inva-<br>lid_schedule | 2402600 |         |         |
| 161   |                                              | sch_un-<br>kown_schedule  | 2402603 |         |         |
| 162   |                                              | sch_failure               | 2402602 |         |         |
| 163   | GET /api/scheduler/calendar-List/:objectId   | sch_Invalid_Cal-<br>endar | 2402601 |         |         |
| 164   |                                              | cal_unkown_cal-<br>endar  | 2402606 |         |         |
| 165   |                                              | sch_failure               | 2402602 |         |         |
| 166   | GET /api/scheduler/calendar/:objectId        | cal_failure               | 2402607 |         |         |
| 167   |                                              | sch_Invalid_Cal-<br>endar | 2402601 |         |         |
| 168   |                                              | cal_unkown_cal-<br>endar  | 2402606 |         |         |

| S.No. | Resource Path                           | Error                        | Id           | Details | Remarks                              |
|-------|-----------------------------------------|------------------------------|--------------|---------|--------------------------------------|
| 169   |                                         | sch_failure                  | 2402602      |         |                                      |
| 170   | GET<br>/api/objects/:designation        | obj_invalid_object           | 2402709      |         |                                      |
| 171   |                                         | obj_failure                  | 2402700      |         |                                      |
| 172   | POST<br>/api/objects                    | obj_object_validation_failed | 2402708      |         |                                      |
| 173   |                                         | obj_object_creation_failed   | 2402703      |         |                                      |
| 174   |                                         | obj_new_object_not_allowed   | 2402704      |         |                                      |
| 175   |                                         | obj_object_already_exists    | 2402705      |         |                                      |
| 176   |                                         | obj_failure                  | 2402700      |         |                                      |
| 177   | DELETE<br>/api/objects/:designation     | obj_deletion_failed          | 2402710      |         |                                      |
| 178   | GET<br>/api/licenses                    | license_failure              | 2403299      |         |                                      |
| 179   | /POST/api/getComamndValidationOperation | validation_missing_input     | 2404000      |         | Valid Property Ids are missing       |
| 180   |                                         | validation_invalid_input     | 2404001      |         | Some of the Property Ids are invalid |
| 181   |                                         | validation_execution_failure | 2400409<br>9 |         | Exception                            |
| 182   | /api/getEditValidationOperation         | validation_missing_input     | 2404000      |         | Valid Object Ids are missing         |



| S.No. | Resource Path           | Error                             | Id           | Details | Remarks                            |
|-------|-------------------------|-----------------------------------|--------------|---------|------------------------------------|
| 183   |                         | validation_inva-<br>lid_input     | 2404001      |         | Some of the Object Ids are invalid |
| 184   |                         | validation_exe-<br>cution_failure | 2400409<br>9 |         | Exception                          |
| 185   | /api/validateCredential | validation_exe-<br>cution_failure | 2400409<br>9 |         | Exception                          |
| 186   | api/multimonitors       | multi-<br>mon_failure             | 240038<br>99 |         | Exception                          |
| 187   | api/multimonitors       | multi-<br>mon_inva-<br>lid_input  | 240380<br>1  |         | Some input Parameters are missing  |

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